

Developing AI/ML-Enabled Diagnostic Medical Devices

Product, Data, Evidence, and Lifecycle Control

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INTRODUCTION

From a Possible Aid to a Product People Can Use

Begin with the laboratory task

A trained laboratory professional has a blood smear to review. A smear is a thin layer of blood spread on a glass slide so that cells can be examined. The professional looks across it, checks cell appearance and context, follows the laboratory's procedures for unusual findings, and helps produce an interpretation the laboratory may release. That is substantial work even before software enters the room.

Suppose a manufacturer proposes an aid for one small step. The software receives a picture of a cell, suggests what kind of cell it might be, and shows the suggestion beside the picture. The professional checks both and continues examining the entire smear. The laboratory still decides what the findings mean for the patient and whether to release a result.

This wholly fictional proposal is **LumaCell Review**, the continuing development case in this book. At the opening, we have a proposed role and a collection of development images. We still need to learn whether this assistance addresses an important task problem, whether a credible simpler approach could address it, and whether the images can support the proposed work. The book follows those questions through development. A plan, a completed test, a manufacturer release, and an observation during use answer different questions at different times.

The picture does not arrive by magic. Another process has already selected a small image cutout, called a *candidate crop*, from the smear and supplied it to LumaCell. The proposed setting is adult peripheral-blood smears already referred for morphology review. The software checks whether a supplied image is usable before offering a cell-category suggestion. It can also show an other/unclassified state when the image does not fit a common category. The professional must be able to see when there is no ordinary suggestion and continue the review. The manufacturer will need evidence that this arrangement helps the stated task; a selected-cell suggestion alone says nothing about cells the upstream process never supplied.

The question that organizes this book is therefore practical: **what must the manufacturer decide, build, observe, and control before this particular assistance can be relied on for its stated role?** The answer includes the pictures supplied, the

quality check, the suggestion shown to the professional, the professional's action, and a way to keep that complete route working. Each part can change what the suggestion means or whether it reaches the intended task.

What development means here

The first manufacturer decision is whether a suggestion at this point could solve a problem that matters. If staff need help with the review of already selected cells, the team can compare the proposed aid with the present way of doing that work. If the actual difficulty lies in finding which cells to examine, the same aid may be aimed at the wrong step. Chapter 1 makes that comparison before asking which computer method to build.

Only then does the team need to choose how to produce a suggestion. Artificial intelligence (AI) is a broad term for computer methods used for tasks such as prediction. Machine learning (ML) is the approach proposed for LumaCell: a training procedure uses examples of cell pictures together with their assigned categories to produce a *model*. If that model is later given a new picture, it produces a suggestion. Applying an already trained model to a new input is called *inference*. This explanation describes the intended mechanism; it does not establish that LumaCell's present images are properly labeled or suitable for training.

LumaCell's proposed inference would run locally, using the same identified model throughout a release. The product would not learn a new answer from each patient during ordinary use. The check for whether an image is usable has a different job from the cell classifier and is supplied as a versioned component. A change to that check could change which images reach an otherwise unchanged classifier. The manufacturer therefore needs to know the identities and responsibilities of both parts, as well as what the professional sees.

The rest of development asks progressively larger questions. Can the team define the exact intended users, inputs, and outputs? May it use particular data, and what do the assigned cell categories mean? Does a model work on separate evaluation images? Does the complete software display the right suggestion for the right image, and can professionals use it as intended? The manufacturer then decides which controlled arrangement it can release and support. Observations during use may send the team back to an earlier assumption. The chapters follow this order so a later test is always tied to the question and product it was meant to examine.

The primary reader is a member of a cross-functional manufacturer team: product and clinical leads, data and ML practitioners, software engineers, human-factors specialists, quality and risk staff, security staff, regulatory colleagues, and people responsible for deployment and monitoring. No single reader must perform every specialty. Each should be able to see the dependency between their work and the next person's

decision. Technical reviewers and consultants can use the same evidence, but the manufacturer remains accountable for the product and its choices.

You need no prior programming or model-design course to start. The book explains the local concepts needed for each decision, including what a model output can establish, why a label needs a reference process, and why a product test differs from a study of people using the product. It is not a guide to producing blood smears, building a sequencing assay, or choosing a regulatory route for a real product. Such work needs the relevant specialists and current sources. The examples do not prescribe sample sizes, performance thresholds, retraining intervals, or a jurisdiction's authorization.

Each chapter leaves a manufacturer work product that carries an answer or an explicit open question into the next decision. Book 1A gives additional depth on software and review, Book 1B on security, Book 1C on diagnostic and quantitative evidence, Book 2 on European regulatory assessment, and Book 3 on consulting practice. Those books are useful companions, not prerequisites needed to decode this one. A real jurisdictional or specialist decision still needs the relevant product facts, qualified people, and current sources.

A principle that travels, and a boundary that does not

Some development questions carry across diagnostic technologies. In digital pathology (DP), tissue on a slide may be scanned into a whole-slide image (WSI), and software may highlight an area or change a pathologist's worklist. In next-generation sequencing (NGS), a laboratory reads many pieces of genetic material and software may help turn those readings into candidate findings for professional review. In both settings, a team can ask what information enters a tool, which action its output influences, what the professional still decides, and what evidence would show the complete work succeeds. A technically correct intermediate output does not settle those questions by itself.

The answers differ by product. PathSight Cloud, a separate fictional case from the series, assists review of entire scanned tissue slides; a pathologist makes the diagnosis. It is not LumaCell's supplied-crop blood-smear classifier, and this book does not give PathSight a new development history. An NGS workflow deals with a specimen, sequence readings, variant analysis, a laboratory report, and the professional who releases that report. It is not a cell-image classification task. When later chapters use DP or NGS to transfer a principle, attend to the specific input, clinical action, unit of study, and human role. A shared reasoning method does not make the evidence interchangeable.

The same discipline applies before and after release. Before release, the manufacturer can specify a comparison and the observations that would support its purpose. During use, a real observation may reopen the assumption behind that comparison. For example, a changed input pattern, a delayed display, or a burden shifted to a

professional would be investigated through the product's actual records before being labeled a model defect or a clinical benefit. The later chapters teach how to do that with LumaCell's controlled fictional states. Separate hypothetical DP and NGS transfers in Chapter 1 illustrate the question without becoming part of LumaCell's history.

How the questions build

The first reading follows the chapter order because each decision gives the next one its starting facts. Four questions recur: what decision is at stake, what ordinary operation should look like, what observation would test it, and what the result would permit the team to say. A later fact may change an earlier brief or plan. That return is part of development: an earlier open question and a later supported answer belong to different moments.

The **Reviewer Lens** asks what the supplied evidence actually establishes for the defined product and use. It can credit adequate evidence and close a bounded question as well as identify a gap. The **Consultant Lens** uses the same facts to help the manufacturer choose a prospective work product, compare routes, assign expertise, and state what would close an open decision. A consultant can assess and advise without silently becoming the clinical lead, data-rights authority, manufacturer, or regulator. Neither lens is a substitute for the main explanation.

Manufacturer tables collect connected decisions. Read their left column as the question or record field and the other columns as the evidence, interpretation, owner, or next action that gives that field meaning. Follow a row from purpose to observation before using a completed table as a conclusion. In this book, the denominator also matters: a crop, a person, a package, a user, an attempted input, and a received input answer different questions. A favorable percentage of selected crops cannot silently become a claim about every smear received or every patient served.

The decision pauses put a question before the experienced reading of its evidence. Some present an incomplete record first and a worked interpretation later; a new exhibit can change the conclusion while leaving earlier favorable observations intact. The examples teach a way of reasoning, rather than phrases to copy. Chapter 16 applies the route to an unfamiliar fictional product; its assessment remains separate until after the reader's initial decision. This introduction supplies no resolution. The working terminology and abbreviations help retrieve ideas after their chapters teach them. Source and Citation Navigation explains the chapter-local trails, and Contents preserves the learner reading order.

The connected journey through sixteen chapters

Part I – Define the product (Chapters 1-4)

Chapter 1, “Is AI the Right Intervention?”, asks a question that should precede the choice of a model: what consequential task needs help, and is AI a plausible way to help compared with a credible alternative? The manufacturer creates an intervention brief linking the task, candidate role, comparator, data feasibility, possible failure, continuing support, and a bounded next decision. That brief is the input to Chapter 2. An image folder or a plausible classifier alone cannot fill it.

Chapter 2, “Intended Use, Claim, AI Role, and Workflow”, takes the candidate intervention and makes its proposed promise testable. Who receives what output, for which adults and inputs, in which setting, and with what professional action or fallback? A trained professional's presence changes the workflow description but does not by itself demonstrate safe use. The intended-use and role contract tells the next chapter which functions and dependencies belong inside the product account. Jurisdiction-specific qualification remains a separate decision.

Chapter 3, “The Complete AI/ML Product and Supply Boundary”, follows an input through its quality gate, preparation, classifier, rules, display, infrastructure, supplier links, and human action. It asks who can change each part and which combination is being discussed. This prevents “the model” from standing in for the entire product. The resulting map of components, responsibilities, and configuration gives Chapter 4 something concrete to plan and later chapters an identity to test.

Chapter 4, “Development, Risk, Requirements, and Evidence Plan”, turns that map into a sequence of work. Which requirements follow from the role? Which possible failures need controls? What must be settled before collecting data, fixing a reference process, evaluating a candidate, or testing human use? The integrated plan names dependencies, owners, review points, and evidence that could change a decision. It leads directly to the data questions: a plan cannot execute responsibly until the team knows which records it may use and what they represent.

Part II – Control data and model realization (Chapters 5-8)

Chapter 5, “Data Provenance, Rights, Privacy, and Lineage”, starts with a tempting but insufficient fact: data are available. It asks whether a specified activity may use them, where they came from, how a source image relates to its crops and derived records, who may access them, and what must happen at the end of the permitted period. The data charter separates permission from technical adequacy. That separation matters when Chapter 6 asks whether the labels attached to those data are meaningful.

Chapter 6, “Reference Standards, Labels, Annotation, and Adjudication”, asks what the correct comparison is supposed to mean. A label typed beside a crop is not

automatically a reliable reference. The team must define the target category, reader instructions and qualifications, how disagreement is handled, and when uncertainty stays unresolved. Its reference plan supplies the meaning of the categories used to train or evaluate the model. Chapter 7 then asks whether the resulting records have been divided without allowing development decisions to contaminate evaluation.

Chapter 7, “Dataset Partitions, Leakage, Representation, and Coverage”, assigns different jobs to data used for fitting, tuning, and independent assessment. It follows related records through patient, specimen, acquisition, site, and time so that a superficially new crop does not become a false independent test. It also asks which relevant conditions are represented and which remain unsupported. The dataset manifest defines what an experiment may learn from and what evidence must remain held apart for later decisions.

Chapter 8, “Model Selection, Experiments, Reproducibility, and Dependencies”, uses those controlled inputs to compare a simple baseline with candidate models. The result is more than a winning score: the team records preparation, code, data membership, settings, supplier dependencies, and what “repeat” means for this experiment. An exploratory candidate is not the finished released product. The model-development record carries a reconstructable candidate into the next part, where its success criteria must be specified before decisive results are interpreted.

Part III – Evaluate the complete product (Chapters 9-13)

Chapter 9, “Metrics, Operating Points, Calibration, Uncertainty, and Acceptance”, asks which quantity the evaluation is trying to estimate and what counts in its numerator and denominator. A score can guide a category or a non-result; a threshold changes which errors and workload appear. The team defines measures, conditions, uncertainty, and decision rules prospectively so a favorable result cannot redefine success after the fact. Chapter 10 challenges whether that plan is robust across the conditions and groups the proposed use actually includes.

Chapter 10, “Bias, Subgroups, Robustness, OOD, and Degraded States”, examines where an aggregate result could hide a meaningful difference. It connects a plausible harm to subgroup, input-shift, out-of-distribution (OOD), missingness, and recovery tests. A difference is a reason to investigate its mechanism and consequence, not an automatic verdict about fairness or defect. The challenge plan identifies which test should be run and what a justified response would look like. Chapter 11 then checks whether the constructed software and system actually perform the specified behaviors.

Chapter 11, “Verify the ML Pipeline, Software, and System”, traces requirements through transformations, packaging, interfaces, identity, display, security-related functions, and regression checks. *Verification* asks whether the identified implementation meets specified requirements under the test conditions; it does not alone establish the value of the clinical role. The verification trace ties a test result to

the exact assembled product and identifies failures, corrections, and renewed checks. Chapter 12 can then ask whether that functioning product has evidence for its broader claim.

Chapter 12, “Integrate Analytical, Technical, and Clinical Evidence”, places model and system results next to the intended clinical use. Selected-crop agreement, input yield, complete-workflow behavior, and any claim of benefit have different units and need different evidence. The chapter shows how to retain an adequately supported narrow conclusion while leaving a larger claim open, and how a product change can make a previous study only partly applicable. That integrated evidence picture still needs a study of the intended people working with the product.

Chapter 13, “Validate the Human-AI Team”, follows the trained professional who sees a suggestion, checks identity and source, makes a judgment, and uses a fallback when the ordinary route is interrupted. It asks whether instruction, interface, workload, reliance, override, and recovery work under representative conditions. A model can pass its own tests while the product is difficult to use correctly. Human-use observations may therefore require a design change and renewed evidence before the release decision in Chapter 14.

Part IV – Release, operate, change, and retire (Chapters 14-16)

Chapter 14, “Software/Model Release Baseline, Deployment, Transparency, and Monitoring Design”, asks which exact combination of model, gate, software, interface, information, and site conditions is ready to leave development. The manufacturer records what was evaluated, what users need to know, how installations will be checked, what fallback remains, and how incoming observations will be counted. A manufacturer release is a defined product event; it is not by itself regulatory authorization. The monitoring protocol gives Chapter 15 a fair way to interpret later signals.

Chapter 15, “Drift, Incidents, Retraining, PCCP, CAPA, and Retirement”, begins when observed use or a proposed change tests that release argument. It asks what changed, what the observations actually count, whether a cause is known, what evidence must be revisited, and who can authorize a response. A retrained model, a supplier gate revision, rollback, support end, and retirement require different decisions. The chapter distinguishes a predetermined change control plan (PCCP) from permission to make any desired change, and corrective and preventive action (CAPA) from an automatic label for every monitoring trigger. Its response method brings the book's early assumptions back into view.

Chapter 16, “Sealed Development Capstone and Advisory Handoff”, brings the entire route to a separate, unfamiliar fictional product. Its claim, configuration, evidence and open questions support a manufacturer decision and a bounded advisory handoff, considered before any worked interpretation appears. Its instructor assessment

is separate from the learner path. The point is to decide what the supplied record supports and what would change that conclusion, rather than recognize the answer to a familiar case.

Across all sixteen chapters, a well-supported narrow result can be valuable. A failed test can lead to a specific correction and a successful new check. A discrepancy can close with a clear reconciliation when the product and evidence really do match. Equally, an unresolved clinical or use question remains open even when a model metric looks strong. By the end, you should be able to connect a proposed task to a controlled product, a relevant observation, an accountable decision, and a defined next action—and explain the limit of that conclusion to someone who was not in the room.

CHAPTER 1

Is AI the Right Intervention?

A manufacturer has a folder of blood-cell images and a proposal: software could suggest a cell category beside each picture. A laboratory professional would check the suggestion. It sounds like a task a computer might help with. But which difficulty in the laboratory would that help remove?

That is the opening question for **LumaCell Review**, this book's wholly fictional product proposal. At this point, the team has a candidate role and an image collection. The important need, the alternative against which the aid should be compared, and the collection's readiness remain unresolved. A useful next decision starts by following the professional's work far enough to see where the proposed suggestion could matter.

Start with a task and a decision

Imagine the proposed suggestion appearing beside a small picture of a possible white blood cell. What would the professional still need to examine? The picture came from a **blood smear**—blood spread thinly on a slide so its cells and their context can be examined. The trained professional reviews the smear, follows the laboratory's process for unusual findings, and contributes to the interpretation the laboratory may release. White blood cells are among the cells examined. Checking one suggested category occupies only part of that work.

The small picture is a **crop**. Another process supplies candidate crops to LumaCell; LumaCell does not choose every cell on the slide. That distinction determines what a successful suggestion could establish. Even if every supplied crop received an appropriate category, a clinically relevant cell never supplied to the aid would remain outside that result. The professional's full smear review therefore has work left to do.

There is another question before a category can appear: is this crop suitable for ordinary classification? The proposed route uses a supplied, versioned **quality gate** to decide whether the image meets defined conditions. For an accepted crop, the manufacturer's **classifier** would suggest one of five common white-cell categories: neutrophil, lymphocyte, monocyte, eosinophil or basophil. It could instead provide an **other/unclassified route**. The gate and classifier do different jobs and have different owners: the manufacturer trains and controls the classifier, while the supplied gate has its own version and owner. These are proposed operations, with no performance result supplied yet.

The category must then reach the right place on the display, beside the correct crop, so the professional can check it and continue the full review. That connection is why the product cannot be described adequately by its list of possible cell names. A category suggestion for a selected crop neither diagnoses nor excludes disease. It also does not establish a complete automated differential, which measures the amounts or proportions of different white-cell types in the blood specimen. A bounded aid may still be useful; its value would come from helping a real part of the professional task while the complete workflow remains appropriate for that use.

The team can now explain the proposed role in ordinary words: help trained staff check supplied adult blood-cell crops. That sentence locates a possible intervention. Whether staff need this help, and how much or what kind of success the manufacturer could claim, remain separate questions.

Reviewer Lens. A convincing demonstration naturally draws the eye to the suggested category. The experienced reading starts one step earlier and ends one step later: where did the crop come from, and what does the professional do after seeing the suggestion? Those connections tell the reviewer what the demonstration could answer. An early description can be adequate when it makes the boundary intelligible and leaves the need open. Treating that honest uncertainty as a failed product test would demand a conclusion before the proposed use has even been investigated.

Consultant Lens. A clinical lead and product owner can often learn more at this point from following one ordinary task from smear arrival to release than from choosing a model family. The useful observation is where a suggestion might change an action, together with the work it would leave untouched. That gives the company a concrete investigation to consider before it commits to a model program. The adviser can recommend that investigation; the product owner decides whether to pursue it.

Retrieve the foundations

Suppose a team presents a favorable analytical result. Before attaching it to a product promise, what would connect that result to someone doing clinical or laboratory work?

1. At what point could information from a specimen or image influence a clinical or laboratory action?
2. What could make an apparently favorable analytical result irrelevant to that action?
3. How do a measurement, an interpretation and a decision differ?
4. How could software change a patient-facing workflow while leaving the final diagnosis to a person?

Decision

Which actual action and possible consequence would make your example more precise than “improve care”?

How a reviewer connects a result to an action

A result may be favorable for the wrong population, input conditions, endpoint, product configuration or use setting. Its quality does not make it evidence about a different action. In the crop example, a finding about the suggested category has to be connected to what the professional actually reviews and does afterward.

A measurement is an observation made under a method. Interpretation gives that observation meaning in context; a decision selects an action. Following those steps exposes where software could matter without signing a diagnosis. It could change priority, visibility, routing, rejection handling or delay. “Improve care” hides the step on which the claim depends; naming the action makes it possible to ask what evidence would show a difference.

This distinction also explains two testing words that return later. A test may show that a specified component or system met its defined requirements under the test conditions: **verification**. The defined product may still need evidence that its use satisfies the intended need under relevant conditions: **validation**. A correctly implemented crop classifier could pass a calculation test while the team's original task problem remains unsolved. Chapters 11–13 develop those different objects of evidence.

Reviewer Lens. The useful first question about a favorable test is where observation stopped. Did the test observe the professional action, or an earlier software output? The earlier result deserves credit for what it establishes. Keeping its stopping point visible allows the reviewer to close that question without silently closing the product's value question as well.

Consultant Lens. Need, proposed role and measurable success can easily collapse into one sentence in an early presentation. Keeping them distinct makes the next work specific. If the evidence reaches only the model output, an observation of the relevant task may change the development decision more than another output test. Calling the existing result “clinical validation” cannot create that missing connection.

What the AI label does and does not tell you

When a new crop arrives, how could the software suggest a category without already knowing the answer? The proposal uses **artificial intelligence (AI)**, a broad label covering several ways a computer can perform a task. Here the proposed approach learns behavior from examples: **machine learning**. Following one development image explains what that means and which objects the manufacturer would have to control.

Imagine a crop paired with a category assigned through a suitable independent reference process. A computer procedure uses many such image-and-category pairs to adjust a model toward the supplied categories. That adjustment is **training**. The category attached to the development image guides the adjustment; it is not an answer discovered independently by the model. This is an account of the mechanism, not evidence that LumaCell's current images or labels are suitable for it.

After training, the resulting model is an identified object that can be applied to another crop. It uses the behavior established during training to produce a score and, under the product's decision rules, a category suggestion. Applying the model this way is **inference**. The new patient's crop does not arrive with a confirmed answer that causes the model to retrain during professional review. A separate training run could create another candidate, requiring comparison and control before any later release. Training procedure, saved model, new-input calculation and clinical product are therefore different objects.

The product still needs to decide whether the crop should reach that calculation. In LumaCell's proposed route, the quality gate comes before the classifier. If it withholds the crop, the professional needs a visible route for continuing without an ordinary suggestion. If it admits the crop, the display must associate the classifier's output with the correct image. A passing gate establishes neither a correct category nor a useful fraction of all relevant inputs reaching the classifier. Conversely, good classification among accepted crops does not show how much work the gate withholds. The complete route is what connects either component to the person.

LumaCell's classifier is predictive: its output is bounded to the defined crop-routing task. Inference would run locally, with the model fixed within each release. The separate gate is also versioned. So “our model is unchanged” leaves a practical question: which gate admitted these images? A comparison of tested and deployed products needs to identify both components, together with the surrounding software behavior.

The broad AI label leaves other distinctions open too. AI may perform a released device function, or a supplied model or gate may be a dependency of the released product. AI tools used only during development can affect the credibility of work products even when they are absent from the release. Open-ended text generation and a changing live model provider raise different output and dependency questions. LumaCell has no live

model-provider connection, no online learning, and no large language model or other generative inference. Its fixed predictive example cannot settle evidence needs for those different systems.

There is also a limit to what terminology can decide. The **International Medical Device Regulators Forum (IMDRF)** uses “machine-learning-enabled medical device” for a product that first meets the relevant medical-device definition. Its N67 document establishes terminology; it does not provide a complete development, risk-management, evaluation or jurisdiction-specific classification method. [S1] Knowing how this proposal learns therefore helps describe the technology while leaving its worthwhile use, exact product and regulatory route to their own facts and accountable decisions.

Reviewer Lens. “Locked AI” can sound more informative than it is. The useful evidence shows what actually runs when a new crop arrives, what was supplied, which versions were used and how a new candidate could enter the product. A fixed classifier and an identified gate are a comprehensible starting account. The adjective alone does not identify the gate that generated a particular result. A request for that identity is narrower and more useful than an undirected request to explain all AI technology.

Consultant Lens. Technology selection becomes easier to discuss when the company distinguishes three choices: whether the intervention is justified, which product role is proposed, and how that role might be implemented. A model family belongs to the last choice. The supplier arrangement belongs there too, but its consequences reach into evidence and support. Establishing who can change and support the gate before contracting gives the manufacturer a basis for judging the whole proposed route.

Locate the work the intervention could change

Suppose the laboratory's concern is the time from specimen receipt to report release. Would a faster category suggestion shorten it? The question cannot be answered at the classifier screen. The delay could occur before a crop exists, during professional review, or after that review has finished.

Describe a consequential problem

A report that “review is burdensome” gives the team a reason to investigate. Which people experience the difficulty, during which task, under what conditions, and with what consequence? The answer separates what has been observed from an explanation or hope. A problem stated in terms of its consequence for people or decisions is a **clinical need**. “Laboratory staff need AI” has not yet said what that problem is.

The possible causes matter because they lead to different interventions. Time could be spent in slide preparation, image acquisition, review, reference consultation, staffing, rework or release checks. If the difficulty occurs before crops are available, faster

classification will leave that portion untouched. If staff frequently struggle over ordinary cell categories, assistance at that point may be worth studying. The first useful work could therefore be a task observation or comparison plan, even though the team already has enough images to imagine a training run.

Once the action is located, the manufacturer can say what the proposed product would do, for whom, on which inputs and in which setting: its **intended use**. How well it is asserted to succeed is a further statement, a **performance claim**. The proposed role “help trained staff check supplied adult blood-cell crops” does not establish either a need for that assistance or a measurable benefit. The need explains why the action matters; the use identifies the contribution; the claim needs evidence about its success.

Different materials expose the same dependency. In **digital pathology**, tissue slides are examined as scanned images. A proposed worklist aid could move an image toward the front of a pathologist's queue without making the scanner deliver it sooner. In next-generation sequencing, which reads genetic material, software might help review candidate changes called **variants**, while the report still waits for specimen-identity reconciliation. These are separate hypothetical plans. Each needs to locate the constrained action before choosing an algorithm to help it.

Identify the intervention point

For LumaCell, the useful path follows the supplied crop through the quality check, the category suggestion or routing state, and the display to the professional's action. That joins input, transformation, output, user and action while leaving the earlier crop-supply process and later complete review visible. A model-only experiment would stop before several of those connections.

What if the gate rejects the image? With a visible non-result, the professional might continue ordinary review; rejection might also require reacquisition or extra manual work. Which effect matters depends on the task, display, frequency and handling of rejected inputs. Those observations are not supplied at this opening. Nor does having a person “in the loop” show that the person can recognize the current crop, suggestion and case or recover from missing output. The proposed action needs to be observable in later testing.

Even without a benefit result, this map can do useful work. It identifies the exact input-to-action link a feasibility activity would examine. It can also expose a mismatch: if the important difficulty is finding abnormal cells anywhere on the smear, a classifier limited to supplied crops does not itself perform that search. The manufacturer has a reason to reconsider this proposed role, without concluding that every classification aid lacks value.

Two separate fictional observations show how the same reasoning continues after release. These are anonymous teaching episodes, not LumaCell, PathSight, HelixBridge

or inherited series-case history. In **the pathology queue episode**, a site log shows a deployed worklist aid moving already imported tissue cases earlier in the on-screen queue. Scanner import before that action still dominates total slide-to-review time. The queue action is observed working; its effect on the larger time measure remains open.

In **the sequencing review episode**, a laboratory confirms that a deployed variant-review aid displays suggestions as designed. Reports still wait because specimen-identity reconciliation remains necessary. The observed display function works, while an improvement in report release remains unshown. Together, the episodes locate an upstream constraint and a downstream constraint. They do not turn the blood-cell proposal into a tissue or sequencing product.

Reviewer Lens. The revealing comparison is between the action in the need statement and the action at the end of the intervention map. A selected-input accuracy result may be strong and still stop before the link the claim depends on. The two field observations show the converse habit as well: a broader benefit remaining open does not erase a working queue or display. Enough for planning is a clear bounded map; enough for a benefit claim must reach the consequence actually claimed.

Consultant Lens. A prospective walk-through with intended users and the clinical lead can distinguish an observed difficulty from the team's explanation of it. The useful next comparison is the smallest one that could separate the competing explanations. If the aid acts at the wrong point, the product owner can compare revising the intervention with narrowing its intended role. The consultant makes those alternatives concrete; the company sets the product direction.

Compare a solution with the work it would replace

Suppose a more complex classifier outperforms a simpler one on the same cell-category task. What should the product owner do with that result? It can justify further technical consideration. It still leaves open what the professional could achieve without the proposed aid.

That second question requires a credible alternative addressing the same need: the **non-AI comparator**. It might be the existing professional workflow, a clearer manual-entry screen, an acquisition control, a different allocation of work or a manually specified rule. These are possible routes to investigate, not an already selected LumaCell alternative. The comparison has to reach the action the company wants to improve.

A manually specified rule might place a case in the urgent queue when a specified field has a specified value. The team writes that rule instead of fitting it from labeled examples. This differs from another property: a method is **deterministic** if the same specified inputs and state always produce the same output. Fixed learned inference can

be deterministic too. Knowing that a system repeats an output does not tell us whether the rule was learned or written by hand.

Whichever route is chosen, the comparison needs comparable inputs, users and conditions. Selected easy crops for the AI route and everything received for the alternative would answer different questions. The comparator need not be permanently fixed before exploration, but enough must be stated prospectively to prevent a favorable model score from defining success afterward.

Now the opening result has a precise place. The simpler classifier can be a **technical baseline**: an identified starting method against which another implementation is tested on the same defined task and data. That comparison asks whether complexity adds technical value. The product comparison asks whether the complete aid helps professional work. A large technical improvement may add no useful workflow effect; a modest one may matter when it helps a well-defined task without unacceptable burden.

The transfer episodes make the choice of comparison tangible. Before deployment, manual or rule-based priority could be an alternative for the pathology queue decision. Existing professional review or a fixed report rule could be sequencing alternatives, depending on the task. After deployment, the observed queue and displayed suggestions earn credit for their functions, while the comparison must still include scanner import or specimen reconciliation if the claim reaches total review or release time. LumaCell needs an alternative matched to its own task, rather than borrowing either workflow.

Reviewer Lens. The title “comparative study” is less useful than knowing which comparison occurred. A fair test against a simpler classifier may be enough for the next technical decision. It cannot settle whether the laboratory's task requires the aid. In the field episodes, the narrow next question is the missing relationship between the working function and the promised consequence. That keeps the request focused and preserves favorable observations.

Consultant Lens. The two comparison questions may need different observations and owners. The development team can investigate an implementation improvement; the clinical and product leads need the professional task comparison. Defining both early prevents a technical success from carrying a commercial or clinical promise it never tested. If field experience points to another bottleneck, revisiting the intervention point or benefit claim can be more useful than commissioning a larger model.

Ask whether the evidence route can be built

The manufacturer can point to its **initial development image collection** (DEV-1). Does possession make the proposed experiment ready? It settles only that records are present—their **availability**. Before anyone uses them, the intended activity has to be identified closely enough to ask who may do what with which records.

That is the question of **eligibility**: whether a specified person or organization may use specified records for a specified activity under the applicable permissions and controls. Permission for controlled exploratory fitting, if established, would answer that particular use question. It would not automatically permit every other activity. No such activity-specific disposition has been supplied for this initial collection.

Even an eligible collection needs to be fit for the question. Its **adequacy** depends on whether it can answer the proposed technical or clinical claim. How do adult subjects, smears, images and crops relate? Which relevant conditions are represented? How were the expected categories established? The collection's name answers none of these questions. A narrow collection, or one entangled with development choices, could be usable for exploration while unsuitable for independent evaluation of the final claim.

The expected category deserves particular attention because it guides learning and later judgment. The process or determination used to judge an output is the **reference**. If the labels simply repeat an earlier model's output, high agreement can reward the same assumption without independently testing it. Collecting more of those labels would leave their meaning unchanged. The next useful action might be to redesign the reference process, with a separate route retained for future independent assessment. Chapters 5–7 develop the permissions, source relationships, references and dataset separation in detail.

The needed relationships depend on the material. Pathology development might need to trace an image through slide, scanner, preparation, selected region, reader interpretation and site. Related regions from one patient cannot simply count as independent people. Sequencing development instead needs specimen identity, sequence data, reference method and reportable variant classes. Both need permission and sufficient independence for the proposed assessment; neither inherits LumaCell's crop objects or cell categories.

The claimed outcome also changes which records are adequate. To investigate the pathology episode's time question, the same cases would need links between slide preparation, scanner import, queue placement and completed review. Labels identifying slides deserving attention would not measure time before import. For the sequencing episode, valid variant labels could address classification, while a release-work question needs links between displayed suggestions, identity reconciliation and laboratory release. The existing observations locate the inquiry without becoming a benefit study.

The collection can therefore support a sensible planning entry even while its readiness remains unknown: identify the proposed subset and controlled exploratory activity, the permission and reference questions, their owners, and whether a later independent assessment route could exist. That entry identifies work to resolve readiness; it does not execute the experiment.

Reviewer Lens. Image count attracts attention because it is easy to report. A more informative early account separates presence, activity-specific permission, reconstructable origin and relationships, and fitness of reference and coverage. “Unknown, with an owner and a decision affected” can be an adequate planning finding. The same words would leave a final validated-performance claim unsupported. The standard of sufficiency follows the decision being made.

Consultant Lens. A narrow readiness inquiry gives the data steward, originating site, reference experts and relevant privacy or legal specialists a question they can actually answer. “May this subset be used for this controlled exploratory fitting?” is more useful than a blanket request to clear the data. The product owner can sequence that inquiry alongside work on the task and comparator. Advice can arrange those dependencies and protect a future evaluation source; it cannot grant rights.

Consider failure and continuing support together

Suppose the proposed gate withholds an unsuitable crop. The professional still has a smear to review. What happens next determines whether withholding gives a useful boundary, extra work, or an interruption the person cannot resolve. The failure question belongs inside the ordinary task, where the missing suggestion would be encountered.

One route would make the non-result visible and leave the professional a usable ordinary review path—a **fallback**. A different route could leave a crop silently missing while the person believes review is complete. The difference is observable: what appears on the display, what the person can recognize, and what action remains possible. A statement that the product is assistive cannot establish that either route is safe.

The same reasoning follows an incorrect suggestion. It could distract the professional or invite excessive reliance; an other/unclassified route or rejected crop could add work; an unavailable display could delay a step. Consequences depend on continued full smear review, the interface, escalation and fallback. An invented risk number would conceal those dependencies. A useful account follows the changed output to the possible action and harm, then identifies a control that could interrupt the sequence and an observation that could show the control working. These are possibilities to investigate, not observed LumaCell failures.

The earlier field episodes show why the response must follow the mechanism too. Changing the pathology prioritizer would not directly remove time spent before scanner import; its observed queue behavior can be preserved while the timing route is investigated. In sequencing, identity reconciliation may be a necessary safeguard. Removing it to shorten a timer could change the wrong part of the workflow. Identity and release records would help the manufacturer decide whether the aid, the

surrounding process or the benefit claim needs revision. Retraining is only one possible product action, not the consequence of every disappointing workflow measure.

That response will also require knowledge of what is actually in use. The manufacturer would need to identify the deployed model, gate, software, rules and interface; know who can change the supplied component; notice relevant use problems; maintain a usable fallback; and judge when a change needs new evidence. Initial development and continuing support meet at those same relationships. A version that cannot be identified later makes a future comparison difficult even if today's test was well conducted.

IMDRF's 2025 **Good Machine Learning Practice** principles connect intended use, multidisciplinary work, data and references, human-AI use, relevant testing, user information and deployed monitoring across the lifecycle. They are high-level, nonbinding principles. They prescribe neither a universal business case nor a fixed retraining schedule. [S2] Their relevance here is to keep the proposed route connected through development and use, rather than treating the first favorable model score as the end of the job.

Continuing support can therefore alter the initial choice. A smaller role, a different supplier arrangement, a non-AI route or deferral may be sensible. So may a precisely limited feasibility activity. Declining or postponing this intervention can be a competent development decision when it follows the decisive uncertainty; proceeding requires an equally clear account of what the next activity would establish.

Reviewer Lens. The phrase “the professional remains responsible” often sounds like closure, but it does not show whether the professional can see the condition and act. A visible non-result and usable fallback give later testing something concrete to observe. If a later field record shows that bounded route functioning, the reviewer can credit it without claiming every fallback condition was tested. This makes both an open planning finding and a favorable observed finding possible at their proper scopes.

Consultant Lens. Ownership is cheaper to establish while the gate, classifier, interface and fallback are being chosen than after support fails between organizations. A useful early discussion includes who controls each part, who receives monitoring information and what would trigger reconsideration. The next activity can then be chosen for its ability to change the decision, with stop or revision conditions made explicit. That lets maintenance influence product design while choices remain available.

Manufacturer artifact — the intervention brief

Suppose the team answers “what problem matters?” with “we have many cell images.” The statement may be true, but it has moved to another question. Keeping the connected questions in an **intervention brief** makes that substitution visible. The brief distinguishes known facts, proposed work and open conditions so another qualified reader can understand what the manufacturer is considering.

The table is useful when deciding what a proposal actually supports. Its rows move from the consequential task to the intervention and comparison, then to whether the proposed activity is possible and supportable. The last row has to fit those answers.

Part of the decision	What the brief needs to establish
Clinical need	Which consequential problem is observed, and which explanation remains a hypothesis?
Workflow consequence	Which action, delay, workload or interpretation could change?
Proposed AI role	What bounded input-to-output function is proposed, and how would a person rely on it?
Alternative without AI	Which credible route addresses the same task?
Data feasibility	What is known about permission, origin, reference meaning and coverage?
Failure cost	What could happen with wrong output, non-results, unavailability or misuse?
Continuing support	What must be monitored, controlled, supported and changed over time?
Decision and open conditions	Which activity could proceed, who decides, and what would change that conclusion?

The first row cannot be filled by the image-folder statement: that is an available resource, not a consequential problem. “The review task may be burdensome; the location and consequence need observation” instead identifies an honest open question. The next row asks where an aid could act; repeating “burden” would not locate it. The rows need not be equally mature. Their value comes from showing the missing fact, its owner and the decision waiting on it.

A renewed brief can make the same distinction after release. The pathology episode gives the workflow row an observed queue action and leaves the total-time question open; staged timing records would distinguish import from queue and review. The

sequencing episode supplies correctly displayed suggestions for the role row, while the need and workflow rows still have to address work that includes identity reconciliation. Those records preserve a working function without making it answer a larger question.

Reviewer Lens. The decision row is where an attractive proposal can outrun its support. An unresolved comparator or permission need not stop every planning conversation. It does prevent a claim that this product is already preferable or that a data experiment is ready to execute. The focused review asks whether the proposed activity depends on that particular open row. This is more useful than treating every blank as an identical failure.

Consultant Lens. Bringing the clinical, product, data and quality or risk owners through the chain together can expose dependencies that separate presentations miss. A useful decision sentence names the activity, its conditions, its owner and the observation that would reopen it. A model feature inserted into an open need or permission field makes the document appear fuller while making the next work harder to see. The adviser helps make the decision reviewable; the relevant manufacturer owners resolve it.

LumaCell — the initial question

We now return to the complete supplied case state. LumaCell Review is wholly fictional. Its proposed users are trained laboratory professionals examining candidate white-cell images from adult peripheral-blood smears already referred for morphology review. For a supplied crop, it would offer a common-cell suggestion or other/unclassified route. The professional would check suggestions and complete full smear review; the laboratory would interpret and release the patient result. A versioned supplied predictive quality gate is part of the proposed route, and the manufacturer's classifier would run locally with inference fixed within each release.

The initial development image collection exists. Activity-specific permission, a credible independent reference, suitable coverage and an evaluation cohort are not established by its existence or name. No observed comparative workflow benefit is supplied. The proposal makes no claim of exhaustive cell detection, disease exclusion, a complete automated differential, proven time savings or patient benefit. These are the supplied boundaries, not observations that the aid has failed.

Decision before the worked answer

For each of the brief's eight fields, what can this record support now? Where a field remains open, which observation or permission is missing, and which manufacturer function could resolve it?

Decision

Which bounded disposition is defensible: a specified feasibility proposal, a revised intervention, deferral pending decisive information, or declining this route? What is the strongest competing reason the aid might miss the need, and what result would make you reconsider? This decision cannot approve clinical use.

How the early proposal becomes a bounded decision

Worked reasoning after commitment

The crop-to-professional route is clear enough to make a bounded investigation plausible. It has not established a need for a common-cell suggestion, its usefulness, or preference over an alternative. A defensible decision therefore identifies the information needed for the next development choice without making a product commitment. The **proposed intervention brief** (BRIEF-1-P1) below is an author-designed example. It is neither an adopted nor an executed LumaCell record, and it grants no missing permission.

The completed table shows how each question receives its own bounded answer. In particular, the data row states an inquiry whose result the decision row still needs.

Part of the decision	Proposed completion
Clinical need	Establish the specific morphology-review problem and consequence; images alone establish neither time pressure nor diagnostic benefit.
Workflow consequence	Examine candidate review in relation to full smear review, escalation and patient-result release.
Proposed AI role	Consider common-cell preclassification with trained review and explicit non-result handling.
Alternative without AI	Identify the existing review route and a credible workflow or deterministic alternative for the same task.
Data feasibility	Assess the initial collection's permitted exploratory use, origin, reference meaning and relevant coverage; preserve a separate future evaluation route.
Failure cost	Examine wrong suggestions, unclassified or non-result handling, missed upstream candidates, reliance and fallback consequences.

Part of the decision	Proposed completion
Continuing support	Identify responsibility for the supplied gate, version control, relevant monitoring, and response to changed inputs or components.
Decision and open conditions	Consider a bounded feasibility activity after its permissions and conditions are confirmed; no benefit or product-release claim follows.

“Assess permitted exploratory use” in the data row cannot be read as “exploratory use is permitted.” That is why the last row proposes an activity without authorizing its execution. Similarly, the need and workflow rows stay tied to the full smear task. Success in classifying supplied crops cannot silently become evidence of complete review or patient benefit. The table's blanks in knowledge are part of the decision, not omissions to cover with confident language.

One defensible disposition is a revised brief identifying the observed task, a credible comparator, the exact eligible activity and prospective criteria for deciding whether the role deserves further development. The product owner coordinates clinical, data and quality or risk review and decides whether that activity is ready. The clinical lead supplies the task and comparator rationale; data and site owners resolve the proposed record use. This is a positive next step because it can discriminate whether development is worthwhile while preserving the conditions that must be met first.

What would make further development justified? Evidence of a consequential difficulty at the crop-checking point, a credible alternative that does not adequately address it under comparable conditions, and an eligible data and reference route could support that choice. A better internal classification score alone answers another question. If the observed constraint instead lies in acquisition or complete-smear assessment outside the proposed role, revising or declining the intervention may be appropriate. Neither decision requires invented benefit or a finding that the untested proposal is defective.

Figure 1.1 — AI Suitability Is a Controlled Decision. The decision joins clinical consequence, comparator, data feasibility, failure cost, and maintainability. Use the figure to find which question your brief has left open.

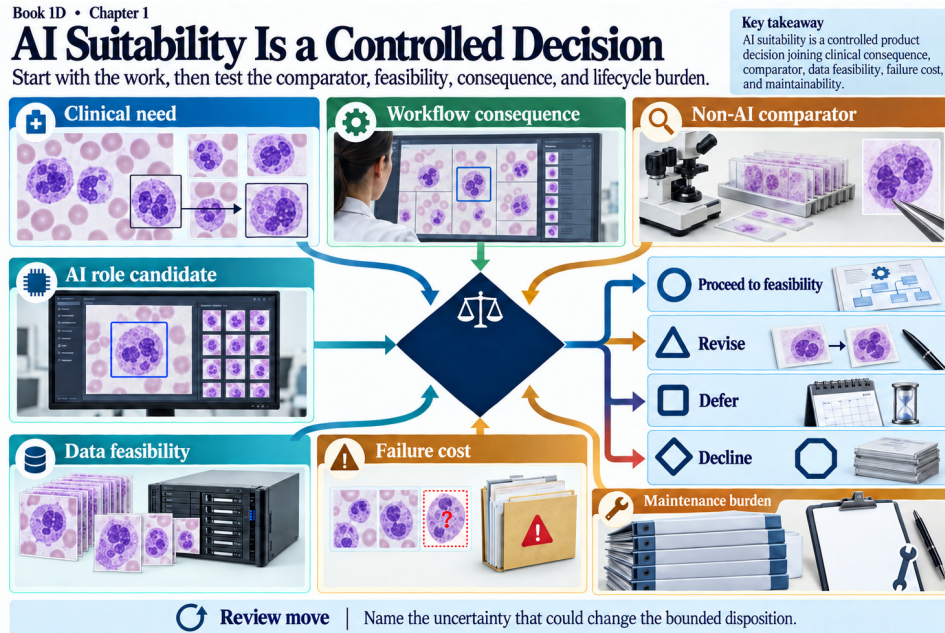


Figure 1.1. AI Suitability Is a Controlled Decision.

The consultant's suitability challenge

A reviewer can establish what the brief supports now. The consultant has a related task: make the next manufacturer decision answerable. Useful **suitability advice** identifies which missing record would change which decision, who can supply it and what alternatives depend on the answer. The adviser assesses and recommends; product responsibility and authority remain with the manufacturer.

How the consultant challenges the same proposed brief

The example suitability advice (ADVICE-SUIT-1) considers the same proposed intervention brief and initial image collection. These five fields help distinguish a disputed assumption from a missing record, then connect the possible response to someone who can decide it.

Advice field	What the example concludes or requests
Assumptions tested	Selected-crop assistance is proposed. Consequential need and preference for this intervention remain hypotheses; that does not show the laboratory needs no aid.

Advice field	What the example concludes or requests
Missing evidence	A workflow account naming the affected action and credible existing alternative; separately, a permission and origin disposition for the exact proposed data activity. Neither is supplied.
Alternative routes	Bounded task observation and eligible technical exploration, a revised intervention, or deferral—whichever resolves the decisive uncertainty. Model complexity alone settles none of these choices.
Consequences	Crop classification cannot establish full-workflow value. Missing use eligibility can block an experiment despite a compelling need; unclear reference meaning can defeat interpretation. No financial or clinical benefit is supplied.
Accountable decision owner	The product owner chooses the development step. Clinical, data, relevant privacy, legal and site owners resolve their questions; quality or risk and software functions address control and support. The adviser recommends.

The first two rows must be read together. An untested claim of value does not prove there is no useful aid to develop. Yet the permission request is independent of that value question. A strong task account could establish a problem while leaving data use unresolved; a rights record could permit exploration while leaving need and comparative value open. The response earns credit for the particular question it answers.

The recommendation is therefore to avoid unconditional product commitment and return to the named owners for a revised brief. The new records could support limited feasibility work, a revised role or a justified stop. Later evidence may reopen the initial rationale, but it cannot retroactively authorize an activity that was not ready at this earlier point.

The field episodes show where reviewer and consultant judgments can diverge without conflicting. For pathology, the reviewer can close the narrow observed queue-function question while the consultant seeks staged timing records before proposing a queue redesign. For sequencing, the next advice depends on whether the intended help concerns variant interpretation or report-release time. That choice places the next work either around the suggestions or around the identity-and-release route. No new event is needed to make those different questions visible.

Reviewer Lens. A useful memo says precisely which claim lacks support and which observation could settle it. When the manufacturer returns an actual task account and a separate permission disposition, each should be judged on its substance. Another document bearing the originally requested title adds nothing if the question is already

answered. This prevents the evidence request from becoming a paperwork requirement unrelated to the decision.

Consultant Lens. “Obtain more validation” leaves the company to guess what would change the conclusion. “Observe the task before making a comparative benefit claim” and “resolve rights for this activity before fitting with the collection” locate two different dependencies. Options presented that way let the product owner choose a bounded route and identify who will decide readiness. The memo makes the choice possible without implying that the adviser has authorized it.

Exercise the boundary

How does “worth investigating” differ from “demonstrated to improve the intended task”? Which manufacturer role owns the decision to invest in that investigation?

Decision

What limited activity, unresolved evidence and accountable owner would make that distinction concrete?

How the boundary appears in a decision

“Worth investigating” needs an identifiable activity and the uncertainty it could resolve. “Demonstrated to improve the intended task” needs evidence of that improvement under relevant conditions. Words such as “promising,” “innovative” or “clinically validated” cannot supply the missing connection. The product owner makes the development choice, drawing on clinical, data and quality or risk advice. The consultant's assessment informs that choice without making it for the company.

Transfer — a different constraint

Consider a counterfactual LumaCell state, separate from its actual early record. The manufacturer can document an eligible exploratory dataset and a reproducible baseline experiment. It still has no evidence identifying a consequential workflow problem or a credible comparator.

Decision

Does the disposition change? Which uncertainty has closed, which remains, and what revised decision follows?

How new evidence changes only the question it answers

The technical route is more feasible. The case for the intervention's value has not been established by that progress. The manufacturer might permit a narrower technical investigation while requiring task observation and comparison before product commitment. Progress on the easiest measurable question cannot answer a different one.

Now reverse the counterfactual: the consequential need is well documented, but there is no eligible data or credible reference route. That need cannot create the evidence path. The team may require a different intervention, another development strategy or a pause. In either direction, the brief retains what has been learned while showing which remaining condition affects the decision.

What carries into Chapter 2

A candidate role now needs boundaries precise enough to test: population, inputs, output, user, reliance, exclusions and fallback. Chapter 2 makes those parts of the proposed use explicit while carrying forward the unresolved intervention decision. A more precise description cannot establish benefit, permission or adequate performance by itself.

For further software and AI review depth, Book 1A Chapters 1, 2 and 7 are useful companions. Book 1B Chapters 1–2 develop connected-system and security-harm reasoning; Book 1C Chapters 1–4 expand the need-to-evidence foundation. If the manufacturer chooses a European Union route, Book 2 Chapters 1–3 add intended-purpose and qualification analysis, subject to current controlling sources and case facts. Those readings cannot supply a missing local answer or decide device status from an AI label.

Reasoning check

Can the proposed problem be explained without saying “AI”? Which alternative addresses the same task, and which consequence matters to the professional? What does a selected-crop result establish, and what remains outside it? What permission would the proposed experiment need, and which manufacturer decision can the present record support? A missing link belongs in the brief before the brief supports a development program.

Decision

Which link in this brief most constrains the next activity, and what would resolve it?

Source trail

S1: IMDRF N67, *Machine Learning-enabled Medical Devices: Key Terms and Definitions*, final document dated 6 May 2022, especially Scope, Sections 4, 5.1, and 7.1.1. It supports the terminology and the limit of that document's scope. The official PDF was checked on 24 September 2026. It does not determine LumaCell's jurisdictional status or provide a full development method.

S2: IMDRF N88, *Good Machine Learning Practice for Medical Device Development: Guiding Principles*, final document dated 27 January 2025, Principles 1–10. The official PDF was checked on 24 September 2026. These are international, nonbinding lifecycle principles, not a universal business rule or a mandatory retraining schedule.

The intervention brief, BRIEF-1-P1, ADVICE-SUIT-1, LumaCell dispositions and counterfactual, and the two clearly separate DP/NGS transfer episodes are fictional teaching constructions. They do not report a real product's performance, establish a clinical benefit, or confer legal authorization.

CHAPTER 2

Chapter 2: Intended Use, Claim, AI Role, and Workflow

A trained laboratory professional opens a small image of a white blood cell. Beside it, the proposed **LumaCell Review** software shows a category suggestion. What must happen before this piece of work is complete? The professional could inspect the image, change the suggestion if needed, and continue the full smear review. But the phrase “assists the professional” leaves almost all of those actions unstated. It also leaves open what the person should see when there is no suggestion.

Those missing actions matter even if the classifier is fixed. A suggestion attached to an uncertain specimen, an image withheld from classification, and an unavailable function create different work for the laboratory. The product definition needs to connect what the software presents with what a person must do next. The laboratory's responsible professional will interpret and release the patient result; the manufacturer must define the contribution on which that person is being asked to rely.

Chapter 1 left the clinical need, comparator, and readiness of the initial development image collection unresolved. The present task is still a proposal. Defining it more precisely supplies neither permission to run a data activity nor evidence of benefit.

A definition that can become a requirement

Suppose a developer is asked to make software that “helps diagnosis safely.” That sentence gives no answer when an image cannot be classified. An evaluator would face the same problem: which observed action would count as fulfilling the promise? The manufacturer needs a definition precise enough for both people to examine the same product.

That record is the **intended-use and AI-role contract**. Here AI means artificial intelligence, and *contract* means an internal, reviewable product definition, rather than a legal agreement or final labeling. It connects the proposed medical role and outcome to the population, users, inputs, outputs, division of work, exclusions, misuse, fallback and unresolved regulatory questions. A statement is testable when it can become a requirement and when evidence could support or contradict it.

The consultant examines where that definition and the proposed claim diverge. A **claim-and-role boundary assessment** identifies the unsupported step, the information

needed, the development or market-route consequence, and the decision the manufacturer must make. Advice can clarify the choice; the manufacturer chooses its intended use and owns the regulatory determination through its responsible functions.

Return to the proposed intervention

Chapter 1's brief asked what consequential task difficulty the aid might address, what current or non-AI route could address it, and what would have to be settled before a bounded feasibility activity could proceed. Those questions remain beside the product definition. How would the proposed software output and the professional's use of it be described separately, without letting “assist” stand in for the action?

Decision

**What would an output sentence say, and what would an action sentence say?
Which Chapter 1 uncertainty would each leave open?**

How a reviewer distinguishes an output from a role

“Proposes a category for a supplied cell image” describes what appears. “Supports a trained professional's review of candidate-cell images within a defined morphology workflow” describes what that output is for. The second sentence now points toward a person and task, but still needs operational detail about the review.

Neither sentence establishes that all relevant cells were supplied, that the aid improves the task, or that reliance on a wrong suggestion is safe. A sentence that implies one of those results has crossed into another evidence question. The useful progress is a more precise candidate role, with the need, comparison and feasibility conditions still visible.

Follow the output to the action

Suppose an accepted crop receives “neutrophil.” What, exactly, has happened? The classifier has assigned an available category to that supplied image. The professional can compare the suggestion with the image and the laboratory's fuller classification practice. The medical purpose lies further along that route: supporting morphology review in adults already referred for it. The work the product is for, the output it generates, the meaning of that output and the action it supports must fit together.

Following the crop backward reveals why the word *supplied* matters. Another process selected the candidate image. The proposed **quality gate**, the supplied component that withholds unsuitable material from ordinary classification, acts before the manufacturer's classifier gives a suggestion for an accepted image. The display then has

to associate that suggestion with the right work item. The professional examines the candidate and continues full-smear work. At this stage, the gate's exact input representation and the rule connecting crop responses to a package state remain open.

Now follow the route forward again. A change in which images reach the person, or in how missing suggestions appear, could change the person's action even with the same classifier. Describing the trained model therefore leaves part of the proposed product unexplained. The contract must reach both the upstream supply of images and the downstream professional interaction, while distinguishing the manufacturer's own function from the work on which it depends.

Chapter 1's **pathology queue episode** and **sequencing review episode** show the consequence of stopping too early. The first aid moved imported tissue cases earlier in the queue, while scanner import still dominated total slide-to-review time. The second correctly displayed variant suggestions, while specimen-identity reconciliation still constrained report release. These remain separate fictional teaching episodes, not LumaCell, PathSight or HelixBridge events. The working queue and display are positive observations. Each answers a narrower question than the proposed total-workflow benefit.

Before release, the pathology team would need to say which imported case receives the priority and which pathologist action it aims to change. The sequencing team would need to say which specimen a suggestion belongs to and which laboratory decision it informs. After release, those commitments tell the teams what their observed queue or display behavior can close, and which linked timing or identity records remain necessary. The same discipline applies to the opening crop: its category must be read as one contribution to a defined action.

Reviewer Lens. A favorable output is a useful place to start because it gives the claim something concrete to attach to. The next move is to follow the claim's verb. If the pathology claim is about queue placement, the observed queue action matters directly; if it is about total review time, linked import-to-review records are needed. For the sequencing aid, a correct suggestion display can be credited while the connection to specimen identity and laboratory release remains under review. Enough evidence for the narrower function should not disappear merely because the broader question is still open.

Consultant Lens. The inexpensive early decision is which professional action the company intends to change. Queue position to pathologist review and candidate suggestion to laboratory review lead to different study endpoints and record owners. If that choice is postponed until a favorable model result arrives, the team may discover that it has measured an output while promising a workflow outcome. Advice is more useful when it makes that choice and its missing links visible before the study is selected.

Define whose material can reach the function

The proposed LumaCell input is a package of candidate-cell images. How could the team tell that a particular package belongs within the intended use? It must know more than whether the file opens. The proposed users are trained laboratory professionals; the patient population is adults aged **18 or older** whose peripheral-blood smears have already been referred for morphology review. A package represents material from that context, rather than standing in for the patient or the entire smear.

The manner in which those images were made matters too. The **proposed acquisition profile** (AP1) names the preparation and acquisition conditions the team intends to support. At this point it is a profile label, with no established instrument or image values supplied. A format-compatible image may still come from a preparation method, instrument or site condition for which performance has not been established. Opening the file answers a technical compatibility question before it answers a clinical-use question.

Even material from appropriate conditions needs an unambiguous destination. The **identity-field requirements** (IF-1) specify the order, specimen, acquisition run, profile, locator, crop, timestamp and image hash needed to identify the work item. The locator connects the item to its source location; the hash is a value used to identify the image content. Each received package would carry its own values for those fields. The specification's name is not itself a package identifier. Detailed values, checks and implemented linkage still need development.

Why keep selection in the account if LumaCell receives only crops? A classifier cannot recover a relevant cell that an upstream selector never supplies. That dependency does not make every upstream component part of LumaCell, but it requires a stated selection and preparation assumption, an accountable actor and a boundary where evidence for the complete workflow can connect. Otherwise the narrow input could conceal work on which the proposed role depends.

The same boundary limits what a later study could establish. Selected adult crops could support a conclusion about handling those crops under defined conditions. They would leave separate questions about how often usable crops arrive from eligible smears, how abnormal material is selected, and how complete review proceeds. Those are reasons to plan the necessary links, rather than silently discarding the questions because the classifier receives only images.

For the pathology worklist, the corresponding link is from scanned slide and import record to imported case and queue entry. For the sequencing aid, it is from specimen and analysis identifiers to the candidate variant and release check. The existing observations begin at queue action or display; a broader timing claim needs the earlier import and later identity records for the same work. Neither product inherits a

blood-cell input specification. Each needs its own record joining the bounded input to the professional task.

Reviewer Lens. A named profile and an interface specification can look reassuring because they resemble finished engineering records. Their maturity matters. Here the reviewer can ask whether the proposed fields could distinguish the crop and specimen and whether supported conditions have been defined, while leaving the actual values and checks open. That is a useful planning judgment. Treating the names as passed interface tests would convert a definition into evidence the case has not supplied.

Consultant Lens. Agreement between product, laboratory and software owners is cheap before requirements are fixed. If they discover later that the study cannot join an image to its preparation or specimen, that information may be difficult to recover. An early work-item walkthrough can reveal the missing join. The options are to obtain the linkage before pursuing the broader claim or to narrow the planned claim appropriately; a later mismatch would reopen the affected interface and evidence without automatically diagnosing a classifier defect.

Give every result and non-result a route

A crop arrives, but no ordinary category is available. What should the professional do? That question cannot be answered until the reason for the absence is visible. The following route describes decisions the product needs to support; it does not settle their final software sequence.

Identity is the first dependency to examine. If a required identity value is missing or conflicts with the active specimen or package, even a plausible category would belong to uncertain work. The item needs a visible reconciliation state, with missing or mismatched identity kept out of ordinary inference. The next useful action concerns association of the image, rather than interpretation of a reassuring suggestion.

An item with sufficient identity may instead be withheld by the **original quality gate (Q0)**. Its proposed job is to keep unsuitable image material out of ordinary classification. Rejection means the laboratory still owes work: reacquisition or its manual morphology route. It does not mean that the cell or smear has a negative finding. Which representation the gate examines, whether its decision concerns a crop or a package, and how mixed crop responses become a package state are still planning questions. The contract can require visible unfinished work while those design and supplier decisions are being resolved.

If an image is accepted, the classifier may offer a common-cell suggestion or **other/unclassified** routing. These are both outputs, with different review implications. The professional checks the actual image and may classify it using the laboratory's richer vocabulary, then continues the full smear review. By contrast, if the function is unavailable, the item needs an accountable manual route. A function that did not

supply assistance has left a different condition from one that supplied an other/unclassified output.

The practical reason to retain these distinctions is what the display would otherwise hide. A blank could mean that identity needs reconciliation, quality prevented classification, or the function was unavailable. A reassuring “no finding” could make those different forms of unfinished work appear complete. **Fallback** is the route that allows the laboratory to continue when the input or function cannot support the intended work. The contract need not select every screen detail, but it must say what distinction the person needs and which work remains. If a later proposal adds a confidence display or an “uncertain suggestion,” that new meaning and response will also need definition; neither feature is assumed here.

The proposed common suggestions are neutrophil, lymphocyte, monocyte, eosinophil and basophil, alongside the other/unclassified route. Those five categories do not exhaust the laboratory's morphology vocabulary. High agreement within them would therefore leave abnormal and non-target material to be examined on its own terms. Reference categories and evaluation groups come later; the contract preserves these items and their human route now, so that an evidence plan cannot make them disappear by counting only easy-to-classify material.

For the separate pathology aid, the equivalent design question is how an imported case without a usable priority stays visible for review. For the sequencing aid, it is how a missing suggestion or unresolved specimen identity remains visible. These are proposed non-result questions, not additional findings from the field episodes. Earlier queue placement and correct suggestion display did not test every such route. The work on either side of the observed function still needs an owner.

Reviewer Lens. The useful review object is a state joined to a next action: identity reconciliation, quality rejection, accepted other/unclassified, ordinary suggestion or outage. Distinguishing them is enough to begin scoping requirements. Showing that an implemented product preserves them will need evidence later. Keeping those two decisions separate allows a reviewer to credit a coherent proposed route without overlooking the still-open gate design.

Consultant Lens. Teams can be tempted to choose the screen first and decide what its blanks mean afterward. The clinical and software owners can avoid that rework by agreeing on state transitions and resolution responsibility before display details or evaluation denominators are fixed. The supplier discussion then has a concrete question about gate input and package meaning. After use begins, disappearing or accumulating pending work would warrant comparison with those specified routes before anyone attributed the pattern to the model.

Describe reliance as visible work

The contract says a professional reviews LumaCell's suggestions. What would someone observing that review actually need to see? The person must encounter the relevant candidate, inspect the image, compare it with the suggestion, recognize when a change or further action is needed, and know when the work is complete. The term **reliance** becomes useful at this point: it concerns the relationship between the assistance presented and the person's actual decisions and actions.

That relationship includes the states in which ordinary assistance is missing. Blocked and unclassified items must come to the professional's attention, as must the route for unavailable work. The proposed LumaCell role requires review of supplied candidates and these unfinished items, followed by the full smear review. An interface that makes unchecked acceptance the easiest route could conflict with that intended interaction even if its label says "for assistance only."

The ability to change a suggestion is part of the design; recognizing a reason to change it is part of use. These questions meet in representative work, where the professional needs image, identity, suggestion and unfinished-work status in an intelligible relationship. A wrong category, a hidden non-result and a result associated with the wrong image would have different causes but could each change the professional's action. The contract thus points toward both implementation checks and observation of human use.

The laboratory's responsibility for interpretation and patient-result release does not remove the manufacturer's responsibility for this designed contribution. It does explain why the planned evidence must distinguish the software's calculation, the professional's review and the final laboratory decision. Naming the report signatory alone would leave the intervening interaction unspecified.

A training term can obscure that separation. In machine learning, **supervised learning** means learning from examples whose labels guide the fitting process. It does not mean that a clinician oversees each new prediction or specify the software's autonomy in clinical work. The International Medical Device Regulators Forum's terminology document, IMDRF N67, distinguishes these ideas. [S1] LumaCell's model would remain fixed during inference within a release while still calculating a category without a person making that calculation. Professional review is an additional, deliberately defined action.

For the pathology and sequencing episodes, this means that observed queue movement and correct display belong beside a separate question about people. What must the pathologist inspect after encountering the case? How must laboratory staff check a suggestion and its specimen link before release? Before release those are interaction requirements; after release they require observations of the actual user work. Neither

supplied field observation establishes completion of those actions or achievement of the broader time promise.

Reviewer Lens. A documented control can answer whether an action is available while leaving its use uncertain. A reviewer following one ordinary suggestion and one non-result looks for where the necessary context is visible together and where recognition and correction could be observed. This keeps a request focused on the interaction. It avoids both dismissing a useful control and allowing its existence to stand in for evidence about people.

Consultant Lens. The company can distinguish actions required for every item from actions triggered by a particular state before designing a human-use study. Clinical and human-factors colleagues can then plan work that reveals unnoticed suggestions, non-results or identity problems, with an owner for completion. The same distinction matters later: an operating function record and a user-action record answer different questions, so advice about a benefit claim should connect them before recommending a design change.

Make the claim no wider than the proposed action

Suppose the evidence concerns suggestions on supplied images, while the proposed headline promises shorter report time. Where did the extra promise enter? Between the image and the report lie review, possible rework, identity checks and release. A claim about the completed task needs those connections to be examined. A favorable image result cannot establish them by changing the wording around it.

The claim's unit helps locate the missing link. A category for a crop concerns an image. Helping a professional review candidates adds a workflow role. A comparative time claim adds an outcome across a defined span of work. A whole-smear or patient conclusion adds selection, reference and consequence questions beyond the selected images. This progression explains changing evidence needs; it is neither a regulatory class ladder nor a required marketing sequence. A manufacturer may choose to pursue a bounded role because that is the role that addresses the need.

The proposed LumaCell boundary excludes exhaustive cell finding, a complete automated differential of white-cell types, red-cell or platelet interpretation, leukemia diagnosis or exclusion, and autonomous patient reporting. Pediatric use is outside the supplied proposal. No time-saving or patient-benefit result has been established. These limits must agree with the headline and the actual workflow. They cannot erase a dependency: if full review depends on upstream cell selection, the intended interaction still has to account for that selection even though LumaCell makes no exhaustive-detection claim.

The transfer episodes make the next choice concrete. A faster pathology-review claim would need a comparison spanning time before and after queue placement. A faster

sequencing-release claim would need specimen reconciliation as well as suggestion review. Each team can retain its observed bounded function, investigate the surrounding workflow or revise the proposed benefit. The actual queue and display successes remain in the record whichever choice the manufacturer makes.

Reviewer Lens. The easiest claim expansion to miss is a change in the unit: an image becomes a specimen, or an output becomes a completed report. Following that change points to a specific selection, timing, identity or user-evidence question. The reviewer need not call the narrower result defective to reject the unsupported extension. A bounded positive conclusion is often the most precise way to show what still needs work.

Consultant Lens. A broad headline can create an expensive evidence program before the product owner realizes that a narrower role was available. Putting the function and its prospective evidence beside the wording exposes that choice early. The owner can retain the bounded role or pursue the broader outcome with the necessary linked study. Advice should make both routes understandable without allowing an attractive sentence to decide the product strategy by default.

Give regulatory colleagues the same product

A team may now have a coherent description and still ask, “Does calling it AI decide the regulatory route?” The description supplies facts for that assessment. The technology label does not supply the conclusion.

IMDRF N67 begins its medical-device terminology with products that already meet a medical-device definition. It does not classify all machine learning as a device or provide a national qualification or classification method. [S1] Likewise, “decision support,” “triage” and “research” leave the actual function and purpose to be examined. Regulatory colleagues need the proposed purpose, image-analysis function, users and professional reliance, together with the markets under consideration, to assess the current controlling law and guidance there.

No class, clearance, certification, exemption or AI Act status is assigned to fictional LumaCell here. Book 2 develops the European Union route; the contract gives that inquiry a stable object. The pathology queue and sequencing suggestion would require the same sequence at their own boundaries. Their observed operation can support a bounded functional statement without settling legal status or permitting a broader claim.

A later change to purpose or user role would also change the facts against which an earlier assessment was made. That is why the contract and route assessment need to refer to the same proposed product. A dated decision about an earlier role can remain valid as a record of that decision while requiring an applicability check for the revised one.

Reviewer Lens. The narrow question is whether the assessment addresses the purpose, outputs, users and reliance now under review. A familiar technology label can distract from a changed role. Comparing the assessment's factual description with the current contract exposes that mismatch without requiring the reviewer to infer a route from the name “AI.”

Consultant Lens. Regulatory colleagues can give more useful advice when the company supplies a short account of actual function and workflow and identifies the markets it is considering. Asking which changed fact would reopen their assessment also gives the product team a practical reason to control its wording. The adviser helps prepare that factual account; choosing the manufacturer's claim and making the responsible regulatory determination remain assigned decisions.

Build the contract from the action route

Could two developers build different user behavior while each claimed to follow the preceding description? The contract is useful when it makes such a disagreement visible. Its fields connect the action the manufacturer hopes to support with the conditions and work required to realize it.

The following table provides questions for testing that connection. The rows belong to the same workflow; they are useful together, rather than as independent slogans.

Contract field	Question it must answer
Intended outcome	What task result is sought, and which benefits remain unproven?
Intended use or purpose	What bounded medical role is proposed?
Users	Who uses it, with what competence, and who interprets the final result?
Population	Whose material and clinical context are included or excluded?
Inputs	What source, preparation, acquisition, identity and acceptance conditions apply?
Outputs	What does each result or non-result mean, and at what level?
Clinical role	Where does that output enter the diagnostic work?
Autonomy	Which actions occur without a user decision?
Reliance	What must the person inspect, verify, change or escalate?

Contract field	Question it must answer
Exclusions	Which uses and claims are unsupported, and which dependencies still matter?
Misuse	Which foreseeable departures from the interaction need examination?
Fallback	How does unfinished work continue when the input or function cannot support it?
Regulatory questions	Which market-specific decisions, owners and open conditions remain?

The *inputs* row leads directly to *outputs* and *fallback*. Missing identity cannot simply produce an ordinary category: the work needs a visible reconciliation route. The *users*, *autonomy* and *reliance* rows resolve another possible ambiguity. **Autonomy** concerns what happens without a user decision. “No automatic patient report” constrains one action, but leaves unanswered whether every suggested image must be inspected. That answer belongs in the *reliance* row.

Finally, the intended outcome must agree with exclusions. A promise about complete review cannot quietly coexist with an excluded detection task on which that promise depends. The team must make the actual role and boundary intelligible. A coherent table is adequate for planning requirements; it does not show that the specified behavior has been implemented or that the intended use performs adequately.

For the separate pathology episode, the imported-case input and queue action can now be placed beside the intended total-time outcome. The observation fills the bounded output/action part while the earlier import stage stays in the broader question. In the sequencing episode, suggestion display can fill an output row while the specimen link and release step remain consequential to the workflow claim. Revising such a contract after use need not discard the function that worked.

Reviewer Lens. The table earns its space when one row changes how another is read. Input identity changes the permitted output; a visible output changes the available professional action; unfinished work changes what fallback must preserve. A reviewer can credit that consistency in the proposed LumaCell contract while leaving profile values and interface tests open. The stopping point is a clear planning boundary, rather than completion of every future evidence task.

Consultant Lens. These dependencies also sequence decisions. Clinical and product leads settle the intended action; software and laboratory owners resolve identity and non-result handling; the appropriate teams then plan evidence for any broader benefit. If field use later shows a working queue or display without the intended wider

improvement, the owner can choose to study surrounding steps or narrow the claim. A larger model study is not the only possible response.

The proposed LumaCell contract

The following decision returns to the same early fictional case. LumaCell Review and its manufacturer are fictional. Its intended users are trained laboratory professionals reviewing supplied candidate white-cell images from stained peripheral-blood smears of adults already referred for morphology review. The proposed acquisition profile names intended preparation and acquisition conditions; the identity-field requirements specify fields whose actual values belong to each package. Neither name establishes completed implementation.

The original predictive quality gate would admit accepted material to ordinary classification and leave rejected work visible for reacquisition or manual morphology. Its input representation and crop-to-package aggregation remain unsettled. Missing or mismatched identity does not enter ordinary inference. For accepted images, the classifier proposes one of the five common categories or other/unclassified routing. The professional checks candidates and blocked or unclassified items, may use the laboratory's full taxonomy to reclassify, and conducts full-smear review. The laboratory interprets and releases the patient result.

No autonomous differential count or patient report, leukemia diagnosis or rule-out, red-cell or platelet interpretation, pediatric use, exhaustive cell finding, or demonstrated time saving or clinical benefit is supplied. It is not known that every relevant cell has been provided to the classifier. The initial development image collection still has its unresolved permission and readiness conditions. These are the facts available for the **proposed intended-use contract (IU-1)**, an internal proposal to carry into requirements and evidence planning.

Which wording could the contract support?

Three statements have been proposed:

1. "LumaCell identifies normal blood smears."
2. "LumaCell suggests categories for supplied candidate-cell images for trained professional review."
3. "Because the reviewer can override a suggestion, the human-control question is already resolved."

The useful judgment reaches beyond choosing the least problematic sentence. What would the complete contract have to say about population, inputs, outputs, required review and fallback? For each statement, what is supported, which step remains

unsupported, and what design decision or evidence would be needed next? Which dependency needs a manufacturer owner before the record can move forward?

Decision

Which statement could anchor the proposed contract, what qualifications would it need, and which unresolved questions would remain after the wording was corrected?

How a reviewer connects the wording, workflow and open decisions

Worked reasoning

The second statement is closest to the supplied role. It still needs the adult referred population, input conditions and actions following the suggestion. The first statement takes an image-level function and asks it to establish something about an entire smear. The third takes an available screen control and asks it to establish how people recognize and correct errors. Each mismatch points to a different missing relationship. None proves that the bounded aid is ineffective.

The proposed intended-use contract can therefore begin with this bounded role statement:

Note

LumaCell Review is proposed to assist trained laboratory professionals reviewing supplied candidate white-cell images from stained peripheral-blood smears of adults aged 18 or older already referred for morphology review, within the proposed preparation and acquisition profile. For an accepted image it proposes a common-cell category or other/unclassified routing. The professional verifies or changes suggestions within the laboratory's full morphology workflow and handles blocked, unavailable and unclassified items. The laboratory's responsible professional interprets and releases the patient result. The proposal does not establish complete-cell detection, normal-smear status, leukemia rule-out or clinical benefit.

The paragraph says what role is proposed. The table makes the operational commitments and open conditions inspectable, so that polished wording cannot conceal a different product.

Contract group	Bounded worked completion
Outcome, purpose and role	Support candidate-image review; usefulness and intended-use performance still need evidence.

Contract group	Bounded worked completion
Users and population	Trained laboratory professionals; adults already referred for morphology review; laboratory professional owns the patient result.
Inputs	Supplied candidate package under the proposed profile and identity-field requirements; actual values, checks, preparation, acquisition and selection dependencies still need specification or assessment.
Outputs	Common-cell suggestions, other/unclassified and visible non-results; no silent conversion to a patient-level negative.
Autonomy and reliance	No automatic patient report; professional verifies or reclassifies and handles non-results; recognition and action remain to be evaluated.
Exclusions and misuse	No pediatric, exhaustive-cell-detection, normal-smear or leukemia-rule-out claim; foreseeable mistaken reliance still needs examination.
Fallback	Reacquisition or manual morphology through the laboratory's pathway, with accountable handling of each work item.
Regulatory questions	Actual function and purpose assessed in each proposed market; no fictional route outcome inferred.

The outputs row tests the other commitments. If blocked work vanishes, neither the reliance nor the fallback row describes what happens. The precise package state for mixed crop responses still needs a design decision. The inputs row supplies another test: because candidate images arrive from elsewhere, evaluation of accepted crops cannot be offered as proof that this product located every relevant cell. The contract now provides a coherent boundary for requirements and studies. It remains an internal proposed record, not proof of performance or an adopted execution permission.

Reviewer Lens. A useful positive conclusion is available here: the proposed role paragraph and its input, output, reliance and fallback rows agree on candidate-image assistance. That closes a product-definition question at a planning depth. Profile values, identity checks, package-state behavior and actual human action remain open for different kinds of evidence. The reviewer can keep each question attached to the record that could resolve it.

Consultant Lens. The next manufacturer work can now be assigned: clinical leads confirm the role; software and supplier teams resolve gate and identity behavior; human-factors staff plan observations of professional action. The product owner decides when the revised record is ready for a specified development activity. Making

the paragraph read well cannot authorize use of the initial image collection or settle the earlier need and comparator question.

Figure 2.1 — Bind the Claim to the Whole Workflow. Read the contract from intended outcome through users, inputs, output, role, reliance, exclusions, misuse, and fallback. An isolated model output cannot carry a broader workflow claim by itself.

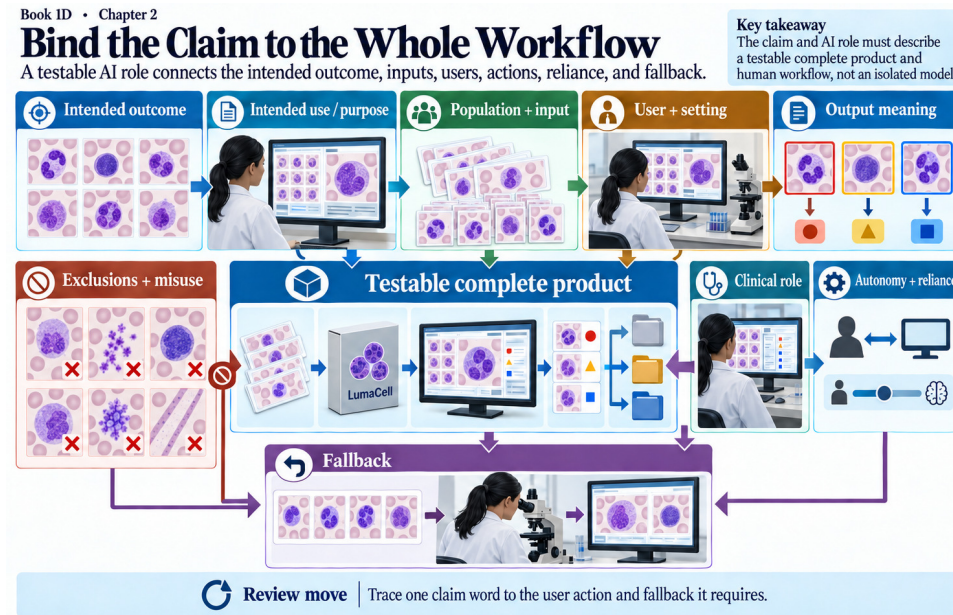


Figure 2.1. Bind the Claim to the Whole Workflow.

Who can turn the proposal into a decision?

A coherent **proposed intended-use contract** still leaves an organizational question: who can make its commitments real? “Other/unclassified remains unfinished work” has to become a specified route that someone implements and checks. It cannot be left to a developer’s default display choice.

The product owner, with clinical and other relevant functions, establishes the role and claims to pursue. Regulatory colleagues assess the actual function and purpose in the proposed markets. Software, machine-learning, quality or risk, and human-factors teams translate the contract into requirements and evidence questions. The laboratory retains interpretation and release of the patient result; the manufacturer retains responsibility for its designed contribution. The contract identifies what would be pursued while Chapter 1’s feasibility permissions and readiness remain open.

The transfer episodes expose why ownership must meet at the interface. The pathology manufacturer defines its queue behavior, while the site accounts for scanner import and

pathologist review. The sequencing manufacturer defines the suggestion and specimen link, while the laboratory retains reconciliation and report release. Their observed bounded functions can be credited. To resolve wider benefit, each manufacturer and its users need linked records showing the whole route and the step needing attention. A site's acceptance of its own responsibilities cannot establish an unsupported manufacturer claim.

Reviewer Lens. An owner is useful when that owner can resolve the particular relationship. An unassigned gate-to-package decision is a planning gap, not an observed LumaCell failure. The focused request is for who specifies that behavior and which evidence will test it. In the transfer episodes, the corresponding request concerns the remaining total-workflow link while retaining the supported queue or display action.

Consultant Lens. A meeting of every team is less useful than a meeting around a shared decision. Product, clinical, software, site and supplier owners can identify who specifies inputs and non-results, who observes professional action and who can authorize a changed claim or development step. The adviser makes the dependencies and closure conditions visible, without taking the laboratory's release authority or the manufacturer's product decision.

What would close the wording concern?

The manufacturer could remove an unsupported headline and still retain requirements that imply the broader role. How would a consultant know the mismatch had actually closed? The **claim-and-role boundary assessment** compares wording, product route and available evidence, then identifies which manufacturer decision the discrepancy requires.

Its central question is where a sentence crosses from image to package, specimen, patient or complete workflow. The adviser identifies the record that would support that crossing and the owner who can choose whether to retain the bounded role or pursue the larger one. Agreement among wording and function can close a definition issue while performance and jurisdictional questions remain open. The adviser supplies neither an invented purpose nor a class decision.

The role advice

The **role advice** (ADVICE-ROLE-1) examines the proposed “identifies normal blood smears” wording against the proposed intended-use contract. This is a worked assessment of the proposal, rather than an adopted change. Its six fields show how the concern becomes a decision and a checkable closure condition.

Review field	Worked finding
Wording and supplied basis	The statement concerns an entire smear; the proposed function supplies selected images, category suggestions or other routing, and professional review. Complete-cell detection and normal-smear evidence are absent.
Unsupported assumption	Classification of supplied images is being extended to absence of relevant findings across the smear. Neither that extension nor ineffectiveness of the bounded aid is established.
Evidence consequence	A whole-smear claim would reopen selection or detection, reference, population, failure and reliance questions. An override does not establish users' recognition and correction of errors.
Information and route	Product and clinical owners reconcile wording with the proposed function. Required-review and non-result paths need owners. Human-factors colleagues plan recognition/action evidence; regulatory colleagues assess actual purpose in identified markets.
Manufacturer decision	The product owner retains the bounded candidate-image role or explicitly pursues a changed role and evidence program. Removing the unsupported expansion must reach behavior as well as the headline.
Closure and re-review	Revised contract, requirements and workflow agree on supplied candidates, suggestion/non-result outputs and assigned full-review/escalation responsibility; they imply no normal-smear or patient-negative conclusion. Use evidence remains tracked; a broader proposal has a distinct change assessment.

The unsupported assumption and evidence consequence rows belong together. The concern is an inference from selected images to the whole smear, so another adjective in the claim would not supply the missing selection and reference argument. The manufacturer decision row offers a positive route to closure: a consistent candidate-image role is defensible as a proposal. The last row then tells the adviser what to compare before closing that particular mismatch. It does not mark actual performance or a market route as established.

Reviewer Lens. A revised sentence is easy to inspect; its effect on the underlying role takes one more comparison. If the contract, required-review path and implementing requirements all retain candidate-image assistance without implying normal-smear status, the definition concern can close. Asking for the same wording correction again would add little. Actual use and performance remain separate questions with their own evidence.

Consultant Lens. The owner needs a usable bounded sentence, the records that must agree with it and the alternative evidence program for a broader role. Clinical and

software owners can reconcile the chosen option across the contract and requirements. Reporting that reconciliation makes the advice actionable while preserving the manufacturer's authority to choose; advice itself is not approval.

A short boundary question

A proposal states: “The label says assistive, so the clinical-risk question is closed.” What statement would make the remaining observation or evaluation question clear, and which decision would still belong to the manufacturer?

Decision

What would have to be examined about the professional's actual task before the claim could be closed?

How a reviewer reads the assistive label

One defensible correction is: “Assistive wording bounds the proposed role, but the manufacturer still needs to examine what users see, how they act, and how incorrect or missing outputs can affect that task.” The label helps define the proposal; it supplies no observation of that interaction. The consultant can identify those questions and their owner in the plan. The final risk decision remains with the manufacturer's responsible functions.

When the proposed role changes

Consider a hypothetical proposal, not an event in LumaCell's history. The technology still classifies images, but packages rated unremarkable would now be removed from the ordinary worklist instead of shown for review. Which contract fields would change? Which earlier evidence might cease to answer the relevant question, and which manufacturer decisions would have to reopen?

Decision

What would this change do to the relationship between the output and the professional's work, even if the model family stayed the same?

How a reviewer follows the changed route

Clinical role, output meaning, autonomy, reliance, failure consequence and fallback all change. A wrong “unremarkable” assignment could now affect whether the package is encountered for review at all. Evidence that users inspect and correct a visible suggestion stops before this new question: it has not examined a suppressed work item.

The manufacturer would need to reassess purpose and requirements, relevant error and human-use questions, and each applicable regulatory route. That is a changed product decision even though the model family might be unchanged. The hypothetical introduces no completed change, failure or outcome into LumaCell's early proposal.

Further reading at this boundary

The local contract stands on its own. Book 1C offers deeper clinical-need-to-claim reasoning. Book 1A Chapters 2–3 develop software promise and risk translation, and Chapter 14 explores human reliance. Book 1B Chapters 1–3 extend misuse, connectivity and threat consequences. For a European Union proposal, Book 2 Chapters 2, 3 and 14 examine intended purpose, software qualification, classification and AI Act applicability against current sources. These readings deepen the relevant question; they do not supply a missing LumaCell decision.

Return to the image beside the suggestion

The professional in the opening needs to know which image is being reviewed, what its displayed state means and what work remains. Can each result and non-result in the proposed contract be followed to that meaning and action? Does “assistive” identify what the person actually reviews? Do the headline, exclusions and fallback describe one consistent role? Are unresolved input, evidence and market-route questions assigned to people who can resolve them?

Decision

Where could two implementers still build different behavior while both claimed to follow this contract? Which sentence, manufacturer decision or evidence would resolve that conflict or open condition?

Chapter 3 follows the proposed contract into the preparation, supplied gate, manufacturer classifier, display, platform, versions and people needed to realize it; otherwise even a precise role could lead development and evaluation toward different products.

Source trail

S1: IMDRF N67 (2022), scope and sections 5.1, 7.2–7.3: bounded AI-medical-device terminology, including the distinction between supervised training and clinical autonomy. It is not a national qualification or classification decision.

IU-1, ADVICE-ROLE-1, the contract fields, and the changed-role exercise are fictional teaching constructions. They are not evidence of LumaCell performance or a jurisdictional decision.

CHAPTER 3

The Complete AI/ML Product and Supply Boundary

A trained laboratory professional is about to open a supplied image of a candidate white blood cell. The proposed LumaCell Review software would place a cell-category suggestion beside it. Before that suggestion can be useful, what must connect the image on the screen to the specimen and the work the professional is meant to complete?

Chapter 2's proposed intended-use contract gave that work a boundary: the professional checks supplied candidates, handles unfinished work and continues the full smear review; the laboratory interprets and releases the patient result. The manufacturer now needs a description that developers, evaluators and laboratory staff can all follow. A model name leaves each of them guessing about a different part of the route. The need and comparator remain open, and the initial development image collection still has unresolved permission and readiness questions. A clearer product description will help plan the next decision; it will not authorize the work.

Retrieve the role before drawing the system

What does LumaCell actually receive, what does it propose to return, and what must that output never silently imply? How does the proposed output connect to the professional's remaining activity?

Decision

Between receiving that input and completing the human activity, where might an ordinary suggestion be withheld? Which record would let someone identify the affected work item afterward?

How a reviewer follows the unfinished item

The useful starting point is a supplied candidate crop: a small image selected from the smear before LumaCell receives it. Preparation, acquisition and candidate selection determine what arrives. Package identity keeps that image attached to its work. Transformation, quality acceptance, classifier calculation and routing determine whether a suggestion can appear. A retained work state, display, professional review and fallback determine what happens next.

A quality rejection is one place an ordinary suggestion might be withheld. The affected crop and package still need a record linking them to the specimen and current work item so that the professional can follow the appropriate route. Other answers may identify a missing identity value or unavailable processing. The common requirement is accountable work, including when calculation produces nothing usable. A candidate-image suggestion cannot establish that an entire smear is normal. An omitted crossing in an initial account identifies a planning question; it does not prove that this unreleased proposal has failed.

What does “outside the product” mean?

A factory ships a batch. A clerk matches its shipment record to an invoice, and an accountant reconciles the ledger. Suppose the supplier can show that the invoice calculator works. What remains to be explained before the office can trust the posting? The calculation needs to belong to the right shipment and be handled under the right matching rule. Following that ordinary transaction gives us three different ways to draw a boundary.

First, the office must post the right transaction. For LumaCell, the corresponding question is which work the proposed product performs: accepting an identified candidate image, withholding unsuitable material, suggesting a category for eligible material and showing an accountable current state. This is the **functional boundary**. It keeps the laboratory's downstream full-smear interpretation visible without turning that interpretation into an autonomous software output.

To realize those functions, particular parts and settings must cooperate. The office relies on its matching rule, invoice program and current records. LumaCell would rely on the trained classifier, supplied quality gate, image preparation, routing rules, display, runtime and supported site conditions. Those choices form the **configuration boundary**. At this early stage they are proposed identities and interfaces, with gaps still to resolve, rather than a selected release baseline.

Once the parts are visible, the next question is who can supply, check, accept, operate or change them. In the office, those actions belong to the warehouse, clerk, supplier and accountant in different ways. In LumaCell, the laboratory controls preparation and patient-result decisions, the supplier maintains its gate, and the manufacturer must judge the gate's fit in the proposed product. This is the **responsibility boundary**. “External” might describe the supplier's organization while the supplied behavior remains consequential to the configured product. The office analogy explains that split; financial reconciliation cannot establish medical safety or replace image and professional-use evidence.

The pathology queue episode and sequencing review episode from Chapter 1 provide a useful check on those boundaries. Earlier placement of already imported pathology cases supports the queue function; correctly displayed sequencing suggestions support the display function. Their records also locate the remaining constraints: scanner import before the queue and specimen-identity reconciliation before report release. These are separate fictional aids, not LumaCell, PathSight or HelixBridge. Mapping the observed function, its configuration and its owners helps preserve what worked while directing the wider benefit inquiry to another crossing.

Reviewer Lens. The word “external” often hides the question that matters. A reviewer can resolve it by following the contribution into one user action: which product behavior depends on it, under which settings, and who can accept the evidence? Enough at this point is a clear planning boundary with explicit open decisions. Treating every outside contribution as irrelevant would miss a dependency; treating the drawing as a legal device-boundary determination would ask it to answer a different question.

Consultant Lens. A component diagram is inexpensive to draw but can be expensive to defend if nobody can explain what crosses its arrows. Following one normal item gives the product owner something usable: an item, a state, a contributor and a next decision at each crossing. In the pathology and sequencing examples, that conversation reaches the import and reconciliation owners before the team spends effort changing a functioning queue or suggestion component. The manufacturer chooses the boundary and development route.

Follow the input until someone can act

The boundary becomes concrete when we follow the supplied crop from its source to the professional's next action. At each crossing, the question is what the next person or program needs in order to proceed.

What does an image file tell us about its source?

A crop may open perfectly and still leave its clinical suitability unexplained. Before it existed, a laboratory prepared a slide, an upstream process acquired images, and a selector chose candidate cells. Color, focus and selection affect what the classifier gets to see. A Portable Network Graphics file—PNG—identifies the image format. It establishes neither those source conditions nor that every relevant cell was selected.

The proposed acquisition profile from Chapter 2 names the preparation and acquisition conditions the manufacturer intends to support. Its acquisition and candidate-location contributions have planning names, but their detailed behavior, numerical imaging values and adequacy are not established here. The map needs the handoff between laboratory preparation, selection and supported input because that is where a later test's applicability begins. Naming the dependency does not give the manufacturer

control of every laboratory act. It does expose a limit that matters: the crop classifier cannot reconstruct a relevant cell the selector never supplied.

Even a suitable crop must remain connected to the right work. The identity-field requirements, the required connections from the image to its work item, call for order, specimen, acquisition run, profile, locator, crop, timestamp and image-hash fields. A received package would contain particular values for those fields. The specification is the list of required connections; it is not a token identifying one package, and it is not evidence that the received values were checked.

One field deserves special care. An **image hash** is a calculated identifier for image data that helps detect whether records refer to the same bytes. Matching bytes can support continuity through a handoff, but cannot prove that the image was originally attached to the correct patient. A reviewer therefore needs both the required fields and a planned trace from source specimen to received package. A familiar filename or a valid format cannot replace that trace.

The same question selects different records in the other domains. For the pathology worklist, the linked item is a scanned slide becoming an imported case; an analysis of total time would join scanner-import, queue and completed-review times for the same cases. For sequencing, it would join suggestion display, specimen reconciliation and report release for the same specimens. Those joins are prospective evidence work. The earlier field observations have not already supplied them.

Reviewer Lens. An interface specification is valuable design evidence, but its neatness can make actual linkage feel established. Reading it beside one planned source-to-package trace keeps the distinction visible: could these fields connect the crop to its specimen, and what would show that the values were correct? The supported profile needs its own comparison with preparation and acquisition conditions. Those two requests are narrower and more useful than asking whether the image is simply “valid.”

Consultant Lens. Source records are easier to preserve before collection than to reconstruct after an attractive result appears. Clinical, imaging and software owners can agree on the minimum source and package fields early; the site supplies preparation or specimen records, while the manufacturer defines the compatibility it will accept. If a condition cannot be supported, the practical choices are a narrower intended input or a targeted compatibility study. A format label does not resolve that choice.

How does the received image become a visible suggestion?

The image on the screen is not necessarily the numerical representation the classifier receives. Software may resize an image, reorder values or apply a rule to make values comparable under the intended conditions. Such a **transformation** can affect the classifier's input even when the original file stays the same. **Normalization** is the latter operation: applying a defined preparation rule to the values.

LumaCell's identified preparation procedure proposes deterministic normalization: the same specified input and settings would produce the same transformed representation. "Identified" here means that the proposal has a name. The sequence, parameters and executable form still need specification, implementation and acceptance. They also need a link to the images used to examine the candidate model. Otherwise two experiments can use the same crop files while feeding different values into the calculation.

An eligible transformed crop would reach the planning classifier, the manufacturer's experimental model. Its output still needs a rule that turns it into one of the five common-cell suggestions or the other/unclassified route. The selected score-to-action rule or threshold is the **operating point**. No numerical point is supplied. The display then has to attach the selected state to the correct current item. Model calculation, routing and screen behavior are connected work, but evidence about one does not silently establish the others.

Where does the original quality gate belong in this sequence? The case does not yet settle what representation it examines, whether it acts before or after normalization, or whether its decision concerns a crop or a package. We can state the logical dependencies: an identified item needs an eligible representation and quality decision before an ordinary classifier result can be used; classifier output and routing rules then need to become a visible current state. Putting the boxes in a single ordered row would supply an implementation choice the records have deliberately left open.

The pathology and sequencing episodes make the final crossing easier to see. A queue rule turns a computed priority into a position after import; a sequencing rule turns candidate analysis into the suggestion a reviewer sees. Before release, each team would identify that rule and its displayed state. After the recorded observations, the rule actually used would help distinguish computation from routing. The observed queue movement or correct display alone does not identify every rule behind it.

Reviewer Lens. When two records name the same classifier, the productive comparison is often just upstream and downstream of it: what values entered, and which rule converted its output into action? For LumaCell, the gate's input and unit also remain material. A difference at one of those crossings justifies a focused applicability explanation. It does not justify assuming either that the complete paths are equivalent or that every prior result is useless.

Consultant Lens. These choices have an order even while execution order remains open. Specifying preparation and the gate interface lets the team choose a candidate operating rule for a defined task and design tests of the resulting visible route. Software, supplier and clinical owners need to settle the gate's representation and crop-to-package handling together. Leaving that conversation until after experiments encourages people to interpret results against different imagined products; the manufacturer must accept the actual design.

What does the blank space leave unfinished?

Suppose no ordinary suggestion appears for a received item. The next action depends on why. Missing or mismatched identity requires an identifiable item for reconciliation. A quality rejection requires the professional to see what was blocked and whether reacquisition or manual morphology is the route. An accepted other/unclassified output belongs to an eligible image and its review route; it is different from material rejected by the gate. If calculation or display is unavailable, pending work still needs an owner and a route back into laboratory work.

These distinctions have to survive interruption. The information retained about what has happened to an item is its **state**. Receipt, completed inference and completed professional review are different events. A repeated delivery might be a retry of the same item; reacquisition might create a new linked attempt. The manufacturer has not selected the retry or package-state implementation. The planning requirement is to retain enough identity and history that an unresolved attempt cannot look reviewed and a prior item's suggestion cannot look current.

That requirement also explains why a screen alone cannot establish completion. The pathology team would distinguish a slide waiting for import from an imported case without a usable priority, and both from a reviewed case. The sequencing team would distinguish a displayed suggestion awaiting identity reconciliation from a released report. These are prospective state distinctions, not reported failures. The two positive field observations did not test these branches, so absence of a suggestion cannot be counted as either a negative finding or finished work on their authority.

Reviewer Lens. A normal path can look persuasive while all exceptional paths disappear into one blank screen. Following one non-result to an accountable next action exposes what that picture hides. At this stage a coherent proposed distinction among identity problems, rejected quality, accepted other/unclassified and outage is enough to scope requirements. The implemented transitions and human response still need their later evidence.

Consultant Lens. States should be agreed before test cases and monitoring denominators, because each count must refer to something the product actually distinguishes. Software and laboratory owners need to define what finishes work, while supplier and product owners settle how crop decisions combine. Assuming every

rejected crop rejects a whole package would invent a rule here. Once the rule exists, interrupted and repeated deliveries can be challenged against an expected result.

What has to be present for the route to run?

The classifier file cannot execute itself. Product code loads it, reusable numerical code performs calculations, physical equipment runs them, and site settings determine how packages arrive and states appear. The software environment executing the code is the **runtime**; reusable code it relies on is a **library**. To attach evidence to a future released baseline, the manufacturer would identify the relevant combination and supported variation. No actual runtime version, complete user interface or accepted local installation is supplied yet.

LumaCell proposes local packaged inference with a model fixed within a release. It does not call a live model provider through an **application programming interface**, the specified request-and-response connection between programs. That narrows one source of outside behavioral change, but leaves other software dependencies to identify. “Standard hardware” also needs a stated range or an explanation of why a variable cannot affect the conclusion in question. The result belongs to the conditions examined.

A changed condition therefore raises an impact question. The International Medical Device Regulators Forum's terminology document IMDRF N67 distinguishes product changes from changes in the data environment. That distinction helps locate the question; it does not answer it for LumaCell. [S1] The manufacturer can compare a proposed setting with the tested setting, identify a possible effect on identity, numbers, availability or display, and use a focused check or a documented unaffected relationship to justify what evidence travels.

For the pathology aid, scanner import and queue software may operate under different site conditions. For sequencing, specimen reconciliation and suggestion display may depend on different laboratory systems. Before release, their teams would identify supported conditions alongside their tests. Applying either field observation to another site or changed installation would then require the corresponding comparison. A site inventory or compatibility test can address it more directly than “the software still runs.”

Reviewer Lens. A changed platform does not automatically reopen every conclusion. The useful question is how the changed element could reach the claimed result or user action. A documented absence of that dependency can justify retaining a bounded conclusion; a changed numerical or display path may need a targeted bridge. The record must identify the actual settings on each side so that this is an explanation another reviewer can check.

Consultant Lens. Deployment choices create future evidence costs. Defining supported variation and preserving execution conditions with results makes a later comparison

possible. Software and deployment owners also need a way to investigate a changed or unavailable dependency. Comparing a site's actual setup with the proposed supported range can reveal a focused compatibility question before a generic local-package description is mistaken for readiness.

What must the supplier's gate evidence connect to?

The original quality gate has a known version and interface in the planning case. Manufacturer acceptance and change-notice arrangements are not approved. Its job matters even though it never names a white-cell category: by accepting or withholding material, it can change which images reach the planning classifier and therefore which population an evaluation describes.

That consequence tells us what to ask of a supplier record. What representation does the gate receive, and what states does it return? A benchmark or certificate may characterize the component within specified conditions. It cannot establish that a LumaCell package retained the right identity, that integration supplied the intended representation, or that a blocked item remained visible to the professional. Those questions require manufacturer evidence about the assembled route. The quality-system standard ISO 13485:2016, through its purchasing and design-change provisions, supports control of relevant supplied contributions and changes; it does not prescribe this chapter's map. [S2]

The invoice analogy returns at precisely this crossing. An outside calculation service could supply a correct amount while the office still needs to match it to the shipment and posting rule. In LumaCell, agreed supplier evidence can be useful while the manufacturer's acceptance remains open. The manufacturer must judge fit to this crop route and control for the stated activity. The analogy clarifies the ownership split; it supplies no medical-safety evidence.

A pathology queue might similarly depend on an import or image-processing contribution, and a sequencing display on upstream analysis or specimen-link information. Those are interface questions, not newly assigned suppliers in the fictional episodes. Each team would specify the contribution and what integration must verify; a later change would reopen the affected input, output and evidence path. Neither observed timing constraint identifies a supplier change or its cause.

Reviewer Lens. The decisive reading places the supplier claim beside the manufacturer's interface requirement. A detailed component file may still answer a smaller question than package-state acceptance. Useful evidence deserves credit at its actual scope; the remaining request should concern the representation, output states, version or change information needed for the manufacturer's decision. The goal is an assessable crossing, not a dossier of indefinite size.

Consultant Lens. Purchasing can request information only after software, clinical and quality/risk owners know what they need to decide. Agreeing acceptance for a stated development activity and the change information needed afterward makes that request concrete. The adviser can connect those owners and frame alternatives; the supplier's name and the adviser's recommendation do not accept the component for the manufacturer.

Who can close the interface question?

Suppose the supplier has delivered the agreed gate test. Who may now say its contribution is acceptable? “Shared responsibility” leaves that question unanswered. Someone produced the record, someone must examine what it establishes, and someone must decide whether it is enough for the manufacturer's stated activity.

These are three roles even when one person holds more than one job. The **evidence producer** creates the specification, record or test. The **reviewer** checks it against the question being asked. The **disposition owner** can accept, reject or require change for the stated activity. A sound supplier test can therefore coexist with an open manufacturer integration decision. A laboratory's acceptance of its procedure cannot make that manufacturer decision either.

Following the crop makes the allocation tangible. The laboratory produces preparation and specimen records; the manufacturer defines supported conditions and the compatibility evidence it accepts. The gate supplier supplies component information; the manufacturer specifies and evaluates its integrated contribution. Software and human-factors teams develop and examine the displayed work state; the professional performs clinical review. A site deployment owner checks its installation against agreed local conditions, while the laboratory keeps patient-result release.

The proposed record joins in the transfer cases need the same separation. A pathology site could supply import and pathologist-review times, the software team queue events, and a workflow reviewer the case-linkage check. For sequencing, the laboratory could supply reconciliation and release records, the product team display traces, and a clinical reviewer the specimen-linkage check. Each manufacturer would then decide what benefit claim survives. These are future assignments, not records already supplied by the episodes.

Reviewer Lens. A named component owner can sound reassuring while acceptance remains unassigned. The narrower question is who can dispose of this evidence for this activity. A clear allocation is a useful planning result even before the tests exist. What would be inadequate is a chain that ends at “supplier responsible” without reaching the manufacturer's integration decision.

Consultant Lens. The handoff becomes executable when it specifies what record goes from which producer to which reviewer, for which decision and with what closure

record. That conversation belongs before decisive evaluation, while missing information can still be requested. A consultant can make an absent owner visible and recommend an assignment; becoming the missing acceptance authority would leave the manufacturer's responsibility unresolved.

Connect the route to its records

A classifier result now has somewhere to go: into a description of the route that produced it and the decisions needed to use it. The **proposed system map** has two linked views. One follows execution and configuration; the other follows evidence and responsibility at those same crossings. Keeping the views linked prevents a convincing route diagram from hiding an unassigned acceptance decision.

The following field guide helps locate what each view must account for. A **user interface** is how the professional sees and acts on the item. A **version baseline** is the connected set of applicable component, setting and procedure identities to which a requirement or evidence result belongs. The table is useful when a record names one component and the reader needs to know what else the claimed result depends on.

Part of the account	What needs an identity or description
Functions	Intended behavior, including work without an ordinary result
Model	Candidate or release, weights and the record defining them
Image changes	Acquisition and software transformations, their sequence and settings
Rules and operating point	Thresholds and processing that turn calculation into action
Display and workflow	Visible states, review actions, escalation and fallback
Platform and hardware	Runtime and supported execution conditions
Hosting or program connection	Supply arrangement, control of behavior, availability and investigation route
Supplied components	Versions, interfaces, evidence and change responsibilities
Site setup	Permitted local variation and acceptance dependencies
Version baseline	The connected combination covered by requirements and evidence
Responsibilities	Who supplies, verifies, reviews, approves, operates and responds to change

The model row identifies the planning classifier; the baseline row asks which preparation, gate, rules, runtime and display accompany it. Some of those identities are procedures or profile states, so software version numbers alone will not complete the connection. The hosting row also has a useful answer for a local product: live provider inference is absent, and the packaged supply arrangement is identified. An explicit absence prevents a later reader assuming a hosted model.

In the pathology example the map would connect imported-case identity, priority rule, worklist, scanner interface and pathologist review. In sequencing it would connect specimen, analysis, suggestion, display, reconciliation and release. The observed functioning queue or display can occupy its actual place, while a separate linked record remains needed for the broader time outcome. The map lets a positive narrow conclusion and an open broader question coexist.

Reviewer Lens. These eleven fields do not add up to a score. They help test whether a favorable observation can be traced to a particular input, rule, display and execution setting. If that chain is clear, the supported conclusion can close. A missing link should be requested for the claim that needs it, rather than turning every open field into a reason to disregard the whole record.

Consultant Lens. A useful planning map develops in dependency order: the role and ordinary behavior explain the input and supplier interfaces; those explain execution and site needs; together they define the evidence-linked baseline. Unknown settings need an owner and next decision. Guessing values makes the table look finished while making the future test less reliable.

What the planning case actually supplies

The fictional LumaCell proposal still concerns adult supplied-candidate morphology review, trained professional checking, full smear review and laboratory-owned patient-result release. The case key below ties meaningful names to the engineering labels used in the underlying records. It distinguishes supplied planning facts from implementation and acceptance evidence.

Planning element	Supplied now
Proposed intended-use contract (IU-1)	The adult role and exclusions established in Chapter 2
Proposed acquisition profile (AP1)	Named acquisition and candidate-location contributions (ACQ1/LOC1), and proposed deterministic normalization (P1); selection and complete-slide adequacy remain separate obligations

Planning element	Supplied now
Identity-field requirements (IF-1)	Required order, specimen, acquisition, profile, locator, crop, timestamp and image-hash fields
Planning classifier (M0)	An experimental manufacturer classifier
Original quality gate (Q0)	A supplied predictive gate with known version/interface; acceptance and change-notice decisions are unapproved
Initial development image collection (DEV-1)	Development-only images grouped by patient
Professional route	Trained review and reclassification, visible non-results and laboratory manual morphology fallback
Supply arrangement	Local packaged inference without a live provider model connection

The profile row does not establish execution order or accepted numerical values. The identity row gives requirements, while an actual package would carry values to check. The gate row still leaves its input representation, placement relative to normalization and combination of crop decisions into package states open. No particular runtime, complete interface implementation, accepted local deployment or activity-specific permission for the image collection is supplied. These are conditions for planning an unreleased proposal, not observed failures of a deployed product.

Decision pause — where should this evidence stop?

The team proposes: “The planning classifier is the only learned classifier issuing cell suggestions, so the evidence boundary can stop there.” The earlier teaching gives us the general boundary lesson. The harder question is how to apply it to these particular records without guessing the missing design.

Decision

What useful conclusion could classifier evidence support, and where would that conclusion stop? For the proposed professional route, which crossings need an input, an output or non-result, and a controlling record? How would the same crossings appear in an owner view showing the evidence producer, reviewer and manufacturer decision owner? Which unresolved conditions could change that decision?

How the two map views make the decision reviewable

Worked reasoning after the pause

Evidence about the planning classifier may answer a useful model question. To answer the proposed intended-use question, however, an item must reach it with the intended identity and representation, pass the chosen quality conditions, and become the correct visible state for a trained professional. The gate can change eligibility; the rules and display can change what the person encounters. The exact additional evidence depends on the manufacturer's chosen route and claimed conclusion.

The proposed system map, recorded as MAP-1-P1, is an author-designed planning artifact. The product owner owns its disposition, with quality/risk coordinating review. The decision is whether the boundary is clear enough to plan the next work. No accepted implementation, clinical performance or release follows from that planning disposition. The E-labels below are shared row references between the two views, so the responsibility for one crossing can be found without mistaking another crossing's evidence for it.

View 1 – what crosses each interface?

The first view answers where the crop and its identity go, what comes back, and what remains to do when an ordinary suggestion cannot proceed. Preparation leads to a received item. Representation and quality then determine eligibility for classification, but the representation and quality rows do **not** establish which runs first. Routing joins their information into a current display; professional review leads to the laboratory's separate result decision. The execution-environment row applies across the software route, rather than occurring after laboratory release.

Crossing	What enters and controls it	What reaches the next step	If ordinary work cannot proceed	What remains open
E1 — preparation to package	Specimen, laboratory preparation and proposed acquisition/profile/locator	Candidate crops with acquisition/profile identity reach receipt	An incompatible preparation or profile cannot count as supported input	Local compatibility and selection evidence; proposed profile acceptance
E2 — receipt to identified item	Package values for all required identity fields	An accountable item and linked crops reach preparation/quality and the state record	Missing or mismatched identity remains visible for reconciliation	Implemented checks and rules for retry or reacquisition

Crossing	What enters and controls it	What reaches the next step	If ordinary work cannot proceed	What remains open
E3 — image to classifier representation	Source crop and proposed deterministic normalization	Identified transformed values reach the classifier; any gate use needs specification	Failed transformation or unsupported representation becomes a visible non-result	Sequence, parameters and executable identity
E4 — quality decision to eligibility and state	Versioned gate; its input and crop/package unit need definition	Quality decision controls classifier eligibility and package state	Rejected or unavailable quality needs a defined route; missing status is not acceptance	Supplier acceptance, notice arrangements and mixed-crop aggregation
E5 — classifier to suggestion rules	Experimental classifier and eligible identified values	Candidate inference reaches category/routing interpretation	Unavailable or uninterpretable inference reaches non-result handling	Candidate evidence applicability and operating/routing rule
E6 — routing/state to current display	Received identity, quality status, classifier output and identified rule	Current accepted, blocked, unavailable or resolved state with its history reaches review	Interruption, duplication or resumption must retain the item and avoid another item's suggestion	Aggregation, state implementation and numerical operating point
E7 — display to review or fallback	Current item, suggestions and visible limits/no n-results	Confirmation, reclassification, escalation or manual morphology reaches laboratory review	Unresolved work retains an owner	Interface identity, task sequence and effectiveness; visibility is not proof of action
E8 — laboratory review to patient result	Complete review obligations and responsible-professional procedure	Laboratory interpretation and release, outside autonomous product reporting	Incomplete review cannot become a normal or negative software result	No whole-smear or clinical-benefit evidence is supplied

Crossing	What enters and controls it	What reaches the next step	If ordinary work cannot proceed	What remains open
E9 — local environment across E2–E7	Packaged code, runtime/libraries, hardware assumptions and site settings	Relevant result and state records identify execution conditions	Unavailable dependency/runtime needs accountable recovery and non-result handling	Actual runtime, supported variation and site acceptance; no live provider inference

The receipt row illustrates why this is a planning map. If an identity value is missing, its route to the state record preserves an accountable item instead of displaying a plausible suggestion against uncertain source material. That gives an engineer an intended behavior and a tester a future challenge. It does not assert an implemented passing check or an event that has occurred in LumaCell.

The quality row makes another dependency visible. Rejected material may prevent ordinary inference, but the right item's non-result still has to reach the state record and professional. If a package contains different crop responses, expected behavior must be chosen before a test can judge that package. The table has not selected the crop-to-package rule. Nor does the preparation row's familiar procedure name supply its executable sequence or parameters.

The initial development image collection sits beside this clinical execution route. Its images, patient grouping, permissions, references and transformations need links to a model experiment. Possession does not grant permission for a particular activity or make the collection an independent evaluation set. None of those development records should be mistaken for a patient's runtime input.

The transfer cases show how this view selects the next evidence. For pathology it would distinguish pre-import slide, identified imported case, priority rule, queue state and pathologist review. Queue timestamps alone cannot measure preceding import, so a total-time question needs linked events for the same cases. For sequencing it would distinguish specimen identity, candidate analysis, displayed suggestion, reconciliation and release; a display trace needs a separate linked release record to address report time. These proposed joins add no new field observations.

Reviewer Lens. One crossing is a manageable review question: what reached its next consumer, including on the non-result branch? A classifier result without a trace through routing and display remains creditable model evidence while professional-use evidence stays open. Clear identity and quality plans with explicit unknowns can be adequate boundary work. Calling them passing implementation tests would remove the reason for the next development step.

Consultant Lens. The expensive ambiguity is the one a test designer must resolve by guessing. Preparation and gate representations come first as decisions, then crop/package meanings, then current-item display and fallback traces. Software, supplier and clinical owners can agree those dependencies before test execution. In the transfer cases, the corresponding advice identifies the event and identity joins needed for the broader workflow claim; the manufacturer chooses whether to pursue that claim and study.

View 2 – who can accept the evidence at that crossing?

The second view uses the same row references to ask who makes each interface reviewable. A proposed record name is an assignment, not a finished report. The last column matters because it states both the present uncertainty and what later change would bring the decision back.

Crossing and proposed record	Who produces evidence	Who checks it	Who makes the decision	Open condition or reason to return
E1 — preparation/acquisition compatibility	Site supplies preparation/acquisition records; manufacturer defines compatibility studies	Clinical and imaging specialists, with quality review	Manufacturer accepts supported-profile evidence; laboratory separately accepts its local preparation/use	Selection coverage and local compatibility; preparation, locator or profile changes
E2/E3 — identity and preparation interface	Software/model team supplies specifications, transformation history and tests	Software verification and data/reference reviewers	Manufacturer software-baseline owner under the development plan	Executable identity, sequence and record checks; changed schema, normalization or identity handling
E4 — gate acceptance and change record	Supplier provides component/version information; manufacturer supplies integration evidence	Software/security and quality/risk leads	Manufacturer accepts the gate for a stated activity; supplier does not accept the assembled device	Representation, aggregation, evidence and notice terms; changed gate behavior, version or support

Crossing and proposed record	Who produces evidence	Who checks it	Who makes the decision	Open condition or reason to return
E5/E6 — classifier and routing /state trace	Model team identifies experiments; software team supplies rule/state evidence	Model/statistical, clinical and verification specialists for their respective questions	Designated manufacturer candidate and system decision owners	Candidate, threshold, aggregation, retry or state-rule changes may affect different conclusions
E7/E8 — interface, essential user-task and fallback record	Human-factors/clinical team supplies use evidence; laboratory supplies procedure and workflow context	Human-factors and clinical specialists with quality/risk review	Manufacturer accepts product-use evidence; laboratory retains patient-result and local-use decisions	Changed users, display, task sequence or fallback reopen the relevant use question
E9 — environment and site configuration	Software/deployment team supplies runtime inventory and compatibility evidence; site supplies actual settings	Software/security and site deployment reviewers	Manufacturer accepts supported conditions; laboratory accepts its installation/use	Unknown runtime /build/settings; dependency, platform or local-setting change needs an applicability decision

Reading the gate row across both views explains the ownership split. The supplier describes the component. The manufacturer must still specify the representation and package rule, assess the integrated behavior and accept it for a stated activity. Useful component evidence can be adequate on its own terms while that decision remains open. Once the required questions are answered and planned checks pass for the stated scope, the bounded acceptance question should be closable. An indefinite request for a “more complete” supplier file would have no corresponding decision left to serve.

The preparation and professional-use rows separate two other decisions. The laboratory can supply its workflow and preparation records and accept its own procedure or installation. The manufacturer decides which profiles and professional interactions the product supports. Site signoff cannot stand in for manufacturer release, and manufacturer acceptance cannot stand in for a laboratory's patient-result decision.

Likewise, the candidate/state row does not give unlike model, clinical and implementation questions one common pass status.

For pathology, a site could produce scanner-import and pathologist-review timestamps, the software team queue records, and a clinical/workflow reviewer could check that they concern the same cases. The manufacturer would decide the time claim; the site would retain its own scanner and review operations. For sequencing, the laboratory could produce reconciliation and release records, the product team display traces, and the clinical reviewer could check specimen linkage. Correct display would remain supported without making reconciliation disappear. These are assignments for a new joined record, not completed evidence.

All eleven field-guide questions now have locations: functions and professional work span preparation through laboratory review; the classifier has its own crossing; transformations sit at acquisition and preparation; quality and routing have separate rows; supplier contributions and site conditions meet their respective owners; local execution applies across the software path. The baseline connects those applicable parts and procedures. A marketing version alone would not do so. A later controlled baseline would replace open settings with accepted records; this planning map must leave the current gaps visible.

Reviewer Lens. A producer's claim, its specialist review and the accountable disposition should concern the same crossing and scope. A component certificate may leave package behavior open; a successful defined user action may leave patient benefit open. Enough means that the named question is answered, with the next question retained at its own level. This permits positive closure without allowing one signature to drift across the whole table.

Consultant Lens. The final column is a practical agenda. For the gate it leads to representation, crop/package rules, component evidence, integration checks and change-information arrangements. For the transfer cases it leads to record linkage before broader benefit advice. The consultant can organize that work and recognize when the named owner has an adequate answer. The manufacturer makes and records the acceptance decision.

Figure 3.1 — The Released Product Is More Than the Model. Model weights are one controlled component inside a versioned product, supply, deployment, and responsibility boundary. Use the figure to check whether your two views show the path and owners around the model; the drawing does not resolve the open Q0 or runtime choices.

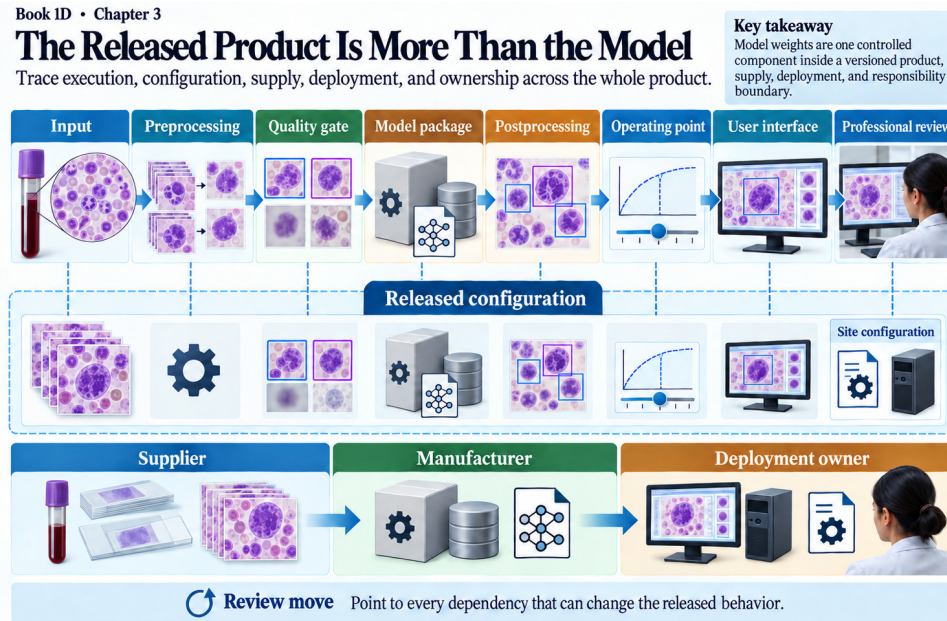


Figure 3.1. The Released Product Is More Than the Model.

Which conclusion does a change actually touch?

Suppose an experimental score were measured on the initial development crops through the planning classifier. The map would let the manufacturer locate a useful model result and ask what a proposed intended-use conclusion additionally depends on: input profile, reference process, preparation, gate selection, operating rule and professional route. A result can remain valuable while the broader claim needs another record.

Now change one thing hypothetically: the model weights remain fixed, but input preparation changes. Different bytes or numerical values could reach the classifier. If the gate uses transformed data, its input could change too; its unresolved interface prevents assuming either effect or no effect. Comparing the old and proposed paths tells the team which outputs and earlier conclusions depend on that preparation. Those relationships may need a focused bridge or new testing. Unchanged weights do not supply equivalence, and an unaffected dependency does not demand an unrelated repeat study.

The same test works backward from a field record. In the separate pathology episode, the queue trace supports placement while import constrains total time. A hypothetical import-interface change would call for renewed timing and linkage assessment, without automatically calling for another priority-model study. In sequencing, a hypothetical reconciliation-procedure change could affect the release question while correct display

remains supported. Neither change occurred in the supplied histories; each shows how a record chooses the evidence to revisit.

Reviewer Lens. The most useful change explanation names an element, the mechanism by which it could affect an output or action, and the prior conclusion at stake. A trace and test can show that a particular conclusion has no dependency on the change. A reasoned no-repeat decision is then a positive review outcome for that scope, while other claims retain their own evidence status.

Consultant Lens. Comparing configurations interface by interface makes the options concrete: which identities stay fixed, which behavior might change, which evidence still applies, and which targeted observation would resolve uncertainty? Specialists can examine the affected mechanism. The resulting bridge should serve a named manufacturer decision rather than become a generic promise to retest everything later.

Whose signature accepts the assembled role?

A supplier test, a local installation decision and the product owner's disposition could all be adequate and still answer different questions. The supplier provides the agreed gate, its version/interface information and change information. The manufacturer's clinical, software, machine-learning, quality/risk, security and human-factors functions must make the contribution testable and judge its fit. The clinical lead needs to recognize the proposed work; software and model leads need an identifiable behavior; quality/risk needs the trace from requirement to evidence and decision. These remain planning responsibilities, not completed approvals.

The laboratory keeps preparation and local-use decisions under its procedures, full-smear review, patient interpretation and release. The manufacturer specifies the conditions its product supports and later accepts evidence for that product. If the site's condition falls outside the proposed profile, the manufacturer needs a compatibility decision and the site needs its local-use decision. One party's general signature cannot answer both.

That division preserves positive evidence in the transfer cases. Pathology site operations can supply the import and review records, and the sequencing laboratory the identity and release records, while product teams judge their wider benefit claims. The observed queue and display functions need not lose their supported status while those joined workflow questions remain open.

Reviewer Lens. A chain of signatures can obscure the point at which product acceptance stops. Reading each decision sentence for its object and activity makes the missing disposition visible. A focused request for that decision is more useful than challenging an adequate supplier or site record merely because it does not purport to accept the whole product.

Consultant Lens. An ambiguous acceptance criterion is often a reason to bring functions together before writing more evidence. A useful decision sentence names the activity, evidence, reviewing functions, accountable manufacturer owner and closure condition. The adviser may prepare that sentence and recommend a route; it cannot approve the gate, product release or patient result for the responsible parties.

Consultant assessment — a gap someone can close

A detailed diagram can still omit the crossing that decides what the professional sees. The consultant's assessment therefore starts with the proposed system map and connects a specific boundary or responsibility gap to the development decision it prevents. The useful output identifies the unbounded contribution, its uncontrolled dependency, the missing ownership, the information needed and the manufacturer closure decision.

Which route do the records actually describe?

Suppose a route diagram says the gate receives source crops while an experiment description says it receives normalized crops. This is a hypothetical disagreement to preserve, not a choice the adviser should settle by drawing the tidier arrow. Neither representation is established at this point. Following the proposed item through both descriptions exposes exactly what needs reconciliation.

The same reading separates required identity fields from the values in a particular package, and absent future runtime details from a defect observed in use. A trace that marks what is known and open gives the reviewer something to compare with the proposed intended-use contract. It gives the manufacturer the decisions to settle before choosing decisive tests.

What does the missing rule prevent?

“The gate comes from a supplier” does not explain a planning consequence. The missing representation and crop-to-package rule do: a verification case cannot know the expected display for a mixed package until those choices exist. A later monitoring count might also use a denominator that differs from the implemented work unit. A package-state specification answers that dependency; a thicker supplier dossier might leave it untouched.

The next crossing is human work. “A professional can reclassify” identifies an available action but leaves open which items appear, who retains unresolved work, what must be checked and what records completion. A warning or override button cannot substitute for that connected route. These questions give human-use evidence its object.

Who needs to decide what?

The supplier can provide agreed behavior, version and support information. Software/security and quality/risk staff can examine integration and change implications. The product owner or designated manufacturer function can dispose of that evidence for the stated development activity. The adviser tests whether this path is executable and recommends an owner where needed, without becoming the acceptance authority.

A site mismatch similarly needs two decisions. The laboratory controls actual preparation and patient-result work; the manufacturer defines supported preparation and use. The adviser can identify the mismatch and the corresponding compatibility and local-use decisions. Laboratory ownership does not remove input relevance, and describing that relevance does not give the manufacturer control of every laboratory step.

What would a useful finding say?

The table follows the unresolved quality decision into a specific information request and closure condition. It can be used when a finding says “more detail needed” but nobody can tell which decision the detail will enable.

Question	Bounded proposed finding
Which behavior is unbounded?	Package behavior when the gate does not accept some or all input material
Which dependency lacks an accepted arrangement?	Gate acceptance and change information for the planned product activity
Which responsibility needs clarification?	Producer, reviewer and decision owner for the aggregation specification
What information could resolve it?	Controlled specification, applicable input/output states, testable requirements and assigned roles
What would close the planning gap?	Manufacturer-reviewed behavior and ownership linked to verification, workflow and later monitoring plans

The information-request row is actionable because its specification would say whether quality decisions concern crops or packages, how mixed responses are represented, and what reaches the current display. Software staff can derive expected behavior and tests; human-factors staff can examine what the professional sees; quality/risk can assess whether those records answer the concern. If the manufacturer accepts that connected record for the bounded planning step, the gap can close at that level. Later performance

remains a separate question. No gate malfunction or failed acceptance is asserted by the finding.

Does the offered reply answer that finding?

“The classifier weights are unchanged” may be true and useful, yet leave the requested gate input, package rule and display unspecified. Comparing the paths shows whether the affected interface matters to the earlier conclusion. The answer could be no new work for a model-only result, a focused integration check, or a broader reassessment. The claim and evidence choose among those outcomes.

A supplier's compatibility statement needs the same reading. It may establish that a file or call can still be exchanged while leaving the meaningful response under intended conditions unexamined. The adviser can credit the exchange proposition and identify the manufacturer check still needed for behavior. This distinction prevents both automatic acceptance and automatic repetition of every study.

How does the finding end?

A bounded advisory conclusion could read: “The proposed functional path is identifiable. Package-state meanings, supported runtime conditions and gate acceptance/change responsibilities need the named manufacturer decisions before evidence can be scoped to a stable implementation.” Re-review would compare chosen specifications, linked plans or tests and owner dispositions against those three questions. Each can close when adequate for its purpose; planning closure supplies neither clinical-performance evidence nor release permission.

Reviewer Lens. A finding with a consequence, a requested record and a closure test gives the reviewer a stopping point. When identity requirements, gate interface and state records agree and the manufacturer has reviewed the planned behavior, the boundary question can close. Actual implementation, performance and use retain their distinct evidence questions. Requiring the planning finding to resolve all of them would obscure what it originally asked.

Consultant Lens. The product owner needs choices with dependencies: settle the interface and aggregation rule, choose an evidence route for the assembled product, and assign the reviews. On return, the adviser can state which manufacturer decisions are actually recorded and which remain open. A recommendation becomes useful by making acceptance possible, without purporting to accept supplier risk or configuration for the company.

Decision pause — the same job through a different supply route

Consider a counterfactual, not a LumaCell event. The quality-gate function would now be supplied through a remotely operated application programming interface whose provider can update its behavior. The manufacturer knows the interface name but has no evidence of a behaviorally pinned version. The intended diagnostic role stays unchanged.

Decision

Which crossings in the execution view and which evidence questions or owners in the responsibility view would change? What is the next manufacturer decision? Which part, if any, of the earlier local-package control argument could still support it?

How a reviewer follows the changed supply arrangement

A similar function can now depend on a different configuration and division of control. The manufacturer needs to establish what identifies the actual behavior, how provider changes become known and assessed, which earlier evidence remains applicable, and how unexpected responses can be investigated. Service unavailability also needs a route that keeps the affected item accountable. Data flows, security and applicable permissions need their own specialist assessment.

The old local-package control argument cannot simply travel with the interface name. The manufacturer could establish adequate controls for the remote arrangement, narrow supported conditions, select another dependency or decline this supply model. The next decision concerns whether and how that arrangement can support the intended activity with reviewable evidence and responsibilities. The exercise neither rejects all remote interfaces nor adds a changed supplier to LumaCell's history.

Where a specialist question can go next

A specimen or image dependency can be explored further in Book 1C. Book 1A Chapter 4 deepens software architecture and reviewable boundaries; Book 1B Chapters 2, 4 and 5 deepen data flows, security architecture and supply chains. These are optional sources of depth for assigned work. The local proposed system map still needs to show where an interface's security question belongs and who will decide it.

For a European Union assessment, Book 2 Chapters 5, 6 and 14 deepen technical documentation, suppliers and software boundaries. The engineering map supplies facts

for that assessment; a box's position does not determine legal device boundaries or a market route.

The next planning decision

Chapter 4 uses the proposed intended-use contract and system map to sequence requirements, risk controls, permissions, supplier work and evidence, including the still-open gate, package-state, runtime, input-compatibility and professional-work decisions.

Source trail

S1: IMDRF N67 (2022), sections 7.1.1–7.1.2, distinguishes product changes and changes in the data environment. This is international terminology, not an acceptance decision for LumaCell or a complete change-control method.

S2: ISO 13485:2016 official identity, sections 7.4.1–7.4.3 and 7.3.9, informs purchasing-control and relevant design-change responsibilities. It does not prescribe MAP-1-P1 or establish that a supplier record alone accepts an assembled product.

MAP-1-P1, its interface tests, the factory/accounting-office analogy, the DP and NGS transfer plans, and the API variation are fictional teaching constructions. No runtime, site installation, Q0 acceptance, clinical-performance result, or changed supply arrangement is asserted as LumaCell history.

CHAPTER 4

Development, Risk, Requirements, and Evidence Plan

A package of candidate white-cell images reaches the proposed LumaCell Review workflow, linked to a laboratory order. Some material may receive a suggestion; some may need a visible explanation of why ordinary output is unavailable. A professional must still check the current work item, complete full smear review, and leave interpretation and patient-result release to the laboratory. This is the proposed route from the preceding chapters, with no verified implementation supplied yet.

Now imagine that processing stops or quality rejects the package, and its work state disappears. Someone reviewing the remaining work could believe the missing item had been completed. What would the manufacturer have to settle before a test of that route could mean anything? It needs an expected state to compare with the observed one, a way to keep the received item identifiable, an action the professional can recognize, and an owner for the resulting decision.

The proposed intended-use contract from Chapter 2 and proposed system map from Chapter 3, linking product interfaces to evidence and decision owners, locate that work. A development plan connects their open questions to the activities that can answer them. An image collection, component name or scheduled experiment cannot supply the missing answer merely by appearing in the plan.

What makes a date meaningful

“Test the package route next month” sounds actionable until someone asks what the test would permit afterward. Could it support another experiment, acceptance of a component for one activity, or a conclusion about intended professional use? The proposed work needs four connected answers: what activity is being considered, what must be true before it starts, what evidence it will produce, and which decision that evidence could support while other decisions remain open. Dates become useful once these dependencies are clear.

For the disappearing-package possibility, those answers form a trace from a plausible diagnostic consequence through requirements, controls and evidence to a bounded review decision. The consultant can examine that same trace in a readiness matrix, showing a missing condition and what would resolve it. That advice cannot approve the

manufacturer's plan or risk decision. The earlier intervention brief, intended-use contract and system map remain the starting records, with their early limits intact.

Retrieve before reading on

The proposed package route gives four earlier distinctions a practical use:

1. How does a proposed clinical benefit differ from an output a model can calculate?
 2. Which two product elements other than model weights could change whether evaluation evidence applies?
 3. If a requirement has an acceptance test, what further question remains before concluding that the device meets its intended use?
 4. When a supplier performs an activity, which manufacturer decision still needs an accountable owner?
-

Decision

What specific specification, evidence review, acceptance decision or response to change would make “the manufacturer remains responsible” concrete here?

How the earlier distinctions become planning decisions

A calculated output does not establish that the intended user can achieve the proposed clinical benefit. Input acquisition, image preparation, decision rules, interfaces and workflow can each affect the route from calculation to use. Verification addresses specified requirements; validation must address intended use in the relevant product and use conditions. A requirement's acceptance test has value without answering all of those questions.

Purchasing an activity or component likewise leaves the manufacturer's product-acceptance decision to be made. The useful account identifies the contribution required, who evaluates the applicable evidence, who accepts it for the stated activity, and what happens when it changes. A general responsibility statement gives none of those steps by itself.

Those distinctions let a planning review ask what can be prepared and what remains unready for authorization, without demanding that the eventual product already be finished.

Locate planning in the product lifecycle

Suppose the next proposed test concerns whether a non-result package remains visible and accountable. Why should it? Because the professional must not silently lose work. The requirement turns that purpose into expected behavior; the software team then specifies and implements how the state will persist. A risk record follows what could happen if persistence or display fails. A software test can compare implementation with the requirement, while a human-use evaluation can observe whether a professional recognizes the state and acts. The decision record identifies what may follow from those findings. Each step needs the previous connection; a stack of similarly named documents would not make it.

The sources have different roles in this chain. The quality-management standard ISO 13485, published by the International Organization for Standardization, connects development stages, reviews, verification, validation, responsibilities and resources. The software-lifecycle standard IEC 62304, from the International Electrotechnical Commission, connects software work with the wider system; its scope does not itself cover final medical-device validation and release. The medical-device risk standard ISO 14971 concerns the risk process. They contribute to connected work within their scopes, rather than offering three interchangeable approvals. [S1, S2, S4]

For LumaCell, the connection must reach beyond its artificial intelligence (AI). The proposed learned classifier and supplied predictive quality gate meet the order/specimen interface, non-result routing and professional action. Each contribution needs an evidence producer, a reviewing function and someone authorized to make the manufacturer's bounded disposition. Otherwise a successful component test may have no identified decision to support.

Those records can be connected without becoming uncontrolled copies. In the Chapter 3 factory analogy, shipment, invoice and ledger records can refer to one another while each original remains authoritative. A requirement, risk-control record, software configuration and test result can be linked the same way. The index points to the controlled originals and their actual states. The accounting analogy explains the linking operation; it supplies no evidence that a clinical control works.

Reviewer Lens. A plan can look comprehensive because it names every expected document. Following one non-result through requirement, possible consequence, implementation, evidence and decision reveals whether those documents actually connect. Enough at this stage means the links and unknowns can be inspected. Counting document titles would miss the relationship that determines whether a later result is interpretable.

Consultant Lens. One ordinary package and one consequential non-result give clinical, software, data and quality or risk leads a shared object to discuss. Each can identify the record it will produce and the later decision that needs it. Where two functions expect

the other to resolve a link, the product owner has an actual planning choice to make. The consultant can expose that choice without substituting an advisory approval for it.

From a schedule to an executable plan

The calendar might put “complete validation” beside a date while leaving the rejected package's expected state undefined. The missing link becomes visible when the proposed next decision is stated precisely enough that another person could disagree with it on evidence.

Begin with the decision, not the document name

What would “complete” mean for this activity? The answer needs an identified configuration and use, a method, prospective criteria and a bounded next step. A decision sentence could ask whether this configuration meets these requirements in these conditions using the stated method and criteria. At an early point, the question might instead be whether an acquisition plan preserves the information a later evaluation will need. That inquiry does not decide whether the eventual device meets its intended use.

A review at which an authorized owner records whether evidence supports a named activity is a **review gate**. The word distinguishes it from the image quality gate inside the product. Design review, internal collection readiness, component acceptance, candidate evaluation and device release are different gates because they permit different work. None of these internal records can grant external site, ethics, privacy, consent or legal permission.

The activity also determines which evidence has to be collected. In the pathology queue episode, a prospective study of placement after import would ask a different question from a study of total slide-to-review time. The latter needs scanner-import timing. In the sequencing review episode, suggestion display and specimen-linked report release need different records too. Their existing observations help locate a next question; they do not supply an unreported benefit result.

Reviewer Lens. A useful test of the decision sentence is whether removing its configuration and criterion would leave the apparent authorization unchanged. If so, “validation complete” may be doing the work of an undefined verdict. The reviewer needs enough precision to know how a favorable result would be interpreted before it arrives. This keeps the later finding from selecting its own question.

Consultant Lens. The practical replacement for a vague milestone is an activity-specific question linked to an evidence producer, reviewing functions and manufacturer decision owner. For the queue example, the timing start matters before a study is arranged. For the sequencing example, the intended display or release endpoint

determines the linkage work. Those choices make the schedule usable without expanding its authority.

Give inputs and outputs enough identity to be reviewed

The package is not supposed to disappear. How would an engineer know what counts as meeting that expectation? Receipt must remain connected to later state changes, duplicate delivery, retry, resolution and display. A **design requirement** specifies needed behavior in a form that can be examined. The proposed role and mapped interfaces supply **design inputs** to that work. The specified or implemented result is a **design output**, which can then be checked against the input. ISO 13485 connects intended-use needs and relevant risk outputs to reviewable inputs and assessable outputs. [S1]

The identity-field requirements from Chapter 3 illustrate why these stages matter. They name the fields that should connect the received image to its source. One actual package contains values in those fields. A field list can be reviewed as a design input; it cannot show that an implemented check verified a package's values. The plan therefore needs both the specification and the later test of the chosen behavior, with someone responsible for each.

The test also needs to know which product path it concerns. The proposed acquisition profile, identified preparation procedure, planning classifier and original quality gate are current planning identities. The preparation procedure is a proposed deterministic normalization—given the same specified input and settings, it should produce the same transformed representation. The gate's input representation, position relative to that preparation, crop-level decision unit and aggregation into a package state remain open. A single model version cannot identify all of these choices.

Those unknowns have to reach the people making the next dependent decisions. The plan can assign a data charter to establish permitted use and source relationships, a reference plan to specify label production, and a dataset manifest to identify partitions and coverage. Their names locate future answers, owners and timing. They do not turn those answers into completed records.

Reviewer Lens. The revealing comparison is between the proposed requirement and the role or interface that produced it. Does the identity specification supply enough to plan a check, and would an unresolved gate input or package rule change the expected outcome? A plan may be clear about the missing choice and adequate for the next planning discussion. It cannot claim that an implementation passed before that choice and its evidence exist.

Consultant Lens. Specifications have an order too. Product, clinical and software owners can settle the behavior needed before a meaningful test is written; the data steward and clinical lead can establish the source relationships and

reference-production method before the image set is treated as evaluable. Filling an open setting from a convenient experimental script would hide the manufacturer's still-unmade choice. An explicit dependency makes that choice easier to resolve.

Assign competence as well as names

Suppose the plan identifies a reviewer for both morphology labels and interrupted software states. What would let that person challenge either method? A job title does not supply knowledge of the cells, the intended action or the software's failure routes. The work needs **competence** appropriate to its actual question, together with the resources to carry it out. ISO 13485 addresses competence, relevant training and its effectiveness, and participation by appropriate functions and specialists in design reviews. [S1]

For the same package, the clinical lead can define what the professional must recognize; software/security can specify and later test receipt and persistent state. The data steward preserves source and grouping links. The human-factors lead plans observation of the intended user's action, and quality/risk coordinates the risk review. These are coordinated functions, not a requirement for five full-time posts. The plan needs their skill, resource, producer, reviewer and decision authority, even when one person holds more than one role.

The predictive-learning risk guidance in ISO/TS 24971-2, an ISO Technical Specification, recommends multidisciplinary consideration and planning for monitoring, updating and retraining where relevant. If monitoring performance is considered necessary for safety, its methods, processes and supporting data need to be planned. The specification does not impose continuous retraining on every fixed-release classifier, and it excludes devices using large language models or generative AI. LumaCell's proposed local fixed inference is not such a service. [S5]

This returns competence to the question the person must answer. A workflow reviewer for the pathology episode needs to distinguish import time from queue time. A reviewer of the sequencing release question needs to understand specimen reconciliation and the laboratory's release tasks. Assigning that expertise prepares the next inquiry without withdrawing credit from either episode's earlier bounded observation.

Reviewer Lens. Attendance and competence are different evidence. The important question is who can challenge the proposed reference method, state test or human-use scenario, and what supports that ability. A realistic allocation can be credited while future results remain open. A list of names with no relevant skill or available resource leaves the work unplanned despite a full meeting roster.

Consultant Lens. A missing skill is easier to resolve when described as a specific contribution needed by a particular date. The adviser can connect each evidence task to its producer, independent or otherwise appropriate reviewer, decision owner and

resources. Data independence and review arrangements need their own rationale; a generic list of professional titles does not close either question.

Keep risk policy, criteria, controls, and evidence distinct

Imagine the interrupted package again. Its state is lost or displayed as finished; a professional assumes the work is complete; necessary review or action might then be delayed or omitted. This is a possible sequence to analyze, not a LumaCell incident. The clinical and risk team still needs to examine exposure and possible harm before assigning a risk level.

Where could that sequence be interrupted? Persistent package accounting could keep received work visible, and a distinguishable non-result could tell the professional what action remains. Such a measure is a **risk control**. Naming it in the plan identifies intended protection. It does not establish that the measure has been built or that it will interrupt the sequence.

The first evidence question asks whether the specified state and display behavior actually exist in the identified configuration: **implementation verification**. The next asks whether the measures interrupt the analyzed sequence under justified conditions: their **effectiveness**. Where the control depends on a person, observation must reach that person's recognition and action. A warning can meet its display specification and still be misunderstood. Conversely, one sensible user action cannot fill an untested required software state. The informative report ISO/TR 24971 explains this implementation/effectiveness distinction and supports integrated, evolving risk plans; it supports the risk standard rather than replacing it. [S3]

How will the resulting findings be judged? The manufacturer needs a governing basis for acceptable risk, its **risk policy**. Device-specific **risk-acceptability criteria** apply that basis to the assessments and must be set before the corresponding results are known. Neither policy nor criteria are product controls. Under ISO 14971, the risk-management plan connects scope, responsibilities, activities, reviews, criteria, control verification and production and post-production information work. [S2] Those connections make clear who will assess the results rather than choosing a convenient rule afterward.

After controls, **residual risk** remains to be assessed. The manufacturer must make the applicable acceptability decision through its process, accounting for unresolved findings. A planning table cannot prewrite that result as “acceptable.” Nor is zero unaccounted packages in a selected test suite equivalent to overall risk acceptance or a guarantee about field frequency. The suite asks an implementation question, which is only one link in the risk argument.

Other proposed sequences require their own links. A stale pathology worklist position could call for a queue-state control and observation of whether the pathologist sees the current case; scanner-import timing would answer another question. A wrong-specimen

linkage in sequencing could require identity control and a human check before release; correct suggestion display would not test that control. These are prospective choices, not new failures in the two episodes.

Reviewer Lens. A passing interface test is easy to stretch because it produces a definite result. An experienced reading locates it in the chain: policy and criteria, analyzed sequence, selected control, implementation evidence, effectiveness evidence, remaining-risk analysis and authorized disposition. The implementation result can be accepted for its scope while the human-dependent question remains open. That is a precise finding rather than an undifferentiated demand for more testing.

Consultant Lens. Choosing the test before locating the interruption point can produce an impressive result about the wrong link. Clinical and quality/risk leads can first state the possible harm and how the proposed control is meant to prevent the sequence. Software and human-factors methods can then examine their respective parts, with reviewers and a residual-risk decision owner identified. The adviser makes the missing decision visible; it cannot accept the risk for the manufacturer.

Plan supplier acceptance at the product boundary

Suppose the supplied gate rejects an image before the raw material is retained for evaluation. Its decision would shape the dataset as well as the product's visible output. Would a general supplier approval establish that this use is ready? The question depends on the contribution the manufacturer intends to rely on in this activity.

The supplier can provide version and interface information, component evidence and change information. The manufacturer specifies what it needs, assesses which evidence applies, verifies the assembled route where needed, and accepts or holds reliance for the defined activity. ISO 13485 addresses supplier selection and monitoring, purchasing specifications, verification proportionate to product effects, and applicable advance change notification. Those controls give the supplier contribution a reviewable place in the product. [S1]

For the original quality gate, the unresolved representation and decision unit are therefore practical questions. What does it receive? Does it decide per crop or package? Can it change which material is kept? If it selects material before raw retention, that selection behavior, acceptance basis and change control may be prerequisites to acquisition itself. A narrower design that truly retains all eligible raw material without gate selection and preserves necessary source and reference information could be assessed separately. Such independence has to be substantiated; it cannot be inferred from the label “raw collection.” Later acceptance of the gate for product use would remain a separate decision.

Component metrics or a certificate can contribute evidence within their scope while leaving assembled identity, routing and professional use unresolved. “Third party” also

does not settle security, licensing, service continuity or classification as software of unknown provenance. Those questions need their applicable specialist analysis. The planning judgment concerns which unresolved contribution this activity would actually use.

Reviewer Lens. Putting the purchasing/interface requirement beside the supplied evidence often exposes the exact gap: perhaps a known version is covered, but the relevant representation or package-selection rule is not. That is more informative than “supplier evidence inadequate.” An unapproved change-notice arrangement identifies a control decision still to make; it does not prove poor performance by the gate.

Consultant Lens. The first supplier question for the next activity is whether the gate changes selection or retention. Software/security and quality/risk can then specify the needed information, change terms and manufacturer integration check. A general request for a larger supplier dossier may delay work while missing that dependency. The product owner makes the bounded reliance decision after the applicable review.

Make evidence milestones prospective

A software test could inject missing identity, rejection, interruption and duplicate delivery, then compare the intake, stored state, display and resolution records. What would a pass show? Specified behavior in the tested system. A human-use exercise instead could observe whether intended users recognize the current non-result and perform the specified fallback in a representative task. The observations complement each other; neither method can silently replace the other. [S1, S4]

That is why an **evidence milestone** needs more than a completion date. It identifies the question, configuration, method, prospective criterion and rationale, producer, reviewer and decision. The account must be specific enough that a result has an interpretable meaning before anyone sees it. A strong model experiment can remain useful model evidence while the professional-use question awaits another method.

Some choices therefore have a last safe time. A proposed reference procedure must be settled before it produces decisive labels. The candidate and analysis method must be fixed before results are interpreted as confirmatory. Formal human-use evaluation needs an identified representative user, task, interface and fallback configuration. In the current planning state, the reference procedure is proposed, its handling of unresolved cases is not approved, and collection of the independent evaluation dataset has not begun. That exact status is more informative than “validation in progress.”

The International Medical Device Regulators Forum's Good Machine Learning Practice principles provide a broad check on expertise, representative data, training/test independence, suitable references, human–AI interaction, clinically relevant testing and monitoring. They do not supply LumaCell's protocol, sample size or authorization. Data

independence concerns relationships and exposure of the data; independence of test personnel is a different question. [S6]

For the pathology episode, a prospective time-benefit protocol would specify when timing begins and how imported cases are linked before the favorable queue observation is interpreted against it. For the sequencing episode, a release study would specify the same-specimen joins among suggestion, reconciliation and release. Those definitions also help after use begins: a new observation may match the planned outcome or reveal a new question, without being retrospectively promoted into a trial result.

Reviewer Lens. The endpoint, eligible unit, reference, configuration and criterion need to fit together. A method may be sound yet incapable of supporting the proposed decision. Asking what a pass could establish identifies that mismatch early. It also permits positive closure at the narrower level the method actually examines.

Consultant Lens. Reference, dataset, candidate and use-protocol decisions should be placed before the work that depends on them. Clinical, data, model-development and human-factors leads can then produce compatible records. In the field examples, the event join and acceptance rationale belong in the proposed study before a queue or display observation is offered as benefit. Sequencing these choices preserves the meaning of later evidence.

Specify what reopens the plan

Suppose the classifier stays fixed while the non-result display changes. The category calculation may be identical, yet the professional may encounter another signal. Which earlier conclusion should be revisited? The answer follows the changed dependency: a new profile can change input eligibility and reference applicability; changed gate behavior can change selected material; a changed display can change action.

A controlled impact record connects that mechanism to affected requirements, risks and prior conclusions. It identifies evidence that still applies, a focused check needed to bridge an uncertainty, and the manufacturer decision. ISO 13485 addresses identification and review of design changes, their product, risk and regulatory effects, and verification, validation as appropriate, and approval before implementation. It does not turn every edit into a requirement to repeat every study. [S1]

The episode applications are hypothetical changes, not added history. A scanner-import process change could reopen the pathology total-time comparison and timestamps without automatically invalidating the queue rule's analytical result. A specimen-reconciliation change could reopen sequencing evidence about release while the observed display behavior remains supported. The plan needs both possibilities: a reasoned unaffected conclusion and a targeted reopened question.

Reviewer Lens. “Unchanged model” is useful evidence about one object. The impact question asks whether the changed element can affect the item, state, reference or action on which a particular conclusion depended. A traceable explanation that it cannot may justify no repeated work for that conclusion. A claim that it can has to point to the evidence needed to examine the effect.

Consultant Lens. A comparison of old and proposed configurations becomes actionable when it names the affected question, owner, targeted checks and decision. That record prevents two costly habits: repeating unrelated studies because any part changed, and overlooking a meaningful change because the model filename did not. The company can retain applicable evidence while resolving the actual dependency.

The connected development plan

A plan row might promise that the professional will never mistake unfinished work for completed work. Where could another reviewer find the basis for that promise? Package-state and display requirements identify proposed controls; software and use methods examine different links; an accountable owner decides only after the relevant review. The plan has to point to those controlled records and their current states.

The following author-designed table helps locate a missing link when a proposed activity is being considered. It is a working index, not a form reproduced from a standard. Each row answers a question that the next parts of the plan depend on.

Plan field	What another reader needs to find
Lifecycle plan	Bounded activities, dependencies and the decisions they support.
Roles and competence	Owners, required skills, resources, review participants and decision authority.
Design inputs and outputs	Intended-use inputs, derived requirements, identified outputs and acceptance interfaces.
Risk activities	Scope, criteria, analysis and controls, reviews, and information needed through the lifecycle.
Requirements trace	Links from need and risk through requirement, implementation and evidence.
Supplier controls	The specified contribution, selection and acceptance evidence, owner and change arrangements.
Review gates	Evidence needed for each named activity, its decision and authorized owner.

Plan field	What another reader needs to find
Evidence milestones	Question, method, configuration, prospective criterion, rationale and limitations.
Artifact index	Controlled locations and versions connected by references, rather than uncontrolled copies.
Change control	Triggers, affected conclusions, reopened evidence, assessment and approval.

The requirements-trace row does more than locate a requirement document. For the missing package, it must connect the possible consequence to expected state persistence, the chosen control, and the evidence that will examine it. The index then leads to the authoritative records, whose status may still be “proposed.” A copied test result that later diverges from the controlled original would break that connection even if the table stayed full.

A single trace can expose the dependencies across the other rows too: the sequence leading to harm, requirement, control, implementation and effectiveness evidence, prospective criteria, remaining-risk question, reviewers and decision gate. Those relationships let the reader move from planned activity and capable people to design and risk, supplier reliance and evidence, and finally a bounded disposition. Equal-looking cells do not imply equally mature evidence.

Reviewer Lens. Starting at the proposed decision and working backward is a useful check against apparent completeness. Can the relevant configuration and actual evidence be found? Does the record say whether that evidence exists? A populated cell without a controlled record may be a perfectly legitimate work assignment, but it cannot close an evidence prerequisite.

Consultant Lens. A productive planning review asks the owner of each consequential cell which record will substantiate it and what later work depends on it. Beginning with the next bounded activity keeps the discussion from demanding every eventual artifact at once. The consultant can return a readiness finding or concrete planning option; the manufacturer retains its signed plan and decision.

Controlled case inputs

The fictional manufacturer LumaCell Devices now brings the following planning packet for LumaCell Review. These are supplied case inputs, not observations about a real device or evidence that the proposed configuration works.

The role remains assistance for trained laboratory professionals reviewing white-cell images from stained peripheral-blood smears of adults already referred for morphology

review. The proposed outputs are neutrophil, lymphocyte, monocyte, eosinophil and basophil suggestions, plus the other/unclassified route. The professional may reclassify using the laboratory's complete morphology taxonomy. Other/unclassified does not mean normal or negative.

The product does not autonomously issue a differential count or patient report. No leukemia rule-out, complete-cell-detection, pediatric, red-cell, platelet, time-saving or clinical-benefit claim is stipulated.

At this review, the **proposed acquisition profile** (AP1) adds numerical preparation and image settings that Chapter 3 had not yet established. The input is adult peripheral whole blood collected with **ethylenediaminetetraacetic acid (EDTA)**, an additive preventing clotting. A **Romanowsky-family stain** makes cellular structures distinguishable on the smear. The proposed blood-smear preparation procedure (SOP-PB1), acquisition contribution (ACQ1) and candidate-location process (LOC1) identify work to specify and evaluate. None of those labels establishes complete candidate selection or suitable images.

What would the receiving software see? The proposed crops have red, green and blue color values and dimensions of **256 by 256 pixels at 0.25 micrometre per pixel**. They use the lossless Portable Network Graphics format, or PNG. A proposed **color profile** tag (CP1) identifies how the color values are interpreted. The identified preparation procedure (P1) is proposed deterministic normalization: the same specified input and settings should yield the same transformed values. Resolution, color interpretation and transformation all affect the representation reaching the classifier. These numerical settings and labels are fictional planning specifications, not recommended diagnostic settings, verified conditions or a fixed position for the quality gate.

The **identity-field requirements** (IF-1) call for order and specimen fields, acquisition run, profile and locator version, crop and timestamp, and an image hash. Together they are intended to connect a received crop to its source and processing route. A hash identifies image bytes without proving the patient link; an actual package supplies field values, rather than being the specification itself. Missing or mismatched values are to produce an explicit non-result. The acquisition profile also does not establish that the locator finds every relevant cell. Candidate selection and complete-slide adequacy need separate evidence.

A proposed **reference procedure** (REF-1) would use qualified morphology reviewers to assign comparison labels. Reviewers would be **blinded** to the candidate output, so that output does not guide their judgment. Disagreement and unclassifiable images would need an adjudication or unresolved route, rather than a forced category. A planned reference label is not automatic ground truth.

The table locates the present state of each record. “Selected for planning” gives the team a boundary within which to work; it does not grant data use or report a successful study.

Record or contribution	What is supplied now
Proposed intended-use contract, acquisition profile and identity-field requirements	Bounded use and proposed input/interface identities selected for planning.
Initial development image collection	Development-only images grouped by patient; no confirmatory evidence claim.
Planning classifier and original quality gate	Experimental classifier and supplied predictive gate version labels, not validated releases.
Proposed reference procedure	Qualified, blinded morphology review; disagreement and unclassifiable images retained for adjudication or an unresolved state.
Original experiment schedule	Experiments and dates; no defined reference-process freeze gate, supplied-component acceptance/change-notice decision, or human–AI validation configuration.
Independent evaluation dataset	Collection has not begun.

The reference row describes how a comparison is proposed to be made, while the last row says there is no collected independent evaluation dataset yet. Reading them together prevents a future procedure's name from sounding like completed reference evidence. The **original experiment schedule** (PLAN-0) is likewise a present record with the specific missing decisions shown in its row.

Two proposed requirements give the team concrete behavior to plan. The **package-accounting requirement** (REQ-017) calls for every received order-linked package to remain accounted for through accepted, blocked, unavailable or resolved states. Its proposed **accounting test criterion** (VC-017) is **zero unaccounted packages in a prespecified suite** injecting interface and failure conditions. Coverage of that suite still needs review. Zero here concerns the selected suite's implementation check; it is not a patient-risk acceptability criterion.

The **recognition-and-action requirement** (REQ-018) calls for trained staff to recognize and act on non-result states. No patient report is to be exported automatically. The two requirements connect machine state to professional action while preserving the laboratory's patient-result authority.

The original quality gate (Q0) has a known version and interface, but its input representation, placement and crop/package decision semantics are still to be settled. Acceptance evidence and a change-notification agreement are not approved. Software/security and quality/risk leads are assigned these decisions; assignment does not mean completion. The planning classifier (M0) remains experimental, and the initial development image collection remains the resource identified in the preceding chapters.

The product owner requests an **internal readiness decision to begin confirmatory collection under the original experiment schedule**. No independent evaluation collection has begun. The request supplies no site, ethics, privacy, consent or legal permission, and no later evaluation or release result. The question concerns readiness for this proposed activity, not approval for clinical use.

Make the decision

Three connected questions can be answered from this packet without predicting a future performance result:

1. What may proceed, what cannot yet proceed, and what is not decided by the requested disposition?
2. What belongs in each of the ten plan fields at planning depth? Where a result does not exist, which record, owner and decision dependency are needed?
3. How would one trace connect the package-accounting and recognition-and-action requirements to their controls? Which evidence would establish implementation, and which would address effectiveness in professional work?

Decision

What disposition, plan and trace would make the next activity controllable while distinguishing evidence preparation from evidence already possessed?

How the manufacturer can bound this planning decision

Worked reasoning

Bound the disposition

The original experiment schedule does not support an unconditional internal go-ahead for confirmatory collection. The reason appears at the first physical act of collecting: which adults and sites are eligible, which candidate images will be retained, how each image will remain linked to patient group and acquisition conditions, and which reference information must be captured now because it cannot be recovered later. Dates and experiment names do not answer those dependencies. Holding this request is a readiness judgment, not a finding that the proposed device fails.

The gate may change the answer. If it selects or changes material retained, its selection behavior, acceptance basis and change controls are prerequisites to that acquisition design, and they remain unresolved. A different design might genuinely retain all eligible raw acquisitions without gate selection while preserving adequate source and reference information. The manufacturer could assess that narrower activity, but no such design has been supplied here. It cannot be treated as an existing escape route from the packet's gaps.

The same dependency test determines how much human-use design is needed now. A final display or formal summative use protocol is not automatically a prerequisite to unrelated raw acquisition. If current human actions determine what is captured, those tasks and conditions do matter before collection. Combining activities brings their relevant prerequisites forward rather than making every future requirement equally urgent.

This hold concerns the blanket request. Segregated exploration can receive its own readiness assessment. “Collect now, document later” remains unsound where the missing decision changes selection or loses irrecoverable information: later signatures cannot restore discarded images or missing source links. Even a positive internal readiness conclusion leaves external site, ethics, privacy, consent and legal permissions to be established separately for the activity.

Reviewer Lens. The useful stopping rule is the proposed activity and what it could irreversibly lose. That supports a hold on this blanket request while leaving a narrower, substantiated acquisition design open to assessment. Demanding every eventual interface, human-use and release record now would confuse later dependencies with current ones and could obscure the information collection really needs.

Consultant Lens. Drawing the exact acquisition and retention route lets clinical, data and software owners locate where a gate decision or human action changes the material kept. The focused request is then for the source, grouping, permission and

irrecoverable-reference records that route needs. The product owner receives options with closure conditions, rather than an informal “go” based on a generic assurance that raw images will be saved.

Complete the plan without fabricating completion

The **development plan** (PLAN-1-P1) below is an author-proposed repair to the original schedule, not an adopted manufacturer plan. The product owner owns its disposition; quality/risk coordinates the linked technical reviews. Its scope remains the proposed intended use, acquisition profile and identity requirements, experimental classifier, supplied gate and proposed reference procedure.

The table shows what each field would contribute to a future readiness decision. The opening rows locate the activity, capable people and configuration. The middle rows connect lost-work risk, requirements and supplier reliance. The remaining rows place evidence before the right decisions and preserve its controlled identity through later changes.

Plan field	Proposed completion
Lifecycle plan	Separate exploration with the initial collection and planning classifier, reference and collection readiness, later candidate verification/evaluation, and eventual release; identify their dependencies.
Roles and competence	Product owner decides the bounded gate; clinical lead owns reference/workflow requirements; model-development lead owns experiments; data steward owns data records; software/security owns configuration and implementation evidence; human-factors lead owns the use plan; quality/risk coordinates review.
Design inputs and outputs	Link the proposed role, acquisition profile and identity requirements to the package-accounting and recognition/action requirements. Identify classifier, gate and user-workflow outputs without calling the experimental components a validated device.
Risk activities	Follow lost or misleading package state to a possible incorrect diagnostic assumption. Define applicable criteria, control checks and review of residual questions.
Requirements trace	Link identity and persistent state to package accounting; link visible non-results, action and fallback to recognition/action. Keep the evidence questions distinct.
Supplier controls	Software/security and quality/risk define selection and acceptance requirements, applicable evidence, change-notice arrangements and impact assessment before reliance on the gate for the activity.

Plan field	Proposed completion
Review gates	Distinguish acquisition, reference/data designation and analysis, and human-use evaluation. Relevant clinical, data, software/supplier and human-factors owners provide evidence; quality/risk reviews dependencies; product owner records the bounded decision.
Evidence milestones	Set acquisition-dependent conditions before acquisition, reference-assignment rules before reference production, candidate/analysis rules before decisive evaluation, and representative-use configuration before human-use evaluation.
Artifact index	Link role, profile, identity requirements, collection, reference proposal, original schedule/revision, requirements, risk/supplier records and future software/use protocols. Preserve status as well as identity.
Change control	Reassess affected evidence after changes to intended use, acquisition/crop profile, reference protocol, gate behavior/version, interface/fallback or data/site coverage. Retain earlier decisions and why they reopen.

The risk row proposes analysis and criteria; it contains no accepted-risk result. The milestones row says when a method must be ready; it contains no passing study. “Define” and “identify” are work assignments whose records and actual states must remain findable. That is how the plan exposes an unfinished prerequisite instead of hiding it behind a scheduled date.

Reviewer Lens. Following the supplier row to its promised output, reviewers and affected activity gives a focused check. If the promised selection record is absent, the finding concerns the decision that relies on selection. It does not establish that the entire product is unsafe. The plan makes the scope of both the gap and its possible closure inspectable.

Consultant Lens. A vague “resolve supplier issues” row can become a task to decide selection/retention, assess applicable component evidence, plan the manufacturer integration check and establish change arrangements. Similarly, acquisition-time reference information and approval of the later labeling method may need different owners and timing. These distinctions turn a row into work someone can actually finish.

Construct the risk-to-evidence trace

The proposed **risk-to-evidence trace** (TRACE-PLAN-1) follows a rejected or interrupted package that is lost or misrepresented. A professional treats work as completed or normal, and necessary review or action is delayed or omitted. Incomplete or misleading diagnostic information is the hazard under examination. Exposure and possible harm still need clinical/risk analysis. No occurrence probability, actual injury, numerical risk-acceptability criterion or residual-risk decision has been supplied.

The two requirements protect different parts of that sequence. One keeps received work accounted for through retries and outages; the other concerns the professional's recognition and action. The **accounting test proposal** (VER-017-P1) and **use-test proposal** (USE-018-P1) are proposed protocol identities, not completed reports. The table shows what each would have to examine, and where their conclusions meet.

Trace question	Package accounting	Recognition and action
Which need or risk does it address?	Received work must not disappear or be falsely represented as resolved.	A visible non-result must not be treated as a completed negative or normal finding.
Which configuration does it concern?	Identity-field requirements, proposed state rules, gate/non-result interface and identified execution environment.	Those same state definitions plus intended users, interface, training, laboratory task order and fallback.
What control is proposed?	Persistent received-attempt identity, explicit state, linked retry/reacquisition records and reconciliation.	A distinguishable current non-result, action information, assigned reviewer and laboratory fallback/escalation.
How would implementation be checked?	Follow every received item through missing identity, quality rejection, interruption, duplicate delivery, dependency unavailability and resumption; compare intake, stored state, display and completion records.	Verify presentation of the specified current item, state, reason/action information and permitted review operations. This checks implementation only.

Trace question	Package accounting	Recognition and action
What would count as an implementation pass?	Zero unaccounted received packages in the prespecified suite. Every item needs a reconstructable disposition; a false resolved/normal state is a mismatch.	Identity, state, action content and enabled operations match approved specifications in every selected state-transition test. Specification or implementation mismatches remain open.
Why would the selected tests cover the concern?	Cover each intake/interface and failure transition, including interruption before and after persistence and acknowledgement. Review the rationale before execution; many ordinary paths cannot replace an omitted failure transition.	Cover ordinary review and consequential non-result, interruption and recovery states, including confusion with previous work; retain links to the critical tasks being controlled.
What effectiveness question remains?	Does implemented accountability prevent the specified information-loss sequence under justified conditions? Human-dependent protection also needs use evidence.	Can intended users identify current work, distinguish non-result from negative, start required review/fallback/escalation and keep unresolved work accountable?
What use method and disposition are proposed?	Carry software findings into the common use/risk assessment. The suite cannot estimate a field failure rate.	Observe tasks without coaching in justified representative scenarios. Omission or incorrect execution is a task failure. Analyze failures and safety-relevant difficulties, mitigations and residual risk; resolve safety-relevant open issues before accepting the control. Final scenario, sample and acceptance rationale need approval before formal evaluation.
Who produces and reviews evidence?	Software/security produces specifications and implementation evidence; verification and quality/risk reviewers assess applicability and coverage.	Human-factors lead produces protocol/evidence with clinical input; qualified human-factors/clinical and quality/risk reviewers assess findings.

Trace question	Package accounting	Recognition and action
What evidence exists now?	Requirement and fictional criterion supplied; coverage proposed; no implemented configuration or result supplied.	Required action supplied; detailed interface, use protocol and effectiveness results not supplied.
Who decides, and when is it needed?	Product owner decides readiness after technical review. Needed before acquisition if it controls collection accounting/retention; otherwise before reliance during evaluation or human-use work.	Collection-affecting human tasks are needed before acquisition. Complete use configuration/protocol is needed before human-use evaluation. Manufacturer accepts the evidence.
What limits or changes reopen it?	Suite passage supports tested implementation only. Intake, aggregation, identity, gate, persistence or runtime change reopens affected checks.	No perfect-human-control inference. Changed users, display, training, task order, workload or fallback reopens the relevant effectiveness question.

Consider a package received with a bad specimen link. The proposed accounting control retains its received-attempt identity and blocked state. Merely storing something is insufficient: a reconstructable but false “resolved” state would still mismatch the requirement. The zero-unaccounted criterion asks whether every item in the justified suite has the appropriate reconstructable disposition. It does not turn the existence of a database row into successful accounting.

The second column then asks what the professional encounters: this item's non-result, distinguishable from a negative finding, with a usable route to laboratory fallback. A passing count in the accounting column cannot answer recognition or action. If a later hypothetical software test found no unaccounted items while a use exercise found a reviewer treating a blocked item as negative, the first result would not cancel the second. The manufacturer would analyze the finding, mitigation and residual risk before accepting the control. This variation is not a later LumaCell event.

Reviewer Lens. Zero is persuasive only when the unit and suite are clear. The suite must have a prospective rationale covering the consequential transitions; the use method must observe the relevant recognition and action. Each future result then supports its own question under its tested configuration. Neither is automatically a field rate or an overall risk-acceptance decision.

Consultant Lens. Software/security and human-factors leads need common package-state definitions so their methods concern the same control, while retaining separate protocols and qualified reviews. Quality/risk needs to know how a software failure or use difficulty returns to the analyzed sequence, mitigation decision and remaining-risk assessment. Establishing that route is useful planning without pre-accepting its eventual findings.

Put each prerequisite at the gate that needs it

The trace identifies what must be examined. When would starting too early make that evidence unusable? At acquisition, information can be discarded or never recorded. Before reference production and decisive comparison, the rules determining what labels and results mean must be stable. Before human-use evaluation, the system and task being observed must represent intended use.

The following three review points are author-designed distinctions: **acquisition review** (G-A), **reference/evaluation review** (G-E) and **human-use review** (G-H). They are not universal regulatory milestones. The table is useful for deciding which unfinished item constrains the proposed activity, including what may legitimately remain open.

Proposed activity	What must be ready for that activity	What may remain open	Current state and route to closure
Acquire and retain specified evaluation material	Eligibility and permission routes; population and sampling frame; profile/acquisition identities; source/group links; retention/accounting; irrecoverable reference information. Gate selection behavior and acceptance/change controls if it selects retained material; current human tasks/conditions if they determine recorded observations.	Final weights, later routing/display and formal use protocol only if they cannot alter collection eligibility, selection or recoverable information. Future evaluation independence must be preserved.	The original schedule does not establish these dependencies, so its unconditional request is held. Clinical/data/software owners specify the acquisition route. Product owner may decide a narrower activity only after its actual dependencies and external prerequisites are satisfied.

Proposed activity	What must be ready for that activity	What may remain open	Current state and route to closure
Produce/freeze references, designate independent data and analyze a fixed candidate	Reference method approved before assignment, with qualifications, blinding, adjudication; data rights and lineage; partitions and exposure history; versioned candidate, transformations, selection and rules; prospective endpoints and analysis criteria before decisive results.	Formal human-use outcomes if the activity claims only the specified technical/reference comparison.	No independent dataset or fixed evaluation package is supplied. Clinical/reference, data, model-development and statistical owners provide controlled records; manufacturer decides the permitted comparison.
Evaluate intended human–AI work	Identified production-representative interface/system, users, tasks, training, critical scenarios, fallback, observation method, prospective acceptance/risk-disposition rationale and implementation readiness for the tested configuration.	Later release and local-deployment decisions; the use study does not confer either.	The use-test proposal remains a proposal. Human-factors/clinical and software owners supply readiness evidence; product owner records whether this bounded evaluation may start.

The first row's permission to leave later work open is conditional. A model, display or human task that changes eligibility, selection or irrecoverable information belongs in the acquisition decision. A protocol combining collection with other activities moves the relevant prerequisites earlier. Conversely, if all eligible raw acquisitions can genuinely be retained independently of the gate, with adequate source information, that narrower acquisition may be assessed without pretending later gate acceptance has passed. The original schedule does not supply that alternative design.

The transfer episodes use this dependency test rather than LumaCell's gate labels. A pathology study of total time has to preserve import and completed-review timestamps for the same case before they are lost; a later queue method might be settled separately if it cannot change capture. A sequencing release study needs specimen-to-suggestion-to-reconciliation linkage, while the final display can be evaluated under its own identified human-use configuration. Local protocols and permissions still need decisions. The common rule is to place a prerequisite before the information or action it can irreversibly affect.

Reviewer Lens. “What could no longer be recovered if this starts now?” is a more discriminating question than “is the entire program complete?” A missing acquisition

link can block collection while an unrelated unfinished display does not. Reference rules must nevertheless precede decisive labels, and a representative use route must precede formal human-use evaluation. The different answers follow the different activities.

Consultant Lens. A dependency record needs the required information, its owner, the decision it supports and the last safe time to settle it. That gives the product owner a basis for revisiting the original request once retention and selection are actually specified. A later positive acquisition decision would still leave reference evaluation, human-use assessment and release at their separate open states.

Figure 4.1 — One Development Program, Connected by Trace. The development plan connects intended use, requirements, risk controls, implementation, verification, evidence, decisions, monitoring and change through accountable records rather than parallel document piles. The figure organizes the trace just worked; it does not assert that a proposed LumaCell gate has passed.

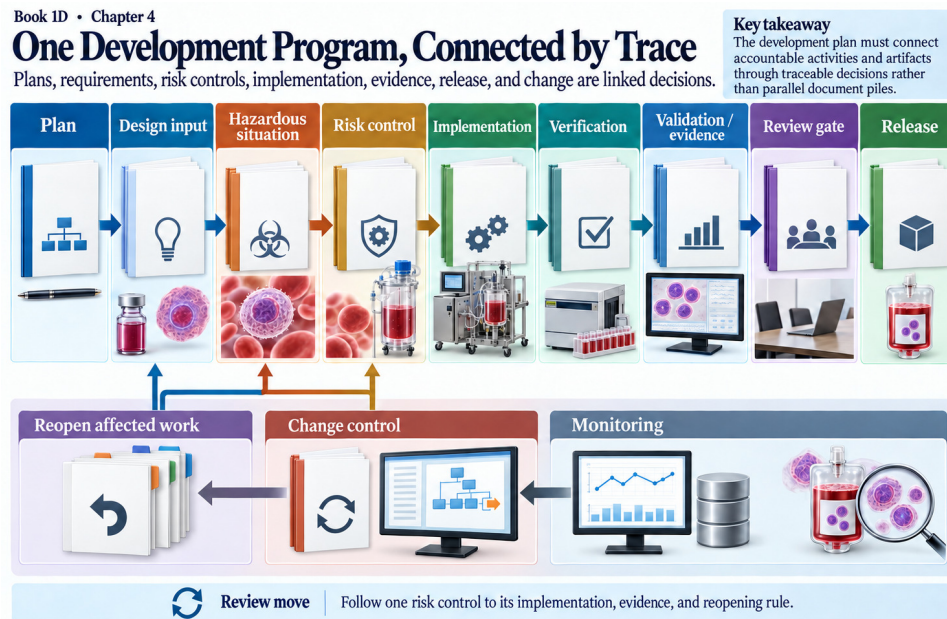


Figure 4.1. One Development Program, Connected by Trace.

Manufacturer accountability

Suppose a controlled acquisition plan eventually becomes ready for its stated activity. Whose decision has closed? The manufacturer coordinates the development program through its clinical, software, data, human-factors and quality/risk functions, and the product owner records the bounded disposition under its process. Coordination does not turn specialist judgments into one person's unsupported assertion. Supplier reliance, remaining-risk acceptance when evidence exists, later product acceptance and release still have their own manufacturer decisions.

The laboratory retains preparation, local workflow and patient-result release. The gate provider can be accountable for its supplied contribution without accepting the assembled diagnostic product. The consultant can identify a gap without approving the plan, risk or patient result. These boundaries allow a genuinely useful positive finding: one acquisition activity can be ready while later evaluation remains open.

The same clarity is needed for information that will arrive after use begins. A future rate or pattern of blocked, missing or unresolved packages could prompt review of the state control and professional action. This is a prospective monitoring question, not a current LumaCell event or automatic instruction to retrain the planning classifier. The plan needs to identify who produces the information, who reviews it and who decides which conclusion it reopens.

In the pathology and sequencing episodes, linked timing or reconciliation records could help the respective manufacturers distinguish a product change, site-process action or narrower benefit claim. An organizational chart alone would not tell them which record answers the unresolved question. A producer, reviewer and decision owner attached to that record would.

Reviewer Lens. Read the scope of a favorable decision before treating its signature as approval of the next one. Readiness to retain material, acceptance of a supplied component for a stated activity, residual-risk acceptance and release can each be supported at different times. Crediting one within its scope preserves progress while leaving the remaining questions visible.

Consultant Lens. An owner-and-evidence handoff is useful both for open gates and future monitoring. It identifies which manufacturer function receives a signal and which earlier conclusion may need another look. Preparing that handoff does not make the adviser the signatory for a supplier, risk or product disposition.

Bounded consultant deliverable

The consultant receives the same packet and proposed development plan. The question is readiness for the requested confirmatory collection. An advisory finding should make that decision answerable without redesigning the product in a margin note or approving a different activity. What would an evidence request need to say so the recipient knows exactly how to close it?

How the consultant turns this packet into focused readiness findings

A useful first finding locates the split left unresolved by the original schedule: which source and reference information must survive acquisition, and which method choices must be fixed before later label assignment? An applicable, reviewed acquisition-specific record can close the first question. A final use-evaluation interface is not automatically required for unrelated raw acquisition.

Build the readiness matrix

A **readiness matrix** connects six ideas: expected artifact, observed state, material gap, consequence, accountable owner and closure condition. The table combines state and gap in one column, giving five columns. This is an author-designed review structure, not a standard form. It is useful when a general request such as “update the document” would leave the decisive dependency unclear.

Expected record	What exists, and what is missing	Consequence for evidence or risk	Accountable owner	What would close this finding?
Reference and collection readiness	Reference procedure proposed; original schedule does not separate acquisition-dependent information from later assignment/analysis requirements.	Irrecoverable reference information could be lost at acquisition, or labels produced under an unstable method.	Manufacturer clinical lead, with quality/risk review; product owner decides readiness.	Approved collection-dependent content before acquisition; approved method, qualifications, blinding and adjudication/unresolved rules before reference assignment.

Expected record	What exists, and what is missing	Consequence for evidence or risk	Accountable owner	What would close this finding?
Supplied-component acceptance and change control	Gate identity known; acceptance evidence and change-notice agreement unapproved.	The gate could select retained material or affect evaluated behavior without a controlled basis.	Software/security and quality/risk leads.	Establish whether the gate affects selection/retention. Close applicable controls before that use, or substantiate an independent collection design; later product acceptance remains required for its own activity.
Human-use evaluation plan	Recognition/action required; user/task/interface/fallback configuration undefined.	Collection-affecting tasks could be uncontrolled, or interface tests mistaken for effectiveness evidence.	Human-factors lead with clinical lead.	Identify collection task dependencies now; complete representative-use configuration, protocol and prospective disposition rationale before human-use evaluation. Unrelated final display details need not precede every acquisition.
Integrated plan and trace	Original schedule lists experiments and dates without these linked decisions.	Work could begin without conditions that determine evidence meaning.	Quality/risk coordinates; product owner owns the bounded gate.	Revised linked plan identifies inputs, evidence, responsible functions, criteria, controlled locations and change triggers.

The reference row turns “method not ready” into two timed questions. Information that cannot be recovered belongs before acquisition; rules assigning decisive labels belong before that assignment. The supplier row offers another conditional route: establish whether the gate selects retained material, then either close controls for that reliance or substantiate an independent acquisition design. Independence at that one activity would not waive later product acceptance of the gate.

The human-use row follows the same logic. A human task that changes collection needs definition at collection; a representative later use protocol has its own readiness point. These rows can therefore close at different times. The consultant identifies evidence for re-review, while the manufacturer chooses and authorizes the product action.

Reviewer Lens. A finding needs its observed record, proposed activity and consequence. Where a requested later artifact cannot affect the present activity, narrowing the request improves the review. Where the absent information could be lost permanently, postponing the question would destroy the opportunity to answer it. The matrix makes those judgments visible.

Consultant Lens. A concrete re-review request concerns the acquisition-specific retention and permission record, the gate's selection and acceptance disposition, and any human tasks affecting collection. Each requested item belongs to a named dependency. The product owner then records the internal gate after competent review; the advisory matrix is not that approval.

Distinguish three review statements

The gate's identity is known and its acceptance and change-notice decisions are unapproved: that is the observation. An activity depending on its selection therefore lacks an established controlled basis for that contribution: that is the interpretation. Software/security and quality/risk should establish the applicable selection and acceptance basis so the product owner can reconsider the affected activity: that is the recommendation. None of the three statements establishes poor gate performance.

If an omitted acceptance report later arrives, the reviewer still compares its version, input representation, output unit and activity with the proposed route. Receipt of the report can close a missing-record finding while a configuration mismatch remains. Conversely, an applicable report and manufacturer check may close the focused finding without rewriting every row. Partial closure describes evidence accurately; it need not be forced into one overall pass or fail.

Prioritize by consequence and dependency

A missing source/group identifier at acquisition may prevent a later independent comparison. An unfinished display may be irrelevant to raw retention and essential before human-use evaluation. The priority follows the next activity and what can be irreversibly lost. This packet provides no occurrence, severity or effectiveness evidence from which to invent a clinical risk score or assign identical urgency to every open cell.

The episode applications ask for records that answer the current decision. For pathology's total-time question, linked scanner-import and completed-review timestamps would be useful before recommending a queue-model change. For sequencing, specimen-linked suggestion, reconciliation and release records would precede advice on a new display claim. This is a focused post-use evidence request, rather than another retelling of the earlier favorable function.

Define what you will re-review

The proposed development plan, risk-to-evidence trace and revised activity-specific records give re-review a concrete scope: content, configuration applicability, competent review and the consequence for each acquisition, reference/evaluation or human-use decision. A substantiated retained-raw acquisition method could close one acquisition dependency while reference-method approval remains open for label production. That is the conclusion to report. The product owner records the internal activity decision, and external permissions remain separate.

Test your advisory boundary

Consider this proposed decision sentence: “The consultant approves the original experiment schedule once the supplier sends its certificate.”

Decision

How would the sentence change to preserve an evidence-based re-review and the manufacturer's authority? What else besides receiving a certificate matters to closure?

How the authority and evidence boundaries change the sentence

The consultant can reassess the specified readiness finding after the manufacturer supplies an applicable supplier-acceptance record and the remaining required readiness evidence. The manufacturer retains approval of the plan and bounded activity. A certificate may be one input, but the packet does not show that it is sufficient, and the supplier finding is not the only open condition. The revised sentence therefore identifies a review the adviser can perform without assigning it a decision the adviser does not own.

Cross-book handoffs

A specialist question in this plan still needs its local object, method, prospective criterion and decision. Book 1C offers optional depth on that question—method—acceptance—decision relationship and evidence applicability; the LumaCell gate remains a manufacturer planning question here.

Book 1A Chapters 3–6 develop risk, architecture, controlled development and verification. Book 1B Chapters 3–6 add threats, security architecture, supply chains and security testing. A referral can help the assigned specialist, but an identified security

dependency still needs an owner and evidence route in this plan. Reading another book does not complete a control.

Book 2 Chapters 5–7 and 14 offer optional European Union quality-management-system, technical-documentation, performance-evidence, supplier and change depth. No European Union route is selected in this case. An internal fictional gate grants no jurisdictional permission and cannot replace the applicable legal assessment.

Changed-decision transfer

Keep the supplied case facts unchanged and consider a different, counterfactual request: a segregated exploratory experiment using the initial development image collection, planning classifier and original quality gate, with no clinical use or confirmatory evidence claim. This is a reasoning variation, not a new LumaCell event or evidence of permission.

Decision

What disposition fits this narrower request? Which conditions still need checking, which holds remain on confirmatory work, what record would separate exploratory output from confirmatory evidence, and who decides?

How the narrower activity changes the readiness question

The requested activity has changed, so its dependencies need their own assessment. Missing final validation does not automatically prohibit every exploratory experiment. Existing permissions for the collection and the proposed segregation still have to be confirmed. Any allowed output remains development evidence, identified by its configuration and purpose. The original confirmatory-collection gaps stay open until their own conditions are satisfied.

The manufacturer's accountable owner decides whether this activity may proceed; neither the consultant nor this answer authorizes it. Repeating the original hold without examining the activity would overgeneralize. Allowing all collection because this task sounds narrower would erase the boundary. The useful decision retains both the possible limited route and its actual prerequisites.

Restore the whole trace

The interrupted package now has a route through the planning record: possible diagnostic consequence, requirement, control, implementation and effectiveness methods, prospective criteria, residual-risk review and authorized decision. Could a reviewer also follow it backward from the decision to the particular evidence and configuration needed? If the chain ends at a proposed plan or the existence of the image collection, that later decision remains open.

The immediate gain is a controlled next question. Its evidence has not been created by naming it, but someone can now tell which work would produce it and when that work becomes meaningful. Figure 4.1 organizes the same connections; the text and trace carry the actual decision logic.

Forward bridge

A date becomes useful when it names bounded work with known dependencies and an accountable decision. The proposed development plan and trace distinguish acquisition readiness, reference and candidate comparison, human-use evaluation, remaining-risk decisions and eventual release without claiming that these activities are complete.

The next dependency concerns the images themselves. Even a reproducible experiment may rely on data whose permitted use, source or transformation cannot be reconstructed. Chapter 5 develops a data charter and lineage record for a bounded activity, giving this plan an answer to that specific dependency.

Reasoning check

- Can another reader identify the next permitted activity without inferring that later gates passed?
- Are criteria prospective, with their rationale and decision scope visible?
- Does every human-dependent control have an effectiveness question distinct from implementation testing?
- Can each supplier dependency be traced to specified acceptance and change arrangements?
- Are unfinished records explicitly unfinished, with owners and downstream consequences?
- Does the consultant's conclusion preserve manufacturer decision authority?
- Which change could reopen each important conclusion?

Decision

Where is one strong link, one open link, and the next owner action in this plan?

Source Trail

This chapter's forms, decisions and fictional case are author-designed teaching constructions. The source packet records thirteen bounded propositions and their limitations. Standards and international guidance are not presented here as automatic legal obligations or as evidence that a real product is safe.

S1 — ISO 13485:2016: 6.1–6.2, 7.3.2–7.3.9 and 7.4.1–7.4.3; resources/competence, planning, inputs/outputs, verification/validation, change and purchasing controls. Licensed full text consulted. Official source identity.

S2 — ISO 14971:2019: 4.3–4.5; policy/criteria distinction, risk planning and traceable risk records. Licensed full text consulted. Official source identity.

S3 — ISO/TR 24971:2020: 4.4.1 and 4.4.3–4.4.7; informative planning guidance, prospective criteria, authority and implementation/effectiveness distinctions. It supports rather than replaces ISO 14971. Official source identity.

S4 — IEC 62304:2006+AMD1:2015, Edition 1.1: clean final text, 1.2 and 5.1.1–5.1.9; software planning and its scope boundary. Licensed full text consulted. Official source identity.

S5 — ISO/TS 24971-2:2026: Scope and 4.3–4.4; predictive-ML risk-planning considerations and conditional monitoring provision. LLM/generative-AI devices are excluded. Licensed full text consulted. Official source identity.

S6 — IMDRF N88 Final:2025: Principles 1–2, 4–5, 7–8 and 10; high-level good machine learning practice, not a prescribed development-plan form. Primary document.

CHAPTER 5

Data Provenance, Rights, Privacy, and Lineage

A developer is offered an image from a stained adult peripheral-blood smear and wants to extract a candidate white-cell crop for an exploratory LumaCell classifier. The pixels are readable. What would let another person reconstruct whether this particular image could be used for that work, and what happened to it afterward? A folder name cannot connect the image to its source, patient group, acquisition conditions and permission. Removing a printed patient name cannot supply those relationships either.

Chapter 4 made those records prerequisites for particular activities. Here the proposed source image gives them a concrete job: its permitted use and its path into a crop, transformed image, annotation, dataset and experiment must stay intelligible. The manufacturer is deciding about development material and the evidence it could support. The laboratory still owns interpretation and release of the patient result.

Which earlier decision needs these data?

LumaCell's proposed role remains adult supplied-candidate-image assistance under the proposed acquisition profile. The proposed system map showed how input preparation, the original quality gate and other components could affect evidence. Chapter 4 separated the prerequisites for a narrow activity from later evaluation and release. The data question starts with that activity, without borrowing a later outcome.

Decision

Does permission to possess a dataset establish permission to train a model?

Does permission to train establish suitability for independent evaluation? Does removal of a name establish that the remaining record cannot identify a person?

How a reviewer separates the three decisions

Each question needs a different account: purpose-specific authority, adequacy and independence for the intended evidence, and privacy in the actual context. A favorable answer to one does not supply the others. The data charter will connect those decisions so that they can support one bounded activity without collapsing into “data approved.”

Four different answers about one image

The developer's first proposed step is to bring the source image into a controlled working area. Successful receipt would show that the team can obtain and read it: **availability**. The transfer and subsequent work still need authority. The data steward therefore compares the requested crop-and-fit activity with the permission record: does it cover this actor, subset, activity and environment? That is **use-specific eligibility**.

A permission match would make the next question meaningful, rather than answer it. The clinical and imaging team needs to compare preparation and acquisition conditions with the proposed profile, and examine image quality, patient mix and reference information for the intended question. This is **adequacy**. A file that opens correctly may still be unsuitable for the planned comparison.

Even a suitable image raises another question if the team wants to use it for evaluation. Did this patient, specimen, image or related information already influence fitting, tuning or candidate selection? **Independence** concerns that relationship to development choices. A transformed crop or a related image may carry the same prior influence under a different filename. The four findings can therefore differ for the same resource.

Changing the activity changes the comparison. Exploratory analysis, annotation, fitting, tuning, independent testing, monitoring, investigation, publication and transfer are specific uses, rather than interchangeable meanings of “use the data.” A decision needs the resource and the operation it covers. The International Medical Device Regulators Forum's good machine learning practice principles, IMDRF N88, emphasize relevant data, appropriate references and training/test independence. They supply nonbinding technical principles, not data rights or a case-specific legal basis. [S1]

The pathology queue episode and sequencing review episode from Chapter 1 are separate fictional teaching episodes, not LumaCell or named-series product history. Before a pathology timing study, the team would need to know whether scanner-import records and queue experiments describe the same cases. A sequencing specimen can generate several candidate-variant and display records. Those relationships matter before records are assigned to an independent test. After the observed queue or display behavior, purpose-specific access and event links would still be needed for further study. Calling the records a new dataset would not create independence. This is a planning inference, not another event in either episode.

Reviewer Lens. A useful review retains four separate findings. When the permission record matches the requested activity, that is a positive eligibility conclusion; it deserves to remain positive while the imaging and reference questions are examined. A single approval label would hide which relationship was actually checked and make the next request harder to understand.

Consultant Lens. The team's verb is often more useful than its folder title. Once the operation is precise, the authority and privacy question can go to the responsible owners, the fit question to clinical and imaging colleagues, and the prior-influence question to data and machine-learning colleagues. That division avoids repeatedly asking for permission when the unresolved issue is technical, or asking for better images when the intended use is outside the supplied authority.

What would the source record have to tell us?

The proposed acquisition profile says what LumaCell intends to support. How would the developer know whether the offered image was produced that way? The originating laboratory, adult patient group, specimen, acquisition run, preparation and image settings, time, original purpose and accompanying reference information provide the comparison. Known facts and explicit unknowns belong in this account. **Provenance** is the source record's origin and production history; a declared profile is not evidence of an actual acquisition match.

The relationship to the patient need not require giving the manufacturer the patient's direct name. The originating site can keep the direct linkage key and supply a controlled **grouping token**. Related crops can then remain associated with the same patient group, with specimen and acquisition tokens preserving more specific relationships. This supports a later independence check while leaving privacy and permission to their own assessments.

That check also needs more than the label “external.” Material from another site could have been selected by the development team, and a supplier could hold related images used in its own fitting. Where those relationships matter to the proposed claim, the source account must expose them. If acquisition settings cannot be reconstructed, the limitation belongs beside the affected comparison. A familiar image-file format cannot restore missing preparation history.

The same logic distinguishes stages in the transfer episodes. A pathology scanner-import timestamp identifies entry into the worklist route; it does not identify completed professional review. A sequencing specimen and upstream analysis record establish what a displayed variant suggestion could refer to. Those source facts tell the team which records to join in a later inquiry. They prevent a successful display or queue action from being treated as a measurement of a different stage.

Reviewer Lens. “Same profile” can conceal an unsupported step between a readable file and a clinically relevant input. The focused question is which source settings support the match. A reviewer can credit the narrower available record while leaving that comparison open, instead of treating the missing settings as proof that the images are poor.

Consultant Lens. The minimum source fields and their correction route are easier to agree with the site before export. The manufacturer needs someone who can resolve a mistaken group or acquisition link, as well as someone who can deliver the image. For the transfer episodes, that means reaching the producer of the import or specimen-analysis record before promising a later benefit study.

How would a crop lead back to that source?

Suppose the image is suitable and its proposed use is permitted. A later experiment may consume a transformed crop rather than the original image. How could a reviewer connect the result to the source that was checked? The crop needs its source identity, extraction coordinates and extraction version. The transformed image needs that crop as its parent and an identified implementation and settings for the preparation procedure. The annotation needs its source or crop attachment, instructions and reviewer state. Dataset membership then connects these versions to the experiment and any candidate it produces. This connected history is **lineage**.

Each step records its parent, operation and version, time or event as appropriate, and output identity. The dataset also needs an actual membership list and its purpose and permission. These relationships make it possible to reconstruct relevant work and identify descendants affected by a change. They do not require every intermediate file to be retained forever; the applicable retention plan may constrain what can remain. Nor does describing this planned LumaCell chain establish that it has run.

Could matching file contents replace those links? A **content hash**, the image-byte identifier from Chapter 3, can help establish that files contain the same bytes. It cannot establish their patient, permitted use or correct annotation, and different files may come from the same acquisition. The patient, specimen and acquisition grouping therefore needs to travel through the descendants. A new crop, transformation or filename does not create another independent patient.

Now consider a possible later correction. An authorized annotator changes a label. A reconstructable record preserves the prior judgment, reason, reviewing authority, revised version and the datasets and experiments affected. Replacing the old cell in a spreadsheet would otherwise make a past experiment appear to have used a label it never saw. Chapter 6 examines whether the annotation is a suitable reference; Chapter 7 uses these relationships to examine leakage. Lineage supplies the history those questions need.

For the pathology episode, the analogous chain would connect import, queue events and completed review; for sequencing it would connect analysis, suggestions, reconciliation and release. Defining those joins before use determines what needs logging. If a source link were later corrected, the chain could identify which timing or

release analyses used it. Neither episode supplies an actual linkage error; this is a proposed method for handling one.

Reviewer Lens. The useful test goes in both directions: from a derived object back to source, grouping and authority, then forward to the datasets and experiments that consumed it. A missing arrow matters through the conclusion it blocks—independence, reconstruction or compliance with a restriction. “Incomplete lineage” becomes actionable when that consequence is named.

Consultant Lens. Parent and membership fields cost little to design before a pipeline starts producing outputs. Retrofitting them later may leave uncertainty about which version an experiment consumed. A correction route should therefore identify affected experiments, rather than merely improve today's spreadsheet. The manufacturer can then decide how the break affects the proposed use.

Which authority covers the next operation?

A crop can have an excellent source history while lacking permission for model fitting. Which record would settle that question? A **rights register** connects a resource and requested use with its source agreement or data authority, subset, purpose, actors, environment, recipients, retention terms, restrictions, responsible review and decision. “Provided to the manufacturer” describes delivery; it leaves the next operation unexplained.

The proposed work is therefore compared with the documented scope: these people, these records, this crop, annotation or fitting activity, in this place, with these transfers. A different recipient or purpose returns to the responsible authority. A supplier's rights in a component do not by themselves establish the manufacturer's rights in associated images, annotations or fitted candidates.

“Consent obtained” cannot substitute for that contextual assessment. Legal bases and institutional duties depend on jurisdiction and circumstances; none is selected for LumaCell here. In the European Union, the General Data Protection Regulation addresses processing purposes and does not treat pseudonymisation as automatic anonymity. The cited provisions support those narrow distinctions, rather than a conclusion that this dataset is eligible or that research is always exempt. [S2]

A clear register also makes closure possible. If the requested operation matches the documented scope, the reviewer can record that match and turn to technical readiness. If it does not, the next question names the missing scope and its owner. A mismatch for one activity need not make the entire resource “bad data.”

For example, local permission to inspect pathology queue events for monitoring would need assessment before export for a new manufacturer study. Access to specimen-linked sequencing release records for a local review would not settle onward

disclosure. These are hypothetical purpose-and-recipient questions around the earlier observations, not statements about the sites' actual agreements.

Reviewer Lens. The strongest permission finding is specific enough to compare with an operation: resource, actors, environment, recipients and time all match. That finding does useful work. It leaves acquisition fit, reference quality and independence to evidence about those relationships, rather than weakening a supported permission conclusion with a generic unresolved label.

Consultant Lens. A focused referral explains the operation and the field that fails to match. Site/data and privacy/legal owners can then decide the relevant scope. Derived annotations and candidates deserve their own retention and transfer question; component delivery should not silently answer it. The adviser helps frame the referral without granting a right or choosing a legal basis.

How can related images stay together without sharing names?

Removing the patient's name from the manufacturer's copy can reduce exposure, yet dates, image metadata and source context can still matter to identifiability. Removing every relationship creates another problem: the team may no longer know that several crops came from one patient, or that evaluation repeats a development patient. The design has to preserve the relationships needed for evidence while addressing access and identification risk in context.

A purpose-limited **grouping token** can say that crops share a patient without handing the developer the direct identification key. The originating laboratory may retain that key; specimen and acquisition tokens preserve the narrower links. Token creation, key access and corrections need an owner and a permitted route. A working token is neither an anonymity certificate nor a reference label saying what cell is present.

Which links are necessary depends on the question. Multiple slide images and queue events may describe one pathology case. Multiple variants, displayed suggestions and reconciliation actions may describe one sequencing specimen. Defining the minimum grouping and event links before study allows later import-to-review or suggestion-to-release comparisons without demanding direct patient names by default.

The privacy specialist assesses remaining identifiability and access; the evidence team specifies what grouping is needed to avoid treating related records as independent. The charter will connect those decisions at their shared interface. Calling the technique pseudonymisation—using replacement identifiers while additional linking information remains controlled—does not complete either assessment.

Reviewer Lens. A token may be entirely useful for grouping while still requiring controlled access and a privacy assessment. Looking at the grouping task beside the

documented privacy conclusion prevents one successful mechanism from being credited for a different question. The reviewer can then identify whether the missing fact concerns repeated patients, remaining linkage or both.

Consultant Lens. The site, steward, privacy/security and evidence owners can agree on a minimum token design and a correction route. The person who controls the key and the person who verifies grouping may have different roles. Making that interface explicit allows a corrected association to reach affected datasets without routinely widening access to direct identifiers.

What changes when the image moves—or time runs out?

The source-to-crop chain also crosses organizational boundaries: originating site, controlled export, manufacturer environment, authorized worker and derivative record. Each crossing needs an account of what moves, who receives it, the purpose, and the applicable protections and agreements. A hosted annotation tool adds a recipient and processing environment even if the classifier code is unchanged. That new route needs assessment against the permission.

Within an authorized environment, access still follows the task. An annotator may need the image and instructions; a machine-learning analyst may need identified fitting partitions. For independent evaluation, the data and results must be protected from influencing fitting, tuning or candidate selection before the comparison. Role-specific access or blinding may help, but independent data do not automatically require separate personnel. Physical access to a folder does not establish authority for every use. Actual legal, agreement and site conditions govern transfers; this chapter supplies no universal geographic rule.

Time adds a further boundary. **Retention** concerns how long records and artifacts may or must remain under their applicable conditions. A source image may have produced crops, annotations, dataset versions, findings and a fitted candidate before its permitted window ends. Deleting the source file does not erase its influence from the candidate. Conversely, a candidate's existence does not grant indefinite retention or use. The responsible authorities and manufacturer need a timely disposition for further use, restriction, deletion or investigation of each affected class of record and artifact.

The same question could arise with pathology queue logs or sequencing suggestion/reconciliation records used to investigate their known workflow constraints. Monitoring, transfer, re-analysis and retention purposes would have to match the applicable records. No new site agreement is supplied in those episodes. The proposed flow makes the decision dependencies visible; it cannot issue a legal disposition.

Reviewer Lens. Following every recipient and retained derivative exposes where a permission can stop even though processing remains technically possible. An expiry trigger that considers source files alone may overlook the candidate created from them.

A complete flow supports this review while still leaving the specific authority or technical-adequacy question open where appropriate.

Consultant Lens. The project-start event, expiry trigger, derivative inventory and decision owner belong in the plan before a run creates hard-to-untangle descendants. When a new service or public recipient is proposed, showing the changed arrow gives the authority owner a concrete request. The cheaper early decision is how to track those consequences, rather than promising a particular disposal remedy before the applicable restriction is known.

Join the decisions in a data charter

Could another person assess the proposed crop-and-fit activity without relying on the developer's memory? A **data charter** provides the joined account. It indexes controlled source, rights and lineage records for a bounded resource and activity, with their current decisions and unknowns.

The eleven fields below help that reader follow the same image from origin to permitted use and onward responsibility. Their purpose is to expose missing relationships, rather than to multiply copies of controlled records.

Charter field	Question the record answers
Data sources	Where did the records arise, under which site, patient, specimen and acquisition context?
Authority and permissions	Which documented basis covers the activity, actors and environment?
Consent or legal-basis questions	What has the responsible person assessed, and what remains unresolved?
Privacy and deidentification	What is minimized, what linkage remains, and who has access?
Licenses	Which purposes, derivatives and onward uses are permitted or restricted?
Transfer and localization	Which actual recipients, services and restrictions govern the flow?
Retention	What time or condition governs holding, restricting or deleting records and their evidence consequences?
Provenance	How can origin and production conditions be reconstructed?

Charter field	Question the record answers
Lineage	Which operations, parents, annotations, versions and later uses are linked?
Supplier and site duties	Who supplies, checks, corrects, notifies and authorizes?
Version control	How are dataset changes, review, decisions and affected experiments identified?

For the opening image, provenance identifies the source conditions; lineage connects the crop to them; authority identifies the permitted operation; retention identifies its endpoint. One object can legitimately have different unresolved questions across these rows. “Unresolved” is a useful entry when it exposes the next decision. “Approved” without scope would conceal it.

Reviewer Lens. The intended use of the resource determines which path through the charter matters. Independent testing would require authority, source and group relationships, reference information and exposure history. If only development is supported, that can be credited while the independent-evidence question stays open. Completing the table is not itself completion of the proposed work.

Consultant Lens. Controlled source and authority records should populate the charter, with explicit unknowns and owners. A request for the missing field and its decision consequence is more useful than a request for a larger “data package.” It tells the steward what would actually change the readiness finding.

Three offers for different uses

A new fictional event now occurs after Chapter 4's planning review. The **restricted exploratory subset** (D-A) is an identified part of the initial development image collection. The new event, DATA-AUTH-1, concerns that subset alone. Earlier permission questions were unresolved at their earlier point; this record does not retrospectively clear the whole collection, finish the confirmatory gates or authorize clinical use.

The other resources are separate offers: the **monitoring-only offer** (D-B) and the **supplier image export** (D-C). The table supplies the facts for a purpose-by-purpose judgment. The last column describes what the team wants to do, rather than what the resource name permits.

Resource	Supplied record	Requested use
Restricted exploratory subset	Completed site/data-authority and manufacturer privacy review plus contractual permission for the specified exploratory activity; subset membership, source and patient-group links available; no independent-evaluation designation.	The exact segregated exploratory activity.
Monitoring-only offer	New site records with documented operational-monitoring permission; no training permission supplied.	Addition to classifier training.
Supplier image export	Training-use license supplied; original patient-group links and some acquisition provenance unavailable.	Designation as independent evaluation.

The first row requires its full scope to be read. The **exploratory authorization record** (AUTH-DA-1) records the completed fictional reviews and permission for controlled annotation, input characterization and exploratory candidate fitting, identified as EXPL-DA-1. The subset manifest MAN-DA-1 identifies the only covered material. Processing is restricted to the local environment ENV-DA-1, for the assigned data steward, machine-learning analysts and reference annotators. The originating laboratory retains the patient-linkage key; the manufacturer receives controlled grouping tokens. Public disclosure, third-party processing and independent confirmatory designation are excluded. No actual jurisdiction, consent instrument or legal basis is specified; the narrow completed review is a case premise.

The contractual **exploratory retention schedule** (RET-DA-1) allows source images, derived image data and annotations to remain for **90 days from the recorded project-start event**. The proposed activity falls within that window. Further use or retention beyond it, including the fate of any fitted candidate artifacts, requires an explicit further disposition. The window is fictional and contractual, rather than a legal rule or perpetual data/model license. No exploratory result has yet been produced.

None of the supplied records establishes full intended-use coverage or a suitable reference for decisive evaluation. No model result is supplied for any offer. The problem is what each resource could contribute to the next activity on this record, and what still needs checking.

What can the manufacturer decide now?

For each resource, what may proceed, what cannot yet proceed and what remains unknown? Which supplied fact supports the conclusion, who owns the missing decision, and what additional record could change it? How would availability, use-specific eligibility, adequacy and independence differ across the three rows?

For the restricted exploratory subset, what would the eleven charter fields and one rights-register entry need to say? How would source, crop, transformation, annotation, dataset and experiment remain linked through patient, specimen and acquisition grouping and the permission record? Which objects and operations are still planned?

Decision

What are the three resource dispositions, and what would make the restricted subset's charter, rights entry and planned lineage adequate for a bounded readiness decision?

How a reviewer preserves the positive permission and the remaining questions

Worked reasoning

The restricted exploratory subset is available with identified membership. Its data-use eligibility for the specified exploratory activity is stipulated, provided execution matches the actors, local environment and window. This is a positive answer to a real question. Execution readiness still needs review, and representative coverage, reference adequacy and confirmatory independence remain unestablished. Chapter 4's hold on the blanket confirmatory-collection request therefore remains at its own scope.

The monitoring-only offer is available, but its documented purpose differs from requested classifier training. That proposed use needs to be held while the originating site/data authority and privacy/legal owners assess the new purpose. If eligibility is established later, scientific suitability will still need examination. The existing monitoring permission is useful within its scope; it cannot silently become training permission.

The supplier image export poses both an authority and an evidence question. Its training license does not establish the requested independent-evaluation authority. Even if that use were later permitted, the missing patient-group links and acquisition history would still prevent demonstration of relevant separation and acquisition-profile applicability on this record. This is an absent basis for the claim, rather than evidence that leakage occurred. Neither a renamed export nor a larger image count reconstructs the relationships.

The disposition table keeps these decisions separate. Its use column answers the requested authority question; its evidence column identifies what remains supportable; its final column identifies the next accountable work.

Resource	Use disposition	Evidence consequence	Next accountable action
Restricted exploratory subset	Specified exploratory work matches the stipulated authorization; actual execution must match it.	Development evidence only; coverage and reference questions remain.	Steward binds authorization, membership and lineage; experiment owner binds the run; product owner records readiness.
Monitoring-only offer	No inferred permission for requested training.	Excluded from that activity pending eligibility.	Site/data and privacy/legal owners assess the new purpose; manufacturer records the decision.
Supplier image export	No independent-evaluation designation on this record.	Purpose scope, grouping and provenance gaps prevent the proposed independence/applicability claim.	Recover relevant history and authority or choose another resource; preserve the limitations.

For the first row, the next action has moved beyond asking whether any permission exists. The team now has to show that the proposed execution matches the permission and that its technical prerequisites are ready. That progress does not extend to the other offers or the rest of the original collection, and it supplies no model-performance finding.

Reviewer Lens. A positive permission finding should survive the discovery of an unrelated technical unknown. The restricted subset has a stipulated scope to credit; the execution match remains a check. For the export, a missing group link prevents the asserted independence conclusion without demonstrating actual overlap. Those are different findings and lead to different requests.

Consultant Lens. The three useful next actions are correspondingly distinct: bind the restricted subset to its authority and execution; refer the monitoring offer's new purpose; and recover the export's purpose and source history or find another evidence resource. The adviser can make that agenda precise without granting rights or designating independence.

A charter for the restricted exploratory activity

The proposed exploratory charter (CHAR-DA-P1) covers the identified subset and activity at a point after authorization and before execution. The steward prepares it; the product owner decides readiness using the relevant site, privacy, technical and quality reviews. The charter links the completed permission instead of issuing it again.

The eleven entries below follow one image through that boundary. Source and provenance locate it; authority, license, transfer and retention constrain use; privacy preserves controlled grouping; lineage and versions make future descendants reconstructable.

Charter field	Worked entry
Data sources	Identified subset of the initial development collection from its originating laboratory; source and patient-group links available. Site and acquisition records stay linked, without identifying this source as a later evaluation or monitoring site.
Authority and permissions	The new authorization event covers the exact subset, exploratory activity, actors, environment and project window. This is data-use authority, not regulatory clearance.
Consent or legal-basis questions	Narrow site/data and privacy review stipulated complete. No actual jurisdiction, basis or consent instrument supplied; a real charter would link the underlying assessments.
Privacy and deidentification	Laboratory retains direct linkage key; manufacturer uses controlled patient/specimen tokens. Assigned roles only in the restricted environment; no anonymity finding supplied.
Licenses	Controlled annotation, characterization and exploratory fitting only. No public images, independent confirmatory designation, third-party processing or perpetual derivative use.
Transfer and localization	Specified laboratory export into the restricted local environment. New recipient, hosted service or processing environment needs assessment; no national localization rule inferred.
Retention	Source/derived images and annotations: 90 days from recorded project start. Start and endpoint recorded before execution; further retention, use and candidate-artifact disposition addressed before expiry.
Provenance	Originating record, patient/specimen tokens, preparation/acquisition metadata, export identity and available profile information retained. Declared profile differs from verified compatibility.

Charter field	Worked entry
Lineage	Every derivative linked to subset, authority, transformation/annotation version and downstream use. No labels or experiments produced yet.
Supplier and site duties	Site provides and corrects source/group information and routes restrictions; steward checks intake links; privacy/security assess flow changes; machine-learning/reference owners remain within scope; product owner decides activity.
Version control	Manifest and authorization identities preserved. Membership, label, permission and processing changes receive identified revisions and affected-use assessment without rewriting an experiment's original basis.

The retention row illustrates why a planning record can be complete while work is pending. It specifies the start, endpoint and required disposition, rather than claiming that the experiment has run. Similarly, the provenance row keeps a declared profile separate from an observed match, and lineage keeps future annotations separate from existing records. These technical unknowns identify the next work without cancelling the narrow permission.

Reviewer Lens. A hypothetical run tests the charter better than a count of completed cells. A different service or post-window use should be recognizable as a changed boundary. A reported candidate should lead back to its actual source, transformation and annotation versions. At this snapshot the charter describes how those checks will become possible; it is not a completed experiment trail.

Consultant Lens. Binding membership, worker roles, environment and start/expiry events makes a readiness recommendation reviewable. A new recipient belongs with the authority owners; profile or reference uncertainty belongs with technical owners. Keeping those routes distinct avoids a vague “approval pending” that nobody can close.

The rights entry for that activity

The exploratory rights entry (RIGHT-DA-1) lets a reader check permission without inferring it from a dataset title. Its rows separate a known restriction, such as excluded public disclosure, from an unestablished technical fact, such as acquisition compatibility.

Register field	Entry for the requested activity
Resource and request	Identified subset for controlled annotation, characterization and exploratory fitting, before results.

Register field	Entry for the requested activity
Basis and scope	Completed stipulated site/data and manufacturer privacy review; assigned steward, analysts and reference annotators; restricted local environment and contractual window.
Restrictions	No public disclosure, third-party processing, independent confirmatory designation or out-of-window use without further disposition.
Evidence open questions	Actual acquisition fit, annotation/reference adequacy and intended-use coverage need their own evidence.
Reviewer and decision owner	Site/data authority and privacy reviewer own scope decisions; steward and experiment lead confirm execution match; product owner records development readiness.
Disposition	Data-use eligibility established for this activity; execution readiness remains separate. No extension to the other offers or a new purpose.
Closure/re-review trigger	Bind membership, roles, environment, start/end events and experiment identity before use. Reassess a new purpose/recipient, changed membership, restriction notice or approaching endpoint.

If a proposed annotation tool sends images to another service, the restrictions and re-review rows identify an authority issue. If an acquisition field is corrected within the permitted environment, lineage and technical applicability may need revision even though the purpose is unchanged. The entry can accurately record a completed bounded permission while those other decisions remain conditional.

Reviewer Lens. Resource, purpose, people, environment and window are the comparison, rather than the word “eligible” alone. A match supports that permission finding; it does not establish readiness or independence.

Consultant Lens. The product owner needs to see what the authority covers, what the pipeline would do and the exact mismatch, if any. That comparison directs a scope question to site/privacy owners and a technical question to the relevant producer. Advice can check the match; it cannot widen the permission.

A chain whose later steps are still planned

One supplied source record now anchors a proposed chain. The patient token (PT-DA-01), specimen token (SP-DA-01) and acquisition token (AQ-DA-01) must stay associated with its descendants. They provide grouping, rather than names or proof of anonymity. The documentary object identifiers below preserve engineering traceability; none is a downloadable file or proof that a planned operation occurred.

The table follows parent, operation, grouping and authority. Only its source row is supplied as a record. All subsequent objects are planned; no crop, label, dataset run, fitted candidate or evaluation result has been produced in this exhibit.

Object and version	Parent and operation	Grouping, reference and membership	Authority and consequence
Source image · SRC-DA-001-v1	Originating record linked to the acquisition token; intake preserves export/source relationships.	Member of the subset manifest MAN-DA-1; reference assignment not supplied.	Authorization and retention records apply; a source restriction or identity correction flags descendants.
Planned crop · CROP-DA-001-v1	Source image, extraction coordinates and extraction version.	Same patient/specimen/acquisition; no completed crop or label.	Parent and permission retained; changed extraction creates a different child identity.
Planned prepared image · NORM-DA-001-v1	Planned crop through identified preparation procedure P1, implementation and settings; actual execution still to record.	Same group; naming the procedure does not establish compatibility.	Input/output content identities and transform record retained, without substituting for origin or rights.
Planned annotation · ANN-DA-001-v1	Source/crop references through versioned annotation; initial and revised judgments preserved.	Instructions and eventual reference version bound; uncertainty allowed.	Bounded annotation covered by authority; no truth label invented.
Planned exploratory dataset · DS-DA-EXP-v1	Membership joins selected image, transformation and annotation versions.	Exploratory activity EXPL-DA-1; no independent-test designation.	Annotation or membership changes identify affected experiments without replacing historical inputs.
Planned experiment · EXP-DA-001	Consumes the planned dataset with identified candidate, code and environment.	No run, result or fitted candidate exists here.	Future candidate artifacts identified for disposition under the retention schedule; deleting a source alone cannot settle it.

Following the crop row demonstrates the mechanism. Its coordinates and extraction version identify how it would come from this source. The next prepared image identifies the implementation and the input/output contents. The annotation retains instructions, reviewer state and authorized revisions, rather than silently becoming truth. Actual dataset membership then determines what a future experiment consumes. If source identity is corrected, parent and membership links expose which crop, annotation, dataset and experiment need assessment. Renaming or deleting a file cannot perform that impact analysis.

Three different joins now do three different jobs. Patient/specimen/acquisition tokens preserve grouping. Parent/object versions preserve reconstruction. Authorization plus the retention schedule support the use-and-time comparison. None replaces another. The later chapters' small method cohorts are separate exhibits, with no implied membership of this subset. Likewise, the monitoring offer's training gap does not cancel this permission, and the supplier's license does not repair its evaluation-history gap.

Reviewer Lens. Starting at the proposed experiment and following membership backward should reach the supplied source, group and authority. A restriction or annotation correction should be traceable forward to the work it would affect. The present positive conclusion concerns a planned mechanism, not a verified record of execution.

Consultant Lens. Steward and experiment lead can implement and check the group, parent/version and permission/time joins on a sample record before the bounded run. The source site needs a correction route; the manufacturer needs a route for assessing affected descendants. Agreeing on both early makes later decisions possible without inventing a past history.

Figure 5.1 — Data Use Requires a Complete Lineage. The visual links source, transformation and use to purpose, authority, privacy, transfer, retention and owners. It summarizes the worked lineage; copying an image or producing a derivative does not create a new permission or independent patient.

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Data Use Requires a Complete Lineage

Availability does not establish permission, fitness, independence, or retention authority.

Key takeaway

Data fitness is inseparable from provenance, permitted use, transformation, privacy, licensing, and version control.

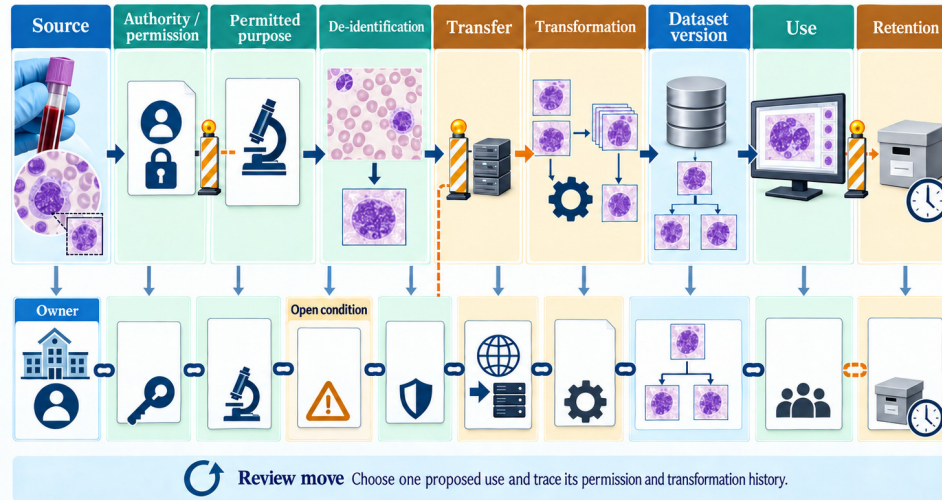


Figure 5.1. Data Use Requires a Complete Lineage.

Who owns the next decision?

The **data charter** gives the steward a record to maintain; it does not give the steward rights held by the originating party. Site/data authority and manufacturer privacy reviewers own their respective scope determinations. Security owns the approved environment controls. Clinical and machine-learning colleagues assess suitability and experiment behavior. The product owner makes the bounded development-readiness decision through the manufacturer's process. Those distinct contributions have to meet at the same proposed activity.

If an exploratory fit later creates a candidate, its source dataset and settings must be identifiable. The endpoint in the **exploratory retention schedule** would then raise the further use and retention question for sources, derivatives and that candidate. A 90-day image window is no perpetual candidate license, and deleting the images is no promise that their influence has disappeared. The responsible authorities and manufacturer must record the applicable disposition before continued use is treated as settled.

Material restrictions or lineage changes also reach quality/risk because they can reopen evidence conclusions. Independent evaluation, clinical benefit and release would each need their own basis. The new authorization event supplies none of them. The consultant can identify a missing relationship without creating permission or certifying anonymity from a filename.

Reviewer Lens. A readiness disposition is assessable when it points to the exact subset, authority, environment, endpoint and technical review. A bounded exploratory authorization deserves credit at that scope. The attempt to elevate it into confirmatory or release evidence is a different claim, with a different missing basis.

Consultant Lens. An owner map becomes useful when it explains what happens upon a permission change, source correction or approaching expiry. It should identify the authority decision, affected descendants and manufacturer re-review. Maintaining that route helps the company act; the adviser does not sign the originating site's or privacy owner's decision in their place.

What would let a consultant close a finding?

“More data approval needed” could describe very different problems in this packet. Is the requested purpose missing from the authority, a source relationship missing from the history, or a technical comparison unfinished despite adequate permission? A **data-governance and provenance assessment** distinguishes documented basis, unresolved rights, lineage breaks, privacy/legal referrals, evidence consequences and checkable closure conditions. Its value is to identify which decision is ready to change.

Start with the operation the team wants to perform

Annotation of the identified subset in the restricted environment, exploratory fitting, addition of the monitoring offer to training and designation of the supplier export for independent evaluation are different operations. The verb changes both the authority comparison and the technical evidence question. “Use the dataset” would conceal the distinction.

For the monitoring-only offer, the observation is that a document covers monitoring. It is not that no permission document exists. The proposed training use creates the mismatch to refer to site/data and privacy/legal owners. For the restricted exploratory subset, the documented scope matches the specified activity; the next questions are execution binding and technical readiness. A precise finding preserves the already closed question and directs effort to the next one.

Reviewer Lens. A missing training scope should not become a claim that the source never had any permitted use. Naming the requested operation and relevant authority passage makes the difference inspectable. The same precision protects the positive permission finding for the restricted subset.

Consultant Lens. The useful one-sentence comparison names what the pipeline proposes and what its authority covers. If they match, the agenda moves to execution and adequacy. If they differ, the referral names the authority owner and changed purpose. Repeating “obtain more permissions” on every row would obscure both decisions.

Which claim does the missing link prevent?

The supplier export's absent patient-group links prevent demonstration that the proposed evaluation units are separate from development exposure. Its missing acquisition information prevents a justified match to the proposed profile. Neither establishes poor images or a leakage event. Each leaves a specific evidence claim unsupported.

That distinction narrows the request to the group and acquisition history needed for the claim, through a controlled interface the manufacturer can verify. If the supplier cannot restore it, another resource or a narrower purpose may be appropriate, with its own authority and evidence questions. A license cannot create a lost patient relationship; a content hash cannot create provenance.

The same method could support a bounded re-analysis in the transfer episodes. If a scanner record or specimen link were corrected after a timing or release analysis, versioned lineage would locate the calculations that used the earlier link. The consultant could request reconciliation, and the manufacturer would decide the effect on the claim. No correction is asserted to have happened in either episode.

Reviewer Lens. Naming the blocked conclusion—patient independence, acquisition fit, source restriction or reconstruction—is more informative than “data quality poor.” It distinguishes absence of support from observed failure and gives the supplier a question it might actually answer.

Consultant Lens. Recovery should have a defined target and verification route. If the necessary history is unavailable, a justified revised-use or no-use decision may close the finding more honestly than promising to restore impossible fields. The manufacturer chooses that disposition.

Which conclusion belongs to the privacy reviewer?

The restricted subset supplies controlled tokens while its originating laboratory retains the patient key. The technical benefit is keeping related images together. What the privacy review covers, who controls the key and tokens, and which roles may use them remain distinct facts in the advisory account.

If a site association later needed correction, could that correction reach affected grouping tokens without exposing direct identifiers to the analysts? That is a concrete interface question joining privacy and evidence needs. It does not justify gathering more personal information merely for convenience, and successful grouping does not settle remaining identifiability.

Reviewer Lens. The documented privacy assessment and the grouping task should be read side by side. A transformation can hide direct names while leaving contextual identification questions. A token can work for partitioning while still requiring controlled access. Neither observation completes the other review.

Consultant Lens. The steward and site need the minimum token and correction interface; privacy/legal owners need to judge remaining identifiability. Advice can say what technical conclusion that interface would make possible without presenting the adviser as an anonymity certifier.

What record would change the finding?

A closure condition must be precise enough for another reviewer to test. For the monitoring offer, “obtain legal approval” leaves the requested training purpose and originating authority unstated. A useful condition names resource, activity, actors, environment, transfers, time and the authority record. For the restricted subset, a new hosted tool would change a recipient and processing path; its existing local-only scope would not move with the file automatically.

For the export, “restore traceability” similarly leaves the decisive relationships unspecified. The condition needs to identify patient grouping and acquisition history and how the manufacturer will verify them for independent evaluation. If those facts cannot be recovered, choosing not to use the export for that purpose can be a defensible closure.

A source restriction or group correction after a future experiment would require a further step: identifying affected datasets and candidates and the owner who assesses whether to rerun, limit or retire a conclusion. Updating today's register cannot retroactively repair yesterday's evidence. This is a prospective response method; no such correction has occurred in the restricted-subset exhibit.

Reviewer Lens. The question is whether a second reader could compare a named record with the proposed use and reach the closure finding. A generic approval label gives that reader nothing to test. A justified no-use disposition may provide a clearer endpoint than an indefinite request to fill an unrecoverable gap.

Consultant Lens. The missing field, producer or authority, verification step and decision owner belong together. Where earlier work could be affected, the re-review also needs the downstream inventory and reconciliation. The adviser makes that route concrete without choosing a remedy before the actual restriction and affected use are known.

The bounded data-use advice

The worked data-use advice (ADVICE-DA-1) reviews the proposed exploratory charter and rights entry at the same point after authorization and before execution. Its seven rows keep observation, interpretation, consequence, recommendation and re-review distinguishable. It records the narrow basis already supplied rather than issuing another permission.

Advisory field	Completed finding
Observation and documented basis	Authorization covers the identified subset and exploratory activity, assigned roles, restricted environment and retention window; no exploratory result exists.
Unresolved rights/privacy route	Additional recipients, public disclosure and continued retention/use are not granted; proposed extensions go to originating data authority and manufacturer privacy/legal owners.
Lineage finding	Source/group links supplied; crop, preparation, annotation, dataset and experiment descendants planned rather than verified completed objects.
Bounded interpretation	Requested activity matches stipulated scope; reference adequacy, acquisition compatibility, independence and execution readiness remain separate.
Evidence consequence	Bounded development evidence only; absent downstream identities would prevent reconstruction of the versions producing a result.
Recommendation and owner	Steward binds membership, roles/environment, endpoint and lineage fields; experiment/reference owners supply activity records; product owner decides readiness.
Closure and re-review	Check exact execution bindings and remaining technical limits before claiming readiness. Reopen on purpose/recipient, membership, restriction or retention change.

The last row would distinguish an analyst, environment or member that differs from the authorized plan and direct the change to the relevant authority or technical review. If the fields match, that focused finding can close while reference adequacy and independence remain open. The other offers retain their own findings. This entry neither approves real processing nor rejects all available material.

Reviewer Lens. The advice offers a supported positive permission observation and a planned lineage mechanism. Those are different evidence states. Reading them separately prevents a proposed record from becoming verified execution or a consultant recommendation from becoming manufacturer authorization.

Consultant Lens. Purpose or recipient change, membership change, a restriction notice and the retention endpoint are specific reasons to return. Explaining which question each reopens and who can close it gives the owner a useful future route. The advisory memo remains an assessment of authority, not an authority record of its own.

A boundary question

“The consultant confirms the dataset is anonymous because names were removed.” What could the consultant actually observe, which judgment belongs to the privacy owner, and which technical grouping question still needs an answer?

Decision

What wording would preserve all three without discarding the value of controlled tokens?

How a reviewer keeps observation and judgment distinct

A defensible wording is: “The consultant found that the supplied record describes removal of names; the responsible privacy assessment must address remaining identifiability and linkage in context. The technical review must also determine whether the retained structure supports the intended lineage and independence claims.”

Name removal is the observation; contextual privacy assessment is the specialist judgment; grouping and reconstruction are the technical questions. Controlled tokens can be useful without being declared anonymous in the absence of the relevant assessment.

A different request after a hypothetical result

Consider a counterfactual, not a later LumaCell event. Suppose the restricted exploratory work has produced a candidate, and the team now wants to publish selected source images in a public demonstration. How does that request compare with the original purpose and recipient scope?

Decision

May publication proceed on the supplied authority? Which charter and register fields change, does the candidate alter the permission question, and what lineage route would matter if a source image later had to be restricted?

How a reviewer follows the new use and its descendants

The purpose and disclosure environment have changed. Public disclosure is excluded by the exploratory authority, so that record does not permit the proposed demonstration. Applicable site/data and privacy/legal assessment and a documented disposition are needed for the new use. A successful candidate supplies neither the new authority nor perpetual rights in its source material and derivatives.

The relevant lineage runs from the source through crop, prepared-image and annotation versions, dataset memberships, experiments and candidate artifacts. That inventory makes an affected-use decision possible. It does not itself decide whether a model must be retrained, a record deleted or a result withdrawn; the authorities and manufacturer must assess the actual restriction. The point of the links is to make that decision evidence-based rather than guessed.

Where the source trail leads next

Book 1C offers optional depth on specimen handling, preanalytical conditions, source identity and applicability. Book 1A Chapters 7 and 13 extend artificial-intelligence/data and generalizability review. Book 1B Chapters 2 and 5 develop data flow, confidentiality, integrity, availability and supplier questions. These readings support the relevant question while accountable privacy, security and rights assessment remains necessary for actual work.

Book 2 Chapters 6–12 add European Union evidence and documentation interfaces. Actual processing questions need qualified privacy/legal assessment; the charter here is an evidence-design tool, rather than a universal permission template.

Chapter 6 follows the label itself: knowing its origin and permitted use does not establish that it represents the cell-category target the product is meant to predict.

Return to the proposed crop

Can the opening image's permitted activity be stated more precisely than “data approved”? Can its transformed descendant be followed back to source and grouping? Which missing fact would be a rights gap, and which would be an evidence gap? If purpose or a label changed, which downstream work and decision owner would have to be found?

Decision

What permission conclusion is now supportable, what technical question remains open, and what is the next accountable action?

Source Trail

S1: IMDRF N88, Principles 3–5, PDF p. 7: data, reference and training/test-independence principles. These principles do not grant rights.

S2: Regulation (EU) 2016/679, Recital 26 and Article 5(1)(b), for the narrowly stated EU privacy/purpose distinctions. No case-specific legal basis, transfer decision or anonymity finding is made.

The data charter, D-A/D-B/D-C exhibit, register and counterfactual are original fictional teaching constructions. The 90-day RET-DA-1 window is a case-specific contract premise, not a legal rule. No licensed data-governance standard is claimed as a consulted technical source merely because it appears in the candidate ledger.

CHAPTER 6

Chapter 6: Reference Standards, Labels, Annotation, and Adjudication

A morphology reader opens a supplied image of a candidate white blood cell from an adult peripheral-blood smear. In this proposed LumaCell study, the reader checks the image's identity, examines the information the instructions permit, and records a category decision with a reason. The current classifier suggestion stays hidden. Another reader may examine the same crop independently. Their records will help produce the comparison answer used to assess later suggestions.

An exported column might call that answer “truth.” The useful question is what happened before the value reached that column: what did the reader see, what question was answered, and what happened when no defensible assignment was possible? The reference is an output of a method that the manufacturer must be able to explain. The ten-image exercise later in this chapter is a separate fictional method example, not the restricted exploratory subset from Chapter 5 or a later evaluation cohort.

What must be known before the reader decides?

The proposed intended-use contract concerns a suggestion on a supplied candidate crop. A trained professional checks it and completes the smear review; the laboratory retains patient interpretation and result release. That tells the reference owner which level of question is being proposed. Chapter 5's lineage, the source-to-crop history, answers a different prerequisite: which specimen, acquisition and transformation produced this image, and which activity its permission covers.

The reader needs the right image and enough permitted information for the task. The manufacturer needs stable source/crop identity and an instruction version to join the eventual decision to a model output. Possessing the initial development image collection does not establish a reference method. The narrow authorization for the restricted exploratory subset also does not turn these separate images into an authorized or independent confirmatory set.

Decision

Can agreement with an image-category label establish a patient-level conclusion? If two readers choose the same label, what does their agreement establish, and what remains unknown about correctness?

How a reviewer reads an agreed label

The image-category comparison concerns the supplied crop. It does not establish a patient diagnosis, a whole-specimen finding or a complete automated differential. Two readers can also share an instruction, a source limitation or an error. Their agreement describes decisions made under a method; it does not independently verify that method against its target. “Expert” identifies a possible source of judgment but cannot replace the explanation of what was judged and why the method suits the question.

Reviewer Lens. The first useful comparison is between the unit in the evidence and the unit in the claim: image, specimen, patient or workflow. A well-supported crop comparison deserves credit at that level. The source-to-crop join and activity permission show what record is being used and under what authority; a broader claim needs its own link beyond that image-level finding.

Consultant Lens. Recruitment is easier to plan once the clinical/reference owner can state the exact task and the steward can demonstrate one source-to-crop connection. If the hoped-for endpoint is broader than that task, the company has a choice to make before collecting judgments: retain a narrower claim or develop a different reference design. The adviser makes the consequence visible; the manufacturer chooses.

Which question should this reference answer?

Suppose the reader is asked which of five common white-cell categories a crop supports, if any. The team first needs to say what property it is trying to assess. That property is the **target construct**.

The study must then obtain an answer to compare with a suggestion. A qualified morphology reading under specified instructions, with an unresolved route, is one possible method. This defined way of obtaining the answer is the **operational reference**. The documented benchmark used as the expected result for that specified comparison is a **reference standard**. Its title does not make the method infallible. [S2]

The distinction matters when the same crop could be accompanied by very different information. A source image, acquisition context, prior laboratory result, later patient outcome and panel judgment may answer different questions. Information known only after the device's intended decision can be suitable for one clinical target while creating a different information pathway into an image-category judgment. Conversely, hiding

context needed to interpret poor-quality material may produce a tidy but weak reference. The owner must specify the information, why it is needed and the inference it supports.

For LumaCell, that inference remains about the supplied candidate crop. Reading it cannot reveal cells that acquisition never supplied, establish complete-smear detection or a whole-specimen differential, determine disease status, or validate the laboratory's final report. The target and information choices therefore determine what a later agreement result could mean.

The **pathology queue episode**, where already imported cases appeared earlier in the queue while scanner import still constrained total time, gives a different target choice. Recorded display order could address placement; a time benefit would need case-linked events including import and completed review. In the **sequencing review episode**, suggestions displayed correctly while specimen reconciliation still constrained release. An independent variant assessment could address suggestion correctness, whereas a release benefit also needs the reconciliation records. These are prospective method choices around separate fictional observations, not new results or LumaCell data.

Reviewer Lens. An endpoint can sound clinically meaningful while the reader's actual question is much narrower. Placing the claimed endpoint beside the permitted information makes that mismatch concrete. For a crop claim, the reference must explain uncertain crops. For a patient or workflow claim, a different unit of evidence is needed. More reference detail at the wrong level will not close that gap.

Consultant Lens. The clinical owner can compare a bounded image question with a broader proposed question before settling the study. Each requires named information sources and an account of how those sources might influence the comparison. This gives the manufacturer a real choice about claim and method, rather than a late discovery that a carefully collected reference answers another question.

What gets lost when fuller morphology becomes one category?

A professional may see more than LumaCell is designed to suggest. Its proposed common categories are neutrophil, lymphocyte, monocyte, eosinophil and basophil, with an other/unclassified output route for review. A reference reader could positively determine that an image belongs outside those five. The reader could instead find the source insufficient to support any defensible category. Both may require attention, but they give the study different information.

The first state says something was determined: an outside-five category. The second says a determination could not be supported. In the exercise these are **outside-five**, **O**, and **unresolved**, **U**. They describe reference judgments. The product's

other/unclassified output route cannot erase that distinction by turning every unresolved reference into a known outside-category answer.

To preserve meaning, the record keeps two layers: the fuller morphology description and uncertainty the reader can support, then the rule translating that record into the comparison category. That rule is the **endpoint mapping**. Retaining both layers also permits later examination of whether an aggregate “other” result hides different outside-category groups. This exercise supplies no such subgroup counts. Where a fuller description is unavailable, the mapping cannot invent it.

The same operation selects distinct endpoints in the transfer episodes. In the pathology queue episode, a worklist order maps to placement among already imported cases; linked import and review times map to an interval. A hypothetically missing import timestamp would leave that interval unassessable, not show “no delay.” In the sequencing review episode, a source-supported variant judgment maps to suggestion correctness; specimen reconciliation and release records support a separate release endpoint. An unavailable required identity record would leave the latter unresolved, not produce a variant category. Neither missing record is asserted to have occurred.

Reviewer Lens. A count of “other” is interpretable only when the source and mapping show which records it contains. A few complete paths through common, determined outside-five and unresolved routes can reveal whether the count means what the claim says. The issue is the comparison's meaning, rather than an aesthetic preference for a particular coding scheme.

Consultant Lens. The clinical/reference owner defines the permissible source descriptions; the evidence lead defines their endpoint mapping before predictions are compared. The steward retains both fields and the mapping version. That early separation makes a later reassessment possible without overwriting the original morphology judgment or pretending the old analysis used a revised rule.

Could another person reconstruct this decision?

Suppose one reader records outside-five and another unresolved. Before deciding how to resolve the difference, the team needs to know whether they saw different morphology, interpreted the rule differently, lacked source quality, or used different permitted context. “Use clinical judgment” leaves all those possibilities open. The instruction must make the route from image to decision reconstructable.

The reader would check the source token, examine the permitted material, choose a state and record a rationale. Recording that decision is **annotation**. The resulting value used in a specified comparison, after any planned review or adjudication, is a **reference label**. For this proposed task, the record would contain the initial common-category, outside-five or unresolved state, morphology rationale, source/crop token, reader

identity and instruction version. An insufficient source or mismatched crop identity also needs an escalation flag.

That record cannot be reliable if the system requires a guessed common category just to fill a cell. The instruction needs a usable unresolved route and a prohibition on inferring disease from one crop. These are proposed contents of the reference protocol, not evidence that a protocol has been implemented. Separate preparation examples can expose ambiguous rules while clinical judgment still has its necessary role. Discussing difficult evaluation crops during instruction refinement, however, changes their exposure history even if no pixel changes. Preparation examples and material intended for independent evaluation must remain distinguishable.

A corrected instruction also changes the meaning of downstream records. If the outside-five definition changes, the old label needs its prior state, instruction and mapping version, source and consuming dataset retained. A new value cannot silently replace what an earlier experiment used. That connection lets the manufacturer judge which analyses need rework.

For the pathology queue episode, a prospective placement instruction would specify imported-case identity, allowed priority information, recorded rank and whether the software's proposed rank is hidden. A timing instruction would separately define import and review events. For the sequencing review episode, the instruction would specify permitted variant evidence and verified specimen context, prevent copying the displayed suggestion into an independent reference, and allow unresolved status when source information cannot be checked. Each record then answers its own target.

Reviewer Lens. A difficult record is often more informative than a page of easy agreements. If its initial decision can be reconstructed from source, instruction, rationale and version, the process is reviewable. If an undocumented conversation or model suggestion is needed to explain it, the finding can name the precise method or independence gap. A plausible final category alone does not repair that missing path.

Consultant Lens. Testing draft instructions on separate preparation material is a relatively cheap way to expose ambiguity before controlled labeling. The clinical lead owns the clinical wording; the steward can make allowed states and source/rationale/version fields operational, including escalation. This is easier to correct before the database contains labels that different readers understood differently.

What does “qualified reader” establish?

A professional title does not tell us whether a reader can apply this crop taxonomy, distinguish a determined outside-five finding from an unresolved one, or document a source-quality limitation. Readiness belongs to the particular instruction and role. A clinical/reference lead could examine documented peripheral-blood morphology competence and work on separate preparation examples before assigning the task, recording who qualified whom and against which version.

Even that readiness would not guarantee that every image is classifiable. A qualified reader may correctly follow the method by recording unresolved when the source is insufficient. Forcing certainty would distort both the individual label and the apparent ability of the reference process to resolve images. The later fixture stipulates qualified readers but supplies neither credential files nor an actual readiness result.

The transfer tasks expose the same boundary between skill and evidence. A pathology priority assessor needs task-relevant worklist judgment; an elapsed-time analyst needs the event definitions and log joins. A sequencing variant reader needs the specified assessment competence, while specimen reconciliation belongs to the authorized laboratory process. A strong variant reader cannot certify a missing identity link. These are proposed role distinctions, not supplied credentials for those episodes.

The International Medical Device Regulators Forum's good machine learning practice principles emphasize suitable expertise and fit-for-purpose references. IMDRF N88 is nonbinding; it does not mandate two readers or a universal adjudication recipe. [S1] Depending on target, information and risk, a justified method could use one reader, multiple readers, an objective measurement or a panel. The manufacturer must explain the chosen method.

Reviewer Lens. The task, readiness criterion, preparation material and authorization record should refer to the same instruction version. Actual evidence of readiness can close that question at its stated scope. A title or teaching stipulation cannot do so, and a legitimate unresolved decision should not be treated as evidence of incompetence merely because it is inconvenient for analysis.

Consultant Lens. A short readiness assessment on non-evaluation images gives the reference owner an observable basis for assignment. A changed instruction may create a reason to reassess that readiness. The adviser can frame the evidence request and observe the process, but cannot confer credentials on the manufacturer's behalf.

What could influence the answer before it is recorded?

An independent reference for testing a classifier should not be produced by a reader who can first see that classifier's suggestion. The suggestion could influence the judgment used to assess itself. Similarly, initial readers should not see each other's states when their separate decisions are intended to reveal disagreement. **Reference blinding**, keeping the candidate output from guiding its own comparison, specifies which information is withheld, from whom and until when.

The information choice still has to serve the target. Source context needed for interpretation can remain visible; removing everything is not a method. A proposed adjudicator might first record a source-only assessment and then inspect the initial rationales. If the method chooses that sequence, the sequence needs a record. The rationale is to identify which judgment preceded exposure to another judgment.

The actual view matters as much as the access design. A platform can usually hide a model field while an exception exposes it in one reading. The activity record must preserve what was presented and the exception. The affected record then needs an assessment of whether it supports the intended comparison; describing the entire run as blinded would erase the relevant history.

There are two distinct independence questions. **Reader independence** concerns how judgments influence one another. **Data independence**, the separation from development influence considered in Chapter 5, concerns whether patient, source image or related information has already influenced model development choices before evaluation. Two separate readers can assess a crop from the same development patient. Conversely, separating patient groups cannot make a weak reference method sound. Source/group lineage and exposure history answer a different question from the reader-view log.

For the pathology queue episode, copying a displayed rank cannot supply an independent expected-rank judgment. In the sequencing review episode, reading with the displayed suggestion may be legitimate assisted review, but cannot silently become an independent reference for that same suggestion. Development and human-use questions may use assisted activity honestly when they name it. Its usefulness does not erase the influence path.

Reviewer Lens. The productive evidence request distinguishes output-influenced labels, correlated initial judgments and reused development material. A representative actual view, access event and source/group membership can locate which problem is relevant. Treating all of them as a generic “independence issue” makes it harder to choose the appropriate disposition and to preserve useful unaffected evidence.

Consultant Lens. Platform and evidence leads can specify allowed context and prohibited model or other-reader information at each step, with a breach record and a

way to find affected dataset members. The steward separately checks patient/source grouping and development exposure. Making both mechanisms explicit early prevents a technically hidden field from being mistaken for proof of an independent evaluation.

What should happen when the readers disagree?

An initial disagreement is useful information only if both states and their reasons survive. A single final value could conceal ambiguous morphology, different interpretations of an instruction, poor source quality or an uncertain boundary. Keeping the initial records gives the reference owner a way to distinguish those possibilities.

Before model results are examined, the owner therefore specifies a review route: which records enter, who examines them, which information becomes visible in which order, what final states are allowed and who may authorize an unresolved ending. This is **adjudication**. Its entry conditions must address uncertainty as well as disagreement, including what to do when both readers say no assignment is possible. The fixture will use a proposed rule; it is not a universal clinical procedure.

The process may produce a supported assignment or document that the available source cannot resolve the question. A majority vote would also be a chosen decision rule, not an independent test of truth. If the rule changes after results become known, a justified correction remains possible, but its reason, affected cases and timing must remain visible. The corrected process cannot be represented as the original prospective method.

This is where the pathology queue and sequencing review episodes pose a sharper question: are two records answering the same target, or recording different stages? Queue placement and scanner import concern different workflow stages. So do displayed suggestions and specimen reconciliation. Their records need linkage and investigation by the relevant workflow owner. Forcing them through a reader-label adjudication rule would mistake different events for competing judgments about one reference question.

Reviewer Lens. The dated entry rule, actual adjudication queue and model-result access history should tell a consistent story. A documented resolution can be credited without becoming “proven correct,” and unresolved endings should survive export. If the rule changed, the review needs the change's consequence for the intended comparison rather than a polished account that conceals when the choice was made.

Consultant Lens. A disagreeing pair, a matching unresolved pair and an access-breach example can expose missing decisions before labeling begins. The reference owner needs authority for rule changes and a route for assessing affected evidence. Leaving the choice to an analyst after seeing which rule favors the classifier would put the evidence question behind the desired result.

Why are these three cells empty?

One reader examined the image and could not support an assignment. Another required reading never happened. A third reading may have happened, but its record cannot be recovered. An **unresolved** judgment, **reading not done**, and **record unavailable** describe different events. The first records a limit encountered by the method; the others identify missing work or documentation. None supplies the positive outside-five determination represented by O.

Those meanings determine how the study accounts for its material. **All entrants** means every image put into the reference process. A **reference-evaluable subset** may include only images with resolved final categories. The latter denominator is smaller when some entrants remain unresolved or lack required records. Counts and reasons must remain beside the subset result so its boundary is visible.

A **sensitivity analysis** can examine how a conclusion changes under stated treatments of uncertain or missing states. It cannot recover their unknown categories by calculation. Neither counting every such record as an error nor automatically removing it is right for every endpoint. Prospective rules must fit the comparison. Choosing the denominator afterward to improve the result while hiding excluded records changes the question being answered.

If a future pathology timing inquiry cannot recover an import timestamp, or a sequencing release inquiry cannot recover a required specimen link, the endpoint remains unavailable with its reason and entrant count retained. The source owner would investigate; a justified correction would create a new version and reconciliation of affected analyses. These are hypothetical missing records. They do not rewrite the observed queue placement or correct suggestion display, and unavailable source material cannot become a determinate endpoint merely to complete a table.

Reviewer Lens. Reconciling entrants to readings, adjudications and exported states makes denominator changes explainable. For each excluded image, the distinction is between observed uncertainty and missing work or record, followed by whether the declared inclusion rule was applied. That trace is more informative than a reassurance that the final dataset has no blanks.

Consultant Lens. The evidence lead can define entrant-to-final-state accounting before comparisons begin, while the steward keeps missing work and missing documentation technically distinct from unresolved and outside-five judgments. If source recovery fails, a transparent limitation or revised analysis remains possible. A filled-in label would make the file neater while destroying the evidence needed to explain it.

Can the plan explain one exported label?

A final category with no source token cannot be traced; a traceable category produced under the wrong information view cannot support the intended independent comparison. The **reference and annotation plan** links those questions rather than giving them separate approval boxes. The table is useful when following an exported value backward to the decisions that made it possible.

Question the plan must answer	What the record needs
What property is being represented?	The target construct and level of the proposed claim
How is the comparison answer obtained?	Information, procedure, rationale and limitations
What does the reader do?	Instructions, states, decision rules, prohibited inferences and recorded fields
Who is ready for that task?	Relevant competence, preparation and readiness assessment
What is withheld, from whom and when?	Blinding rationale, implementation and handling of breaches
How are disagreement and uncertainty reviewed?	Entry conditions, roles, permitted information, decisions and unresolved route
What does each uncertainty state mean?	Distinct meanings and their representation
What if a reading or its record is missing?	Reasons, denominator treatment and analysis implications
Was the process followed?	Checks on consistency, identity, access and decision records
What happens when a rule or label changes?	Versions, reasons, approval and affected evidence
Can the whole path be reconstructed?	Source, reader, protocol, initial/final states, times and dataset membership

The process-checking row is **quality control** at this level: required readings and rationales exist, identities agree, access exceptions have dispositions and final states have earlier records. It does not certify every final category as biologically correct. The next row, **change control**, makes a corrected instruction or label a traceable new version and identifies the experiments affected by it.

Following the last row leads back to Chapter 5's data charter and Chapter 4's development plan. Missing source/group links can prevent the intended join, while scheduling confirmatory work before reference readiness can prevent the proposed evidence step. The earlier narrow exploratory authorization cures neither method gap. A coherent plan shows what remains to be done without granting permission to label or evaluate.

Reviewer Lens. One complete demonstration from plan to exported value tests whether target, source, qualified reader, instruction, permitted view and disposition fit together. When a link is absent, the finding can state the particular claim it blocks. Calling the whole table incomplete loses that precision and can conceal the work that is already adequate.

Consultant Lens. A prospective walk-through brings clinical/reference, data, platform and evidence owners to the same item. Each can identify its produced record and handoff, making a sequence for unresolved readiness decisions possible. A completed plan is then a useful manufacturer planning record, with actual execution and activity authority still to establish separately.

Two readings of each image

The following author-designed fixture contains ten images from ten fictional adults, one image per adult. It is separate from the restricted exploratory subset and from every later method or evaluation cohort. The stipulation does not perform an operational check of identity, permission, acquisition fit or future partition independence. It supplies a small process example, not clinical-performance evidence or a recommended study size.

Two stipulated qualified readers independently assess each image under the same proposed instructions. Neither sees model output or the other's initial state. The table contains two reader records for one image in each row; neither column is a classifier prediction. Category names are written out below. Outside-five corresponds to O and unresolved to U. Reader rationales, credential/readiness files and actual access logs are not supplied.

Image	First reader's initial state	Second reader's initial state
R01	Neutrophil	Neutrophil
R02	Lymphocyte	Lymphocyte
R03	Monocyte	Monocyte
R04	Eosinophil	Eosinophil

Image	First reader's initial state	Second reader's initial state
R05	Basophil	Basophil
R06	Outside-five	Outside-five
R07	Neutrophil	Lymphocyte
R08	Outside-five	Lymphocyte
R09	Eosinophil	Outside-five
R10	Unresolved	Unresolved

For example, R08 has two different initial judgments on the same source image. The table says what each reader recorded; it supplies no independent truth with which to choose between them. No model predictions or patient outcomes are present anywhere in this fixture. The reference process can review disagreements and retain unresolved endings without forcing each record into a common category.

Decision pause — what can this ledger tell us?

The earlier distinctions can now be applied to a particular ledger and a proposed method, with missing protocol details left visible.

Decision

What fraction of images have identical initial reader states, counting unresolved as a recorded state? Why is that fraction different from reference accuracy? What next process action is appropriate for R07–R10, and what would need to survive afterward? How could the eleven-field plan protect the comparison from model influence? Which details would be proposals because the exercise has not supplied them?

Additional dispositions and the next question

One more record: the adjudication dispositions

For this fictional exercise, adjudication resolves R07 as neutrophil and R08 as outside-five. It cannot resolve R09 from the available material. R10 remains unresolved because its source does not support an assignment. The initial reader records are retained and no model output is used. These are stipulated dispositions, not timestamped adjudicator reports or proof that the proposed protocol was executed.

Decision

What fraction of all entrants now have a resolved final reference? Which images remain outside a reference-evaluable subset, and what must still be reported if that smaller subset is used?

How a reviewer connects the two calculations to the method**Two quantities, two questions**

The first numerator counts matching initial states. R01–R06 match on a determinate category. R10 also matches: both readers recorded unresolved. That gives seven matching pairs among ten images, $7/10 = 70\%$ **initial-state agreement**. The denominator is all ten entrants, not twenty readings. Matching inability to assign is part of this state-agreement question, even though it supplies no evaluable common or outside-five category.

The second numerator asks which images have a resolved final reference. The additional dispositions resolve R07 and R08, adding them to the six initially matched determinate records. R09 and R10 remain unresolved for their stated reasons. Thus $8/10 = 80\%$ **have resolved final references**. This is the yield of the fictional reference process. Neither percentage measures classifier accuracy, diagnostic sensitivity, reference correctness or a patient outcome. There are no classifier predictions or independent truth values in the exercise.

Under the proposed rule, the reference-evaluable subset contains R01–R08. A future analysis using it would still need to disclose the ten entrants, the two excluded records, their reasons and the chosen denominator. The present fixture cannot calculate performance even on those eight, and a future result on resolved images would not automatically apply to unresolved material.

Reviewer Lens. Reconstructing the numerator and denominator from the ledger gives both percentages a defensible meaning. They are useful process observations and can be credited as such. Any proposed model-performance statement would need the missing prediction/reference join; changing the heading on a process percentage cannot supply it.

Consultant Lens. An analysis population and process-flow account are easier to agree before anybody sees model results. Retaining the two unresolved records and their reasons in the export ledger allows a later excluded-case assessment. Deleting them now would remove the information needed to judge whether the eventual narrower comparison answers the intended question.

What the proposed method would have to specify

The worked reference and annotation plan, recorded as REF-1-P1, applies the eleven fields to this fixture. Its identity is a plan identity. It is not a record that readers were actually qualified, checks executed or reference adequacy established. Two readers and the checks below are choices for this example, not universal device requirements. The table is useful for seeing which operation and record would have to exist behind the final category column.

Plan question	Proposed method for this exercise
What is the target?	The supplied crop's image-level category: one of the five common categories or a determinable outside-five category. No complete-smear detection, differential-count or disease-status claim.
How is its reference obtained?	Two independent morphology readings of the same source crop under controlled instructions, followed by the adjudication below. No model prediction or patient outcome is supplied. Limited source context and possible shared error remain limitations.
What does each reader record?	Initial state, source identity, reader identity, instruction version and rationale. Outside-five requires a positive determination; inability to determine is unresolved. A fuller morphology description is retained where supportable.
Who is ready?	A clinical/reference lead reviews documented peripheral-blood morphology competence and performance on a separate preparation set. A separately assigned adjudicator made neither initial reading. No actual readiness result is supplied.
What can each person see?	Initial readers see permitted source and instructions, not each other's states. Neither readers nor adjudicator see model output. A recorded breach suspends affected records' independent-comparison use pending impact review; records are retained.
Which records enter adjudication?	Every differing pair and every pair containing unresolved, including matching unresolved. The adjudicator records a source-only assessment first, then reads both initial rationales and records a final disposition, which may remain unresolved. No state is selected to agree with the model.
How is unresolved meaning retained?	Outside-five and unresolved remain different. Unresolved disagreement and insufficient source have distinct reason codes: U-DIS and U-SRC. These identify process reasons, not microscopic diagnoses.

Plan question	Proposed method for this exercise
What if work or documentation is missing?	M-NOT-DONE means a required reading did not occur; M-UNAVAILABLE means its record cannot be recovered. Neither is an observed unresolved judgment. All ten entrants remain accounted for; only resolved common or outside-five states enter the evaluable subset.
How is adherence checked?	Before export, the steward reconciles ten source IDs, twenty initial readings and required adjudications. A separate process checker examines every record's identity, version, permitted state, rationale, access exceptions and final disposition. Identity or unexplained record mismatch blocks export until resolved or explicitly dispositioned. Agreement percentage cannot replace this check.
How are changes controlled?	Instruction and exported-label versions are frozen separately. Corrections create new versions with reason, authorized reference-owner disposition and affected-dataset list. Earlier experiment inputs remain intact. A changed definition triggers reassessment of affected records rather than automatic reuse.
What does each export row join?	Image/patient tokens, original source, both initial readings, adjudicator assessment, final state/reason, instruction and label versions, event timestamps and consuming dataset IDs. Access history preserves exposure relevant to claimed independence.

The adjudication row makes the queue reconstructable: R07, R08, R09 and R10 enter. R10 enters despite matching initial states because matching unresolved judgments do not give a resolved reference. Recording the adjudicator's source-only view before opening the rationales distinguishes that assessment from the later informed disposition. It does not make the adjudicator infallible.

R08 shows what the final export must carry. The first reader recorded outside-five, the second lymphocyte, and the later stipulated adjudication gave outside-five. That final value needs a join to the same image and patient/source group, both initial states and rationales, the method version and adjudication reason. The steward checks the linkage; the authorized reference owner judges whether the morphology disposition is justified. R10 needs the same history even though its final value stays unresolved.

The following compact ledger separates the supplied final dispositions from the fuller reasons that the proposed operational method would require. It is useful when checking that export has not erased the disagreement or an unresolved outcome.

Image	Retained initial judgments	Final reference state	Limited reason supplied here
R07	Neutrophil / lymphocyte	Neutrophil	Adjudication resolves it; the underlying rationale would be retained in an operational record
R08	Outside-five / lymphocyte	Outside-five	Resolved outside the common set; no finer morphology finding is supplied
R09	Eosinophil / outside-five	Unresolved	U-DIS: the available material does not resolve the disagreement
R10	Unresolved / unresolved	Unresolved	U-SRC: the source does not support an assignment

R09's final state is not another name for a category outside the common set. Its row preserves an unresolved disagreement. R10 likewise preserves an insufficient-source ending. These rows supply stipulated dispositions, not actual dated reading or adjudication reports. A missing operational rationale would still be a documentation gap even if a final category looked plausible. Nor does the one-image-per-adult stipulation replace a future audit of original patient/specimen groups, derived crops and development exposure against the intended partition rule.

Suppose an analyst wants to recode R09 and R10 as outside-five so the export has no missing categories. That would change the target mapping: inability to assign would become a positive outside-category determination. Their unresolved states must remain, with treatment governed by the stated analysis population. Justified new information or a corrected method under controlled review could change an assignment; pressure to fill a cell cannot.

The same records explain two tempting misreadings of the percentages. Calling the resolved fraction “improved accuracy after adjudication” invents an independent truth comparison. Calling the final dataset simply eight uncomplicated images hides two entrants and the limits they revealed. Adjudication gained useful resolved assignments for R07 and R08, while the unresolved images remain part of the process evidence.

Reviewer Lens. R08 tests the path from final category back through initial disagreement to a justified disposition under a known version. R09 and R10 test whether unresolved reasons survive a smaller evaluable export alongside the full entrant account. The plan can be coherent at this level while actual qualification, access and execution evidence remain to be produced. Those are distinct review states.

Consultant Lens. A dry run on separate preparation material can exercise one disagreement, a matching unresolved pair, a source-token mismatch and a later label

correction. The reference owner authorizes the disposition rule, the steward demonstrates a versioned export, and the evidence lead identifies affected analyses. This can improve readiness, but its completion must be recorded before it supports a real activity claim.

Figure 6.1 — From target to a traceable reference label. The selected technical figure places qualifications, controlled views, independent initial reads, adjudication and unresolved outcomes between the target question and the final label. A label is an output of that versioned process, not self-evident truth.

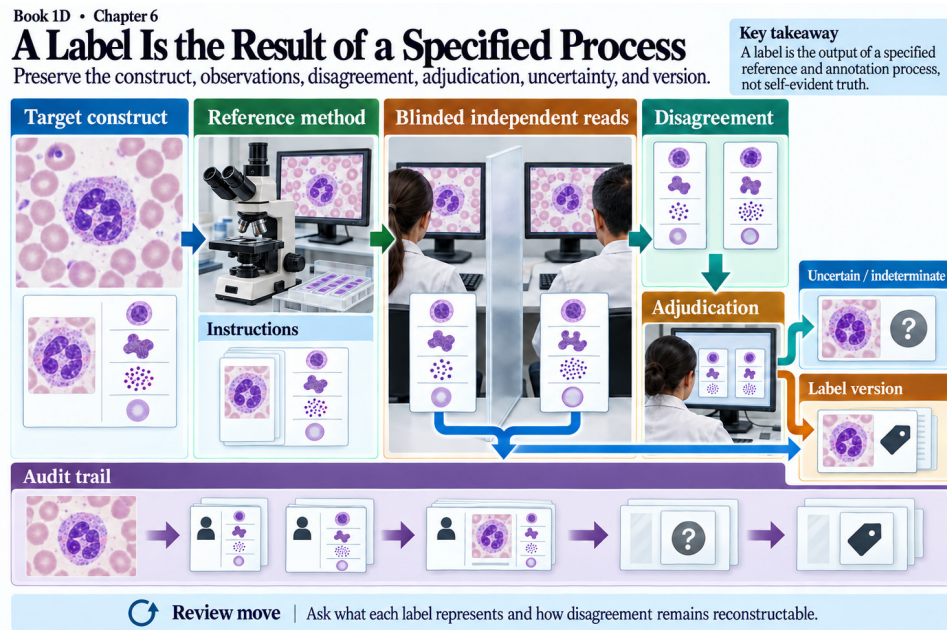


Figure 6.1. A Label Is the Result of a Specified Process.

Who decides that the reference is ready for this use?

A steward can demonstrate that the final label points to the right source without deciding the morphology. The clinical/reference owner specifies the question, readiness and adjudication method, and signs the rationale for final reference decisions. Readers produce initial states and reasons; the adjudicator produces a documented disposition. The steward retains their joins to source/group identity, versions, permissions and consuming datasets. Software/security owners implement and evidence permitted views and integrity controls. Each supplies a different part of the same decision record.

The model team may use a legitimately available development label, but cannot invisibly feed evaluation results back into a supposedly independent reference method. Quality/risk and evidence owners assess what unresolved states and protocol changes

mean for the endpoint. The manufacturer then judges adequacy of the executed method and data for a particular activity. A documented method might be accepted for bounded exploratory analysis while independent confirmatory use remains unready.

Chapter 5's narrow authorization belongs to its exact restricted exploratory subset and environment. It supplies neither rights for these ten method images nor acceptance of the worked reference plan or any future independent set. Permission, reference adequacy and independence remain separate conditions even when the manufacturer reviews them in one readiness decision.

The transfer episodes also need an owner chain when a new inquiry changes a source record or reference method. The authorized owner documents the reason, the steward versions the record, the evidence lead finds consuming analyses and the manufacturer decides whether to qualify a claim or obtain new evidence. A supported narrow endpoint can remain usable while broader time or release claims await stage-specific records. Neither the consultant nor model team can silently backfill the original observations.

Reviewer Lens. The intended-use statement needs to agree with the source, reading, disposition and endpoint records. A bounded claim can close when those records actually support it, while missing execution evidence remains an identified gap. A broader clinical question may require a different reference or workflow measure, rather than more signatures on the same crop label.

Consultant Lens. Readiness work becomes manageable when each function receives its own question: clinical/reference for method and qualification, steward for joins, platform for actual views, evidence lead for denominators and partition rules, manufacturer for the activity disposition. Advice can improve or challenge those records. It cannot create missing truth or approve confirmatory use on another party's behalf.

Consultant assessment — what is behind the final column?

The consultant receives proposed instructions, a sample source-to-label export, reader visibility rules and the intended evidence claim. A class-frequency chart may look complete while the sample path exposes a consequential gap. The useful assessment connects what the method supports now to the open inference and the owner or record that could resolve it. This prospective review differs from a reviewer's question about whether executed manufacturer evidence substantiates a stated claim; neither role takes over the morphology decision.

Which meaning survived export?

A useful sample follows a straightforward common-category judgment, a positively determined outside-five judgment and an unresolved one. For each, the path shows information used, instruction version, reader, changes and remaining uncertainty. Selecting only easy labels can miss the branches that determine whether the endpoint is interpretable.

If the export merges outside-five and unresolved, the finding concerns a lost distinction between a known target category and absence of a resolved reference. The issue is more consequential than inconsistent spelling. Even a perfectly retained label can still be the wrong evidence if an image-level reference is attached to a complete-smear or patient claim.

Reviewer Lens. The source-to-endpoint trace decides the scope that can be credited. A clean class-frequency plot does not establish complete-smear detection. The narrow request is the mapping and records needed to explain what the plotted classes mean.

Consultant Lens. Clinical and evidence owners can compare the source, fuller supported description, mapped category and unresolved reason side by side. A versioned mapping decision resolves an ambiguity that a cosmetic export change would conceal. It also identifies which existing analyses would need reassessment.

What did the reader actually see?

Once the label's meaning is clear, its influence history matters. A protocol saying that readers and adjudicators were blinded needs agreement from the implementation and actual process record. If predictions were visible, the affected records and intended comparison need assessment. The labels need not become useless for every purpose, but their circularity cannot be ignored for the proposed independent evaluation.

Separation between readers also does not establish that their errors are statistically independent. Shared training, instructions and source limitations may produce related errors. Separate readers do not establish patient or source separation across development and testing either. The rationale, grouping and exposure checks need to address these different relationships without assuming independence from a job roster.

Reviewer Lens. Comparing a promised view with an actual access event and source/group record identifies the precise comparison an exposure or overlap affects. A claim-specific disposition is more informative than declaring all uses of the image invalid, and more defensible than calling any two-person process independent.

Consultant Lens. The platform lead can reproduce each role's view, while the steward checks patient/source overlap and development exposure. If a breach appears, its affected identities and proposed evidence-impact route give the manufacturer's evidence owner a concrete decision. Renaming the run cannot change what the readers saw.

When was the resolution rule chosen?

A final category's history must show whether the entry rules, information, authority and unresolved route were specified before model performance became known. A changed method needs its reason, affected records and analysis consequences. A late correction can improve the dataset; preserving a tidy story is no reason to keep an error. The requirement is an honest account of the correction's timing and effect on prospective and independent claims.

In the fixture, resolving R07 and R08 gives legitimate additional process information. It does not turn the resolved-reference fraction into diagnostic accuracy. The initial and final records support the first conclusion without supplying an independent truth benchmark for the second.

Reviewer Lens. The proposed queue rule can be compared with its four entries and retained dispositions. Documented resolution deserves credit, and R09/R10 remain visible. A changed rule needs an impact assessment; it does not erase the earlier process or prove that all corrected labels are wrong.

Consultant Lens. An old-rule/new-rule comparison can link affected images, timing relative to model review and consuming analyses. The authorized reference owner decides any justified reassignment. The manufacturer decides whether earlier evidence remains usable and what needs to be rerun or qualified.

What would close a missing-record finding?

A finding about absent adjudication records does not close merely because every final category is filled. It closes when the method and decisions can be reconstructed, including unresolved endings. That could require restored initial states, adjudication reasons, instruction versions, final uncertainty codes and the prospective denominator rule, followed by reconciliation of affected datasets and analyses.

Some initial records may prove unrecoverable. Their limitation must remain explicit. Inferring the missing reference from model output would manufacture the very comparison used to evaluate the model. A defensible closure can preserve that uncertainty and qualify the affected analysis rather than invent a label.

Reviewer Lens. One affected export row tests closure directly: initial states, final reason, method version and inclusion decision should be recoverable or explicitly limited. Nonblank categories are a poor substitute for this check because they can hide both missing documentation and lost uncertainty.

Consultant Lens. A useful closure condition gives each owner a testable action: restore or disposition missing reading evidence, reconcile the versioned export, then rerun or qualify affected analysis. "Fill all labels" would reward the behavior the finding is meant to prevent.

A bounded assessment

A defensible assessment could say: “The proposed process separates initial reader decisions from adjudication. The export must retain outside-five versus unresolved, initial versus final states, and the two unresolved records. The 70% initial-state agreement and 80% resolved-reference fraction describe this fictional process; neither establishes model accuracy, reference correctness or clinical performance. Endpoint, denominators, adherence and traceability need confirmation before the labels support the stated activity.”

That conclusion leaves room for a positive decision. Once the specified identities, actual views, qualifications and records are checked, the manufacturer may accept the method for its declared image-level activity while withholding broader clinical or confirmatory claims. The open question has a route to closure more precise than a demand for “better ground truth.”

Decision pause – who may resolve a label?

The proposed memo says: “The consultant will resolve ambiguous labels so the evaluation can finish.”

Decision

What can the consultant properly promise, who can authorize a justified reassignment, and what should happen when the evidence still cannot resolve the label?

How an adviser preserves decision authority

The promise can concern identifying unresolved states and assessing whether the manufacturer's qualified reference process and endpoint treatment are adequate. Authorized reference personnel determine any justified reassignment. Where evidence cannot resolve a state, uncertainty stays in the record. The adviser can improve the method and its reviewability without becoming an undocumented source of truth so a schedule can be met.

Decision pause — a helpful suggestion on the reader's screen

In a counterfactual proposal, the platform would show each reader the current model's suggested category before the reference assignment to speed annotation. The team would still call the activity “independent review” because two people work separately. The general influence distinction has already been taught; the question now is which concrete records and comparison would need to change.

Decision

Which information reaches whom, at what point, and which evidence claim does that change affect? What should happen to the reader-view version, exposure history and source-to-label records? What useful purpose could the changed activity still serve?

How a reviewer follows the new influence path

Separate readers can both be influenced by the same model suggestion. Showing it before assignment changes the basis of the reference used to assess that model. The manufacturer needs to assess the activity's purpose, available information and consequence for the intended comparison. Assisted annotation may be useful for development or human-use work, but cannot automatically supply an independent reference for evaluating the same model.

The changed field is the current model's suggested category; its recipients are the reference readers; the exposure occurs before each initial decision. The manufacturer would version the reader-view instruction and retain model/version and exposure information on affected records. It would decide whether the intended evaluation needs a separate, properly independent comparison. Chapter 5's data-flow, access and lineage records must reflect the same change; introducing it through a screen setting does not make its history disappear.

Reviewer Lens. Actual screens and access evidence establish what happened in the comparison under review. If suggestions were visible, the useful request identifies affected records and the claim-specific disposition. The presence of two people cannot establish model-independent references.

Consultant Lens. Choosing between assisted development reading and independent reference generation before changing the interface lets the team configure and log an honest activity. Already exposed records go to the manufacturer for evidence-impact review. Quietly relabeling them would preserve the desired name while losing the information needed for a defensible decision.

Where a specialist question can go next

Book 1C offers optional depth on reference, measurement, interpretation, bias and validation questions. Book 1A Chapters 7 and 11–13 deepen artificial intelligence, data, modalities and statistics; Book 1B Chapters 2–6 deepen annotation-platform access, integrity, supply and threats; Book 2 Chapters 8–12 deepen European Union scientific-validity, performance and study interfaces. The local method and plan provide the working basis here.

Chapter 7 carries source and reference relationships into dataset partition and coverage decisions: correct labels cannot prevent leakage, while independent partitions cannot repair an inadequate reference.

A final reference question

For a future image-level comparison, could one exported label be followed back through its target, instruction version, reader qualification, permitted view, initial judgment, adjudication rule and unresolved route to the source? Could the same record identify its final label version and consuming dataset?

Note

Which owner would check source/group independence, and who would decide readiness for the activity? Do the entrant and evaluable denominators explain the endpoint without forcing unresolved records into an ordinary category?

Source Trail

S1: IMDRF N88, nonbinding principles on expertise, dataset independence and fit-for-purpose reference standards.

S2: IMDRF N67, 6.3 and 7.3. Its reference-standard and validation terminology does not prescribe this chapter's reader procedure or turn a benchmark into perfect truth.

The ten-image exhibit, two-reader/adjudication method and calculations are original fictional teaching constructions, not a recommended clinical protocol or adequate study size.

CHAPTER 7

Dataset Partitions, Leakage, Representation, and Coverage

Three folders of candidate-cell crops sit beside a proposed LumaCell Review experiment. The developer would use one to fit the classifier, another to choose among settings, and a third to challenge the chosen candidate. Every filename in the third folder is different from those in the first two. What would make its challenge new?

A different filename leaves two questions unanswered. The crop might descend from a patient or source image already used in development. Or its results might help the team choose the very candidate later presented as independently tested. Assessing data independence, the separation from development influence, needs a source history and a history of decisions as well as a list of files.

The twelve-image example later in this chapter makes those relationships small enough to inspect. It is a separate, proposed and unused planning fixture: neither Chapter 5's restricted exploratory subset, Chapter 6's ten-image reference exercise nor a later evaluation cohort. No model score or patient outcome has been produced from it.

What does one record stand for?

Before a crop became a row, a specimen was acquired and a source image produced. Chapter 5's lineage connects those events to the patient, the transformations that produced the crop and the authority for its proposed use. Chapter 6 adds the reference instruction and reference label version: how the comparison category was obtained, including any unresolved state. The row gives access to those relationships only if the joins survive.

The pathology queue episode and sequencing review episode raise the same counting question at different units. Two worklist rows could be repeat views of one case; two variant rows could belong to one specimen. A future study would need those relationships before claiming evidence about new cases or specimens. The episodes supply no additional cohort here.

Decision

How many independent patients are represented by twelve crops? Which source and grouping information would make the answer possible?

How a reviewer finds the unit behind the count

Twelve describes the crop rows. The patient count requires the links that show which rows belong to the same person. Even that count leaves other relationships to examine for the intended inference: shared specimen, acquisition, site, period or selection process. The evidence question determines which unit needs to be new; the largest available count does not choose it.

Reviewer Lens. “Independent” becomes assessable when it is followed by the unit and the relationship being excluded. A crop count can be entirely correct while a reader mistakes it for a patient count. The useful request is the source-to-patient grouping and its counting rule, with any further dependence named for the proposed claim.

Consultant Lens. One backward join from crop to patient and source, followed by a forward join to membership and label version, exposes whether allocation is ready to plan. A missing link then has a field and a producer. Resolving that small example before scaling the split is more useful than choosing a percentage for rows whose relationships remain unknown.

What did the data help someone decide?

Suppose a developer fits two candidates, examines a folder called “test,” and chooses the candidate with the better score there. What role did that folder play? To answer, follow what changed at each step.

During **model training**, eligible examples enter a fitting procedure and the classifier's parameters change. Next, comparing candidates or settings can determine which version or operating choice the team carries forward. That selection work is **model tuning**. Once the candidate is fixed, a separate comparison can ask how it behaves on material that did not supply those development choices: **independent evaluation**. The work purposes assigned to parts of a collection define its **dataset partitions**.

The supposed test folder in the opening question supplied the candidate choice, so its result is tuning information for that question. The same is true when a result selects a threshold or another operating choice. A folder label cannot remove that influence. Repeated inspection followed by revision gradually makes the set part of development, even if the files themselves stay unchanged.

The International Medical Device Regulators Forum's terminology document, IMDRF N67, distinguishes training from test data and explains why “validation” needs context: it can refer to data curation, model tuning or medical-device validation. [S1] A record of the actual activity, candidate version, results seen and decision made is therefore more informative than a label alone. For a whole-product conclusion, it also needs the

supplied quality-gate version and input rule; assigning crops cannot identify the complete configuration.

The episode reminders help separate a useful observation from a new comparison. A prospective pathology priority rule could be fitted on past cases, chosen using development cases, then examined on reserved cases. A sequencing suggestion method could similarly use variant examples to fit or choose a candidate before comparing a fixed version with its reference. Their recorded queue and suggestion observations can motivate a new workflow question. Once those observations have influenced a choice, however, they cannot also be described as untouched evidence for that choice.

The development plan places access and decision boundaries before the intended evaluation and provides a route for accidental exposure or newly discovered relationships. The aim is a reviewable history of influence. It does not require every development record to come from a different institution.

Reviewer Lens. Following one reported test result to the next decision often reveals its role more clearly than reading the folder names. If it selected a candidate or threshold, it can be credited as development evidence. The remaining question is which evidence assesses the fixed choice without having supplied it.

Consultant Lens. A dated fit → choose → fix → evaluate sequence lets model-development and evidence leads agree who may see each result and when. Exposure is cheaper to prevent than to reconstruct after several choices. If it occurs, an affected-result and decision record gives the manufacturer a claim-specific question to resolve before further development obscures the path.

Which related records must stay together?

A rotated crop has different bytes from its parent image. Has it become evidence from another patient? Following its creation answers the question. One adult supplies a peripheral-blood specimen; slide preparation and acquisition produce source images; a locator selects crops; transformations create derived files. A rotated copy used in development is an **augmentation**—another form of an existing example, still linked to its source.

The useful chain is patient → specimen → slide or acquisition → source image → crop → descendant. Another specimen or acquisition from the same patient can also matter. Permissions and reference versions stay attached to that chain, so the allocation does not detach the image from the conditions under which it may be used or interpreted.

Which part of the chain must remain together depends on what the comparison calls new. For a new-patient question, that patient's related records must stay out of development when the patient enters evaluation. A question about a new specimen from an already known patient may need a different grouping argument. The group

relevant to that inference is the **unit of independence**. A controlled patient token can support the allocation without showing direct identifiers to every analyst; the token alone establishes neither anonymity nor permission.

In pathology, a case can produce an image, import event, worklist row and transformed view. Allocating those rows separately would still leave one underlying case on several sides of the comparison. In sequencing, different variant rows can share a specimen and run. Their distinct values do not make them new specimens. These are prospective checks of the claim, not evidence that either episode experienced leakage.

Patient grouping also leaves some conditions shared. The development and evaluation material may use the same scanner settings, stains, laboratory practices, sites or periods. That can fit a bounded new-patient question within those conditions while leaving transport to another setting unexamined. IMDRF N88's nonbinding principles call attention to relevant data and sources of dependence, including patients, sites and acquisition; they do not prescribe one site split for every question. [S3] A clear account names what is separated, shared and unknown before calling a set external.

Reviewer Lens. Two apparently different rows can be followed through patient, specimen, acquisition and parent-image tokens before their assigned purposes are compared. If a link is missing, the finding is that the proposed separation cannot yet be substantiated. It would be premature to turn that missing evidence into an asserted patient overlap.

Consultant Lens. The evidence lead's claim determines the grouping key; the site and data steward make that key and its parent-child links usable. Agreeing them before allocation avoids treating whatever identifier happens to survive an export as the correct scientific unit. A later correction to a patient or source link needs a defined route back to membership and affected evidence.

Can reserved data influence a model without entering a training batch?

Suppose a team estimates a normalization setting from every crop, including the reserved test crops, and trains using that setting. The reserved files never enter a training batch, but their values have shaped the preparation applied during development. This differs from a fixed transformation specified without learning from the reserved records. The operative question is which data helped establish the setting.

When information treated as unseen by the intended evaluation influences development, the comparison has **data leakage**. Shared patient or source relatives provide one route. A fitted preparation or selection rule provides another. A third route runs through decisions: someone sees a score, changes a threshold or exclusion rule,

then reports a score on the same records without identifying that influence. Separate pixels cannot close that result-to-choice path.

A dated development record can expose these routes. It identifies the records entering each fitted transformation, the data and transformation versions, who saw results, what changed afterward, and which report used the resulting candidate. Those facts let the reviewer judge the claimed comparison rather than infer independence from the absence of a filename in a training log. The later twelve-image fixture has not been used through any such route.

The same prospective questions apply to the two episodes. Queue results seen before choosing a priority threshold become information for that choice. A held-out suggestion analysis seen before selecting variant-display settings does too. Existing field observations may guide a new timing or reconciliation study, but the new study needs an honest account of which earlier results informed it.

This is a question about the procedure and the inference it supports, rather than a prohibition on every external datum or later analysis. Kapoor and Narayanan describe how methodological leakage in machine-learning-based science can produce overoptimistic findings and reproducibility problems. Their research supports examining these information routes; it does not prescribe LumaCell's medical-device method. [S2]

Reviewer Lens. A clean patient split answers a source-overlap question only when its links are reliable. It leaves fitted preparation and result exposure to their own checks. Separating the three routes makes a finding actionable: which reserved information affected which choice, and which independent-evaluation assertion is now unsupported?

Consultant Lens. Model-development and evidence owners can reconstruct the sequence from input versions, fitted settings, disclosed results and subsequent choices. The useful advice identifies the affected comparison and an evidence route for its fixed question. A generic leakage label would conceal both what remains valuable for development and what needs a different comparison.

What does “no duplicate found” establish?

An image hash compares image bytes. A check for matching hashes can identify identical files, yet miss a re-encoded image, rotated crop or different acquisition from the same patient. That is why **deduplication**—checking for repeated material—needs a stated method and unit. A negative result means little until those are visible.

A visual-similarity screen may identify records worth investigating. It cannot establish patient identity or make a clinical exclusion decision by itself. Controlled source and parent links can show whether a crop descends from a particular image; patient,

specimen and acquisition tokens can connect different images that matter to the proposed inference. In LumaCell, both kinds of link matter to the new-patient question.

The episode parallels show why byte identity is too narrow for some questions. Two exports of one pathology slide can have different bytes while remaining one case. Distinct variant rows can share a sequencing specimen without sharing image-like content at all. “No matching hashes” supplies no answer to those grouping relationships.

The audit therefore records what ran, what it found and what it could not see. A flagged transformed relative is checked against provenance before an authorized allocation or exclusion changes. If import lost the lineage, source recovery or a narrower evidence use may be necessary. An unperformed relationship check cannot become a negative result merely because the exact-file check passed.

Reviewer Lens. A byte-level result deserves credit at that level. The next request should concern the parent/source or patient-group evidence required by the claim, with the specific missing relationship named. It need not default to unrestricted personal identifiers, which are different from controlled group tokens.

Consultant Lens. A reproducible queue of suspected relatives gives the steward an orderly repair path: retain the flag, inspect controlled provenance, obtain a group disposition under the authorized rule, then propagate it to versioned memberships and affected experiments. Silently deleting a suspicious file might remove the visible flag while erasing the explanation needed later.

What can this selection of records tell us?

Suppose the patient groups are properly separated, but every supplied crop came through an accepted-quality filter. What could a comparison on those crops tell us about images the filter rejected? The selection has bounded the question before a score is calculated.

For LumaCell's intended use, the evidence needs a visible connection to the adult referral population, preparation and acquisition profile, site and period, reference categories, image-quality states and professional workflow. A **coverage matrix** places those conditions beside the records and groups that represent them. The proposed acquisition profile describes intended input conditions; source records still have to show whether the dataset matches. Counts include relevant patients as well as crops, and retain whether the reference resolved and whether the product took a non-result route. Presence identifies material that could be examined; it does not measure performance in that condition.

How that material was selected determines another limit. An ordinary-use sample may describe the encountered mix while containing too few uncommon cases to probe a

consequential mechanism. Deliberately adding more difficult inputs or rare reference states creates an **enriched challenge set**. Its value lies in the challenge it makes possible. Its chosen proportions cannot tell how frequently those conditions occur in ordinary use.

The same distinction applies to balanced training data. Selecting equal or otherwise engineered category counts does not establish clinical prevalence, how often a positive suggestion would be right in routine use, or the frequency of a non-result. Those claims need a selection design and denominator suited to them, with justified weighting or separate analysis where needed. The design and intended inference belong in the record before a frequency is reported.

For the pathology queue episode, an ordinary stream of imported cases could inform a placement question, while selected import-delay conditions could challenge the total-time pathway. The latter cannot measure how common those delays are just by including them. For the sequencing review episode, ordinary variant review and a targeted specimen-reconciliation challenge would likewise serve different questions. These are prospective designs prompted by the earlier observations, not new field studies.

The useful matrix row therefore joins its group and record counts, source and reference conditions, endpoint and remaining unknowns. A large total or several site names cannot substitute for those relationships.

Reviewer Lens. Selection explains what a favorable result could mean. A deliberately enriched difficult condition can provide valuable challenge evidence while lacking a basis for routine-frequency inference. Reading that purpose beside the claimed population and workflow permits a positive narrow finding without overextending the selected mix.

Consultant Lens. Clinical and evidence owners can distinguish an ordinary-use question from a consequential challenge before asking for more data. Each needs a group count, reference and non-result states, and an analysis purpose. A missing condition then becomes a choice about targeted collection, challenge or claim scope, rather than an open-ended request to enlarge the image folder.

Which conditions does the holdout change?

Keeping a patient's source family out of development gives a new-patient comparison under whatever other conditions remain shared. That design is a **patient holdout**. If the question instead concerns a setting not used to make development choices, the relevant separation is a **site holdout**. A question about later conditions calls for a **temporal holdout**. The name should identify the changed condition, not suggest that all conditions changed together.

A design can combine these separations, but doing so changes which evidence can be collected and what remains for development. Reserving a whole site might remove an important acquisition profile from development while answering a different question from the intended within-setting claim. There is no universal requirement that every study separate all sites. The actual claim determines the separation argument and the shared conditions that must be reported.

A new-case pathology test within one site could address bounded queue placement; a different scanner site or later period would ask about transport of import and worklist behavior. For sequencing, a new specimen within one laboratory differs from a site with different sequencing or reconciliation practice. People, instruments, profiles, sites and periods may remain shared in different combinations. Their identities explain what each result would reach.

Selection remains relevant within any holdout. A convenient retrospective export of known categories could support a technical classification question while omitting ordinary unsorted work, identity exceptions and fallback actions. An accepted-quality crop collection cannot by itself measure how often the gate rejects supplied crops or whether full smear review becomes faster. Those questions need records of the routes and work the selected collection omits. Sample size, precision and formal analysis follow in Chapter 9, after the question and selection route are clear.

Reviewer Lens. “External” and “later” are useful only when the patient, site, period and acquisition membership support their intended meaning. A focused review identifies the transport question actually examined and the conditions still shared. Applying an all-sites-separate rule regardless of purpose could reject a valid bounded question while missing the real omission.

Consultant Lens. The manufacturer first needs to know which decision requires a new-patient, new-site or later-time answer. The clinical and data leads can then compare holdout options and show which input profile or workflow condition each would remove. That consequence belongs in the design discussion before the split is treated as fixed.

Make the allocation reconstructable

A spreadsheet says a test is independent. Could another reviewer determine which records belonged to it, why they were placed there and whether their results affected an earlier choice? The manufacturer's **dataset manifest** is the controlled account that makes those questions answerable.

The table is useful when one of those joins is missing. Membership and grouping establish what was allocated; the checks establish which relationships were examined; coverage establishes what the set can address; version and use history preserve the basis of the claim.

Field	What the record needs to show
Dataset identity	A fixed identity for this selected membership and purpose.
Unit of independence	The group and assumptions relevant to the proposed inference.
Inclusion and exclusion	Prospective rules, reasons and an account of affected records.
Duplicate checks	Exact-file and related-record methods, results and limits.
Partition logic	Allocation by the relevant groups, with its rationale.
Leakage checks	Overlap, transformation fitting, result exposure and other relevant paths.
Site and time	Which conditions are separated, shared or unknown.
Intended-use coverage	Represented population, input, reference and workflow conditions.
Challenge groups	Prespecified difficult or consequential conditions and their purpose.
Limitations	Unsupported transport, sparse groups, missing provenance and unresolved states.
Version and lineage	Source and parent links, changes, access and experiment or evidence uses.

The identity row has a practical consequence: a membership correction gets a new version linked to the earlier one. Otherwise a report can point to a name whose contents changed underneath it. The grouping row determines what the count means, while the leakage row reaches beyond membership into fitted settings and result access. Coverage and limitations together prevent a complete-looking table from making a claim its sources cannot support.

For one member, a reviewer should be able to follow source → group → assigned purpose → preparation and result history. A script can execute the allocation, but those relationships justify it. An unresolved link belongs in the manifest with its consequence, rather than being hidden by the spreadsheet's title.

Reviewer Lens. Following one member backward to patient and source, then forward to its actual use, tests whether the manifest can be reconstructed. The finding can be narrow: a missing exposure history leaves an independent-evaluation claim open, while a verified membership count remains usable. The table need not lose all value because one claim lacks its link.

Consultant Lens. The steward, model-development lead and evidence owner each contribute part of that path. One complete example reveals their handoffs before the allocation scales. Permission gaps go to the data authority, reference gaps to the reference owner, and coverage choices to clinical and evidence owners; the adviser makes those decisions possible without certifying representation.

A proposed allocation before anyone uses it

The following small fixture concerns LumaCell planning. Twelve fictional image records have been assigned proposed purposes. For this exercise's new-patient question, all records from one patient need the same purpose. No row has been used to fit a model, choose settings or evaluate a candidate. The table is an **original partition proposal** (SPLIT-07-v0), not a clinical dataset or an adequate study size.

One row means one image record, linked to a patient token, site token, coarse period and proposed purpose. The site labels S1–S3 do not establish different scanners or suitable populations. Earlier and Later are relative labels, with no dates or demonstrated chronological study. No specimen or acquisition chain and no image bytes are supplied. The available operation is a patient-token audit; exact-file and source-relative checks would need other records.

Record	Patient	Site	Period	Proposed purpose
I01	P01	S1	Earlier	Training
I02	P01	S1	Earlier	Test
I03	P02	S1	Earlier	Training
I04	P02	S1	Earlier	Training
I05	P03	S2	Earlier	Tuning
I06	P03	S2	Earlier	Test
I07	P04	S2	Later	Test
I08	P04	S2	Later	Test
I09	P05	S3	Later	Test
I10	P05	S3	Later	Test
I11	P06	S3	Later	Training
I12	P06	S3	Later	Test

For example, the I03 row connects an image to patient P02, site S1 and the Earlier period, with training proposed. It says nothing about that image's category, acquisition compatibility or quality. The same limits apply to every row: no image content, reference category, model score, clinical outcome or final sample-size claim is supplied.

Eligibility for this bounded planning exercise is stipulated. It neither transfers the restricted exploratory subset's narrow permission nor establishes real-world use rights or authorizes a run. Any conflict found here is still preventable in the design; the packet does not describe an already contaminated evaluation.

Decision pause — which change would answer the new-patient question?

The point of a repair is a coherent allocation for the stated question. There is no model score to optimize. Five connected questions locate what this packet can establish:

1. How many image records belong to each proposed purpose, and how many distinct patients appear overall?
2. Which patients cross purpose boundaries?
3. What **patient-group repair** would keep each patient's related records together?
4. Would that repair also establish site independence or a wholly later-time test?
5. Which missing information prevents an intended-use coverage conclusion?

Decision

Which allocation could close the visible grouping problem, and which evidence questions would remain open afterward?

How the group repair changes one property of the proposal

Worked reasoning

The original training members are I01, I03, I04 and I11: four records. Tuning contains I05 alone: one record. Test contains the remaining seven. The patient column has six distinct tokens, each appearing twice. Comparing the two purposes for each patient reveals three crossings: P01 spans training and test, P03 tuning and test, and P06 training and test. Those pairs conflict with this exercise's new-patient rule. No run has used them yet.

One repair moves only the records needed to place each pair together. The table supplies a lookup by patient, so every image carrying that token receives the same revised purpose.

Patient group	Revised purpose	Image records
P01 and P02	Training	4
P06	Training	2
P03	Tuning	2
P04 and P05	Test	4

The first two rows put P01, P02 and P06 into training: three patients and six images. P03 supplies one patient and two images to tuning. P04 and P05 supply two patients and four images to test. Every patient's two records now have one purpose. The repair gives a positive, bounded result about this unused proposal; it is not an executed correction to a leaked evaluation.

The site and period columns still matter. S2 occurs in tuning and test, while S3 occurs in training and test. P06 is a Later-period training patient, so development does not lie wholly in the Earlier period before the test. The repaired proposal asks about new patients under some shared site and period conditions. It does not become a held-out-site or wholly future-period comparison.

A design reserving S3 or a later collection could ask a transport question, with another allocation and another check of development profiles and remaining test coverage. There is no universally best split of these twelve rows and no all-sites-disjoint requirement for every intended inference. The choice follows the question.

Nor has the repair supplied the absent categories, acquisition profiles, reference states or adult referral conditions. Six patient tokens cannot establish representation or precision, and two test patients do not establish an adequately powered or clinically relevant real evaluation. What makes this immediate repair possible is non-use: no fitting, tuning or test result has shaped a choice that now needs reassessment. The manufacturer can revise and verify membership before any authorized activity.

Reviewer Lens. Recounting images and patients separately shows exactly what improved. Every patient token can be checked against one purpose, giving the allocation finding a clear stopping point. Shared sites and periods, missing source relationships and unknown coverage retain their own findings; the successful token check supplies no evidence about them.

Consultant Lens. The steward has a concrete action: apply and version the six-patient lookup. The evidence lead checks its fit to the new-patient question, and the manufacturer decides what source or coverage evidence is still needed before use. Describing the initial proposal as failed performance would miss both its unused state and the inexpensive prospective correction still available.

Can another reviewer reproduce the repaired membership?

The **patient-group repair** (SPLIT-07-v1) is an author-proposed successor to the original partition proposal. Both identifiers belong to teaching records, rather than downloadable image sets or verified operational manifests. The worked table connects the repair to the eleven manifest fields. Its identity and allocation rows reproduce membership; its check and coverage rows show where the supplied tokens stop answering the question.

Manifest field	Proposed repaired record
Dataset identity	Twelve records I01–I12. Training: I01/I02/I03/I04/I11/I12. Tuning: I05/I06. Test: I07/I08/I09/I10. No fitting, selection or evaluation use.
Unit of independence	Patient group for the new-patient question; each pair stays together. Four test records represent two patients. Site and acquisition dependence remain separate questions.
Inclusion and exclusion	All twelve stipulated planning-eligible records retained; none removed to improve a score. Actual source eligibility, rights and reference adequacy need their own evidence before use.
Duplicate checks	Patient-token audit complete for this table. Image files, source identities and transformation histories absent; exact-byte and transformed-relative checks unperformed.
Partition logic	P01□training, P02□training, P03□tuning, P04□test, P05□test, P06□training, applied to every member. This explicit lookup needs no random seed. An unlisted patient needs a new disposition, not an automatic test assignment.
Leakage checks	Patient sets are disjoint after repair. Real evaluation still needs source relationships, actual access and development-exposure history, and fitting restricted to the declared development procedure. Non-use is stipulated here; no actual access log exists in the packet.
Site and time	S1 training-only; S2 shared by tuning/test; S3 shared by training/test. Earlier records in training/tuning; Later records in training/test. Only patient-group separation is established.
Intended-use coverage	Patient, site and coarse period visible. Adult status, morphology-referral context, profile compatibility, reference categories and workflow selection cannot be established.
Challenge groups	No difficult condition identified. Acquisition-quality and richer reference-category information is needed to judge possible challenges; site tokens cannot stand for difficulty.

Manifest field	Proposed repaired record
Limitations	Six patients and three named sites establish neither representation, adequate precision nor independent transport. Two test patients cannot support an unspecified broad clinical claim. Unchecked identity relationships remain a restriction.
Version and lineage	Preserve the original proposal and rationale. Only I02, I06 and I12 change purpose; no source-content change is asserted. Steward preparation and manufacturer approval must be recorded before real use; downstream sets and experiments bind to the approved version. Approval is not supplied.

Applying the lookup to each source row reproduces the change without guesswork. I02 moves from proposed test to training with P01; I06 moves from test to tuning with P03; I12 moves from test to training with P06. Every I-record appears once. The resulting image counts are 6/2/4 for training/tuning/test, while patient counts are 3/1/2. No patient belongs to two purposes.

The distinction between those counts is visible in the test: two images from P04 and two from P05 give four rows but only two patients. The lookup verifies the listed allocation property. It cannot verify source-file deduplication, use permission, reference adequacy or an operational execution absent from the packet. An unlisted patient would fall outside the rule and need a disposition before joining it.

A future pathology manifest would connect case, slide and import events, transformed views and queue observations to the priority-rule version that consumed them. A sequencing manifest would connect patient, specimen, run and variant rows to the suggestion method, retaining reconciliation records separately from variant labels. Such records could reveal whether a post-use question requires new data or whether earlier results already shaped the choice. None of those additional records is supplied here.

Reviewer Lens. The three moved rows are a concise way to test whether the stated repair actually occurred. The next comparison is with the check-status rows: a complete patient-token audit is positive evidence, while absent image/source files leave those checks unperformed. Treating both statuses as passed would erase the manifest's most useful distinction.

Consultant Lens. Preserving the old proposal and generating its successor from the explicit lookup retains the reason for each change. The steward can reconcile twelve unique IDs, then attach the still-pending source, rights and reference questions. That gives the manufacturer an inspectable purpose decision before any permitted run; the advisory work itself does not approve the activity.

What does the same membership show about coverage?

The next table views the same twelve records by site and period; it adds no observations. Each numeric cell means **image records / patient groups**, rather than a fraction or performance rate. A 0 / 0 means the complete toy inventory has no listed member with that known site or period. Unknown means the attribute was never supplied. Those states lead to different conclusions.

Coverage dimension	Trainin g	Tunin g	Test	Supported reading
Total membership	6 / 3	2 / 1	4 / 2	Lookup separates patient groups.
Site S1	4 / 2	0 / 0	0 / 0	No S1 test evidence in this allocation.
Site S2	0 / 0	2 / 1	2 / 1	Test shares S2 with tuning.
Site S3	2 / 1	0 / 0	2 / 1	Test shares S3 with training.
Earlier period	4 / 2	2 / 1	0 / 0	No Earlier-period test observations.
Later period	2 / 1	0 / 0	4 / 2	Later test and Later training coexist.
Adult referral population and proposed acquisition-profile compatibility	Unkno wn	Unkno wn	Unkno wn	Tokens do not establish intended-use eligibility.
Five common categories, richer outside-category states and unresolved references	Unkno wn	Unkno wn	Unkno wn	Neither category balance nor a reference-evaluable count is supplied.
Difficult supported inputs and non-result pathways	Unkno wn	Unkno wn	Unkno wn	Robustness and fallback coverage remain unestablished.

The totals reproduce the lookup: 6/3, 2/1 and 4/2. The S1 row gives a known absence from test. The S2 row counts the tuning patient's two images and one test patient's two images, so the site is shared despite patient separation. S3 is similarly shared between training and test. Three sites appearing somewhere in the collection therefore does not mean the test examines a new site.

The period rows answer another question about those same members. Test has only Later records, but P06's Later records are in training. Earlier has no test member.

Adding site and period counts would count the same people and images twice; these are different views, not extra cohorts. Four test crops also remain four records from two patients.

The bottom rows cannot be read as zeros. Adult referral, profile compatibility, category/reference and difficult-input attributes were not supplied, so the table cannot establish their presence or absence. It cannot identify a reference-evaluable denominator or characterize quality-gate and non-result behavior. Recovering the relevant attributes could change those rows; changing the site label alone would not.

For pathology, the corresponding matrix would count cases with linked import and review events and distinguish deliberately challenged import delays. For sequencing, it would count specimens with variant references and reconciliation states. Each could expose a missing endpoint link, while selected difficult cases would still lack a basis for estimating ordinary-use frequency.

The useful disposition is that this patient-group allocation is suitable for further planning. Source integrity, use eligibility and coverage for a stated claim remain open. An attribute request earns its place by showing which claim it could support or rule out, rather than merely filling an empty cell.

Reviewer Lens. Zero and Unknown warrant different requests: the first can show an absent test condition in the listed inventory, while the second calls for the missing attribute and its source. Checking the totals and shared sites/periods gives a reproducible bounded conclusion. Representation needs a further argument from intended-use conditions.

Consultant Lens. The unknown rows identify concrete work for data and clinical owners: source/acquisition profile, adult referral and selection route, versioned reference categories, non-result states and challenge purpose. The evidence lead can explain what each would change in the intended claim. More records lacking the same attributes would leave that decision unresolved.

What would turn those unknowns into an evidence argument?

The LumaCell question starts with adult peripheral-blood morphology review and the proposed acquisition profile. That profile is a planning specification, not a property retrospectively established for these twelve records. A future coverage row needs its permitted source, patient and specimen counts, acquisition condition, reference method/version, product result or non-result, and the claim element being examined.

Chapter 6 supplies an essential reminder for that row. An unresolved reference, U, means the method could not support an assignment; a determinable outside-five category, O, is a supported finding outside the common set. Counting the first as the

second would create apparent category coverage from uncertainty. The product's other/unclassified route is also an output route whose behavior needs its own record.

The five common cell classes leave further questions. Relevant abnormal, immature and other non-target conditions can matter to professional routing. Difficult supported inputs and quality-gate rejections probe different behavior. One included image establishes presence, without characterizing a subgroup. A broad other endpoint therefore needs any supportable richer reference types retained, so a common subtype does not hide sparse evidence for another.

The manufacturer might balance development examples or deliberately enrich a challenge, with the chosen design and group denominators reported. Neither selection nor this twelve-row site matrix supplies the natural-use distribution. A justified next step might be targeted collection, a challenge, a narrower claim or an explicit limitation. The missing relationship in the matrix selects among those options more usefully than the desire for a larger image total.

Reviewer Lens. For a proposed category or fallback claim, the relevant join runs from source profile and patient denominator through reference state to actual product state. A defined challenge can answer a mechanism question. A statement about everyday frequency or clinical benefit needs evidence for that different inference, even when the challenge result is strong.

Consultant Lens. Turning one unknown row into a proposed collection or analysis question makes the choice manageable: an owner, permitted source, grouping key and endpoint. If that condition cannot be adequately supported, a narrower claim can be considered. The manufacturer chooses the evidence use after the relevant review.

Figure 7.1 — Keep related records together. The selected technical figure separates source-family grouping from site/time coverage, duplicate checks and development-result exposure. A random crop-row split does not by itself establish any of those forms of independence.

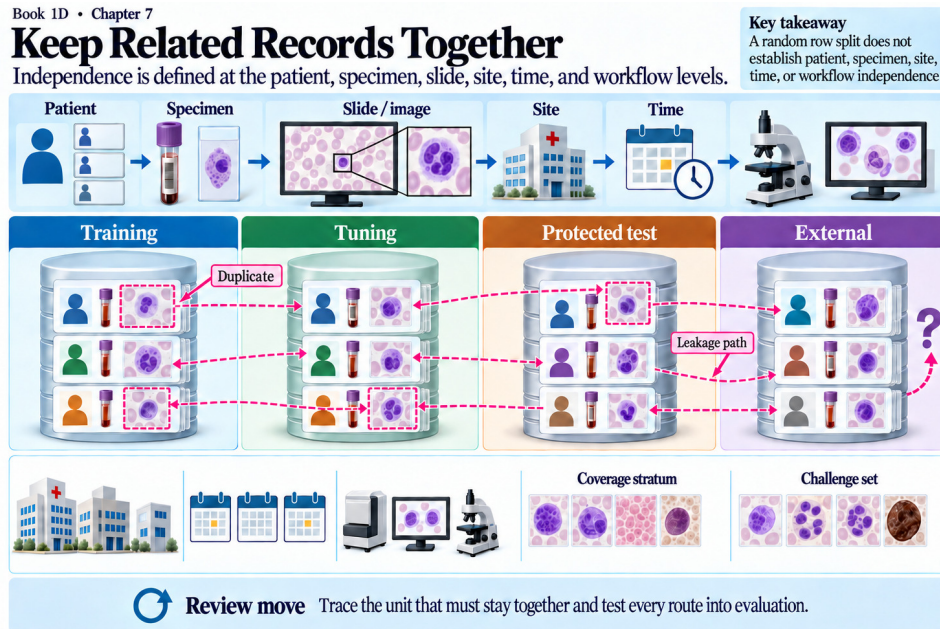


Figure 7.1. Keep Related Records Together.

Who can decide how the evidence is used?

A manifest connects records produced by different owners. The data steward maintains versioned membership and source/group joins. The model-development lead records the data and fitted transformations that shaped each candidate. Clinical/reference and statistical specialists define the relevant population, reference and inference questions, while the evidence lead controls result disclosure and analysis purpose. Quality/risk examines the consequences of overlap, exclusions or changed rules. The manufacturer decides whether that combined history is adequate for a particular evidence use.

For an unused proposal, this permits a bounded sequence: direct an allocation repair, inspect the resulting property, then address source and coverage questions before actual use. A tidy lookup cannot make the complete evidence history independent. A future pathology timing or sequencing release claim would also need its stage-specific events and grouping, since queue rank or a displayed suggestion leaves the wider endpoint unmeasured.

Access settings should support the intended result boundary, while the disclosure and decision log shows what actually happened. A consultant can reconstruct and challenge that history. It does not thereby grant data rights, certify representation or approve the evaluation for the manufacturer.

Reviewer Lens. Purpose, membership version, grouping audit, rights and exposure history need to refer to the same proposed or executed activity. A supported allocation conclusion can close without pretending unresolved coverage has closed too. That distinction gives a favorable technical finding a precise scope.

Consultant Lens. The practical handoff sends allocation and source checks to the steward, use-history review to the evidence lead, and population or reference gaps to their competent owners. Closure conditions tell the manufacturer what it can decide when each returns. An advisory approval stamp would hide those still-separate judgments.

Consultant assessment — which claim can the records support?

The consultant receives the original partition proposal with a request to call its test “independent and multi-site.” The useful assessment would connect that claim to membership, history and covered conditions. What would make a finding specific enough that the manufacturer could close it without mistaking one repaired property for a complete evidence argument?

How the advisory assessment separates the findings

What do the visible crossings establish?

I01 and I02 share P01 but have different proposed purposes; the P03 and P06 pairs show the other crossings. Those are observable conflicts with the stated new-patient allocation. Non-use is also stipulated, so the finding concerns an unused design. It cannot assert that a trained model or completed result has leaked.

The proposal version, assignment method and actual use history determine the next request. In a real packet with missing grouping keys, the reviewer could say patient separation is unsubstantiated, but could not identify a particular overlap without evidence. A recovered patient/source join and a history of results influencing choices would change what the assessment can conclude.

Which relationships did each check examine?

A byte check, transformed-descendant check, source-image check and patient check concern different relationships. A clean hash report can coexist with related crops, while the present token audit can find a crossing without proving that source files are unique. The useful request is the controlled linkage needed for the declared inference, with no default demand for unrestricted personal identifiers.

Development history remains a separate path. Perfect file separation cannot undo a score used to select a threshold. This fixture supplies no score exposure because it

supplies no result at all. That favorable planning condition is stipulated; it is not evidence that an operational access-control test passed.

Which intended-use condition is documented?

The repaired test has two patient groups. It shares S2 and S3 with development purposes and has no supplied acquisition-profile or category/reference facts. The words multi-site and large cannot replace those missing relationships when a new-site, input-profile or category claim is proposed.

“No test member has documented compatibility with the proposed acquisition profile” identifies an evidence gap. It does not say LumaCell fails on that profile: there are no predictions here. Similarly, an enriched pathology import-delay or sequencing reconciliation set could support its selected challenge, while its ordinary-use frequency remained unestablished. A clear finding states which conclusion the selection permits.

What would close the allocation finding?

For this unused proposal, the patient-crossing finding has a concrete closure route: the proposed patient-group repair, all twelve IDs appearing exactly once, one purpose for every patient group, and manufacturer approval before a permitted run. Those records would close the allocation defect for the stated question. Source-relative checks, rights, reference adequacy and coverage retain their own prerequisites.

If the use history were different, a new manifest alone would not close the finding; the affected results and choices would first need reconstruction. A coverage gap likewise calls for its own response: justified additional data, a targeted challenge, a narrower claim or a stated limitation. More records with the same missing-condition pattern would leave that question unanswered.

How narrowly can the conclusion be stated?

A bounded conclusion is possible: three patient groups cross development and test purposes in the original proposal; because it is unused, the manufacturer can revise and verify membership prospectively. The illustrated repair keeps patients together. Site independence, complete temporal separation, intended-use coverage and adequate study size remain separate design and evidence questions.

That conclusion credits progress while preserving the shared sites, Later training and unknown profile/reference facts. Attaching member and version evidence makes the positive finding inspectable. Assigning each remaining question to an owner makes the limits actionable.

Which count belongs in the summary?

The repaired test contains four records from two patients at S2 and S3, both sites represented in development for some purpose. Describing it merely as “four independent test images from two sites” could make the four correlated records sound like four independent patients and the sites sound new. The summary needs both record and group counts and the conditions still shared.

An advisory memo is subject to the same counting discipline as the technical report. Its reader should be able to reproduce the numbers and distinguish membership properties from properties whose checks were absent. The linked membership, duplicate, representation, closure and denominator findings then support a bounded disposition together.

Reviewer Lens. A verified pre-use group repair can be credited once its member/source join and version are clear. Actual exposure history and counted conditions determine what else the evidence can support. The remaining request concerns a named independence or coverage claim, rather than demanding an undefined level of dataset completeness.

Consultant Lens. Sequencing the steward's allocation/source checks, the evidence lead's history review and the clinical coverage decision gives the manufacturer separate closure opportunities. The memo can recognize each completed finding while retaining unresolved limits. It cannot certify the dataset on behalf of those decision owners.

Practice the boundary

Consider the proposed conclusion: “The consultant certifies that the data are representative because the test set is from three sites.”

Decision

What would an evidence-based replacement say about sites, intended-use conditions and the manufacturer's authority?

How the corrected conclusion connects sites to a claim

The consultant can assess the documented relationship between selected sites, records and relevant intended-use conditions, including its limitations. Site count alone cannot establish representation; the manufacturer retains the evidence-use decision. The corrected statement describes a review whose conclusion depends on actual coverage evidence, rather than assigning the adviser a certification power or treating a count as its justification.

Decision pause — the test has already influenced selection

Now change one condition in a counterfactual. The original twelve-record allocation was used repeatedly to compare candidates, and the team chose its preferred configuration after inspecting test scores. It proposes the same group repair and wants the next score on those records described as an untouched independent test.

Decision

Is the group repair enough for that claim? Which historical fact changes the reasoning, and what must the manufacturer now establish?

Why a repaired membership retains its development history

The test results influenced candidate selection. Moving records into new groups does not remove that influence, so the manufacturer must accurately characterize the existing evidence and establish an appropriately independent route for its intended confirmatory question. The remedy depends on the complete development history and study design. Renaming or redistributing these files cannot by itself make them untouched.

This after-use history is a counterfactual, not the history of the original partition proposal. Chapter 6 supplies a related exposure question: if ambiguous reference labels were revised after model disagreements were seen, their versions and that influence would also enter the evidence assessment. A changed label could be justified while the corrected comparison needed a different account of independence.

In the pathology and sequencing episodes, the recorded queue and suggestion observations can guide a future timing or reconciliation study. That future study would identify which earlier results informed its design and reserve suitable evidence for its own fixed question. Historical findings can remain valuable development information without being described as untouched comparison evidence.

Reviewer Lens. The dated score-view and candidate-selection record is decisive here. A renamed partition answers a membership question while leaving that historical path intact. The focused request is an accurate development characterization and a design for the intended independent comparison.

Consultant Lens. Connecting each viewed score to the next candidate choice helps the evidence lead specify a new purpose, grouping and disclosure plan. The manufacturer receives options and affected analyses. The adviser can make the consequences clear, but cannot erase exposure or authorize a replacement study.

What carries into the next experiment?

Chapter 8 uses these controlled roles to compare candidates and reproduce experiments. The manifest gives that work a starting point whose membership, source relationships, actual uses and limitations can be understood. The claim about a score will depend on that history even when a new candidate is tested.

For optional depth, Book 1C develops population coverage, denominators, bias and generalizability. Book 1A Chapters 7 and 13 address artificial intelligence, data and statistical review; Book 1B Chapters 3–6 address integrity, poisoning, provenance and pipeline access. Book 2 Chapters 7–12 connect a chosen European Union evidence route to performance questions. The local allocation does not choose that route.

Reasoning check

For an unfamiliar proposed dataset manifest, which records and independence units are being counted? What fit, choice and evaluation history belongs to them? Which check reaches source relatives beyond identical bytes? Which patient, site and time conditions are actually held out? What can its ordinary or enriched selection support?

Decision

Which missing relationship or exposure would make an independent-evaluation conclusion premature, and which bounded allocation conclusion could still be positive?

Source Trail

S1: IMDRF N67, 6.7–6.9 and 7.3; training/test roles and contextual meanings of validation.

S2: Kapoor and Narayanan, Leakage and the reproducibility crisis in machine-learning-based science, *Patterns* (2023). Used for the general leakage/reproducibility problem, not as a medical-device regulatory method.

S3: IMDRF N88, nonbinding Principles 3–4 on relevant data and train/test independence. It does not prescribe the exercise's ratios or a universal site split.

The manifest, twelve-record exhibit, repair and transfer are original teaching constructions. Counts are deliberately small for inspection, not recommendations for evaluation design.

CHAPTER 8

Model Selection, Experiments, Reproducibility, and Dependencies

Two developers bring the team a classifier score. Both started with adult peripheral-blood cell images; both say they trained the same kind of model. One score is higher. What would make that a reason to prefer its candidate? The answer starts before the score: which images entered the program, how were they turned into numbers, and which saved classifier produced the suggestions?

In the proposed **LumaCell** experiment, fitting uses one collection of examples to adjust the classifier. A different development collection then supports comparison and candidate choice. Following one crop through those operations keeps the question concrete. If its image version, reference or preparation differs between runs, the team has changed something besides the candidate design. A useful result has to remain connected to the work that produced it and the decision it influenced.

The experiment below is a separate fictional development fixture. Its training and tuning collections do not continue Chapter 5's restricted exploratory subset, Chapter 6's ten reference-method images, Chapter 7's unused six-patient allocation, or a later independent evaluation cohort. Permission and lineage are stipulated only for this segregated exploratory use. Those premises supply a bounded teaching comparison, without granting broader rights or product release.

What would make two runs comparable?

The proposed system map locates the proposed classifier inside a larger route. The data-use and lineage records identify which sources can enter an activity; the reference record says what the comparison labels mean; the dataset manifest from Chapter 7 records membership and how records and results have already been used. Which of those links could another person recover from the developers' two scores?

If the runs used different crop members, in what sense would they be repetitions? If a collection's score helped the team choose a candidate, what role did that collection play?

Decision

Which executed conditions and result-to-choice link would settle those two questions?

How a reviewer distinguishes repetition from another comparison

Membership is an experimental condition. Keeping the model family unchanged while changing the images creates a different comparison. It may be informative, but its result cannot simply stand in for repetition on the original members.

The score's subsequent use also matters. Once it informs candidate selection, it is development information for that choice. A folder named "test" cannot make that history disappear. The useful record joins the actual members, reference version and execution conditions to the dated choice, so the reader can tell both what happened and what the result now supports.

The **pathology queue episode** suggests a future comparison of priority rules with chronological ordering on the same eligible worklist stream. Its observed queue action left the import bottleneck unresolved. The **sequencing review episode** suggests a future comparison with defined specimens, references and professional review actions; its working suggestions left identity reconciliation unresolved. These separate fictional observations motivate questions. Neither supplies an experiment result for this chapter or becomes named-series product history.

Reviewer Lens. A persuasive pair of numbers can hide different denominators or measured paths. Matching membership, reference, boundary and use in selection is a narrow first check. When those conditions differ, the reviewer can credit another useful comparison without calling it a repeat. That preserves evidence while correcting its description.

Consultant Lens. Before the next comparison, the data and experiment owners can demonstrate one complete join from a score to permitted members, reference version and selection decision. A missing join then becomes a specific handoff to repair. Waiting until a favored candidate has been chosen makes the same question harder, because the missing history may have shaped the choice already.

What exactly would be repeated?

Suppose the team can recover the images. Someone still has to decide how to prepare them, which model design to try, how fitting proceeds, when it stops and how a candidate is chosen. Together those instructions form the **learning procedure**. A choice made before fitting, such as the number of passes through the examples, is a **hyperparameter**. It directs the procedure; it is different from a weight adjusted using the images.

Running that procedure leaves a particular saved set of fitted values. This is the **trained artifact**. It can be loaded later to calculate suggestions. Another training run may follow the same general procedure but start from a different declared random state and leave a different artifact. The instructions and their saved result therefore need separate identities.

Now place that artifact back into the product map. LumaCell would still need inputs matching the **proposed acquisition profile**, package values meeting the **identity-field requirements**, implemented preparation, the supplied gate, output and unsupported-state rules, runtime, display and professional workflow. The **original quality gate** still has unresolved placement, input representation and crop-to-package aggregation decisions. Replaying a saved classifier does not settle them. Even unchanged fitted values can receive different numerical inputs after a transformation change; even identical suggestions can reach a person through a changed display.

“Reproduced the model” thus leaves three possible operations: loading the same saved object again, running training again to create another object, or testing the assembled product. This fixture includes the first two. The International Medical Device Regulators Forum’s N67 terminology distinguishes the training process from its resulting model; that distinction supplies language, not validation of either the fixture or a device. [S2]

The same separation applies to a future pathology priority procedure, its saved model and the installed worklist, or to a sequencing suggestion procedure, its saved values and the report-review workflow. A post-use observation about import or reconciliation can require a product-level study even if the saved model still returns identical scores.

Reviewer Lens. The useful replacement for an unqualified reproduction claim is an account of the object, inputs, environment, outputs and comparison actually exercised. A complete fixed-artifact check can close that question. Demanding that it also establish the entire workflow would confuse the evidence objects as much as accepting it as workflow validation.

Consultant Lens. Giving procedure, saved artifact and product configuration separate identities early makes later choices easier to frame. The manufacturer can then state

which object must remain fixed for the next activity and which is still being designed. A single “final model” filename leaves both teams and reviewers to guess.

What would justify a more complex candidate?

Knowing which artifact ran does not tell us whether its complexity is useful. A simpler predictive method can face the same technical task, reference, tuning members and measurement rule. It gives the team a **technical baseline** against which to judge what another implementation adds.

That comparison leaves the Chapter 1 question intact: what would the professional task achieve without the proposed aid? The **non-AI comparator** addresses that wider workflow question. A tuning gain over a simpler classifier might justify investigating the complex candidate while leaving the case for the product unresolved.

The reason to try the candidate should therefore connect the image task, available examples, expected input variation and routing constraints to its design. A score can hide costs in latency, memory, maintenance and dependencies. It can also hide a failure to meet an essential input or safety condition. If that condition is required, a favorable average across other properties cannot make it pass. Separate conditions make the choice intelligible: a candidate may improve agreement and still lack an acceptable execution path.

In a proposed pathology comparison, chronological order could provide the technical reference for queue placement. Total slide-to-review time still includes the observed import constraint. In sequencing, candidate suggestion methods could be compared on variant correctness while the existing review method supplies a workflow comparator. Report release still includes reconciliation. These different endpoints locate the additional work; a candidate score has not demonstrated a defect in either workflow.

Reviewer Lens. The first comparison is whether baseline and candidate faced the same task, members, reference and timing boundary. The next is whether the conclusion stays inside that comparison. A clear technical gain deserves credit even while the workflow comparator remains open. Calling both “the comparator” without their questions invites the gain to travel too far.

Consultant Lens. Separate consequential conditions before trials give the product owner real options when a candidate is attractive but incomplete. Recovering its execution record, changing its implementation or pursuing another method are different actions. Folding everything into a single rank conceals those options and can direct effort toward a model when import or reconciliation owns the endpoint instead.

What actually changed between the runs?

Suppose the intended change is classifier design. The team fixes eligible crop membership, reference version, preparation, tuning measure and workstation conditions, then records each design and its fitting settings. That plan makes the comparison possible. What establishes that the planned conditions actually ran?

The **experiment log** connects the question and baseline to changed factors, members, references, transformations, settings, code, environment, outputs, anomalies and conclusion. If a local implementation used another preparation copy or stopping rule, the result belongs to that executed history. A neat protocol cannot retrospectively make the execution match it.

The history also needs the unsuccessful or inconclusive runs that influenced selection. Those results help explain why a later setting was chosen. “Best-model-final” cannot identify that sequence or the immutable content of its winner. If several factors changed together, the result may help choose a practical package of changes; it cannot isolate which factor caused the gain.

For example, a future pathology experiment changing both priority logic and import handling might alter elapsed time without attributing the gain to either alone. A future sequencing experiment changing the suggestion method and specimen-linking display would have the same attribution problem. The earlier observations help pose these experiments; no combined experiment is claimed to have occurred.

Reviewer Lens. A result that cannot be reconstructed from execution records has a different status from a protocol that reads well. The focused check follows the decisive score backward to what ran. When two factors changed, the package comparison can remain useful; the unsupported part is the narrower causal claim.

Consultant Lens. A run log and a dated decision log do different jobs. One reconstructs execution; the other explains why the team moved on. Agreeing how deviations and inconclusive runs enter both records before experimentation reduces the temptation to preserve only the attractive final result.

Which preparation has already learned from the images?

A crop file does not enter the classifier unchanged. Software turns it into the numbers the model consumes. This **preprocessing** can use a fixed rule, such as a specified size or channel-value range. It can also estimate settings from a collection, such as the average and scale used to center features. In that second case, preparation itself has learned something from those records.

Which records supplied that estimate? Fitting scale values only on training members preserves a different information boundary from estimating them over training and

tuning members together. The latter has exposed tuning information to development before the reported comparison. Chapter 7's separation therefore reaches into fitted transformations, before the final classifier is adjusted. The record needs the fitting membership as well as the rule and executable version.

The implemented preparation at later inference must also correspond to the configuration being evaluated. A resize or numerical-conversion change can move a suggestion near a route boundary even when the average score barely changes. Following one affected crop from pixels to numerical input is more informative than the assurance that the saved classifier stayed the same.

Other inputs make this visible too. Converting scanner timestamps to queue-age features would require a source, clock convention and missing-time rule in a future pathology experiment. Variant normalization or reference-lookup versions could change the record presented for sequencing review with unchanged model weights. A later mismatch calls for a versioned input/output comparison; silently “cleaning up” the input would conceal what changed.

Reviewer Lens. A preparation description is sufficient only if its fixed rules, fitted settings, fitting members and executed identity can be distinguished. Missing identity limits reconstruction; an inappropriate fitting scope raises an information-use question. Keeping those findings separate prevents a generic preprocessing concern from obscuring the actual remedy.

Consultant Lens. Preserving specification, executable identity and fitting scope together is easier while the transformation is being built. The software and data owners can also identify which outputs would need reassessment after a change. That makes a later correction an accountable comparison rather than an undocumented improvement to an old run.

Why might another training run differ?

Two runs may begin with different fitted-value starting points or encounter training examples in different orders. A **random seed** helps specify and control those choices. It does not describe every operation or execution condition that can affect the result.

PyTorch's official 2.14 documentation illustrates the limit: equality is not guaranteed across releases or platforms, or between central processing unit (CPU) and graphics processing unit (GPU) execution with the same seed. Controlling random sources and avoiding operations that can vary are separate concerns. [S1] This fixture uses CPU execution and declared changed seeds; it makes no promise about another release or hardware path.

The useful question is therefore what is being repeated. Loading one saved artifact through the same controlled path asks whether its outputs replay. Running the fitting

procedure again creates another artifact, for which a prospective exploratory tolerance might allow bounded variation. Changing the environment or product raises another question. A permitted difference in a development score does not permit an unexplained identity, unsupported-state or consequential routing discrepancy.

An unchanged overall placement summary could hide individual cases moving in a proposed pathology queue. Two sequencing variants could swap displayed rank while aggregate reference agreement stayed steady. Those are reasons to inspect paired records and action context in a future repeat, not claims that either change occurred in the recorded episodes.

Reviewer Lens. The seed is evidence about one condition. The complete repeat account says what reran, which random sources and environment were controlled, what varied and whether individual decisions were compared. A stable total is an observation worth retaining, but it cannot answer a question about unchanged routes.

Consultant Lens. Establishing the repeat target and tolerance before seeing the result keeps a plausible explanation of variation from becoming a convenient acceptance rule afterward. Saved-artifact replay, retraining and platform-change comparisons need their own discrepancy dispositions because they deliberately hold different things fixed.

Where does the execution record—and the clock—start?

Even identical images and stated settings can pass through different computations when source code or a library changes. The execution identity has to join reviewed source, configuration, dependencies, toolchain and hardware to the saved artifact and outputs. A preserved environment helps reconstruct that boundary. It does not establish scientific suitability, security, licensing or what remained unchanged outside it.

A timing result needs a boundary just as precise. Does its clock begin with a decoded crop and end at six scores, or does it follow specimen arrival to a professional seeing usable information? Warm-ups, measured runs, simultaneous work and the workstation also affect what is being compared. The fixture's technical condition concerns a single crop through preparation to scores on a development workstation. Gate processing, transfer, display and professional review are outside that clock.

The separate episodes explain why the boundary matters. Model-only queue timing cannot replace slide-to-review timing when scanner import dominates the latter. Suggestion-display speed cannot stand in for identity reconciliation and report release. If a library or interface changes after use, the manufacturer needs to identify affected inputs, outputs and measured paths before carrying an earlier result forward.

Reviewer Lens. Enough for a timing conclusion includes the code, environment, hardware, workload and exact start and stop. A crop-to-score observation can meet a technical condition without establishing product turnaround time. The focused request

is for the missing boundary or version, rather than an undirected demand for faster software.

Consultant Lens. Saving execution identity alongside the workload protocol gives software and performance owners a basis for later comparisons. Conversion, precision, runtime or dependency changes can then be assessed against the affected path before the manufacturer extends a product claim.

Which supplied component affects this result?

A training library, starting weights from a previously trained model, an annotation service and LumaCell's supplied quality gate contribute differently. Each is a **dependency** whose origin, version, function and available evidence have to be understood for its actual role. Did it support development, or would it operate in the configured product?

That role determines the next questions. Engineering records may reconstruct the run while rights to use or distribute data, code or a supplied component remain unresolved. Both can be satisfactory while the evidence is inadequate for a scientific claim. Exploratory data permission cannot silently supply future distribution, third-party processing or deployed-component rights.

A development-time AI assistant could draft code without becoming the product's inference engine. The manufacturer would still need to review and verify that code. If pretrained weights were proposed instead, unknown prior training exposure could limit a later independence claim even if their name were familiar. The fixture below stipulates no pretrained weights and gives the controlled first-candidate runs a dependency manifest; those are local facts to match to each run, rather than a blanket conclusion about every candidate.

A scanner-interface component could determine which import timestamp enters a future pathology study. A versioned reference or normalization component could determine which sequencing suggestion reaches review. A later supplier change creates an impact question for the manufacturer; it does not establish that either episode's previously observed outcome changed.

Reviewer Lens. Missing version identity, missing permission and insufficient scientific evidence are three different findings. Each deserves a request tied to the conclusion it affects. A familiar component name cannot close any of them, but neither should an unrelated rights question erase a well-documented technical observation.

Consultant Lens. A map of dependencies by function and life stage lets software, security, rights and supplier specialists answer their own questions. Limited supplier disclosure may call for additional verification, narrower use, replacement or rejection of an arrangement. The useful advice explains which option could resolve the actual gap.

What result would count as a successful repeat?

Suppose the team loads the saved classifier on the same inputs in the same environment. Comparing its routes and scores asks about **fixed-artifact replay**. Suppose instead it executes fitting again with an allowed seed change and compares the new artifact's agreement and paired routes. That asks about **repeated training**. A future runtime or workflow comparison changes the question again. Book 1C Chapter 7's measurement-repeatability discussion likewise makes the changed conditions matter; a label cannot replace the actual operation.

The comparison and tolerance need to precede the new result. Aggregate and affected-record differences then answer complementary questions. A small bit-level difference may be irrelevant to one defined purpose, while an unexplained unsupported-state change may matter despite an unchanged total. This is why the criterion has to follow the intended evidence question rather than whichever summary proves easiest to match.

If a repeat fails, the discrepancy stays in the history. Input, code, environment and artifact identities provide the first investigation route; the outcome may require qualification or repetition of the affected result. Correcting a record now can support a later conclusion without making an earlier unsupported statement sound at the time it was made.

A future pathology repeat might compare each case's queue position under one fixed import snapshot. A sequencing repeat might compare every specimen-linked suggestion and its professional review route. The recorded field constraints identify separate follow-up endpoints; they supply no tolerance for the LumaCell fixture.

Reviewer Lens. A prospective criterion and a criterion chosen after observing differences are different evidence. The reviewer can assess both honestly while declining to describe the latter as the planned pass condition. The narrow conclusion names the repeated operation, observed differences and actual rule.

Consultant Lens. A discrepancy record becomes useful when it connects each changed output to its input, procedure and artifact, then identifies an investigation owner, impact question and manufacturer disposition. Another screenshot can document an observation; it cannot perform those joins or decide the consequence.

Join the claim to the run

A manager asks why the team wants to carry one candidate forward. The answer should lead from the claimed technical gain to the comparison, executed conditions and saved object. The **model-development record** keeps those links together.

The table helps locate a missing link. Its first rows define the comparison and preparation; the middle identifies what ran; the last establishes the dependency terms and actual repeat conclusion.

Record field	Question its evidence answers
Baseline comparator	What credible method faces the same technical question?
Preprocessing	Which fixed or fitted transformations ran, and which members fitted their settings?
Candidate rationale	Why try this method for the role, inputs and constraints?
Experiment identities	Which question, conditions, inputs, outputs, anomalies and choices belong to each run?
Hyperparameters	Which settings, stopping rule and selection rule were planned and executed?
Randomness controls	What random sources were controlled, what varied and what limits remain?
Code and environment	Which implementation and execution conditions can be reconstructed?
Compute	Which hardware, resources and workload produced the result?
Tool versions	Which training, conversion, evaluation and dependency tools were used?
Dependency provenance	What supplied each component, for which role, with what evidence or unresolved exposure?
Licenses and terms	Which use, distribution, investigation and maintenance conditions apply?
Reproduction result	What repeated operation met which prospective rule, with what differences and disposition?

For one crop, the preprocessing row has to lead to an implementation and any fitting membership. The experiment row joins that prepared input to the saved output; the reproduction row says what was compared afterward. Repeating “controlled” in those cells would leave the path missing. Dataset memberships, reference instructions and permissions can be linked at their controlled locations rather than copied into another form.

An advancement statement then identifies the artifact, limited purpose, passed comparison and remaining evidence needs. The same record for a future queue or

sequencing study would say whether its result concerned an algorithm endpoint or changed import/reconciliation work. Completion of the form alone establishes neither.

Reviewer Lens. Working backward from one conclusion is an efficient sufficiency test: can another qualified reader recover members, executed settings, output and selection rule? A single unrecoverable link can qualify the claim even when every cell contains text. The request should identify that link.

Consultant Lens. Each link has a producer—data, machine-learning, software, supplier or clinical/evidence owner. An agreed handoff before runs begin gives the eventual decision a source. Reconstructing it from the winning developer’s memory is an avoidable dependency of its own.

Four runs, one bounded development question

The following author-designed fixture concerns LumaCell development, not an executed study of a real product. Each synthetic run suggests one of five common white-cell categories or the other/unclassified route for an accepted candidate crop. Its suggestions are compared with versioned **experiment reference labels** (REF-08). Agreement means matching that specified comparison; the labels are not assumed to be infallible clinical truth.

Which examples did the work?

The **experiment training collection** (TRAIN-08) contains one index crop from each of 1,200 distinct fictional adults, with 200 in each of the six reference routes. Fitting uses those examples. The **experiment tuning collection** (TUNE-08) contains one index crop from each of 200 different adults, with no patient overlap. Its results inform candidate choice. No independent confirmatory set is opened.

Permission and lineage cover only this segregated exploratory work. These are separate records from the earlier restricted subset, ten-image reference fixture and unused six-patient allocation (D-A, R01–R10 and P01–P06). A balanced training collection describes its selected examples; it does not change the role of the separate tuning comparison or confer rights on another collection.

What did the team decide before seeing results?

The **simple classifier baseline** (B0) supplies the technical reference. A candidate can displace it *for further development* only when three conditions are satisfied: tuning agreement proportion improves by at least 0.03; its configuration can be reconstructed; and its 95th-percentile single-crop latency is no more than 80 milliseconds under the fixed development workstation profile (W-DEV). These are separate gates. A higher agreement number cannot offset missing execution identity or an excessive time on the specified path.

The agreement proportion is the number of matching routes divided by the 200 tuning crops. A difference of 0.03 means three percentage points, not a 3% relative improvement. These values are fictional teaching choices, not medical acceptance thresholds. A different proposed endpoint or measured path would need its own justified condition; it could not inherit the apparent pass merely because the candidate name stayed the same.

Two repeat questions have their own prospective rules. Loading the same saved artifact on the same 200 inputs in the same controlled environment must reproduce every route and score exactly for this exercise. Training again with a declared changed seed creates another artifact. That second comparison allows at most 0.01 absolute difference in agreement proportion and at most 8% **paired route changes** across those same inputs. A paired route change occurs when the two artifacts suggest different routes for one crop, including when both routes disagree with its reference.

The repeated-training rule also permits no unexplained input, package-identity, class-map or unsupported-state discrepancy. The class map connects score positions to their intended route names; agreement of a total would not explain a mismatch in that meaning. A permitted summary difference therefore cannot excuse an unexplained record or state change. Neither numerical tolerance declares different suggestions harmless in clinical use.

What does a timing entry mean?

The **first candidate** (C1) and **second candidate** (C2) are exploratory alternatives. The first table distinguishes their original runs from a changed-seed training repeat of the first candidate. “Record state” belongs to each result: it says how far the team can trace what produced that number.

For latency, the protocol sorts 200 measured single-crop times from shortest to longest. The **nearest-rank percentile** selects position $\text{ceiling}(0.95 \times 200) = 190$, where ceiling means rounding up. That observed position supplies the 95th-percentile time. Twenty separate warm-up runs are excluded. The individual measured times are not supplied, so the table reports the synthetic result without inviting a reconstruction of unprovided observations.

Run	Candidate and purpose	Agreement with reference	95th-percentile time	What can be reconstructed?
Baseline run (EXP-01)	Simple classifier baseline B0	172/200	20 ms	Inputs and execution

Run	Candidate and purpose	Agreement with reference	95th-percentile time	What can be reconstructed?
First-candidate run (EXP-02)	C1	182/200	60 ms	Inputs and execution
Second-candidate run (EXP-03)	C2	188/200	55 ms	Local preprocessing copy unidentified; dependency version missing
Training repeat (EXP-04)	C1 trained again	181/200	61 ms	Same controlled data/reference; declared stochastic variation; no unexplained execution difference

The repeat row refers to a newly trained artifact, not another loading of the first one. The second-candidate row reports a number alongside two open identities. Those columns will have to be read together when applying the advancement rule. Before making that judgment, it helps to understand what ran between the input crop and its six scores.

How does the simple classifier turn a crop into scores?

The **experiment preparation** (P-EXP-01) gives the candidates red-green-blue inputs of 256 by 256 pixels and converts channel values to the 0–1 range. There is no random image augmentation and no preparation parameter fitted on the tuning collection. This common specification is the comparison’s intended preparation; the second-candidate run’s unidentified local copy remains a reconstruction gap.

The baseline reduces each of the three color channels to its average and spread, represented by its standard deviation. The feature formulas are fixed. The values used to center and scale those features are estimated from the training collection alone. A **linear classifier** weights the resulting features to score six possible routes. Its simplicity does not remove learning or execution history: its weights are fitted, and its regularization coefficient of 1.0 constrains the fitting under a setting fixed before comparison.

This distinguishes three things in one operation. The average-and-spread formula says which numbers to extract; training-only centering and scaling determine their transformation; fitted classifier weights determine their contribution to route scores. Losing the identity of any one could obstruct reconstruction even if the label “linear classifier” remained accurate.

What changes when the candidate learns image patterns?

The first candidate works with spatial patterns in the prepared image. At each position, learned weights combine values in a 3-by-3 neighborhood across the input channels. This operation is a **convolution**. Its result at successive positions forms an **activation map**, showing the strength of the computed pattern response across the image. Unit stride moves one position at a time; one-pixel padding supplies a border for the edge positions.

The next operation sets negative computed activations to zero while retaining positive ones: a **rectified linear unit** (ReLU). Then **max pooling** keeps the strongest activation from each 2-by-2 local region, reducing spatial detail. Each block therefore has a local weighted calculation, a nonnegative response and a local reduction. The second block applies that sequence to the first block's resulting maps.

The two blocks produce 16 and then 32 **learned feature channels**. These are pattern maps, distinct from the input's three color channels. **Global average pooling** then averages each final map across its whole area before a six-output classification layer supplies route scores. The architecture sets filter sizes, channel counts and operations; fitting supplies the weight values. Merely repeating the architecture name cannot identify which values were learned.

What makes those weights change during fitting?

For a training crop, the reference label identifies the route the procedure should learn to support. Assigning little probability to that route incurs a larger **cross-entropy** penalty. The **Adam optimizer** uses the fitting calculation to update the learned weights. Here a **batch** of 32 examples contributes to each update, and the initial learning rate of 0.001 sets its initial scale.

One full pass through the training examples is an **epoch**. This procedure runs 30 fixed epochs. The team neither watches the tuning collection to stop early nor chooses a favorable intermediate checkpoint using its score. Those alternatives would change the selection history. The given architecture and schedule are executed conditions for this teaching fixture, not recommendations for medical-device development.

The following table connects that crop-to-score explanation to the records needed to repeat it. Source and environment identifiers name controlled records; saved-artifact and output identifiers name different objects. None is a downloadable model or an invented cryptographic digest.

Record element	Executed configuration or observation
Common input preparation	P-EXP-01: red-green-blue 256-by-256 crops; channel values converted to 0–1; no random image augmentation; no parameters fitted on TUNE-08

Record element	Executed configuration or observation
Simple baseline	B0: multinomial linear classifier on fixed channel-mean and channel-standard-deviation features; centering/scaling fitted on TRAIN-08 alone; regularization coefficient 1.0 fixed before comparison
First-candidate design	C1: two blocks, each with 3-by-3 convolution, unit stride, one-pixel padding, ReLU and 2-by-2 max pooling; 16 then 32 channels; global average pooling and a six-output classification layer
First-candidate fitting	Cross-entropy, Adam, initial learning rate 0.001, batches of 32 and 30 fixed epochs; no early stopping or tuning-set checkpoint choice
Code and environment	Reviewed source SRC-08-01; ENV-08-01 identifies Python 3.11, PyTorch 2.14 and the full dependency manifest; CPU-only execution on fixed W-DEV workstation profile
Randomness	EXP-02 uses seed 101; EXP-04 uses seed 102 for initialization and training-record order; other declared conditions fixed
Saved objects and outputs	EXP-02 produces ART-C1-02 and OUT-C1-02; EXP-04 produces ART-C1-04 and OUT-C1-04; execution record checks artifact content identity and matching input/reference membership
Latency method	200 single-crop measurements after 20 separate development warm-ups; decoded crop through P-EXP-01 to six scores; nearest-rank 95th percentile; excludes Q0, transfer, user display and review
Same-artifact replay	Loading ART-C1-02 again in ENV-08-01 yields 200/200 identical routes and zero observed score difference on TUNE-08

The output row makes the repeat operation visible: the changed seed produced another saved artifact and another output record. The replay row instead loads the first saved object again. Their comparisons have different prospective rules because they held different objects fixed.

The latency row starts after the crop is already decoded, includes the experiment preparation and stops at six scores. The gate, transfer, display and professional review remain outside it. A time of 60 milliseconds at the specified percentile cannot guarantee patient turnaround time; a later product timing claim would need the relevant longer path and workload.

Did similar totals mean similar suggestions?

The two first-candidate artifacts may agree with the reference on similar numbers of crops while disagreeing with each other about particular crops. The next table keeps all 200 tuning inputs paired. Each crop has its first-run suggestion, repeat suggestion and the same reference label.

Reference agreement on the same paired inputs	Crops
Both runs agree with the reference	177
Only the first run agrees (EXP-02)	5
Only the training repeat agrees (EXP-04)	4
Neither run agrees	14

The last row needs another distinction before routes can be counted. Eleven of those fourteen crops receive the same reference-discordant route twice. Three receive different reference-discordant routes. A binary “wrong both times” label would hide that difference. The reference groups and actual changes are retained for later clinical and robustness assessment.

The changed-route records have been checked against input and configuration identity. None has an unexplained input, class-map, identity or unsupported-state discrepancy. That is an observation about the declared screen; it does not establish that the changed suggestions are harmless or that the reference labels cannot be wrong.

No independent final test result is supplied. The fixture establishes neither statistical superiority, subgroup adequacy, calibration, human-use performance nor clinical benefit. The two exploratory candidates are not asserted to be the later evaluation or released classifier (M1). The present decision concerns their next development activity.

Which next development decision does this support?

What are the four tuning agreement proportions? How much does the first candidate improve on the baseline? What is the absolute agreement difference between its original and repeated training runs, and what fraction of the paired inputs receives a different route? Why are those last two quantities different?

Which result can now support a bounded advancement recommendation under all the prospective conditions? What should happen to the second candidate’s run, what useful role remains for the baseline, and which claims are still unsupported? What would the twelve-field model-development record need to retain for another reviewer to defend that recommendation?

Decision

Which calculations, execution records and limits support further development without selecting a production model or claiming final validation?

How the paired outputs and execution record support a bounded choice

Worked reasoning

Dividing each agreement count by the common 200-crop denominator gives 86%, 91%, 94% and 90.5%. The first candidate's original run exceeds the baseline by five percentage points, above the prospective three-point improvement condition. Its training repeat remains 4.5 percentage points above the baseline. These are tuning comparisons used for choice, not evidence of statistical superiority or product benefit.

The original and repeated first-candidate runs differ in aggregate agreement by only half a percentage point: 0.005. The paired record explains why that is not a count of changed suggestions. Five crops lose agreement and four gain it; three others receive different wrong routes. Thus $5 + 4 + 3 = 12$ crops change route, or $12/200 = 6\%$. The eleven remaining crops that are wrong both times have the same wrong route and add no route change. A net loss of one agreement can coexist with twelve different suggestions.

Both prospective numerical limits for repeated training are met: 0.005 is no more than 0.01, and 6% is no more than 8%. The specified identity/state screen also has no unexplained discrepancy. This closes the bounded repeated-training condition under the fictional protocol. Exact saved-artifact replay is a separate positive finding: 200/200 routes match, with zero observed score difference, when the first saved artifact is loaded in its recorded environment.

The first candidate has a reconstructable basis for further development: the improvement, 60-millisecond latency on the specified path, execution identity and both repeat checks meet the relevant exploratory conditions. That conclusion does not declare 6% decision variation acceptable in clinical use. Its changed records remain work for later robustness, risk and product-evaluation questions. A new unexplained identity or state discrepancy would reopen the applicable repeat condition even if the numerical limits still passed.

The second candidate has the highest reported agreement, but its local preprocessing copy and dependency version cannot yet be reconstructed. That fails the record condition for advancement now. Recovery and verification may resolve it; if recovery is inadequate, a controlled rerun and assessment are needed. The finding is about the present basis for the decision, not proof that the candidate is poor. A favorable score remains a reported observation while its missing execution history is addressed.

The simple baseline remains the technical reference. Its comparison with the first candidate motivates further investigation; it neither becomes the professional workflow comparator nor replaces independent evaluation. The completed record below preserves those different conclusions instead of reducing all four runs to a rank.

Record field	Worked completion
Baseline comparator	Simple fixed-feature classifier B0: 172/200 agreement and 20 ms 95th-percentile time; retain it as the technical reference
Preprocessing	Identified experiment preparation P-EXP-01, with no tuning-fitted parameters; baseline scaling fitted on TRAIN-08 alone
Candidate rationale	First candidate C1 meets the prospective improvement and latency conditions with a reconstructable record; investigate further. Hold selection of C2 pending recovery of its execution identity
Experiment identities	Retain EXP-01 through EXP-04, including the second-candidate run's unresolved preprocessing/dependency history
Hyperparameters	Retain the specified first-candidate architecture and fitting values; 30 fixed epochs; no tuning-driven checkpoint selection
Randomness controls	Seeds 101/102 identify the permitted initialization and record-order variation between training runs
Code and environment	Link reviewed source SRC-08-01 and execution environment ENV-08-01, including full dependencies and controlled W-DEV profile
Compute	CPU-only development workstation; 200 measured crops after 20 separate warm-ups; no whole-product latency claim
Tool versions	Python 3.11 and PyTorch 2.14 as recorded; no conversion or quantization in these runs
Dependency provenance	No pretrained weights; environment-linked inventory for controlled runs; EXP-03's missing dependency version remains open
Licenses and terms	Stipulated segregated exploratory use is covered; later distribution and deployed-dependency obligations need separate assessment
Reproduction result	Exact replay: 200/200 matching routes, zero score difference. Changed-seed training: 0.005 aggregate difference, 12/200 changed routes, no unexplained identity/state discrepancy. Development only; retain changed-record review and independent-evaluation needs

Reading the randomness and reproduction rows together distinguishes loading the first artifact (ART-C1-02) again from creating the repeated-training artifact (ART-C1-04).

The corresponding output records retain what each produced. In operational work, those names would have to lead to actual content-integrity evidence, members, settings and outputs. A documentary label alone could not establish a match.

The decision is supported by task relevance, controlled identity, separate constraints, observed repetition and a bounded conclusion. A reviewer can verify those links without selecting the manufacturer's production model. A consultant can convert the open second-candidate identities and changed first-candidate records into specific recovery and impact work.

What would have to join the product configuration?

The positive development decision leaves the product map unfinished. A later evidence activity would need to join its selected artifact to acquisition and locator assumptions under the proposed profile, actual package identity values meeting the identity-field requirements, implemented preparation, the original gate, five-category and other/unclassified rules, runtime, display, fallback and professional full-smear review. This crop-only fixture settles neither the gate's placement, representation and aggregation nor whole-product speed.

An artifact can be fixed for a particular activity while other product decisions remain open. "Final model" is useful only when that scope is clear; it neither forbids later change nor approves release. A changed input profile, gate, threshold, numerical precision, runtime or interface can change evidence applicability and needs controlled assessment. The nonbinding IMDRF N88 principles distinguish sound development from appropriately independent, clinically relevant testing. They do not prescribe this architecture or its numerical tolerances. [S3]

For a future total-time pathology evaluation, the configuration and evidence would have to include the versioned import-time input and worklist display. For a future sequencing report-release claim, the suggestion artifact must connect to specimen identity and reconciliation behavior. The observed constraints inform the later impact question. They cannot authorize carrying an algorithm-only result into a changed workflow claim.

Reviewer Lens. The useful match is between the candidate and product configuration named by the claim and the versions actually exercised. The first-candidate record earns its development conclusion. Extending it to gate, runtime, display or human-use paths requires a separate applicability argument, with the particular missing join identified.

Consultant Lens. A candidate-to-product join lets the manufacturer mark what is fixed, what remains unresolved and which verification or independent evaluation follows a material change. That makes further development actionable without treating the experiment as a release package.

Figure 8.1 — A Candidate Carries Its Full Experiment History. A performance result is not reproducible evidence unless the candidate and its complete realization context are controlled.

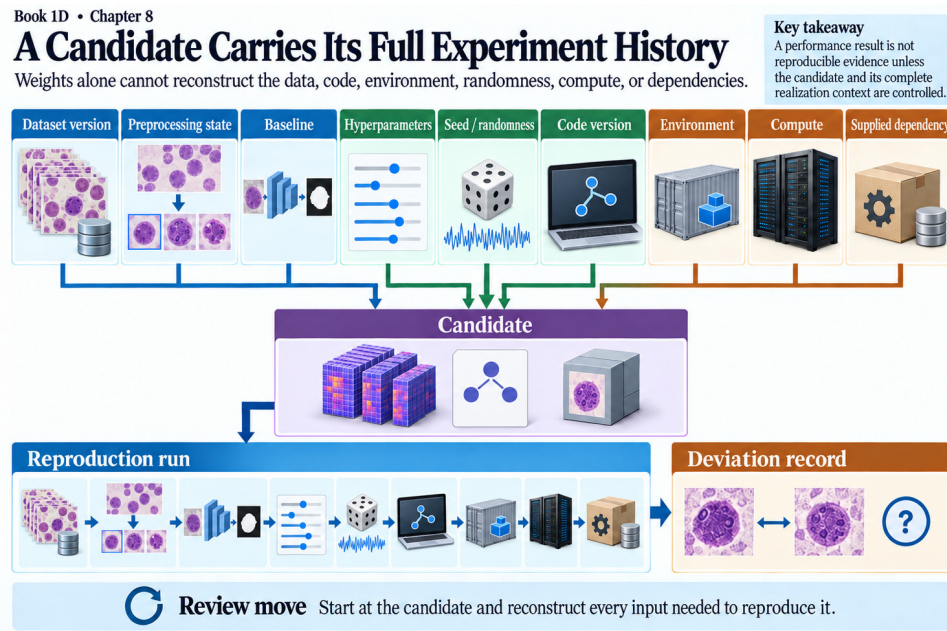


Figure 8.1. A Candidate Carries Its Full Experiment History.

Who can make the next decision?

A decision to investigate the first candidate needs several owners' work. The machine-learning lead produces run, selection and changed-record analyses. Software and security specialists examine implementation identities and dependency controls. Clinical and statistical colleagues assess whether the task, reference, variation and planned evidence address the intended question. Quality and risk owners consider material anomalies and decision traceability; supplier and licensing specialists address their respective evidence and terms.

The manufacturer then decides whether to advance the first candidate, resolve the second candidate's record, change the design, begin a later evaluation or ultimately release a specified product. The decision cites the actual record and unresolved work. The consultant advises on reproducibility and dependencies; the reviewer judges the claim and support. Neither role quietly chooses or approves a production model. In eventual professional use, staff would still check suggestions and complete smear review, while the laboratory owns patient interpretation and release.

How does a consultant turn a score into a closure question?

A screenshot arrives with the claim that the highest-scoring candidate should move forward. A useful **reproducibility and dependency review** follows that assertion backward, compares the actual repeat question and identifies conditions that could change the conclusion. It gives the manufacturer a choice among recovery, rerun and limitation; it does not select the product model.

Which missing identity affects the claim?

The second candidate's reported maximum should lead to its tuning members and experiment reference version, then to executed transformations, settings, code, dependencies, environment, saved output and the later choice it influenced. The screenshot alone cannot reliably supply those links. The two specifically missing identities are the local preparation copy and dependency version.

Their absence matters because reconstructability was a condition for advancement. The result may be genuine; "the candidate is invalid" would overstate the finding. Recovery of those two identities, verified against execution, could change the record's status. A broad request for every development notebook would obscure that concrete closure route.

Which operation did the repeat actually test?

The same distinction guides the positive first-candidate finding. Loading its original saved artifact again, training the repeat artifact and executing on another platform are different comparisons. Only the first two appear here. The declared exploratory tolerance permits the supplied changed-seed training result; it does not promise identical behavior on every future platform or release.

The 0.005 agreement difference and twelve changed routes also identify a next task. The adviser can check the arithmetic and that the specified identity/state screen was performed, then bring the retained suggestions to clinical and risk owners for their next relevance question. A favorable aggregate result cannot supply their clinical judgment. New unexplained identity or state evidence would require revisiting the repeat disposition itself.

What did the team learn while choosing?

A winning candidate can emerge from extensive exploration. To describe its evidence accurately, the review needs enough history to establish which dataset's results supplied which decisions. Repeated tuning is ordinary development work. Describing its resulting selection evidence as an independent final test would assert a different history.

The same information path could arise in a future pathology study if placement results led to repeated priority-setting changes, or in sequencing if comparison results led to revised display thresholds. Those results would be development information for those choices. These examples describe prospective checks, not new events in the earlier episodes. A later evaluation needs an appropriately separated route for its fixed question.

Changes in the acceptance rule belong in that history too. A tolerance revised after the result may have a reason worth examining, but it cannot be reported as the original prospective criterion. The date and reason for the revision let the manufacturer and reviewer judge what the pass means.

What does the manufacturer need from the supplier?

The second candidate's missing dependency version is an execution-identity question. The original quality gate would be a different kind of dependency when integrated into a later product: its terms, interface evidence and change behavior have to suit that role. The first candidate's training manifest cannot answer for the gate.

For any supplied component, version, role, provenance, relevant terms, change behavior and evidence limits should serve the actual proposed use. Some proprietary information may be unavailable in full. That differs from knowing the question has no adequate answer. Additional verification, narrower use, another supplier or rejection of the arrangement may provide different ways forward; the manufacturer still owns the adequacy of its chosen control and evidence argument.

Popularity provides no product-specific acceptance decision. A future scanner-interface change or sequencing-reference component would need the same concrete account of version, affected records and intended question. The relevant issue is the effect on this result or use, rather than whether the supplier is familiar.

What would let the finding close?

For the second-candidate run, closure is a reconstructable, verified execution record or a controlled rerun, assessed against the relevant criteria. Recovery would change the present identity finding. A rerun need not reproduce the favorable score exactly unless that is a justified criterion for its particular question; its observed result must be assessed honestly.

For a reproduction discrepancy, the closing record needs a cause or bounded explanation, its effect on result and product, necessary corrective work and a manufacturer disposition. Another screenshot supplies none of that by itself. The conclusion can therefore be both positive and limited: the first candidate's supplied record supports its stated development advancement; the second candidate's higher score lacks the required reconstructable basis until the listed identities are resolved.

Independent evaluation remains separate, and this advice approves no production model.

Reviewer Lens. Across these checks, the evidence is the original/repeated first-candidate outputs, the still-open second-candidate identity, the prospective rules and the exploratory-use boundary. A qualified positive development conclusion fits that packet. An independent-test or released-product conclusion would require different support. The finding should change when the relevant missing evidence arrives, not merely when a memo acquires a new title.

Consultant Lens. The next work divides by cause: experiment and software owners recover or rerun the second-candidate record; clinical and risk owners assess the changed first-candidate suggestions for their next question; data and evidence owners preserve a separate later evaluation route; dependency specialists resolve role-specific supplier and terms issues. Each return goes to the manufacturer for a recorded disposition. Sequencing the work this way helps resolve the choice without making it for the company.

Practice the review language

Consider the statement: “The model is reproducible because the engineer set the seed.” What has actually been established, and which observed comparison would support a stronger statement?

Decision

How could the claim identify both the controlled condition and the evidence still required?

How a seed becomes one part of a supportable claim

The seed identifies one randomness control. A supportable reproduction statement also identifies the actual repeated operation, data and execution conditions, other sources of varying behavior, and the observed result against a justified criterion. That wording makes the missing evidence visible without promising universal determinism or prescribing a particular software stack.

What changes when execution is optimized?

Consider a counterfactual after a candidate has been selected for further evaluation. To improve latency, the team converts it to a lower-precision numerical representation and changes the runtime. Its architecture and training-dataset names remain the same. Which record identities change, and which earlier conclusions can no longer be carried forward without assessment?

Decision

What comparison would connect the new execution path to the claimed scores, routes and timing?

How unchanged training history can accompany a changed artifact

The converted artifact has a different content identity. Conversion procedure and tool version, numerical representation, runtime and execution environment now belong in the comparison. The original training membership and architecture remain true history, but they do not identify the new executable object.

The manufacturer needs scores, routes and unsupported states compared on suitable controlled inputs, including relevant boundary cases. A small numerical change near a route boundary may affect the decision. A focused assessment can be justified, but affected performance and whole-product checks must address the intended new configuration. The old exact replay and development-workstation latency do not automatically establish its behavior or speed.

Chapter 3's product boundary already includes runtime and transformations. Chapter 7's information boundary applies as well: a comparison used to tune this optimized candidate becomes development evidence for that choice. It is not automatically a fresh independent test.

In a future pathology optimization, the affected worklist positions and their links to imported cases would need preservation. For sequencing, changed suggestion states and specimen links would need examination before a report-review claim continued. A post-use discrepancy would connect the earlier and changed configurations to investigation and disposition. These are prospective analogies; no new event has been supplied for either episode.

Reviewer Lens. The prior evidence belongs to its artifact and execution path. The focused comparison can show what remains applicable, while some independent or human-workflow claims may require further evidence. Shared training names alone do not settle that boundary.

Consultant Lens. Explicit earlier and later artifact/runtime identities, affected decisions, timing boundary and discrepancy owner let the manufacturer scope a meaningful comparison before carrying a claim forward. That preserves useful history while making the changed object reviewable.

What does the next evidence question need?

The first candidate has a reconstructable exploratory history in this fixture; the second candidate's run identity remains open. Chapter 9 asks how metrics, operating points, uncertainty and acceptance criteria fit a later evaluation question. Even a well-recorded tuning number can answer a different question from the one the product claim needs.

Optional depth is available in Book 1A Chapters 4–7 on architecture, development, verification and AI/data review; Book 1B Chapters 4–6 on secure implementation, suppliers and testing; and Book 2 Chapters 6 and 14 on European Union documentation and change interfaces. Those readings supply no local jurisdictional classification or release decision. Unresolved security, rights and supplier questions remain attached to this experiment record and their accountable owners.

Return to the two scores

Can the procedure, trained artifact and product be identified separately? Can another person reconstruct the decisive experiment and distinguish tuning from independent evaluation? What exactly does each reproduction result demonstrate? Why might the highest score still lack a basis for advancement?

Decision

Which missing link would change your conclusion, and which bounded result can already receive credit?

Source Trail

S1: PyTorch 2.14 reproducibility documentation, opening limitations and controls for randomness/nondeterministic operations, checked 2026-09-24. This is a platform-specific illustration, not a universal medical-device standard or recommendation to use that version.

S2: IMDRF N67, Sections 4, 6.8–6.9 and 7.3, training/model and validation distinctions, checked 2026-09-24. It supplies terminology, not validation of this fixture.

S3: IMDRF N88, Principles 2, 4–6 and 8, engineering, independence, reference-fit and testing principles, checked 2026-09-24. Its principles are nonbinding and do not set this chapter's architecture, sample size or numerical conditions.

The experiment record, synthetic scores, latency condition and reproduction tolerance are original teaching constructions. They are not clinical thresholds, study-size recommendations or evidence that any real model is adequate.

CHAPTER 9

Chapter 9: Metrics, Operating Points, Calibration, Uncertainty, and Acceptance

A selected adult peripheral-blood cell image arrives at a fixed **LumaCell Review** candidate. The reference places it in the target category. The product might route it to target, route it elsewhere, or return no evaluable result. Which of those received images will still appear in the denominator when the manufacturer reports performance?

That small accounting decision determines what the percentage can mean. A high agreement among images that received answers can coexist with a less favorable result among everyone entering the route. The manufacturer needs to decide which question matters before the results arrive, then preserve enough records to answer it.

The 1,000-unit method exercise later in this chapter is a separate synthetic construction. It uses none of Chapter 5's restricted exploratory subset, Chapter 6's ten reference-method images, Chapter 7's unused allocation, or Chapter 8's training and tuning collections. It is also separate from later release evidence. Its counts will support a bounded image-routing decision, without supplying disease detection, complete-smear review, human-use or release findings.

Which question would the percentage answer?

Suppose a report calls the opening task “accuracy.” Before any calculation, there are four earlier decisions to recover. Chapter 2's intended use separates the proposed suggestion from the professional's full-smear work. Chapter 6's reference, the comparison used to judge an output, determines what agreement means. Chapter 7 asks whether counted images are independent and whether their results already influenced development. Chapter 8's model-development record joins the trained artifact, the saved fitted values, to its executed conditions and later selection decision; a general model name cannot supply those links. Those decisions give the calculation an object.

What should count as the event above the fraction line, and what population belongs below it? An identity rejection, quality rejection and unavailable output can all interrupt the ordinary route. An accepted other/unclassified suggestion has a different

meaning, as does a reference that remains unresolved. Is the proposed result about what the device returned, what a professional decided, or what the laboratory released for a patient?

Decision

Which numerator, denominator and treatment of those states would answer the proposed image-routing question? Which would answer a different question?

How a reviewer follows the images missing from a headline

The word “accuracy” leaves the target, comparison, counted event and population unresolved. Agreement among answered crops asks how the product behaved when it answered. Successful routing among all received target-reference crops asks whether each image entering that defined route reached the desired output. A product non-result can count as a non-success for the latter endpoint without establishing clinical error, patient harm or failed manual fallback.

An unresolved reference raises another problem: the comparison needed to judge the output is unavailable. That is different from a product that supplied no output even though the reference is known. Keeping both states visible permits a defensible denominator. Neither image-level quantity measures patient benefit.

The **pathology queue episode** offers a useful reminder of the boundary: imported cases moved earlier, while scanner import still dominated total review time. A future queue-placement percentage could count already imported cases; a total slide-to-review endpoint would have to account for cases still waiting to enter the queue. In the **sequencing review episode**, correct suggestion display coexisted with specimen-identity reconciliation constraining release. Counting displayed variants and counting reconciled specimen reports would likewise follow different work. These are earlier fictional observations, not new datasets for this chapter.

Reviewer Lens. The fastest way to understand a strong headline is often to look for the records it did not count. The request can stay narrow: entry unit, reference, returned state and exclusion reason. That reveals whether an image or variant statement has quietly become a claim about the completed workflow. A valid narrower percentage remains useful once its population is named.

Consultant Lens. Product, data and clinical owners can settle a received-item flow before commissioning analysis. Each branch needs a state destination and a person who can explain it. Recovering an omitted branch after a favorable percentage has become the headline is harder: the team must recover both records and the original decision it meant the number to support.

What quantity is the study trying to estimate?

Two analysts can start from the same received-image table and produce different, correct fractions. One excludes images without an evaluable answer; the other retains them. The disagreement is about the target quantity before it is about arithmetic. Statisticians call that quantity the **estimand**: what the study seeks to estimate, rather than the particular number observed this time.

For the opening task, the quantity can be stated through a population, fixed configuration, outcome, treatment of relevant states and summary. **Answered-only agreement** concerns a fixed candidate's suggestion compared with the specified crop reference among images receiving evaluable answers. **All-received target routing** concerns the fraction of every received reference-target image routed to target; a product non-result is a non-success for that endpoint. Both quantities can inform development. Choosing the more attractive one after inspection would change the question the planned study was meant to answer.

The word *image* also holds a boundary in place. A selected crop does not enumerate all cells in the smear, establish a blood differential, or record the professional's response. The professional checks suggestions and continues full-smear review; the laboratory interprets and releases the patient result. The International Medical Device Regulators Forum addresses human–AI performance and clinically relevant testing in IMDRF N88. Those nonbinding principles explain why a model-only quantity leaves further use questions open. [S3]

The same distinction makes a future timing plan concrete. For the pathology queue episode, total time needs specified start and end events, with import delay among eligible case states. The earlier queue observation can still support its narrower placement question. For the sequencing review episode, a specimen and its reconciled report may be the unit of a release endpoint, while a variant suggestion remains the unit of another quantity. The earlier observations motivate those new definitions; they supply no numerical results for them.

Reviewer Lens. A change in the report's denominator may be presented as a minor analysis detail. Comparing the plain-language quantity with the protocol exposes whether population, configuration, outcome or unavailable-state treatment changed. Enough is an explicit account of that change and what the revised result can support. The label “same study” cannot do that work.

Consultant Lens. It is useful to agree on one primary sentence and a separate answered-only sentence early. The clinical owner can then distinguish the desired product question from the professional-workflow question, while data owners identify who supplies each missing state. This preserves useful secondary information without leaving the acceptance endpoint to the analyst's later preference.

Which errors could an overall number hide?

A target-reference image routed to comparison and a comparison-reference image routed to target are different disagreements. A single total can obscure that difference. The first step is to retain two reference groups, then count the relevant route within each group's own denominator. These give **positive agreement** and **negative agreement** when the comparison is an agreement question.

Why not simply call them disease sensitivity and specificity? Those terms require an appropriate reference for the target condition. A comparison method that has not established disease status permits a narrower interpretation. The FDA's final, nonbinding 2007 guidance for binary qualitative diagnostic tests distinguishes accuracy against a suitable reference standard from agreement with another comparator, and recommends reporting the pair of measures with uncertainty. It does not turn this chapter's fictional image reference into a disease standard. [S1]

Even the two-group view leaves something out of LumaCell's proposed task. The classifier suggests five white blood cell categories or the other/unclassified route. Collapsing that structure into target versus comparison helps expose denominator choices, but could conceal a rare category or consequential wrong route within the comparison group. The original reference-by-output counts and relevant class-specific results must remain available for the full claim.

Counts of displayed routes answer a different question from the ordering of scores. A model could rank target-like crops above others, yet the product still has to decide which score causes which visible action. The selected threshold or combined rule is its **operating point**, the score-to-action choice. Good ordering across thresholds leaves performance at the actual chosen rule to be checked. If the displayed score also claims to be a probability, its agreement with observed event frequency adds another question.

For a future pathology priority rule, ordering already imported cases would leave the chosen cutoff and resulting workload unresolved. A sequencing suggestion ranking similarly leaves the display and reconciliation rules to be examined. In either episode, a later changed rank would need a link to the action it changes. Rank movement alone would not measure the patient's outcome.

Reviewer Lens. Overall agreement is easy to read and easy to overextend. The underlying table is usually more revealing: which reference group supplies the denominator, which route counts as agreement, and which smaller category could disappear in the total? That request preserves a useful overall description while testing whether it conceals the error structure that matters to the claim.

Consultant Lens. A small set of questions tied to the claim is easier to defend than a collection of attractive plots. Clinical and statistical owners can choose the routing and class-specific comparisons that decide the question, add ranking or probability

assessments where they serve an intended use, and label the remainder descriptive. That early distinction prevents a secondary plot from becoming the preferred decision route after the primary result arrives.

What turns the score into an action?

Six scores arrive from the crop classifier, but no professional has yet received a suggestion. The product may choose the highest score, require a threshold for a target route, or withhold an ordinary prediction under specified conditions. The operating point binds those numerical outputs to an action. Changing it can alter false target routes, missed targets, manual-review work and the frequency of non-results.

That is why the rule belongs in the declared development procedure. Once selected, its actual configuration is fixed for the independent comparison. A threshold chosen because final evaluation counts look more favorable has used those counts for development. Preserving the original model name would not preserve the original decision rule.

The decision unit must be fixed as well. A threshold on one crop score differs from a rule combining several crops in a package, and both differ from a patient-level decision. Chapter 3 leaves the original quality gate's representation, placement and crop-to-package aggregation open. The later 1,000-unit fixture stipulates a fixed candidate and rule but supplies neither executable identity nor numerical threshold. Counts cannot reveal which implementation produced them or establish a validated setting.

Local adjustment would add another controlled choice. The manufacturer would need to define who may adjust the rule, over what range, and which evidence supports those settings. In a future pathology queue, moving a priority cutoff affects which imported cases move; it does not remove preceding import time. In sequencing, a display cutoff can alter suggestion workload while identity reconciliation still determines release. The earlier episodes identify these dependencies without providing a numerical cutoff.

Reviewer Lens. A ranking curve can make the selected rule seem like an incidental setting. The deployed action occurs at a particular setting and unit, so a focused review follows that rule's identity back to the records that selected it. Evidence for one threshold does not establish every local adjustment or package-level combination.

Consultant Lens. The inexpensive early agreement is a threshold-to-action map that product, clinical and software owners all recognize. Its version belongs with the configuration locked before decisive result access. If later observation motivates revision, the map identifies the changed actions and the evidence question to reopen; it avoids trying to recover the old decision from memory.

What would a score of 0.90 mean to the user?

Suppose the display shows 0.90 beside a crop. A number used only to order images makes a different promise from a “90% probability.” Even that second phrase needs an event. **Chosen-class confidence** concerns whichever category the model selected. A **target probability** concerns one specified target event, whether or not that target is the selected class.

The comparison must match that promise. When scores are presented as probabilities, **predictive calibration** asks whether their values correspond to observed frequencies of the specified event under a stated population and reference. It is distinct from metrological calibration of a measurement procedure in Book 1C, Chapter 7. Establishing either kind does not establish the other.

A route table cannot supply the needed score-to-event comparison if the per-image scores are absent. The main 1,000-unit exhibit has exactly that limitation. The separate fifty-observation illustration later supplies binary target probabilities and matching reference events. It is not a hidden subset of the main table. Guo and colleagues' primary classification-methods paper explains empirical reliability diagrams by grouping confidence scores and comparing them with observed correctness. That methodological account helps distinguish the event being assessed; it does not prescribe the chapter's five bins or establish a medical-device probability claim. [S2]

Grouping also creates a question about evidence within each range. A smooth curve can conceal a small or empty group, and a deliberately enriched reference mix can differ from routine event frequency. If a correction is fitted after seeing those outcomes, the corrected score is a new development choice needing its own assessment. These issues matter before a professional is asked to rely on the number.

For a future pathology display, an “urgent probability” would need a defined urgent event and eligible-case sampling. Correct queue ordering would not establish that frequency. A sequencing variant-target probability would need specimen-linked references and an account of how the professional is expected to use it. The earlier import and reconciliation findings supply neither probability meaning.

Reviewer Lens. The first question is what the display invites the professional to believe. A score event, population and reference can then be compared with the actual assessment. A persuasive-looking curve is less useful than the counts behind its groups when sparse observations or absent scores limit what can be said.

Consultant Lens. Clinical and statistical owners can agree on the meaning and intended reliance before a score becomes a user-interface feature. The corresponding per-unit score and reference records must be retained. If the company later fits a correction, treating its independent check as a separate activity avoids presenting the fitted curve as its own confirmation.

What happened when no ordinary answer appeared?

The main table will contain a row called “no evaluable product result.” What would be needed to call every entry in that row deliberate **abstention**? There would have to be a rule declining an ordinary prediction under specified conditions, and records showing that rule produced the state. An absent answer alone does not establish its cause.

An identity check could reject an unmatched package; the supplied quality gate could reject material; the runtime could be unavailable. By contrast, an accepted image taking the other/unclassified route has received an evaluable answer. An unresolved reference belongs to the comparison data, separate from all of those product outputs. The gate's final placement and aggregation remain planning questions, so a real study needs event records before causes can be assigned.

The endpoint then determines how the observed states enter the calculation. For successful target routing among all received reference-target images, a non-result has not fulfilled the counted event. It remains a non-success in that denominator. That statement alone establishes neither clinical error nor the success or failure of fallback, the route for continuing the professional's task.

An answered-only percentage therefore needs its **answered fraction** beside it: how many received images supplied an evaluable output? It also needs the non-result pattern within the relevant reference groups. Otherwise the reader cannot see which selected subset produced the agreement figure. Manual completion would need a subsequent workflow record; it cannot be recovered from the output/reference table.

The corresponding future pathology question starts before the queue. A case waiting for import belongs to the treatment defined by an all-eligible timing endpoint and its start event, even though the queue algorithm has not seen it. In the sequencing review episode, unresolved specimen identity is different from a missing variant suggestion. The later reconciliation action needs its own record. Neither earlier observation supplies counts for these states.

Reviewer Lens. A report may have a convincing explanation for why a model abstains without showing that all observed non-results came from that mechanism. Reconciliation begins with received units, product states and reference states. The cause breakdown is a separate request, which keeps an observed count from acquiring an unsupported causal label.

Consultant Lens. Interface, data and laboratory owners can preserve reason codes and fallback-completion records separately while the route is being designed. The first explains why ordinary assistance stopped; the second shows what happened to the task. Joining them later is possible only if item identity survives both records, and neither should be substituted for the other in the plan.

How many independent units produced the fraction?

A spreadsheet may contain many crop rows from one adult. Counting those rows does not create an equal number of independent adults. The later exercise instead stipulates one index image from each distinct adult and treats those units as independent for its arithmetic. That source relationship determines what a sampling calculation can use.

A **confidence interval** expresses sampling uncertainty under the specified design and model. Here a two-sided 95% **Wilson interval** will accompany a binary proportion. Its finite-sample adjustment changes the center as well as the width; the worked answer will show both operations. The method is an illustration for the stipulated independent count, rather than a general recipe for every diagnostic study.

If images are paired, clustered, repeated or sampled in strata, the analysis needs to respect those relationships. A category-quota sample also needs a reason before its interval can describe the routine population. Simply adding rows cannot supply that reason, repair selection bias or leakage, establish a suitable reference, or restore an omitted product state.

For a future pathology study, several images or timestamps can belong to one slide case. For sequencing, several variants can belong to one specimen. The earlier episodes help identify which case or specimen record would answer the new workflow question. They supply no new precision estimate, and derived rows cannot turn their shared source into independent observations.

Reviewer Lens. The interval's stated independent unit should be recognizable in the source records. A quick comparison between that unit and actual patient, case or specimen links can reveal a mismatch that the polished calculation hides. Precision deserves credit for the defined quantity under the justified design, with other evidence gaps left visible.

Consultant Lens. Statistical and data owners should settle grouping and selection before study size is chosen. Otherwise the company may collect a large number of related rows and discover that its precision plan counted them as unrelated. Repeated records can remain useful, but their role needs an appropriate design and analysis from the outset.

Which result is allowed to decide?

Suppose the planned routing result is weak but another percentage is strong. Whether the second can establish success should already be settled. An **acceptance criterion** gives a prospective threshold and its justification. A **decision rule** states how the conditions combine and what follows from failure or an unavailable calculation.

The claim, eligible population, fixed configuration, metric, uncertainty method, threshold and rationale therefore belong together. If conditions are joined by *and*, each must pass. If they are joined by *or*, any one may establish the specified success. A missing calculation cannot quietly count as a pass. If target routing and safe non-result handling are both required, a favorable overall agreement cannot compensate for absent evidence on one of them.

Allowing a choice among favorable endpoints, subgroups or thresholds creates additional opportunities for a finding. This design issue is **multiplicity**. One primary endpoint with two mandatory conditions is different from three alternative tests where any one positive result wins. The main exercise fixes one primary endpoint and treats its other percentages as descriptive. A separate three-test illustration later explains the alternative-opportunity problem under its own simplified assumptions.

That distinction also protects future workflow studies. A pathology claim about both placement and total review time could need both kinds of evidence. A newly noticed import constraint can motivate another prospective endpoint without rewriting the earlier queue result. In sequencing, correct suggestions and report release could require separate criteria. The recorded display and reconciliation observation supplies no threshold for either.

Reviewer Lens. A review can acknowledge a useful secondary finding and still apply the original mandatory rule. The decisive question is whether the plan allowed that result to establish the claim. This prevents a genuine descriptive success from being stretched into a replacement endpoint while preserving its value for further work.

Consultant Lens. Clinical, statistical and risk owners can make the joint requirements, missing-calculation disposition and selectable analyses explicit before decisive access. If the rule later fails, advice can offer investigation or redesign for a manufacturer decision. Revising the threshold only because the observed result missed it would conceal the question the original study answered.

Where do these decisions meet in the plan?

A threshold written in one file and a denominator specified in another can leave two teams evaluating different questions. The **evaluation matrix** joins the decisions in one prospective record. The first four rows below connect the claim to the counted population and calculation. The next three connect scores and missing answers to product behavior. The final five determine how that evidence can support a decision.

Plan field	Question it must settle
Claim element	Which technical, clinical or workflow assertion is under examination?

Plan field	Question it must settle
Estimand	Which population, outcome and handling of relevant states define the target quantity?
Analysis population	Which received units are included or excluded, with what reference and output states?
Metric	What are the numerator, denominator, unit and relevant disagreements?
Operating point	Which fixed rule and configuration act, and how were they selected?
Calibration	What does the score claim to mean, and how will that meaning be assessed?
Abstention	Which reasons, rates, group patterns and downstream routes need records?
Uncertainty	Which method matches the unit, design and expected precision?
Multiplicity	Which comparisons can decide acceptance, and how is selection controlled?
Acceptance criterion	Which prospective threshold has a justified clinical or technical rationale?
Decision rule	Must all requirements pass, how are failure or missing results handled, and who decides?
Limitations	Which populations, configurations, endpoints or uses of the result remain unsupported?

The analysis-population and abstention rows must agree about the same received image. If the plan retains it as a non-success in an all-received endpoint, the analysis cannot later remove it because its cause is unknown. The calibration row tests a different relationship: a plan with no retained scores must leave that assessment unavailable. “Not estimable” can be an informative entry. Filling the row does not demonstrate performance.

This matrix and the later numerical criteria are author-designed teaching tools. No cited source prescribes the layout. A pathology version would separate queue placement among imported cases from all-eligible slide-to-review time; a sequencing version would separate variant suggestions from specimen/report reconciliation. Those are decisions about what to count, rather than new field results.

Reviewer Lens. Starting at the decision row and following it backward is a useful test of the record. The claimed pass should lead to one population, fixed setting, output accounting and reference. An empty field matters when it interrupts that path; the review can identify the specific inference that cannot be checked instead of treating every blank as the same defect.

Consultant Lens. Each field needs a version and an owner who can settle it: product, clinical, data or statistical as appropriate. The last point before decisive result access is consequential because a later revision changes the evidence history. A shared planning record makes that boundary easier to maintain than separate undated promises.

One thousand received images

The following separate method fixture contains one index candidate image from each of 1,000 distinct fictional adults. Independence is stipulated for the simple arithmetic. It supplies no complete smears, natural category frequency or disease diagnosis, and none of these images comes from an earlier chapter's collection. All counts and criteria are synthetic.

Its question is fixed before the results: among received images whose resolved reference says target, what fraction does the fixed candidate route to target? The reference divides the images into 200 target and 800 comparison units. Every reference is resolved, which establishes an available comparison without making it an infallible clinical standard. The candidate and operating rule are stipulated fixed before evaluation; their executable identities and numerical threshold are not supplied.

The columns in the table identify reference state. The rows identify product output. It is useful for checking where each received image went before selecting a denominator. The target column accounts for 200 images and the comparison column for 800.

Product output	Reference target	Reference comparison
Target route	153	39
Comparison route	17	741
No evaluable product result	30	20
Total received	200	800

The third row is a product-output state: thirty images have a target reference but no evaluable product result, and twenty have a comparison reference in the same product state. Their references are available. The first two rows are evaluable answers. This binary view does not supply LumaCell's complete six-route confusion structure.

The primary endpoint is successful target routing among **all received reference-target images**. A product non-result stays in that population as a non-success for this endpoint. The fictional prospective rule requires **both** a point estimate of at least 85% **and** a two-sided 95% Wilson lower bound of at least 80%. These are teaching inputs, with no clinical rationale supplied to establish them as adequate medical acceptance thresholds.

Agreement among answered images, the answered fraction and non-result rates are secondary descriptions. The table does not identify the causes of the fifty non-results or show successful manual fallback, professional actions or patient outcomes. Its constructed 200/800 mix is not a statement of clinical prevalence.

Which denominator decides the result?

What fraction of all received images received an answer? Among answered images, what are positive agreement, negative agreement and overall agreement? What is the primary all-received target-routing fraction, and what fraction of each reference group received no evaluable result?

The supplied two-sided Wilson intervals are 84.57%–93.66% for 153/170 and 70.16%–81.84% for 153/200. Each belongs to its own proportion; the prospective rule stays fixed.

Decision

Which calculated quantity determines acceptance, and do both requirements pass? What would an overall answered-only headline leave out of the decision? How would the twelve-field record preserve the supported findings and the information still missing?

How a reviewer keeps the received images in the decision

Follow the counts through the rule

The two answered rows contain $153 + 17 + 39 + 741 = 950$ images. The non-result row contains $30 + 20 = 50$. Together they reconcile to the 1,000 received images, so the answered fraction is $950/1,000 = 95\%$. This count concerns the product output; no unresolved reference has entered it.

Within the answered reference-target group, there are 153 target routes and 17 comparison routes. Conditional positive agreement is therefore $153/(153 + 17) = 153/170 = 90\%$. Within the answered reference-comparison group, there are 741 comparison routes and 39 target routes, giving conditional negative agreement

$741/(741 + 39) = 741/780 = 95\%$. Combining the agreeing answers gives $(153 + 741)/950 = 894/950 = 94.11\%$. All three statements describe their named answered populations correctly. The fifty non-results are outside them.

The primary event has the same 153 target routes but a different population. All 200 received reference-target images remain in the denominator, so $153/200 = 76.5\%$. Removing the thirty without results would answer the conditional question instead. Its 84.57%–93.66% interval cannot become the primary interval by having a more favorable lower endpoint.

Why this interval uses 200 units

For this endpoint, each reference-target adult/index-image unit has a successful target route or a non-success, including a product non-result. The calculation treats the 200 units as independent binary trials with one underlying success probability. The one-index-image-per-distinct-adult condition is stipulated here. It would need justification in a real study.

A two-sided 95% interval method has a repeated-sampling meaning under that model: its intervals would cover the fixed underlying proportion in about 95% of repeated samples. It does not give a patient a 95% probability, or assign a 95% probability to the fixed population proportion after this single sample has been observed.

Let x denote successful routes and n the independent reference-target units. The observed fraction is:

$$p_{\blacksquare} = x/n = 153/200 = 0.765.$$

For a two-sided 95% calculation, $z = 1.96$ comes from the central 95% of a bell-shaped normal reference curve, approximately -1.96 to $+1.96$. This use of the normal reference curve does not assume that the individual binary outcomes are normally distributed. Wilson's adjustment first produces a denominator slightly larger than one:

$$D = 1 + z^2/n = 1 + 1.96^2/200 = 1.019208.$$

The adjusted center moves the observed fraction slightly toward one half. Here that puts it below 0.765:

$$C = [p_{\blacksquare} + z^2/(2n)]/D = [0.765 + 1.96^2/(2 \times 200)]/1.019208 \approx 0.760006.$$

The half-width combines observed binary variation with the finite-sample adjustment:

$$H = (z/D) \times \sqrt{[p_{\blacksquare}(1 - p_{\blacksquare})/n + z^2/(4n^2)]} = (1.96/1.019208) \times \sqrt{[(0.765 \times 0.235)/200 + 1.96^2/(4 \times 200^2)]} \approx 0.058421.$$

Subtracting and adding the half-width gives the endpoints:

$$C - H \approx 0.701585; C + H \approx 0.818427.$$

Thus the rounded interval is **70.16%–81.84%**. Its coverage interpretation belongs to the stipulated independent-binomial model. The calculation does not certify reference quality, selection, clinical applicability or the professional's later action.

What the prospective rule decides

The point estimate, 76.5%, is below the required 85%. The Wilson lower bound, 70.16%, is below the required 80%. **Both requirements fail.** The non-result pattern is also informative: $30/200 = 15\%$ in the target-reference group and $20/800 = 2.5\%$ in the comparison-reference group. These different rates warrant attention without identifying a cause. The table does not separate deliberate abstention, quality rejection or other failure mechanisms.

The supported finding is that this candidate did not meet this fictional prespecified image-routing endpoint. The manufacturer needs to investigate causes and consequences and decide the next justified development or evidence activity. The 94.11% answered agreement remains a useful secondary description; leading with it as acceptance would hide the actual decision. These counts establish neither disease sensitivity, successful fallback, clinical harm nor product release.

What the completed record can honestly say

The worked matrix below keeps the stipulated prospective question separate from its observed result. The first four fields preserve the event and exact populations; the rule and score fields identify what is fixed or absent; the final fields apply the original joint decision. It is an analytical record for the exercise, not evidence that a real protocol was signed before access. The table cannot recover executable files, source files or access history that were never supplied.

Plan field	Worked entry
Claim element	Fixed candidate's image-level target routing under the defined reference. No disease, complete-smear or human–AI benefit finding.
Estimand	Fraction of all received reference-target index images routed to target; each product non-result is a non-success for this endpoint.
Analysis population	All 1,000 received adult/index-image units accounted for; all references resolved. Primary population: 200 target units. Answered secondary populations: 170 target and 780 comparison units.
Metric	Primary: successful target routes divided by all received reference-target images. Secondary: positive, negative and overall answered agreement, answered fraction and group-specific non-results. Full output/reference counts retained.

Plan field	Worked entry
Operating point	Candidate and rule stipulated fixed before evaluation. Executable identities and numerical threshold not supplied; counts cannot reconstruct them. A real plan would bind them before evaluation, and a changed rule would define another evaluation.
Calibration	Not estimable for these 1,000 units: individual scores absent. No calibrated probability-display claim supported. The separate fifty-observation illustration uses additional data.
Abstention	Thirty target and twenty comparison non-results retained in received denominators. Causes and downstream fallback outcomes unavailable; all fifty cannot be called deliberate abstentions.
Uncertainty	Two-sided 95% Wilson interval under the independent-binomial model for 200 target units. The interval using 170 answered target units concerns another quantity. Neither captures reference bias or transport to other conditions.
Multiplicity	One primary endpoint at one fixed operating point. Both primary criteria mandatory. Descriptive secondary results, additional candidates, thresholds or subgroups cannot replace it as the winner.
Acceptance criterion	Fictional prospective point estimate $\geq 85\%$ AND two-sided 95% Wilson lower bound $\geq 80\%$. No rationale sufficient for real clinical acceptance supplied.
Decision rule	Both conditions must pass; an unavailable primary calculation would leave acceptance unestablished. Observed 76.5% and 70.16% fail both. Manufacturer investigates and defines justified further work.
Limitations	Constructed mix and selected images; no complete-specimen, successful-fallback, patient-outcome, calibrated-score, site-transport or clinical-sufficiency evidence. Resolved references do not establish reference and sampling adequacy.

The calibration entry illustrates completeness without a favorable answer. “Not estimable” correctly records the absence of individual scores; “pass” would invent evidence. The abstention entry does similar work by retaining a count while leaving its cause open. The matrix supports a definite primary failure and valid secondary descriptions, with different follow-up records for mechanisms, configuration and workflow.

Reviewer Lens. The worked decision can be checked without guessing the missing executable threshold. Both criteria use the same 200-unit endpoint, and both fail on the

supplied arithmetic. The missing configuration identity limits reconstruction, while the missing cause breakdown limits explanation. Treating these as distinct gaps keeps the review from obscuring a decision that the fixture does support.

Consultant Lens. The follow-up has an order: recover the executable candidate/rule identity and analysis history, investigate the reference groups' different non-result patterns, and define a prospective next action. Product, software, data and clinical owners then have specific work to return. Substituting a favorable secondary result would leave every one of those questions unresolved.

What can one populated score group show?

Now consider a **separate** illustration: fifty independent fictional observations, each with a predicted probability of 0.90 for one specified binary target. Thirty references have target state 1 and twenty comparison state 0. No score or reference is missing. These are new synthetic assumptions, not observations recovered from the 1,000-unit routing table.

The score states a target probability. Should it be compared with the target-state fraction, the fraction of all classifications that agree with their references, or the fraction receiving an answer?

Decision

Which event matches the 0.90 promise, and how far can this one group support a calibration statement?

How a reviewer compares the score with its promised event

The score names the binary target, so its matching observation is the target-event fraction: $30/50 = 0.60$. The observed difference is $0.60 - 0.90 = -0.30$, or thirty percentage points below the predicted probability. This is neither an answered fraction nor a calculation of agreement across all chosen classes.

Using the same Wilson formula with $x = 30$, $n = 50$ and $z = 1.96$ gives adjusted center $C \approx 0.592865$ and half-width $H \approx 0.131053$. The illustrative two-sided 95% interval is **46.18%–72.39%** for the group's event fraction under the independent-binomial assumption. It is not an interval for an individual patient's probability and does not incorporate uncertainty in the reference process.

Why four empty groups cannot complete the curve

An author-proposed descriptive plan could fix five score bins before viewing outcomes: $[0, 0.2)$, $[0.2, 0.4)$, $[0.4, 0.6)$, $[0.6, 0.8)$, and $[0.8, 1]$. The square bracket includes a boundary and the round bracket excludes it. Thus 0.2 belongs to the second bin; the final bin includes 1. Each populated bin would retain its count, mean predicted target probability, observed target fraction and suitable interval. Absent scores would have their own accounting.

Here every score falls in the final bin. Its mean is 0.90, with observed event fraction 0.60. The other four bins have no observations. They cannot mean zero error or perfect calibration, and a line through them would not establish performance across the score range.

The event remains important after grouping. This is target probability, not confidence in whichever class was selected; the latter would need a matching correctness event. A constructed or enriched sample also needs an applicability argument before its event frequency is offered as routine-use probability. Selection and reference meaning remain part of that argument.

No prospective calibration acceptance threshold is supplied. The useful finding is the discrepancy and its limits: this populated group supports identifying a mismatch, while other score ranges and populations remain open. Fitting a correction to these fifty observations could be development work. Its fit on those same observations would not independently confirm the correction.

Reviewer Lens. There is a positive analytical conclusion here even without a whole-range claim: the stated 0.90 and observed 0.60 concern the same specified event, with the group's uncertainty visible. Four empty bins identify exactly where the evidence stops. A smooth curve or patient-level probability assertion would hide that stopping point.

Consultant Lens. The next plan starts with the intended use of the number, followed by retained per-unit scores and references, adequately populated groups and appropriate uncertainty. Score, reference and statistical owners can prepare those records. Any fitted adjustment and its subsequent independent check need separate identities in the development history.

When several opportunities can produce a finding

The main fixture's 90% positive agreement, 95% negative agreement and 94.11% overall answered agreement are descriptive. None replaces the primary endpoint. Its point-estimate and lower-bound criteria are two mandatory conditions on that one endpoint, rather than two chances to select a favorable result.

Consider a different, simplified design: three alternative tests, with *any one* positive result allowed to establish a claim. Each starts from a no-effect baseline, a **null hypothesis**. A **false positive** occurs when a test declares a finding although that baseline is true. Suppose all three baselines are true and each test has a 5% false-positive chance. One then has a 95%, or 0.95, chance of avoiding a false alarm.

Only in this toy, assume the tests are independent. The chance that all three avoid a false alarm is $0.95 \times 0.95 \times 0.95 = 0.95^3 = 0.857375$. The remaining chance, that at least one raises a false alarm, is $1 - 0.95^3 = 0.142625$, or **14.26%**. Allowing three alternative opportunities has therefore changed the chance from the 5% assigned to each test.

The relevant question is which successes are selectable. The main fixture's metrics share observations and are not independent tests; its two criteria must both pass. The 14.26% calculation is consequently neither its false-positive estimate nor a correction for its table. A real plan offering several selectable claims, repeated looks or candidate thresholds would need a justified error-control strategy with stated assumptions before inspection.

Reviewer Lens. Counting columns is less informative than following the permitted route to success. “All conditions pass” and “any attractive result passes” are different rules even if their reports contain the same number of percentages. The simplified calculation illuminates that choice; applying its independence assumption to overlapping image metrics would create another error.

Consultant Lens. A statistician can label mandatory, descriptive and exploratory analyses, with their decision order, while the plan is still prospective. If later monitoring in a pathology or sequencing workflow motivates a new endpoint, the record can preserve that useful new question without calling it a newly discovered way to pass the original study.

What can the narrower interval reassure us about?

The conditional result uses 153/170, the target-reference images on which the product answered. The primary result uses 153/200, all received target-reference images under the specified non-result rule. Each Wilson interval accompanies its own quantity. The conditional lower bound cannot satisfy a condition imposed on the all-received endpoint.

Both calculations describe sampling variation under the simplified independent-binomial model. The fixture provides distinct-adult index images and resolved references. It does not provide site selection, proof of reference adequacy, a cause breakdown for product non-results or later workflow outcomes. A real plan would have to justify source grouping, reference meaning and intended-use coverage.

Narrowing an interval cannot recover an absent fallback record or repair a biased source mix.

That boundary should survive into the report's ordinary language. The 76.5% result concerns image-level target routing against this reference. Calling it disease sensitivity would change what was established. A future narrow interval for pathology queue placement would similarly leave preceding import time unmeasured. A precise sequencing suggestion result would leave specimen reconciliation and report release unmeasured. The prior episodes show why those separate records matter; they add no field precision estimates here.

The useful report keeps received flow, full counts, primary decision, answered-only descriptions, model assumptions and non-result pattern together. Even with a failed criterion, this is progress: the manufacturer can identify the question answered and the specific records needed for its next action. Uncertainty has become intelligible rather than merely small.

Reviewer Lens. An interval is easiest to assess beside its numerator, denominator and unit. The executive sentence should refer to that same object. Comparing the methods and headline is a focused way to catch a crop result turning into a patient diagnosis or completed-workflow claim without demanding that every useful analysis be repeated.

Consultant Lens. A clear failed-primary sentence can sit beside separate supported secondary sentences. Missing fallback and site evidence then become assigned work rather than qualifications buried under a favorable headline. This gives the manufacturer usable options while preserving the decision it actually made prospectively.

Figure 9.1 — Set the Decision Rules Before Seeing Results. Claim, estimand, population, metric, operating point, calibration, uncertainty, and acceptance must form a prospective decision chain.

Book 1D • Chapter 9

Set the Decision Rules Before Seeing Results

Claim, estimand, population, metrics, operating point, uncertainty, and acceptance form one protocol.

Key takeaway

Claim, estimand, population, metric, operating point, calibration, uncertainty, and acceptance must form a prospective decision chain.

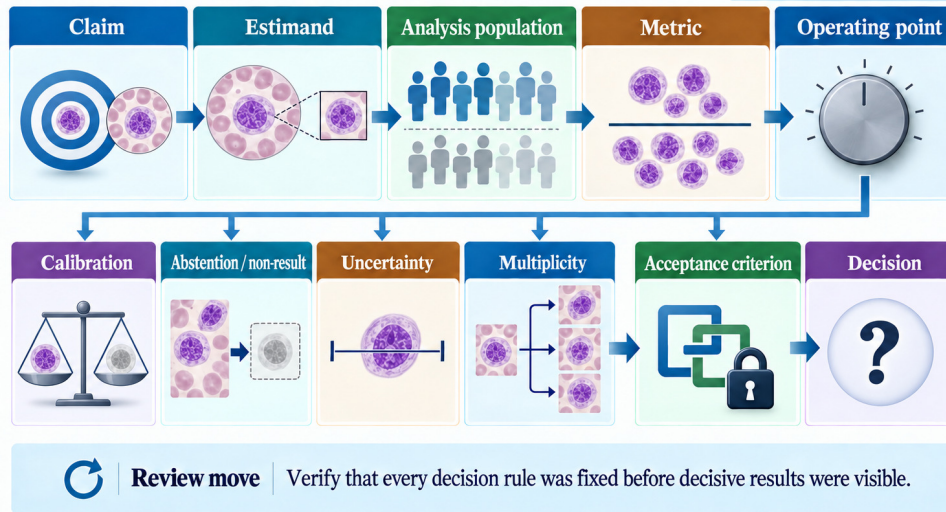


Figure 9.1. Set the Decision Rules Before Seeing Results.

Who can make the next decision?

The failed fictional rule leaves work for several owners. Clinical and statistical colleagues shape the target quantity, methods, criteria and rationale before decisive inspection. The machine learning team supplies the fixed candidate and its development history. Data and reference owners establish received membership, source grouping and reference versions. Software owners bind the executable rule to captured output states. Quality and risk owners connect failures or missing states to the product decision.

Those contributions meet in a manufacturer disposition. Here the prescribed endpoint warrants investigation and a new controlled decision, with conditional agreement and the non-result pattern retained as useful observations. The consultant challenges the plan and interpretation; the reviewer evaluates the submitted claim against its prespecified rule. Neither resets criteria after seeing results or substitutes for the manufacturer's acceptance. The professional still performs full-smear review, and the laboratory retains patient interpretation and release.

What would close the denominator finding?

An adviser could point out the misleading headline and still leave the manufacturer unsure what to change. A **prospective evaluation-plan challenge** makes the proposed decision traceable: it reconstructs counted units, tests the metric and fixed rule, and identifies the denominator, uncertainty and post-result-choice concerns with records that would close them. The same route lets a reviewer assess the eventual report without allowing an impressive plot to rewrite the plan.

Start with the received work

The first join is from entry to analysis. In this fixture, the reference columns reconcile to 200 and 800; the answered and non-result counts reconcile to 950 and 50. All 1,000 references are resolved, while fifty product outputs are not evaluable. A summary that merges these two kinds of missingness has already changed the evidence question.

The count's unit needs the same check. “One thousand images” supports the exercise's independent-unit arithmetic because it explicitly means one index image from each distinct adult. That relationship cannot be inferred from an ordinary image folder. A proposed pathology case-time study would need case-level joins across scanner and queue events; a sequencing release-time study would need specimen-level joins across suggestions and reconciliation. These joins show what the number actually counts.

Follow the rule back to its selection

Once the population is reconstructed, the plan and version history show which endpoint and threshold were fixed before final evaluation and which analyses followed inspection. The fixture's rule is stipulated fixed, but its executable identity and numerical threshold are not supplied. The honest memo retains that absence. It cannot infer the operating point from the observed counts.

Exploratory analyses can still help the manufacturer decide what to do next. Their value depends on preserving their role: a finding used to alter the candidate or rule becomes development information for that choice. A pathology import finding or sequencing reconciliation finding can motivate a new prospective workflow endpoint. Neither changes what the earlier placement or suggestion study measured. The configuration actually intended for use still needs its own applicable assessment; a retrospectively selected point on a ranking curve cannot supply it automatically.

Keep both requirements on the same endpoint

The plan requires a point estimate and lower confidence bound for all-received target routing. The thirty target-reference non-results stay among the 200 received target units for both requirements. A memo stating only “90% target performance” has described answered images instead. The primary value is 76.5%, and this difference determines the specified decision.

If the manufacturer believes its original criterion was inappropriate, it needs a justified and transparent reassessment with consequences for the evidence claim. Lowering it to turn the observed failure into a pass would erase the original decision history. The adviser can explain the options without silently making that substitution.

Locate what the interval leaves unanswered

The uncertainty review now has a definite object. The method must match the unit and design, including related observations, sites, selection, reference uncertainty and missing states. Those questions remain relevant even when the interval is narrow.

Some gaps require another kind of record altogether. Absent fallback outcomes are not a width around image routing; they are an unobserved workflow endpoint, potentially with another unit and producer. Likewise, an image target is not a patient disease endpoint. Executive wording should retain the same boundary as the analysis rather than asking a precise fraction to support a different claim.

Make closure visible in the report and plan

The denominator concern closes when received flow reconciles, metrics have correct labels, the original rule is applied and the affected conclusions change accordingly. Moving the table to an appendix would leave the headline problem intact. Here the primary sentence must lead with 153/200 and the failed joint rule, while 153/170 and 894/950 retain their useful answered-only meanings.

An incomplete acceptance plan may need a prospective protocol before further decisive work. A failed criterion may need investigation and a new manufacturer development decision. These are different closure routes. Advice can identify the required record and accountable owner without guessing an unsupported technical fix from the pattern alone.

A bounded conclusion the manufacturer can act on

Note

The primary all-received target-routing result is 153/200, or 76.5%, with the two-sided Wilson interval 70.16%–81.84%. It fails the fictional prespecified rule. The 90% positive agreement and 94.11% overall agreement concern answered subsets and cannot replace that decision. Non-result rates differ between reference groups; their causes, fallback and patient outcomes are not supplied. The manufacturer should investigate the causes and risk consequences and decide the next controlled action.

This conclusion gives the manufacturer a supported finding and a defined next question. It supplies neither device certification nor proof of clinical harm.

Reviewer Lens. The memo can be checked against a short chain: fixed endpoint, complete 1,000-unit flow, the 200-adult/index-image primary population, Wilson assumptions and the unsupplied executable rule. That is enough to credit the bounded failure and secondary descriptions. It is not a reason to extend either into patient or release conclusions.

Consultant Lens. Closure has a sequence of producers: the plan owner supplies version history, software identifies candidate and rule, data/reference owners supply joins, and clinical/risk colleagues investigate non-result consequences. The revised report then returns to the manufacturer. A new threshold or workflow endpoint belongs in a new prospective evidence plan, preserving the original study's decision.

What is wrong with accepting a narrow interval?

A summary says, “The consultant accepts the study because the confidence interval is narrow.” What would a defensible statement of the consultant's work need to consider?

Decision

Which evidence relationships does precision leave open, and which acceptance decision remains with the manufacturer?

How a reviewer separates precision from sufficiency

A defensible correction is: “The consultant assesses whether the estimand, design, reference, denominator, uncertainty method and prospective rule support the stated conclusion; precision alone does not establish applicability or sufficiency.” The interval answers a sampling question under assumptions. The adviser checks that relationship and identifies what is missing. The manufacturer owns the resulting acceptance and next action.

What changes when the threshold is lowered afterward?

Consider a counterfactual next step. After observing the failed primary endpoint, the team lowers the threshold until more target images receive the desired route. It then offers the revised performance on the same evaluation set as confirmation of the revised candidate.

Decision

What has the evaluation set now contributed to development? Which identities, affected actions and evidence would be needed before the new rule could support its claim?

How a reviewer follows the reused results into development

The observed results selected the revised operating point. They are useful development information for it, but no longer untouched confirmation of that choice. The manufacturer needs versioned old and new rules and their selection history, an assessment of false target routing, other class routes, non-results and review burdens, and appropriately independent evidence for the changed rule and claim.

The rule also remains inside the product's input checks, class mapping, display and professional review. More desired target routes alone cannot establish safe fallback or patient benefit. In a hypothetical future pathology product, lowering the queue cutoff after inspecting a held-out total-time study would similarly make that study development information for the new cutoff. In sequencing, relaxing the display rule could change workload while leaving specimen identity unresolved. These are transfers of the reasoning, not new post-use events.

Reviewer Lens. Two records matter: the claim evaluated at the original point and the observations used to choose the revised one. Keeping them distinct preserves the useful history while preventing reused results from being described as independent confirmation.

Consultant Lens. The manufacturer can retain both rule identities, the affected output and workload comparisons, risk review and a prospectively justified next evaluation route. This makes revision a controlled development decision with an evidence path, rather than an undocumented way of changing the original pass rule.

Where the next evidence question begins

Book 1C develops target quantities, denominators, precision, bias and validation methods. Book 1A Chapters 10–13 add evidence architecture, modalities and statistics. Book 1B Chapter 6 addresses security-performance and adversarial-test metrics, whose questions should remain distinct from clinical metrics. Book 2 Chapters 7–12 cover European Union performance-plan and study interfaces. These optional routes add depth without assigning fictional LumaCell a jurisdictional classification.

Chapter 10 applies the prospective rules to subgroups, shifted inputs and degraded states. A defined aggregate estimate provides a starting point for asking where behavior differs. This fixture's primary estimate fails its criterion, but its transparent accounting still identifies what the manufacturer needs to investigate.

Return to the received image

Can each numerator be followed back to an event and each denominator to a defined population? Which criterion actually determines acceptance? What does the reference permit the metric to be called? Where does a non-result remain visible without acquiring an unsupported cause? What can the interval establish, and which applicability question still needs another record?

Decision

If the opening image receives no evaluable product result, how would its treatment differ between the primary and answered-only calculations? What further record would reveal what happened to the professional's task?

Source Trail

S1: FDA, Statistical Guidance on Reporting Results from Studies Evaluating Diagnostic Tests, March 2007, Sections 3–4.1 and Appendix 7.2, checked 2026-09-24. It is final, nonbinding guidance for binary qualitative diagnostic-test reporting, not the source of this chapter's invented thresholds or a finding that the fictional image reference establishes disease truth.

S2: Guo, Pleiss, Sun and Weinberger, On Calibration of Modern Neural Networks, ICML 2017, Section 2 (PDF pp.2–3), checked 2026-09-24. Its empirical reliability summaries are primary classification-methods research, not medical-device validation or a prescribed five-bin clinical method.

S3: IMDRF N88, Principles 7–8 on human–AI performance and clinically relevant testing, checked 2026-09-24. These are nonbinding principles, not evidence that a classifier endpoint covers a whole workflow.

The evaluation matrix, 1,000-unit table, calibration illustration, Wilson calculations and acceptance rule are original teaching constructions. They do not specify an adequate clinical design or acceptance criterion for a real product.

CHAPTER 10

Bias, Subgroups, Robustness, OOD, and Degraded States

A candidate-cell crop arrives from an adult peripheral-blood smear. Its identity and acquisition fields accompany it; a professional is waiting to check a suggestion and complete the full-smear review. Suppose this crop is harder to interpret than the usual one. Does the product still owe the same response? Now suppose its file is incompatible with the declared input format. Or the crop is valid, but the quality service never replies. Those three situations can all leave work unfinished, through different paths.

For proposed LumaCell Review, the identity-field requirements specify the information accompanying the crop. The proposed acquisition profile names supported acquisition conditions, while the input contract must make the accepted forms and responses explicit. An eligible crop may reach the fixed classifier and receive one of five common white blood cell suggestions or an evaluable other/unclassified route. A trained professional checks it; the laboratory retains patient interpretation and release. The original quality gate still has open choices about the representation it examines, where it runs and how crop decisions combine into a package decision. A challenge plan cannot silently finish that design.

The task is to connect a condition to an observable product state, a possible work consequence and a comparison that informs a decision. The separate 250-adult exhibit later in this chapter lets us practice that connection. It is neither Chapter 5's restricted exploratory subset nor Chapter 9's 1,000-unit fixture, independent product evaluation or release evidence. Its differences will call for investigation without supplying their cause, a fairness conclusion or an observed patient harm.

Which received work does the percentage describe?

Chapter 9 distinguished two questions. Of all eligible inputs received, how many produced a reference-concordant result? Of the inputs that produced an evaluable answer, how many answers agreed? The second quantity, answered-only agreement, omits work that never produced an answer. That omission changes the meaning even when every calculation is correct.

The supported boundary matters before either denominator is counted. A difficult crop can remain inside the promise. A declared unsupported format needs the specified

response to an input the product cannot ordinarily process. Both can matter to safety, but they enter different comparisons.

Two explanations for one observed difference

If one supported group has a lower reported result, what explanation concerns actual product behavior? What different explanation concerns how its cases, references or records were assembled? Which received states would be needed to tell the explanations apart?

Why the difference starts an investigation

The product could genuinely produce more discordant suggestions or more non-results under that condition. The observed comparison could also change with reference quality, case mix, selection, missingness, acquisition or sample size. A difference identifies something to explain. It does not select one causal explanation from those possibilities.

The pathology queue episode makes the denominator issue concrete: already-imported cases reached the on-screen queue earlier, while scanner import still dominated total slide-to-review time. A comparison beginning after import cannot establish the whole timing claim. In the sequencing review episode, correct displayed suggestions did not remove specimen-identity reconciliation before report release. Counting only suggestions that reached the display leaves that release work outside the result. Neither episode supplies a new subgroup experiment here.

Reviewer Lens. A percentage needs an entry population, the output and non-result states, and the reference source beside it. An empty category might mean no such event occurred, or that the analysis excluded those records. That distinction changes the evidence request.

Consultant Lens. Clinical, data and product owners can trace an ordinary supported receipt, a rejected input and an unresolved dependency before choosing comparisons. Each proposed group then has a work question: which promise could change, and which record would expose that change?

What consequence makes a challenge worth doing?

Suppose a supported acquisition condition produces more wrong or unavailable suggestions. A professional might need more verification or fallback work. That conditional path—from input condition, through product behavior, to a possible consequence—is a harm hypothesis. It gives the investigation a reason without claiming that anyone has already been harmed.

What would make the path observable? All received eligible crops need to remain countable. Discordant answers and non-results need separate states because checking a

wrong suggestion and working without one are different tasks. Records of professional review are needed to examine what happened next. Agreement among answered crops alone leaves the fallback question open: whether the intended work can be completed when the ordinary suggestion is unavailable.

A proposed pathology queue study might ask whether a supported scanner-import condition keeps cases outside a priority rule until import completes. A sequencing study might ask whether suggestion availability differs while specimen identity is unresolved. These are future questions about the episodes' actual work boundaries, not additional reported incidents.

The condition also defines the group to compare. A stratum is a group whose membership follows a stated characteristic, such as an acquisition condition. If passing a stratum is necessary to support the claim, its membership and decision criterion must be specified before looking at the result. A group discovered after inspecting outcomes can generate a useful new hypothesis. Its discovery history prevents presenting it as the original confirmatory comparison.

Clinical and risk owners explain why the condition could matter. Data and statistical owners examine representation and precision. Machine learning (ML), software and human-factors owners connect the resulting states to behavior and user response. A list of groups becomes an investigation when those connections are explicit.

Reviewer Lens. Following one proposed group from condition to state to professional action shows whether the comparison can answer its harm question. A post-hoc group with a favorable score cannot inherit the authority of a criterion chosen before the outcomes were known.

Consultant Lens. A short conditional account lets the owners identify the minimum records together. Labeling planned comparisons as confirmatory or exploratory before inspection preserves the difference between testing an existing proposition and discovering the next one.

What does a difference establish, and what needs another argument?

Two observed rates can differ without identifying why. The input may be different, the reference procedure may be less dependable, or the selection process may have assembled different cases. Acquisition, preparation, site and workflow can also be associated with patient characteristics. Calling a group an “acquisition group” does not isolate acquisition as the mechanism.

Three claims therefore need separate support. The measured difference describes the observations. A causal explanation needs evidence linking those observations to a mechanism. A conclusion about unfairness, discrimination or clinical unacceptability

needs the relevant affected people, consequences and evaluative criterion. Evidence sufficient for the first claim may leave the other two unresolved.

A systematic difference in how inputs or groups are represented, predicted or otherwise treated is a technical bias. The International Medical Device Regulators Forum's N67 distinguishes that technical sense from a judgment of unfairness and identifies possible human, data and engineering sources. That distinction helps prevent an observed technical difference from becoming an unsupported judgment about people. The guidance does not decide the fictional product's result. [S1]

The reverse inference is also unsafe: a study can fail to detect a difference because its estimates are too imprecise. To support equivalence, a team would first need a justified equivalence margin—the largest difference small enough for the stated purpose—and a design capable of assessing that margin. A broad uncertainty range that includes both small and important differences cannot establish that the groups are close enough. The coming fixture supplies neither that design nor a fairness study.

In a future pathology review, the evidence connecting an import condition to queue behavior would differ from evidence about patient access. In sequencing, a difference in identity-reconciliation workflow would differ from a finding about variants or patient groups. Actual linked records must bridge each proposed inference.

Reviewer Lens. Separating the observed rate, proposed mechanism and broader conclusion exposes where an argument changes level. The next question is what supports that link, rather than whether the report uses the word “bias.”

Consultant Lens. A requested equivalence or fairness claim needs its population, consequence, criterion and study design before a comparison is selected. Where those are missing, a descriptive result with uncertainty is a useful bounded finding; it does not become an equality-of-safety assurance.

Which groups reveal the promise being tested?

A useful group corresponds to supported use or a plausible failure: patient context, reference category, image characteristic, acquisition system, site, time or workflow condition. The membership rule identifies who or what belongs. The counts identify how many people and inputs are represented. Separate reference and product states show where a comparison is unavailable. Ten related crops from one adult remain evidence from one adult for a patient-level question; Chapter 7's unit of independence follows that relationship.

A broad output name can conceal different reference questions. LumaCell's other/unclassified route is an evaluable output, while the outside-five and unresolved reference states distinguish a supported determination outside the five categories from an image that cannot support a defensible reference assignment. Strong results on one

outside-category type do not establish results on another. Keeping the reference label and its source meaning allows the comparison to remain interpretable after the output categories are combined.

For the pathology queue, grouping by import stage could expose work omitted by counting only cases with queue positions. For sequencing, recording whether specimen identity was already reconciled could clarify a later release-time endpoint. These are prospective measurement choices; neither has a supplied subgroup result.

What if several characteristics intersect? A small group defined by both condition and reference category may matter greatly but yield a very imprecise estimate. The coverage matrix connects those conditions to actual evidence groups. Clinical, data and statistical owners can prioritize intersections by the claim and risk, plan enough evidence where a decisive comparison is needed, and keep sparse cells as explicit limitations. A descriptive table may guide investigation even when it cannot support the required pass/fail claim.

If a subgroup must pass, that requirement also needs a declared relationship to the overall decision. Does every specified subgroup have to satisfy its criterion? The answer must precede the results, because otherwise the favorable overall average can quietly replace the intended promise to each group.

Reviewer Lens. Membership, independent-unit counts and reference distinctions reveal more than the table's number of rows. A broad category or omitted workflow stage may hide the very condition the proposed claim needs to cover.

Consultant Lens. The coverage discussion can give each intersection a purpose: prospective evidence capable of assessing the criterion, exploratory investigation, or an acknowledged limitation. That decision makes the sample requirement answer a claim instead of merely filling empty cells.

Is the crop difficult, unsupported, or unfamiliar?

Consider three inputs arriving at the same interface. The first is difficult to interpret but satisfies the supported profile. Its difficulty remains part of the claimed evaluation. The second has an incompatible pixel layout or a declared unsupported acquisition identifier. A specific contract field fails, so the defined rejection response can be tested. The third passes file and metadata checks but differs from the images on which development relied.

That third relationship is the problem of distribution shift: the conditions supplying inputs differ from development experience. An input described as out of distribution (OOD) is unfamiliar relative to that evidence distribution. It need not violate the input contract. Nor does the description prove that it will fail clinically or that the product can detect its unfamiliarity. Supported scope and familiarity answer different questions.

Each distinction changes the test. Difficult supported crops need the relevant output and non-result comparison. Explicit contract violations need evidence that the specified violation is detected and communicated through the required state. For a novel but valid crop, a proposed detector needs recognizable signals and evidence about both missed conditions and false rejections of valid inputs. Correct metadata alone supplies none of that. The next state also matters: what happens if the signal is missing, positive or wrong?

N67 describes how changes in input quality or format, populations and clinical practice can alter the environment relative to development. [S1] It does not supply a detector for this product. The original quality gate's representation, placement and crop-to-package rule remain open. The plan may specify required observations, but executable checks still need those design choices resolved.

A malformed pathology import record is a contract problem. An unfamiliar yet valid import pattern could instead challenge whether the queue rule works under a new condition. Likewise, a malformed sequencing specimen identifier differs from a valid new variant-review context. Future investigation would preserve which path an item actually took before revising a supported claim.

Reviewer Lens. “OOD handled” is incomplete without the recognition method, its limits and the downstream state. A known field violation can support a precise rejection test; it cannot stand in for every valid unfamiliar input.

Consultant Lens. Interface, gate and clinical owners can specify examples for all three situations, including valid inputs wrongly rejected. Novelty that the proposed mechanism cannot recognize remains a limitation and a monitoring question, rather than disappearing into a generic gate requirement.

What can changing one input property tell us?

Suppose an original crop and a digitally altered copy produce different routes. Keeping the original identifiable lets the team compare the same source under a defined change. That controlled modification is a perturbation. It can expose a state change hidden in an average across unrelated images.

The comparison becomes meaningful only when the change is specified. Which transformation was applied, over what range, and why could that range matter? Does the altered input remain within the supported profile? Does it still depict the target well enough to support the original reference? A digital contrast change does not automatically reproduce a stain or scanner change. If it destroys the relevant information, a changed suggestion has a different meaning from disagreement on a still-interpretable crop.

The parent-to-variant link preserves lineage. Reference blinding keeps the model output from steering the judgment about whether the transformed target remains supportable. A reviewer records an indeterminate transformed reference as such; automatically copying the old label would conceal the uncertainty. Two variants directly derived from each original provide related observations, not new independent adults.

A prospective pathology perturbation might change an import-record field used by the queue rule. A sequencing format change would need its own check that the variant meaning and specimen link survive before comparing suggestions. Neither proposed test has been performed in the episodes.

A finite set of perturbations can reveal a bounded vulnerability or meet specified checks. A new condition found after use may require another justified challenge and evidence decision. Its discovery cannot retrospectively turn an earlier artificial transformation into a realistic simulation of that condition.

Reviewer Lens. The original link, transformed-reference disposition and real-world rationale determine what a changed route means. An unchanged route also needs those qualifications before it can support a robustness claim.

Consultant Lens. Imaging and reference owners can first establish whether the chosen ranges preserve a meaningful target. When exploratory findings shape development, the plan for later independent confirmation must acknowledge that influence instead of treating the same pairs as untouched evidence.

What changes when someone intends to manipulate the result?

An ordinary contrast experiment asks how behavior changes under a specified variation. An adversarial test begins with someone who could act: what can that actor access or change, at which lifecycle stage, with what knowledge and intended result? Those assumptions define whether a proposed manipulation is possible and what the test challenges. A wrong route by itself tells us neither that an attack occurred nor that the cause was benign.

The National Institute of Standards and Technology's adversarial-ML taxonomy organizes objectives, capabilities, knowledge and lifecycle stages. Its role here is to make threat assumptions explicit. It is a general security framework, without supplying clinical robustness criteria or evidence that LumaCell is secure. [S2]

A forged acquisition identifier would require an analysis of who could alter it and how the product uses it. Poisoned training lineage or a malicious service response would require their own access and stage assumptions. Ordinary image contrast variations do not test those threats. For the pathology queue, possible targets might include case

identity or priority metadata; for sequencing, specimen-link or suggestion metadata. These are prospective security scenarios, not reported incidents.

The security owner specifies the threat and test. Clinical and workflow owners connect successful manipulation to what might be exposed, misdirected or delayed. The security result and the clinical performance estimate remain distinct evidence, even when both inform the manufacturer's risk decision. Book 1B supplies optional depth after this local distinction is understood.

Reviewer Lens. “Adversarially robust” needs the actor, access, objective, stage, fixtures and outcomes behind it. Changing any of those assumptions can change which attack the result addresses.

Consultant Lens. A concrete manipulation scenario gives the security owner work that can be scoped. Its possible product states then connect to the clinical or workflow hazard question; an ordinary performance threshold cannot decide an unspecified attack test.

What remains accountable when the ordinary route stops?

A received package may lack its image. The quality response may disappear. Processing may time out, or a completed result may never reach the display. These degraded states interrupt different parts of the ordinary route. None is an evaluable negative result merely because the system has no current positive suggestion.

The first questions concern what the user can see and what the system retains. Is the work item still in the queue? Is its unresolved reason retrievable? Could a previous item's suggestion appear as the current result? Which disposition owner is responsible for resolving the received work? A retry needs to keep the receipt and its history; a new acquisition needs a linked new attempt. Otherwise recovery can erase the failure that made it necessary.

For LumaCell, a proposed blocking requirement can forbid an ordinary usable suggestion while preserving received identity and the unresolved or rejected state. Software verification can check that behavior against the requirement. It cannot show that a trained professional noticed the state, completed full-smear review under workload, or reconciled the laboratory record. Those are human-use observations requiring intended users and relevant conditions. Open quality-gate design choices still have to be resolved before the software checks can be bound to an executable path.

A pathology import interruption would need the case's queue history before a priority promise could be assessed. A sequencing identity interruption would need the specimen and suggestion histories before release could be justified. The episodes do not establish a fallback defect. They show why an interruption would need to be examined at the

work boundary, including accumulated burden when many individually correct non-result messages arrive.

Safe failure therefore names specified behavior and evidence about its consequences. A rejection alone leaves open whether work remains visible and whether the intended response can be completed.

Reviewer Lens. Software state and identity records can establish one part of the response. Observed user actions establish another. A displayed error string cannot supply the missing observation of full review and closure.

Consultant Lens. Software and human-factors owners can share the same interruption scenario while recording different evidence. The state requirement, user task, retry history and unresolved-work owner then remain connected without one test pretending to pass all four.

Could another team execute and interpret the challenge?

A future test team needs more than a list of difficult images. It needs to know why each condition is present, which people and inputs are eligible, which reference can support the comparison, what will be changed and what must be observed. It also needs the decision rule and the boundary of the conclusion. Those linked instructions form a challenge plan.

The eleven fields below follow that chain. The first three connect the possible consequence to the evidence available. The next four distinguish changed inputs, unfamiliarity, missing states and intentional manipulation. The final four connect required behavior and recovery to acceptance and remaining limits. None of the fields declares that a test has run.

Field	Required content
Harm hypotheses	Condition, failure mechanism and potential diagnostic/workflow consequence
Meaningful strata	Patient, reference, input, site and use conditions justified by the claim
Coverage basis	Available units, sampling, reference and known limitations
Shift and perturbation	Defined changes, ranges, realism and target-preservation rationale
OOD conditions	Unfamiliar or shifted inputs versus explicit contract violations; recognition limits and the intended response for each
Missingness	Source, product and reference states with denominator treatment

Field	Required content
Adversarial boundary	Threat assumptions and security-test handoff
Safe-failure requirements	Explicit behavior and evidence needed to justify the response
Fallback and recovery	User/system actions, identity, continuity and re-entry conditions
Acceptance logic	Prospective criteria and their relationship to the overall decision
Residual limitations	What remains untested, sparse, uncertain or outside the claim

Consider a missing quality response across those rows. The hypothesis concerns an unresolved item or an inappropriate usable suggestion. Coverage identifies which eligible receipt is used and what reference information exists. Missingness preserves the unavailable service state. The blocking requirement specifies the display and identity behavior; recovery specifies who completes the work and how retries remain linked. Acceptance says which observation would prevent closure, while limitations retain any unobserved human response. A change in one field can require changes in the others.

“Coverage basis” therefore needs eligibility, sampling and usable references as well as counts. “Acceptance logic” needs a prospective criterion and an authorized decision owner, rather than whichever metric looks most favorable afterward. Another competent team should be able to act on the completed fields and explain the resulting conclusion without guessing either dependency.

Reviewer Lens. Following a single hypothesis across challenge, observation, decision and limit reveals whether the broad word “robust” has an evidential basis. A row with no interpretable outcome is still a planning gap.

Consultant Lens. The gap can be assigned to its clinical, data, statistical, software, human-factors or security producer. Closure advice becomes possible when the dependencies are explicit, rather than when every table cell merely contains text.

One image from each of 250 adults

The next exhibit is a separate author-designed method exercise. It contains one index candidate image from each of 250 distinct fictional adults. The candidate and decision route are treated as fixed for this comparison, although their executable configuration identifiers are not supplied. It supplies no LumaCell release evidence.

Two acquisition-condition strata were specified in advance, and both are within the proposed supported profile. A contains 200 images under an ordinary condition. B contains 50 under a harder but still supported condition. These are acquisition groups; they are not demographic groups or site labels.

The first row below counts evaluated outputs agreeing with the specified image reference. The second counts evaluated outputs disagreeing with it. The third preserves received images without an evaluable product result. All 250 references have a recorded resolution. That positive fact permits the exercise's arithmetic; it does not independently establish the correctness of those resolutions.

Result	Stratum A	Stratum B
Reference-concordant evaluated output	190	35
Reference-discordant evaluated output	6	5
No evaluable product result	4	10
Total received	200	50

The prospective rule for this exercise requires reference-concordant output among **all received** eligible images to be at least 88% overall **and** at least 85% in each specified stratum. A non-result is a non-success for this endpoint: the received image did not produce the required concordant output. That scoring rule does not identify its clinical consequence. The thresholds are invented teaching criteria, not clinical recommendations.

Each column's three states exhaust its received denominator. The joint rule requires the overall comparison and both within-stratum comparisons; one favorable result cannot substitute for another required clause. No fallback outcome, patient harm, protected-characteristic analysis or adversarial cause is supplied. The decision will concern the invented rule under those limits.

Does the candidate meet its stated rule?

The same table supports several calculations, but their denominators answer different questions.

1. What proportion of all received images produced a concordant output overall, in A, and in B?
2. What is the overall agreement conditional on an evaluable answer being available?
3. Does the candidate meet the complete joint rule?
4. What does the observed difference establish, and which explanations or consequences remain unknown?
5. Which next investigation could distinguish those possibilities without assuming a cause?

The supplied illustrative two-sided Wilson intervals are 91.04%–97.26% for 190/200 and 56.25%–80.90% for 35/50. They use Chapter 9's working model: independent adult/index-image units, a binary success state and a common success probability within each stratum. Under that model the intervals describe sampling precision. They do not establish reference correctness, causal explanation or the chance that an individual patient is safe. A different dependence structure or success process would require reconsidering that model.

A decision with its limits

Which denominator belongs to each calculation, which clauses must all pass, and what evidence would still be missing even if the rule were met?

The joint decision and an unexecuted investigation plan

Follow each denominator through the decision

There are $190 + 35 = 225$ concordant outputs. Across all received images, $225/250 = 90\%$. In A, $190/200 = 95\%$. In B, $35/50 = 70\%$. Each denominator retains the received adult/index-image units belonging to the comparison.

The unanswered subset contains $4 + 10 = 14$ images. Excluding them produces the conditional calculation $225/(250 - 14) = 225/236 = 95.34\%$. That number describes agreement when an evaluable answer was produced. It cannot replace the original question about all received work, because the fourteen unavailable results have disappeared from its denominator.

The overall result reaches 88%, and A reaches 85%. B does not reach 85%. Because the rule requires the overall result **and each** specified stratum, the candidate fails the exercise's joint criterion. A passing B result would change that decision if the other clauses remained satisfied; a higher pooled score alone would not. The supplied intervals show that the 50-unit B estimate is less precise than the 200-unit A estimate. They do not add an interval-based acceptance criterion to the stated point-proportion rule.

The observed result is a difference under the specified acquisition conditions. It does not establish a mechanism, patient harm or demographic unfairness. The non-result pattern supplies another observation worth following: B has $10/50 = 20\%$, whereas A has $4/200 = 2\%$. Removing those records from a reported comparison would leave their operational consequences unresolved.

The next investigation can examine actual inputs, reference and product states, quality-gate and transformation behavior, discordant-output patterns and professional fallback. Clinical, ML, software, statistical and risk owners need those records to discriminate possible explanations. Security expertise becomes relevant when the

evidence or a threat model supports that question; the lower B result does not itself imply an attack.

The outcomes of this cohort are now known. Re-examining it can diagnose a pattern or help choose a correction. It cannot provide fresh independent confirmation of choices it helped make.

Reviewer Lens. The decisive reading applies all three clauses to their received denominators, then separates uncertainty from cause. The attractive 95.34% answered-only result is real within its selected subset, but it answers a different question from B's failed requirement.

Consultant Lens. Preserving B's original membership and states keeps the investigation honest. Input, gate, reference and attempt records can then be assigned to clinical, software and statistical owners before deciding which subsequent evidence would need to be independent.

How can the result guide a concrete next investigation?

The author-proposed challenge plan, engineering record CHAL-10-P1, supplies instructions rather than outcomes. Its first three rows preserve the harm question, original membership and evidence base. The middle rows propose different challenges; the final rows state the response, decision and limits. Actual build identifiers, raw images, the input contract and event records must still be bound before execution. The plan has not been executed.

Plan field	Proposed CHAL-10-P1 entry
Harm hypotheses	A difficult supported input could yield a discordant suggestion or a non-result, increasing reliance or fallback demands. An interrupted transaction could expose a stale suggestion or lose an unresolved work item. These are hypotheses; no patient consequence or cause is established by the table.
Meaningful strata	Preserve A and B as prespecified acquisition-condition strata. Retain reference category and actual input descriptors when recovered. Do not replace B with a post-hoc subset that removes its failures, or use A/B as proxies for demographic groups.
Coverage basis	Existing 200 A and 50 B adult/index-image units; all references resolved but method adequacy still requires examination. Use all 250 for the paired technical investigation below. No demographic, rare-category or routine-prevalence coverage is inferred.

Plan field	Proposed CHAL-10-P1 entry
Shift and perturbation	For an exploratory image test, create contrast variants at factors 0.9 and 1.1 around the 8-bit midpoint 127.5, independently from each original. Apply the same formula to each channel, with clipping and rounding as specified below. These are artificial stress conditions; their relationship to real acquisition variation is not yet justified.
OOD conditions	Separate visibly unsupported contract violations from an unfamiliar image that still passes file checks. Exercise malformed input, a declared unsupported acquisition identifier, and an incompatible pixel layout against the frozen input contract. Correct metadata on an unfamiliar image does not guarantee detection of distribution shift.
Missingness	Preserve distinct input-missing, reference-indeterminate, quality-status-unavailable and interrupted-execution states. The original fourteen product non-results retain their received denominators. Do not silently force an unavailable quality status into acceptance.
Adversarial boundary	This suite uses controlled fault injection and benign input variation, not an established attacker model. Refer forged metadata, poisoned training lineage and malicious service responses to the Book 1B threat-led assessment; do not label this suite an adversarial-security pass.
Safe-failure requirements	For each declared blocking condition, no ordinary suggestion may be presented as current and usable. The received work item must remain identifiable, its unresolved/rejected state visible, and the reason retrievable. Passing a message-display check alone does not establish an effective clinical response.
Fallback and recovery	Assign the unresolved work item to the qualified laboratory reviewer; use the retained complete-smear review pathway without relying on the unavailable suggestion. A retry keeps the received attempt identity; a new acquisition creates a linked new attempt. Verify stale-output suppression and reconciliation before returning to an ordinary state.
Acceptance logic	Retain the original overall $\geq 88\%$ AND each-stratum $\geq 85\%$ all-received criterion for its stated evaluation; B remains failed. Exploratory perturbation findings require investigation, not a new clinical pass. Every enumerated blocking-state functional check must meet its expected behavior; any mismatch prevents closure of that requirement. Human-use effectiveness requires separate evidence.

Plan field	Proposed CHAL-10-P1 entry
Residual limitations	No suite execution, perturbation realism review, completed fallback observation or new independent evaluation is supplied. The plan does not establish robustness to every shift, effective recognition of all OOD images, demographic fairness or acceptable residual risk.

Following B through this plan shows why the rows depend on one another. Both discordance and non-result can reduce its all-received output. A changed gate might move an item from discordant output to non-result, or in the opposite direction, without resolving the total work consequence. The plan keeps B's fifty received units, retrieves the relevant input and gate states, and observes software behavior separately from professional fallback. An artificial contrast result cannot create a new clinical tolerance or erase the original failed rule.

The imaging/ML lead specifies the bounded transformations and could execute them only with the actual materials bound and the necessary authorization. The reference lead decides whether each transformed image still supports the target. Software verification records state and identity behavior. Human-factors and clinical owners examine whether intended users complete fallback under relevant workload. The manufacturer integrates these different findings and retains B's unresolved result until there is a justified disposition.

Reviewer Lens. A proposed test earns its place by addressing a plausible mechanism or consequence of the observed failure. Its presence in a completed plan supplies no successful observation.

Consultant Lens. Permissions, raw items, configuration binding and reference review come before interpretable technical execution. Independent follow-up depends on what that work changes. A clean contrast suite alone would leave other explanations of the supported B failure open.

What exactly would the operator do?

For this test only, the proposed image representation is 8-bit red–green–blue (RGB): three color channels, each holding an integer from 0 to 255. This test setting does not resolve the original quality gate's open representation choice. The transformation operates on each original channel value x around the midpoint 127.5.

Subtracting 127.5 gives the value's distance from the midpoint. Multiplying that distance by contrast factor c draws it inward when $c = 0.9$, and pushes it outward when $c = 1.1$. Adding 127.5 restores the intensity scale. Half-up rounding then uses `floor(value + 0.5)`; `floor` means the greatest integer no larger than its input. Finally, the result is limited to the available range 0–255. The complete rule is:

`min(255, max(0, floor(127.5 + c × (x – 127.5) + 0.5)))`.

For an original value of 100, the midpoint distance is -27.5 . Factor 0.9 produces the unrounded value 102.75, saved as 103. Factor 1.1 produces 97.25, saved as 97. Neither needs clipping. If the pre-rounded value falls outside 0–255, the operator records that clipping was needed, then applies the stated rounding and cap. A different representation would need a different specification rather than an unexamined reuse of these constants.

Both variants come directly from the original. Applying one factor to the other's output would test a different operation. Dimensions, parent identity and clipping count remain recorded so another team can reconstruct what happened. The counts are 250 originals + 500 derived image observations = 750 views of the same 250 adult/index-image source units. No new independent patient has been created.

A reference reviewer blinded to model outputs records whether the original target remains supportable in each variant. A transformed reference that becomes indeterminate stays visible in that state. It does not inherit the original label by default. No transformed-reference decisions or product outcomes are supplied here.

The next table separates execution from interpretation. Its first row compares related views. The next four induce particular contract, service or attempt conditions. The last requires observation of people. Each right-hand entry is a decision rule for a future observation, not a claim that the expected behavior has occurred.

Challenge	Execution and required observation	Interpretation rule
Paired contrast variants	Run original and two variants with the same candidate configuration. For each parent, record reference disposition, input/gate state, suggestion or non-result, and any route/state change. Keep A/B membership.	Investigate every new discordance, new non-result and unexplained route/state change. This is a diagnostic trigger, not a clinically justified tolerance or fresh performance acceptance rule.
Unsupported input contract	Present one identified fixture for each declared unsupported acquisition identifier, incompatible pixel layout and malformed file, plus a valid control. Record the exact violated contract field and resulting state.	The applicable blocking requirement must hold without an ordinary usable suggestion. A valid control checks that the test did not merely disable all processing. Passing says nothing about an unfamiliar image that satisfies these checks.

Challenge	Execution and required observation	Interpretation rule
Missing quality response	Suppress the quality-service response for an otherwise valid received item. Record timeout/unavailable status, queue presence and visible output.	Unavailable is unresolved, not accepted. The item must remain accountable; an ordinary suggestion must not be exposed as usable.
Interrupted delivery and retry	Interrupt delivery after processing starts but before the current result is acknowledged, then retry the same received item. Include a preceding item with a different suggestion to test stale-result exposure.	No previous item's suggestion may appear as the current result; retry must not create an unlinked duplicate receipt or erase unresolved history.
New acquisition after a non-result	Create a new acquisition for the same work context and link it to the earlier attempt. Inspect both records and their displayed states.	Preserve the earlier non-result and the new attempt separately. A later answer must not retrospectively convert the first received attempt into success.
Human fallback under interruption	In a simulated-use setting, present a non-result during competing laboratory work; observe assignment, complete review, escalation if review cannot proceed, and closure recording.	Record actual actions and difficulties. This plan supplies no validated response-time threshold or outcome evidence; those must be justified for the intended workflow before a formal human-use acceptance decision.

The valid control in the unsupported-input row has a particular purpose. Rejecting every input would make the bad fixtures appear contained while disabling the ordinary product. A supported control exposes that possibility. Even correct handling of those explicit violations leaves unfamiliar but valid images untested.

The interruption row needs a preceding item with a different suggestion so a stale display can be recognized as belonging to the wrong item. The retry uses the same received item; the new-acquisition row creates another linked attempt. Those different events require different histories. A later successful attempt cannot rewrite the first non-result into success.

A mismatch in one blocking fixture prevents closure of its functional requirement even if hundreds of image comparisons look favorable. Conversely, no mismatch would establish only the specified behavior under the enumerated conditions. The contrast row uses every new discordance, non-result or unexplained state change as an investigation trigger; it does not automatically classify a change as clinical harm. The human-fallback row needs actual assignment, review, escalation where needed and

closure observations. Software logs cannot supply those actions or a justified response-time threshold.

A future pathology queue interruption or sequencing specimen-link failure would require the same distinction between traceable state and completed human reconciliation. Each product would need its own endpoints and records. A single “robustness passed” label would conceal which of these questions had actually been answered.

Reviewer Lens. Parent links, reference dispositions and execution records make each challenge interpretable. The expected-state column must be compared with actual observations row by row before any conclusion is drawn.

Consultant Lens. Operators can bind fixture identities, versions and required records before execution, while the human-factors owner designs the fallback observation and the security owner scopes threat-led work. Each result then has a particular closure question it could answer.

Which response addresses the unresolved condition?

Investigation may support a product improvement, an input-control change, additional justified evidence, a narrower supported profile or a decision not to pursue the claim. Those options change different things. Selecting among them requires knowing which mechanism or consequence remains unresolved.

A narrower profile changes the product promise. Its detection, communication, workflow and applicable regulatory consequences need assessment along with the definition. A condition that failed while supported cannot disappear silently from the record after the result is known.

More ordinary-condition A observations add information about A. They also change the pooled A/B mix and can move the aggregate percentage. They do not directly resolve B's result. Any argument about improved overall precision or relevance to a target population needs a justified sampling and weighting basis. The extra observations should answer the evidence question being claimed.

Similarly, rejecting more B inputs might improve agreement among the remaining answers while increasing fallback work. The received denominator and whole-product consequences expose that trade. A failed criterion therefore leads to an investigation and an accountable choice; it does not automatically prescribe retraining.

The pathology queue episode suggests why the remedy must follow the bottleneck. Import delay could motivate a focused timing investigation or workflow change; better queue ordering alone cannot establish an improvement in total slide-to-review time. In sequencing, correct displayed suggestions alongside slow identity reconciliation could motivate work on specimen links and release. Fitting another classifier would not by

itself answer that question. Each broader benefit would need its own prospective evidence.

Reviewer Lens. A proposed remedy may change the failed state, alter who is counted, or transfer work to the professional. Keeping the original result beside the revised claim and evidence boundary makes those differences visible.

Consultant Lens. The manufacturer can compare design correction, input-control or claim revision, focused evidence, and no advance by the owners, records and independent checks each requires. The favorable aggregate alone selects none of them.

Figure 10.1 — Challenge What Could Change Harm. Robustness and subgroup evaluation must follow harm hypotheses and intended-use conditions, with safe failure and residual limits explicit.

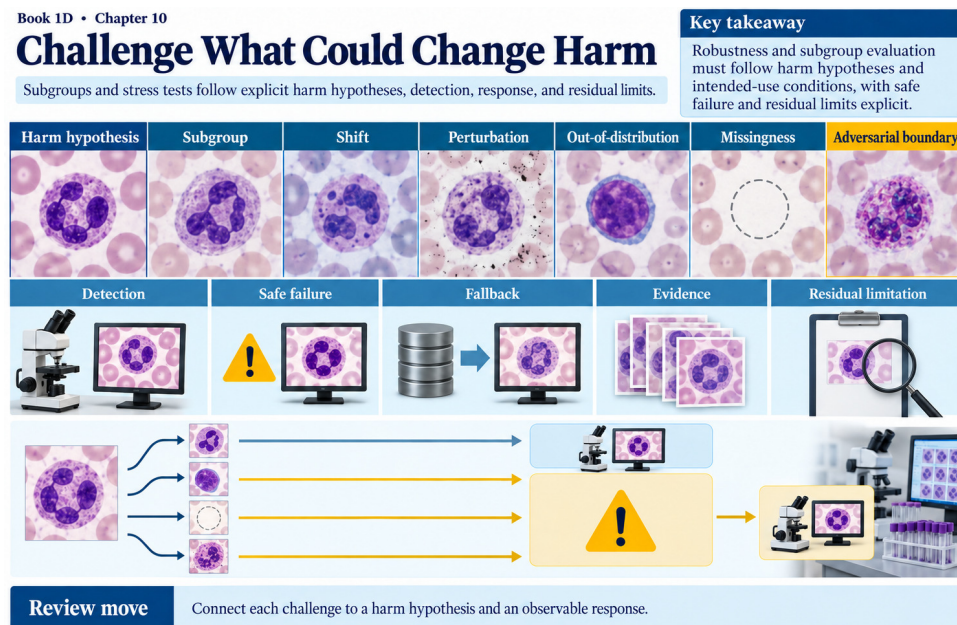


Figure 10.1. Challenge What Could Change Harm.

Who can turn the evidence into a disposition?

The manufacturer defines supported conditions and the evidence required for its decision. Clinical and statistical specialists help choose meaningful strata and interpret uncertainty. ML and software teams examine behavior. The supplier's gate evidence must join manufacturer-controlled interface and workflow records, because a supplier response alone does not establish what the user received. Human-factors and deployment owners assess fallback and recovery; security specialists address relevant threats.

The proposed challenge plan in the worked disclosure makes these contributions explicit while leaving execution and results unsupplied. A supplier test, a contrast comparison and a software blocking-state check can each answer a useful question. Their different objects of observation prevent treating them as interchangeable approvals.

The risk owner coordinates consequences and residual risk, the risk remaining after the proposed controls. The accountable manufacturer decision owner records whether a claim or design may advance under the actual criteria. A consultant can challenge an unsupported conclusion and propose closure work. The adviser does not accept the manufacturer's risk, infer discrimination from a table or authorize clinical use. For the pathology and sequencing episodes, actual transaction histories and the original claims would likewise constrain any revised product decision.

Reviewer Lens. A closure needs a trace from raw group and non-result records through specialist interpretation to the authorized manufacturer decision. An unexplained jump between those stages remains a gap even when each specialist report looks polished.

Consultant Lens. The owner-and-evidence map connects supplier/interface, reference and human-use questions, with security work where relevant. The requested endpoint is a reasoned manufacturer disposition with its evidence, rather than a replacement headline that merely sounds favorable.

How can advice be decisive without overstating the evidence?

An advisory assessment connects the claim-relevant strata, coverage and precision, failure modes, possible unsupported fairness claims, residual-risk questions and closure conditions. Its sequence starts at the original promise, follows the actual evidence and ends with a manufacturer decision to be made. This can identify a material problem without assuming the adviser's authority to accept risk.

Apply the advisory sequence to the two-stratum result

Was the condition supported before its result was known?

The first record is the supported profile used to plan the evaluation. Calling a condition “out of distribution” afterward does not determine whether it was promised before the results. A genuine proposal to revise the boundary needs to be examined as a change. It cannot silently remove a failed condition from the original decision.

Here A and B were both explicitly supported. B's 70% fails the invented 85% all-received requirement, even though the overall result is favorable. That finding is already material under the stated rule. The adviser needs no speculative clinical mechanism to report it accurately.

Which conclusion does this record support?

The observations concern acquisition conditions and image-level outputs. They supply no demographic fairness study or patient outcomes. A broader claim would need those data and an appropriate design. At the same time, a missing or sparse subgroup analysis does not establish that no problem exists: its absence leaves a question about the intended claim unanswered.

“Not established for this condition” identifies that limit when it is what the evidence supports. “Proven unsafe” would require evidence for that stronger conclusion; “no concern” would require evidence that the missing comparison does not leave a material issue. Neither follows merely from an incomplete table.

Where did the unanswered work go?

High answered-only agreement can coexist with a concentration of unavailable results. B's non-results remain received work even when excluded from a conditional metric. Gate-state and actual input records, the discordant-output pattern and relevant fallback observations are therefore part of the evidence request.

The adviser cannot presume that manual review always succeeds or that every non-result causes harm. Both would skip the human response that needs observation. Closure must address evidence for the product's response, including the work left to the professional, rather than stopping at a revised classification score.

What real condition does the artificial challenge represent?

A perturbation report needs the operation, range and rationale, plus the supported-profile and target-preservation judgments. A collection of artificial corruptions may be useful exploratory stress testing. Without a justified relationship to intended conditions, its breadth does not establish comprehensive clinical robustness.

The analogous question for an adversarial test concerns the actor's capability, exposure and intended consequence. Passing one attack configuration supports that bounded result. Failing an unrealistic configuration does not automatically establish a clinically relevant vulnerability. Appropriate specialist assessment connects the assumptions to the product and the possible consequence.

What would resolve the failed supported condition?

The manufacturer needs a disposition for B and its evidence consequences. Depending on the investigation, the next defensible path could be revised design with independent evidence, a justified changed claim, or no advance. More A observations cannot directly close B, and a denominator that hides B non-results would obscure its work consequence.

If the profile is narrowed, the intended-use contract, input controls, user information, fallback, evidence and applicable regulatory analysis must be reconciled. A changed sentence leaves actual exposure unchanged unless the product and workflow can recognize and handle the now-unsupported condition. That practical premise is what makes a scope change an engineering and evidence question.

The same advisory sequence can guide a future pathology review: eligible-case entry and import history belong beside any total-time claim. Earlier on-screen placement remains evidence for its narrower endpoint. For sequencing, specimen-link and reconciliation history belong beside any report-release claim; correct suggestions remain a narrower observation. If a discrepancy is found after use, the original transaction and version are preserved, an owner is assigned, and new evidence is identified for the changed claim. Earlier observations are not rewritten to match the desired conclusion.

What would a bounded conclusion say?

Note

The overall all-received result is 90%. B is 70%, so the exercise's required per-stratum criterion is not met. Agreement of 95.34% among answered images does not replace that decision. Non-results are more frequent in B. The manufacturer needs an investigation and disposition for the failed supported condition. The record does not establish its mechanism, patient harm or demographic fairness.

The conclusion identifies the decisive rule, preserves the additional non-result question and states the limits. It gives the manufacturer a next action while leaving residual-risk acceptance with the accountable owner.

Reviewer Lens. The original supported profile, actual A/B membership, received denominators, uncertainty and proposed closure evidence constrain the advisory conclusion. Both an unsupported fairness judgment and a cosmetic exclusion of B would break that trace.

Consultant Lens. Source, gate, reference and fallback records can be retrieved in an order that discriminates explanations. Named investigation and risk owners can then specify the changed-design or changed-claim decision and the independent evidence needed for closure.

Does failure to detect a difference establish equal safety?

Consider the statement: “No statistically significant subgroup difference means the model is equally safe for everyone.”

Which claim is actually available?

How should the statement distinguish the study's finding from an equivalence or equal-safety claim? Which design and uncertainty information would constrain the correction?

A correction that preserves the evidence level

Statistical significance asks whether the observations meet a specified comparison rule against a no-difference model. Failing to meet that rule can leave important differences unresolved, particularly with limited precision. The study's design, sample sizes, uncertainty and prespecified comparison determine what can be concluded about subgroup differences. Failure to detect a difference does not by itself establish equivalence or equal safety. A study designed to assess a justified equivalence margin would answer a further question; a safety conclusion would also need the relevant consequences and evidence.

What changes if the boundary is redrawn only on paper?

Consider a counterfactual proposal made after inspecting the two-stratum results. The manufacturer removes B from the supported profile, leaves the input checks and user information unchanged, and reports only A as the final evaluation. The case presents this proposal for analysis; it has not established an implemented restriction.

Follow the next B input

What would recognize it, tell the user what changed, and keep the affected work accountable? Which product, evidence and decision controls are still missing from the proposal?

Why a narrower claim still needs operational controls

Narrowing the claim may be a legitimate product choice. Here, unchanged checks and user information leave recognition and handling of B unestablished. The manufacturer must assess actual exposure, product behavior, communication, fallback and the evidence needed for the revised boundary. The original result and the reason for changing the claim remain part of the record.

Chapter 2's principle still applies: exclusions do not remove foreseeable consequences. Chapter 9's denominator principle also applies: a revised analysis population is not the original prospective endpoint. A working control for the revised boundary could change the future product assessment, but would not change what the original comparison showed.

From a challenge plan to an implemented response

Book 1C offers optional depth on bias, denominators, coverage and uncertainty. Book 1A Chapters 7, 12 and 13 add artificial intelligence (AI)/data, pathology and statistics; Book 1B Chapters 3, 4 and 6 add adversarial threats and security testing; Book 2 Chapters 9–11 add European Union (EU) performance and human-use questions. The local method already distinguishes ordinary variation from intentional manipulation. Relevant security assessment remains part of the product work without becoming a prerequisite reading assignment.

Chapter 11 asks whether the specified controls and execution path are actually implemented and verified. A challenge plan can identify what should protect the received work. Until that behavior is implemented and evidenced, the plan alone cannot provide the protection.

Can the conclusion be traced back to the condition?

Why does each stratum matter to the claim? What separates an observed difference from its cause? How does the proposed perturbation relate to a real use condition? Where do non-result consequences remain visible? What would still be uncertain after a finite challenge suite passed its specified checks?

For each question, which item of evidence would support the answer, or which unresolved prerequisite prevents it?

Source Trail

S1: IMDRF N67, 6.1 and 7.1.2, for bounded terminology and sources of variation.

S2: NIST, Adversarial Machine Learning: A Taxonomy and Terminology of Attacks and Mitigations, NIST AI 100-2e2025, published March 24, 2025. General security taxonomy, not clinical robustness or a medical-device compliance method.

The challenge matrix, two-stratum exhibit, Wilson intervals and thresholds are original teaching constructions. The source documents support terminology and framing, not this fixture's outcomes, perturbation realism, acceptance values, fairness finding or security conclusion.

CHAPTER 11

Chapter 11: Verify the ML Pipeline, Software, and System

A candidate cell image has arrived, but the professional has no current suggestion. Did the classifier produce nothing, did a quality decision block the route, or did a result fail to reach the screen? The blank space alone cannot tell us. Following that one transaction is the starting point for verifying the product.

For LumaCell, another workflow selects and supplies the crop with its work-item and acquisition identity. The product must account for the received package. Image preparation and a response from the supplied quality gate are required at their specified interfaces; their relative order and input representation are not assumed here. The fixed classifier runs only when the route permits. Its ordered scores must become the intended category names, attached to the correct current work item. A trained professional then checks the suggestion and performs the full-smear review. Patient interpretation and release belong to the laboratory. Finding every relevant cell or producing an automated differential is outside the proposed role.

Chapter 10's challenge plan connected a condition to required behavior and its observation. Here that connection must reach the implemented build. A saved calculation could repeat perfectly while a changed label map or delayed message changes that transaction. Conversely, faithfully performing the prescribed steps would leave the adequacy of the model and clinical workflow to be established. We will follow the small operations into a connected test, then consider an initial result and a separate correction record. This chapter introduces a new fictional build milestone. It does not promote Chapter 8's exploratory candidate into a device or supply the later crop-evaluation and human-use results.

What would locate the missing suggestion?

The absent suggestion raises three questions before any test count can help:

1. Which parts of the product could change its output while the model weights remain unchanged?
2. Why did Chapter 8 distinguish a reproducible experiment from a deployable device?
3. What evidence would distinguish “no suggestion was produced” from “the suggestion was produced but not delivered”?

Pause — which record would make each answer observable?

Following the result beyond the calculation

Image preparation, the score-to-category map, operating point, quality gate, runtime, package and presentation can all change the result experienced by the user. Reproducing an experiment addresses the operation that was repeated. It supplies no automatic answer about installation, interfaces, failure handling, security controls or clinical performance.

To locate the missing suggestion, receipt must connect to processing, processing to delivery, and delivery to the current display and review state. A processing record with an output but no corresponding delivery points to a different boundary from a completed processing record with no permitted ordinary output. A screen image alone cannot make that distinction.

The pathology queue episode provides a familiar reminder: imported cases moved earlier on screen, while preceding scanner import still dominated total review time. Import, queue calculation and display need separate observations. In the sequencing review episode, suggestions appeared as designed while specimen-identity reconciliation still constrained report release. Computation, display and release therefore cannot share one undifferentiated “completed” label. These are the earlier fictional observations, not new tests in this chapter.

Reviewer Lens. A result is easier to assess once its object is clear. I would first locate the tested product baseline and the records joining receipt, computation, delivery and review. Asking for those links is more useful than treating an absent screen item as proof of a particular component failure.

Consultant Lens. The inexpensive early decision is to agree what each handoff will record and who produces it. A route drawn around one candidate image exposes missing identities and observations before the team has built a large test suite that cannot explain where a transaction stopped.

What would count as a failed requirement?

Suppose the team offers a screenshot of a blocked package as proof that rejected inputs are handled. It shows a screen the product can display. It does not show that a real rejection reaches that screen or clears a previous suggestion. The missing link determines the test.

A usable design requirement specifies a condition that can be created, the response expected, and the boundary where that response must hold. Here a fictional requirement is that rejection of a required crop produces a visible blocked state with no ordinary suggestion for the current package. “Process images accurately” would hide

the distinction between classifying an image and preventing an old result from appearing on a blocked item.

The test must therefore receive the package, induce the gate response, follow the state into the interface, and observe whether prior output remains usable. A manually selected screenshot bypasses that route. A gate unit test checks the component's response; an integrated observation checks whether the product obeys it. If the gate rejects correctly but the interface presents an ordinary suggestion, the component result stands while the integrated requirement fails.

Where should other questions go? Independently calculated pixel values can test a transformation. Actual identities from receipt through display can test a system handoff. Intended users recognizing and acting on a blocked state require human-use evidence. Whether this role is clinically adequate requires clinical evaluation. These observations connect, but each needs the condition that could disprove its own claim.

For example, a proposed pathology worklist rule could keep imported-case identity joined to queue state after a retry. That would not establish total slide-to-review benefit. A proposed sequencing rule could separate a suggestion from an unreconciled specimen's release state. Checking that separation would not establish how often reconciliation succeeds. These are possible requirements to justify, not additional observed defects in either episode.

The same boundary explains the cited standard. International Electrotechnical Commission (IEC) 62304 addresses software lifecycle processes. Its Scope 1.2 excludes validation and final release of the medical device as a whole, including a device consisting entirely of software. Thus even complete software evidence cannot, by its scope alone, answer the whole-device release question. The missing step is evidence and decision-making at that level; calling the device “all software” does not remove it. [S1]

Reviewer Lens. “Passed” becomes meaningful when I can find the induced condition, expected result, actual observation and requirement level. The question that needs narrowing is often the conclusion, rather than the test: a sound component result may simply be offered to support a claim it never examined.

Consultant Lens. Software, clinical and human-factors owners can divide a broad promise into questions before testing starts. That lets the manufacturer plan which observation will support implementation and which later decision still needs people using the product. Discovering that distinction after a suite passes leaves a much harder evidence gap.

Could the expected answer be wrong in the same way?

Consider a tiny image with two deliberately different pixels. The left pixel has red–green–blue values (10, 20, 30) and the right has (40, 50, 60). If an operation is meant to preserve them, the first output should still be (10, 20, 30). If it intentionally exchanges red and blue, the first output should be (30, 20, 10). Because the positions and colors differ, an unintended flip or channel swap has somewhere to show itself. This is a constructed fixture, not a statement of LumaCell's actual preparation settings.

Why start so small? Before classification, software decodes stored bytes into pixel values and may reorient, cut or resize the image. Resizing can calculate new values by interpolation. Color handling, channel order, scaling and normalization may change what reaches the model. These are possible transformations, not a supplied list of operations in the manufacturer path. The reviewer needs to know which specified operations actually execute.

A small fixture lets the expected answer be calculated independently of that execution. Running the product function to produce both “expected” and “actual” values could repeat the same fault twice. A deliberately awkward image size can expose rounding or padding at an edge, provided its expected dimensions and edge values also come from an independent calculation. The useful record joins the input, specification, expected values, observed output and exact tested implementation.

Would passing that comparison show that clinically relevant information survived? It would establish conformity to the specified operation. A readable compressed image could still lose useful detail; altered color or crop boundaries could change the information available for the task. Clinical suitability of those choices and their fit to the data distribution need their own assessment. If the implemented operation changes, its conformity test reopens; if the operation stays the same but proves unsuitable for the clinical task, correct implementation does not solve that problem.

The technique transfers without importing a new incident. Distinct scanner-status values could test how a future pathology implementation produces a queue state, followed separately by a workflow-timing assessment. Distinct variant and specimen fields could test a proposed sequencing parser and identity join, followed by a separate assessment of release work.

Reviewer Lens. The strongest small fixture is one whose answer I can inspect without trusting the implementation being tested. I would then check that the fixture ran through the actual product path. A correct calculation in a convenient demonstration function leaves the installed transformation unexamined.

Consultant Lens. Imaging and software owners can choose asymmetric, boundary and invalid-input fixtures together. The clinical owner has a different question: whether a

correctly executed preparation preserves the information the role needs. Keeping both questions visible avoids asking either group to certify the other's evidence.

Can a repeatable fitting pipeline preserve the wrong relationship?

Suppose two development crops have different reference labels and the image order is shuffled. Does each label follow its image, or stay in its old row? A fitting run could repeat exactly with the same wrong joins. Repeatability would preserve the error.

The route that produces saved model values selects eligible records, joins crops to reference labels, assigns related records to permitted partitions, prepares images, forms batches, updates values and saves a checkpoint linked to the experiment. Each relationship can be challenged on a tiny development-only fixture. Two crops with the same invented patient identifier should expose a forbidden crossing between fitting and reserved evaluation. Distinct labels should follow their images through a shuffle. An excluded record should remain excluded even if a cached copy is available. A saved checkpoint should point back to the actual experiment and selection decision, the connection Chapter 8 required.

These checks use controlled development material so the protected final test set does not become a debugging resource. They verify what the software does with supplied identities and labels. They do not establish that a source patient identifier is true, that the reference label is clinically sound, or that the checkpoint performs adequately. Those source and performance questions remain necessary precisely because correct software can preserve an incorrect input faithfully.

The manufacturer must also identify the role of this software. Software used within a production process may need an applicable quality-system assurance analysis distinct from software incorporated into the device. The relevant question is what the fitting software contributes and how it is controlled; runtime-device evidence should not silently stand in for that separate analysis.

Reviewer Lens. I would ask which controlled record made a forbidden partition, displaced label or mismatched checkpoint visible. That is a narrow software question. Source-identity and reference-quality records then show whether the correctly maintained relationships represent what the study claims.

Consultant Lens. Source checks, pipeline fixtures and saved-artifact binding can be sequenced without spending the final evaluation material. A team tempted to debug on that material needs a small substitute fixture, so an implementation repair does not quietly change the evidentiary role of its evaluation set.

How small a difference can change the action?

Imagine a rule that changes action at 0.5. A reference implementation produces 0.4999; the deployed implementation produces 0.5001. The difference is only 0.0002, yet the action changes. This is an illustrative threshold, not LumaCell's operating point. It shows why comparing numbers and comparing decisions are separate tasks.

First, the team must identify which operations should correspond. During fitting it may deliberately create altered development images, an approach called augmentation. Routine inference on a patient crop need not, and often should not, repeat those training-only changes. The comparison should establish the common deterministic preparation, the intended absence of fitting-only operations, the calculation and the interpretation of its output. "Training and inference match" is too broad until those boundaries are stated.

For the calculation itself, one controlled setting may require exact bitwise equality. Another may justify a numerical tolerance because its implementations perform the calculation differently. The allowable difference and rationale belong before the result; this chapter supplies no actual LumaCell tolerance. That numerical comparison answers how close the values are, not whether the user-facing action is preserved.

The action comparison therefore includes category, ordered scores, abstention or non-result, and displayed state as applicable. Boundary inputs, ties, missing or unexpected numbers and unsupported input shapes can expose differences that ordinary easy images hide. If close values cross the action boundary, a numerical pass cannot substitute for the failed action criterion. If the declared comparison requires both and both hold on justified challenges, it supports that bounded correspondence.

The same reasoning applies to a prospective pathology queue rule: a small score change near its cutoff may change which imported case is first. In sequencing, an unchanged top suggestion need not preserve its confidence or the specimen-linked action. Neither comparison supplies a new outcome for the earlier episodes.

Reviewer Lens. I would locate the prospective numerical criterion and the comparison of consequential actions on the same runtime and configuration. A table of close scores is persuasive only for the question it answers; it is weakest exactly where a small difference can send a work item down another route.

Consultant Lens. Machine-learning, software and clinical owners can agree which intermediate values and visible actions must remain equivalent before selecting fixtures. That prevents an easy-image comparison from becoming the definition of equivalence after the difficult boundaries have been missed.

Who says what the first score means?

A model returns $(0.8, 0.2)$. With an ordered map $[A, B]$, the first value supports category A. With $[B, A]$, the same numbers support category B. Nothing in the unchanged score vector repairs the exchanged names. This two-category example is illustrative; it does not specify LumaCell's actual category order or scores.

For a six-output classifier, all six positions need the same kind of explicit connection to meaning. The package manifest binds score positions to intended names and ties the other required parts to identified content. Learned parameters alone are not that agreement.

The controlled package therefore includes ordered labels, preparation expectations, supported input profile, operating point, supplied gate version, local runtime and dependencies, and interface behavior. A version nickname is a pointer to those contents, not proof that installation loaded compatible ones. The evidence must reach the combinations actually installed and operated.

That suggests a paired test: a correct combination should load and display the correct names, while a missing, mismatched or unsupported combination should be rejected with a recognizable blocked state and retained reason. Otherwise a loader could “solve” bad packages by rejecting every package, including the valid one. The continuation also needs consideration. Quietly substituting another component is not automatically a safer fallback, and stopping alone does not establish that the professional can continue the required work safely.

For a proposed pathology worklist, the package would bind priority labels and the queue rule to the version in use. Earlier on-screen placement provides no certificate for that packaging. A proposed sequencing package similarly needs the variant meaning and specimen-linked rule, beyond a suggestion that happens to display correctly.

Reviewer Lens. I would compare an actual content-bound manifest with valid and invalid loader observations through to the screen. Matching model weights leaves the meaning boundary untested if the package can attach those weights' outputs to the wrong labels.

Consultant Lens. Product and supplier-interface owners can agree compatibility checks and recoverable mismatch handling before installation. It is tempting to keep a permissive loader for convenience, but a convenient load that changes output meaning leaves the manufacturer with a much harder clinical and configuration question.

Is this response still for the current item?

A correctly named category can still be wrong on the current screen if it belongs to a previous package. Follow actual values from receipt and crop source through gate response, execution, delivery, professional review and export. The identity-field requirements say which fields must exist; a test needs the values those fields carried in this transaction.

Three distinctions make delayed responses understandable. One received acquisition package is an attempt. Sending its result again is a delivery retry of the same attempt, not a new clinical input. Starting a new processing/result instance creates an execution generation, allowing an earlier response to be distinguished from the currently active computation. A genuinely new acquisition receives a new attempt linked to the earlier work. The required relationships are specified here; actual identifier formats are not supplied.

Chapter 3's factory and accounting office give a limited analogy. The shipped batch has a shipment record and invoice to match before a ledger posting. Re-sending the invoice does not create a second physical batch. Re-running the match is new processing of the same transaction; a replacement batch is a new linked physical transaction. That explains why identities must distinguish retry, processing and replacement. It does not establish clinical appropriateness, user recognition of a blocked state or complete smear review.

To test the relationship, a response must sometimes arrive late, twice, or after a switch to another item. Interruption after computation but before delivery, reconnect, correction and reacquisition all test different transitions. Observing only the final screen could miss an earlier interval in which an old suggestion appeared current. Receipt, processing, delivery and review records also preserve different denominators: they let the manufacturer reconstruct what arrived, what ran, what became visible and what a professional completed.

The same questions could guide a future investigation of the pathology queue episode: join import, queue attempt and current displayed case without counting a retry as another case. In the sequencing episode, join the current specimen and reconciliation state so an earlier suggestion cannot become the apparent result for a new specimen. These are proposed investigations, not newly observed failures.

Reviewer Lens. A late response after an item switch is especially informative because every component may have produced a locally valid record. The test is whether their actual identities and states still describe one current transaction. A final screenshot cannot recover a transient mismatch that was never recorded.

Consultant Lens. Interface and clinical owners should agree the interruption and correction transitions together with the audit fields needed to reconstruct them. That

makes it possible to choose a fixture that exposes a wrong association rather than merely proving that a message eventually arrives.

What does the installed product do under pressure?

Chapter 8 timed a specified path from a decoded crop through experimental preparation to six scores. A professional waiting for a current usable result experiences more steps. Changing the runtime, dependency, hardware limit or interface endpoint may change that wait even if model parameters stay fixed.

A performance claim therefore needs its supported workload and exact start and end events. If it gives a percentile, the record must identify the measured events, sample and treatment of failed attempts. A fast set of completed calculations can coexist with interrupted deliveries that never enter a reported completion-time sample. The claim changes when the clock covers those additional steps or the handling of failures changes.

Now put the product under its specified image-size, concurrent-workload or storage limit. Does an interrupted current item become visibly unavailable? Does it keep its identity and recover in the defined way, without exposing earlier output as current? Resource testing is also testing state. These observations belong to the installed environment and exact build, not a generic model family.

The earlier pathology observation makes the clock boundary tangible: faster placement of already imported cases left scanner-import time untouched. The sequencing observation likewise separated visible suggestions from the remaining specimen-reconciliation work before release. A changed implementation needs measurements at the affected boundaries before either whole-workflow conclusion changes.

Security controls belong to the same installed configuration. Authentication, access permissions, package integrity, dependencies and recovery affect availability and whether output can be trusted. If a speed comparison disables a control, it compares a changed product configuration. Functional and security evidence must therefore connect, while retaining their distinct questions; neither supplies clinical performance merely because it passes.

Reviewer Lens. I would first align the timing interval, workload, failed-attempt treatment and security settings with the proposed installation. The important judgment is whether a difference can affect the claimed behavior. The model's unchanged name cannot answer that question.

Consultant Lens. Runtime, deployment and security owners can define a small supported-environment matrix and meaningful interruption/recovery fixtures. This

focuses effort on conditions the manufacturer actually proposes to support and leaves distinct clinical and workflow questions with the right evidence owners.

Can one requirement be followed to its decision?

A report says an interface passed, but its screenshots came from one build and its message records from another. The individual observations may be genuine while the claimed connection remains unproved. A verification trace joins the requirement, exact configuration, relevant operations, observed behavior and disposition so the manufacturer can reconstruct the decision.

A correction adds another question: could it disturb behavior that previously passed? Repeating justified earlier checks alongside the targeted correction checks is regression testing. Its scope follows the possible consequences of the change. A large rerun count without that connection can still miss the affected transition.

The following twelve fields are a tool for following one problem across owners. Each asks what its evidence contributes to the same decision; the fields are not twelve independent certificates.

Part of the trace	What the record must make inspectable
Requirement	Which requirement, at which level, with what acceptance condition and risk connection
Image transformations	The actual operations, independent expected values, boundaries and observed results
Fitting pipeline	How selection, partitions, labels, saved checkpoints and records were controlled
Inference comparison	The reference and deployed implementations, with numerical and decision criteria
Package	Identified contents, output meaning, compatibility, integrity and incorrect-package behavior
Interfaces	Actual identity, state, messages, retry, interruption and recovery behavior
Installation	Supported environment and relevant installation or operating conditions
Security	Linked requirements, tested configurations, results and open findings
Timing and resources	Workload, measured events, limits, failures and recovery

Part of the trace	What the record must make inspectable
Repeated checks after change	Why the chosen challenges cover affected and neighboring behavior
Anomaly decision	Observed deviation, impact investigation, correction or justified disposition
Configuration	Exact tested baseline and its relationship to a proposed release

For example, the interface field should lead to the actual delayed message and observed current state on the named build. “See test report” is insufficient when several configurations appear in that report and the observation cannot be located. The configuration and anomaly fields then preserve which build failed, what changed and which tests were repeated. A later pass does not rewrite an earlier build's result, and the trace should make clear which clinical or security questions the software correction did not answer.

A future pathology queue change or sequencing suggestion-interface change would need the same joins between parsing, displayed identity and workflow state. The trace would locate implementation evidence; it would not replace the post-use timing or report-release analysis with a checklist.

Reviewer Lens. I would follow one consequential requirement all the way through rather than start by counting populated fields. A pass borrowed from another build often becomes visible only when the raw observation and the claimed closure must refer to the same configuration.

Consultant Lens. A walkthrough of one changed or failed path tells the company which exact records to retrieve from each owner. Another inventory of document titles may look complete while leaving the manufacturer unable to explain the decision that matters.

The candidate reaches a verification milestone

The fictional episode now introduces a candidate for later chapters. Its fixed evaluation classifier is identified as M1; the original quality gate remains Q0. The identified preparation procedure, P1, is the manufacturer's image-preparation path. The initial presentation, U1, provides the professional display, and the identified runtime, R1, supplies local execution. The proposed acquisition profile, AP1, identifies the candidate's declared supported input conditions. Together these give the engineering configuration M1/Q0/P1/U1/R1 under AP1. They identify roles; actual content and environment identities are still needed to identify a tested build.

This is a separate development milestone from Chapter 8's exploratory candidates. It also retains the externally selected-crop role: LumaCell does not locate every relevant cell in a smear.

What if one required response is missing?

Here, for the first time in the case, the manufacturer specifies how crop-level gate responses combine into a package quality disposition. Start with the required crop identities. A known rejection is already sufficient to prevent ordinary output. Missing evidence from another crop cannot undo that observed rejection, although it still needs investigation.

If there is no known rejection, completeness becomes decisive. Incomplete membership or identity, an empty required set, or a missing or unavailable required response leaves the package unresolved. “Every returned response was accepted” would omit the absent evidence. Only a complete, nonempty identified set with every required crop accepted permits accepted quality. The empty-set rule prevents a package with no required crops from passing merely because none was rejected.

The table is used in order: first look for a known rejection; only if none exists move to completeness; only then can all-accepted evidence close the quality question.

First condition that applies	Package quality disposition	Facts retained alongside it
Any identified required crop has a gate rejection	Rejected; visibly blocked, with no ordinary suggestions	Rejection reasons and any additional missing or unavailable responses
No known rejection, but required membership or identity is incomplete, the set is empty, or a required status is missing or unavailable	Unresolved; no ordinary suggestions	Missing identities or statuses and their accountable investigation
Complete, nonempty identified required set, with every status accepted	Accepted	Processing availability, current-output validity and human review completion remain separate

Thus accepted + rejected + missing gives rejected, while preserving the missing-response fact. Accepted + missing, without a rejection, gives unresolved. These are illustrations of the rule, not additional received case records. Completing a missing response with acceptance would not remove a different crop's rejection; completing all missing evidence with acceptance could change an otherwise unresolved package to accepted, provided the required membership and identity were complete.

Accepted quality does not mean an ordinary suggestion has arrived. Inference must have completed and the output must have valid current identity. A runtime interruption remains a processing problem even if every crop passed quality. Professional review and laboratory release are later actions again. Conversely, a rejected package can have both a rejection to respect and a missing response to investigate. Keeping those facts separate prevents a single quality label from standing in for the whole transaction.

This aggregation does not supply the actual input representation of the gate, its placement relative to the preparation path, or the remaining crop/package processing details. Those implementation questions remain open here.

What was challenged?

The receipt rule applies the earlier identity distinction: the first received acquisition package has an attempt identifier; a delivery retry retains it; reacquisition receives a new linked identifier. Processing generations and arriving responses must be reconciled with the active attempt. Those rules will also matter for monitoring receipts, without supplying a later monitoring result now.

The verification record contains 120 prespecified functional and integration challenges selected to examine requirements. They are not a random sample of clinical use. The first execution reports 118 passes and two failures. The following record identifiers locate particular tests, not patients or crops. The numerical comparison uses a declared tolerance whose value is not supplied.

Test record	Stipulated observation
V17 — ordered category map	A deliberately altered map is accepted at package loading; two displayed category names exchange places
V44 — interrupted display	A rejected package can briefly display the preceding package's suggestion after interrupted delivery
V61 — calculation and decisions	Numerical outputs meet the declared tolerance and every tested decision matches the reference implementation
V85 — duplicate delivery	A retry retains the original attempt identifier
V96 — missing quality status	With no known rejected crop, a missing required status remains unresolved and produces no ordinary suggestions
V112 — resource interruption	The specified unavailable state and recovery record are produced

The calculation row answers a real question, but matching numbers does not establish that they receive the correct category names in the first row. Similarly, the missing-status row establishes a non-result response for that condition; it does not

establish what the display presents after an interruption. Each row needs its own requirement boundary.

For this exercise, correct class mapping and absence of stale suggestions are stipulated release-blocking functional requirements. No evidence of actual patient exposure or harm is supplied.

What does the initial result permit?

Should 118 passes out of 120 support passage of this verification milestone? Which two product boundaries are affected, what evidence would correction require, and which conclusion is outside this test set?

For the interrupted-display finding, what joined account would connect the requirement, observed deviation, affected configuration, risk question, correction evidence, regression scope and accountable owner? The task is to explain the observed failure without inventing a clinical incident.

Pause — what would your milestone decision and interrupted-display trace say?

From two blocking failures to bounded correction evidence

Why the initial milestone is held

The arithmetic is $118 \div 120 = 98.33\%$, to two decimal places. It describes the challenged checks. The milestone nevertheless fails because the two failed requirements were stipulated as blocking. The count is not a rule allowing other passes to compensate for those failures, and the selected challenge set supplies no patient or field failure-rate estimate.

The altered-map finding, V17, crosses the package-to-display meaning boundary. The model may calculate the same numbers while exchanged names change what the professional sees. Correction requires the loader to reject a mismatched ordered map and to display the valid map correctly. Both paths need observation on the actual package.

The interrupted-display finding, V44, crosses the attempt-to-current-display boundary. The crop gate may correctly reject while an ordinary suggestion from earlier work briefly appears usable for the current rejected item. Another gate-only test cannot resolve that observation. The investigation needs the active attempt, execution generation, display clearing, late responses and recovery history. It remains a constructed verification failure; actual patient exposure or harm has not been supplied.

Reviewer Lens. The 118 passes deserve their recorded scope, but they do not answer either blocking requirement. I would request the failing build's raw loading/display and

interrupted-state evidence, keeping the two boundaries separate. Otherwise a correction to one can be mistaken for closure of both.

Consultant Lens. The useful next step is to agree correction and regression questions with software, interface, quality/risk and clinical owners. A milestone recommendation becomes reviewable when the manufacturer can compare the corrected behavior with each original failure on identified builds.

Follow the interrupted transaction through the complete record

The joined verification trace is TRACE-VER-1. Its initial tested build is BUILD-V0; the corrected build is BUILD-V2. Both retain the component-role names M1/Q0/P1/U1/R1, but their executable content differs. The old names therefore cannot make old and corrected observations interchangeable.

The proposed interface correction clears the current display when a new active execution begins. An arriving response becomes usable only when it matches both the active attempt and execution generation. Rejected and unresolved packages cannot expose ordinary suggestions. Separately, the loader compares the ordered category map with the model/package manifest before accepting the package. These are the specified fictional controls, not an inferred implementation of the gate itself.

The following completed trace shows where each field contributes to investigating and closing the initial failure. The package-accounting requirement (REQ-017) provides current-work accountability; the recognition-and-action requirement (REQ-018) provides the non-result recognition connection. For this functional challenge the observable criterion is no stale ordinary suggestion for the blocked current package.

Part of the trace	Entry at the initial interrupted-display disposition
Requirement	Current-work accountability and non-result recognition, joined to no stale ordinary suggestion for a blocked package; no such exposure in the prescribed challenge
Image transformations	The display failure does not measure preparation accuracy. Retain the unchanged preparation-path identity and its separate tests
Fitting pipeline	No retraining is part of this correction. Selection, partition and checkpoint evidence stays linked separately; the display finding supplies no fitting-pipeline pass
Inference comparison	The calculation record meets its numerical and decision criteria on the tested implementation; it cannot establish current-display identity after interruption

Part of the trace	Entry at the initial interrupted-display disposition
Package	The altered ordered map failed on the initial build. Correct manifest comparison and check both valid and invalid combinations through loading and display
Interfaces	Interrupted delivery of a rejected current package briefly exposed the prior suggestion. Correct active-attempt/generation matching, display clearing and ordinary-output suppression
Installation	The recorded local runtime and test environment apply. A runtime or platform change needs an impact assessment; its role name is insufficient content identity
Security	Link integrity, authentication, message and recovery requirements to security evidence. Functional matching does not establish control of malicious messages or all dependency threats
Timing and resources	The resource record shows the specified unavailable state and recovery. Recheck clearing and matching under interruption and relevant resource pressure; no whole-workflow timing benefit follows
Repeated checks after change	Challenge delayed and out-of-order responses, accepted/rejected/unresolved neighboring states, duplicate retries, reacquisitions and valid recovery. Rerun the justified 120-challenge set on the corrected build and retain the coverage rationale
Anomaly decision	Initially hold the milestone for both failures. Software produces correction evidence; verification reviewers assess it; clinical and human-factors owners assess changed presentation; quality/risk reviews implications; the manufacturer owner decides progression
Configuration	Retain the initial build's failures and bind correction results to the corrected build. A later report with the same candidate name cannot change which executable failed

The interface row locates the observed transition. The preparation and fitting rows prevent that transition from being used to claim unrelated passes, while the calculation row preserves its own narrower success. Packaging carries the separate map failure. Installation, security and resources connect the correction to the environment and message conditions in which it must work. The last three rows explain what gets repeated, who decides and how the failed history remains visible. That is one causal record, rather than twelve boxes whose completeness proves success.

A targeted check can now be exact. Begin with a prior attempt's ordinary suggestion on screen. Start a different rejected attempt, interrupt its delivery, and then deliver the

prior response late. Observe identity and visible state throughout interruption and recovery, not just at the end. The expected behavior is no prior suggestion presented as current usable output, with both attempts' dispositions retained. Relevant timing and transition variants follow the documented coverage rationale. This tests the control; it does not measure how often patients would encounter it.

The next fictional development record supplies the changed evidence. Corrected packaging and interface implementations undergo targeted checks and a rerun of all 120 prescribed challenges on BUILD-V2. All 120 meet their prespecified criteria. The manufacturer records VER120 revision 2 and retains the initial V17/V44 deviations and closure evidence. Later use of the component-role names refers to this corrected candidate baseline, not the earlier failed executable and not a released product.

The disposition table separates closure of each identified failure from passage of the complete prescribed suite. Each conclusion depends on the corrected build and stated scope.

Finding	Evidence in the corrected verification record	Supported disposition
Altered category map accepted — V17	Targeted valid/invalid package checks meet loader and display criteria on BUILD-V2	Functional finding closed for the tested build and scope; retain BUILD-V0's failed record
Prior suggestion exposed after interruption — V44	Targeted interruption/recovery checks and the rerun meet the no-stale-suggestion requirement on BUILD-V2	Functional finding closed for that scope; intended-user interpretation still needs Chapter 13's evidence
Complete prescribed challenge set	All 120 meet their criteria, with targeted corrections and coverage rationale linked	Manufacturer records passage of this verification milestone; no clinical validation or release conclusion

The second row changes because the corrected implementation now meets the observable requirement under its targeted conditions. It does not change because no harm was reported or because the other checks passed. The third row adds the stipulated complete-suite result. It adds neither a clinical sample nor a human-use study nor a security threat assessment.

This is a positive result within the fictional exercise: the identified corrected build meets the 120 specified implementation challenges, including the targeted corrections, and the manufacturer records milestone passage. Clinical suitability, human use, security, residual risk and applicable regulatory questions keep their own evidence and decision requirements. The result is neither clinical accuracy nor a field failure rate.

The earlier episodes illustrate the boundary without gaining new incidents. A future corrected pathology queue message could be verified as current for an imported case while the total-time limitation remains. A future corrected sequencing specimen-link display could be verified as identity-consistent while report-release benefit still needs evidence. Neither is assigned a defect or a result from this verification record.

Reviewer Lens. The corrected build earns the exact positive result supplied. I would preserve the failing build's history and check that each closure points to the same corrected content. The hard boundary is between crediting implementation and importing its success into clinical, human-use or security conclusions.

Consultant Lens. A useful closure map shows the two anomaly records, the corrected-build observations that close them, the manufacturer progression owner and the downstream questions still open. It helps the company move forward with the evidence it has without treating that movement as release.

Figure 11.1 — Verify the Deployable System End to End. Notebook performance is not end-to-end verification of the packaged, interfaced, secured and deployable product.

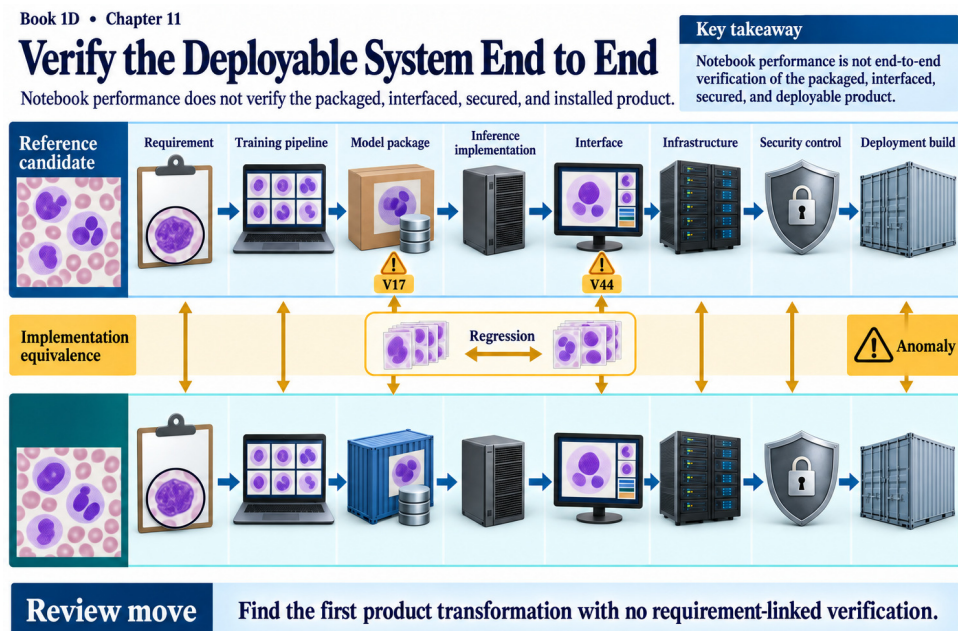


Figure 11.1. Verify the Deployable System End to End.

Who can close each part of the finding?

A supplier can show that its gate rejected a crop, while the local interface still displays an earlier suggestion. The manufacturer's question is how the supplied contribution behaves inside its product. The version integrated, the crop-to-package aggregation, the runtime and the display need connected records. The supplier's component pass cannot verify a display path outside that component.

Engineering produces and reconstructs the software/system observations. The map failure needs model/package and display evidence; the stale-state failure needs interface and recovery evidence. Quality, risk, clinical, security and human-factors functions contribute according to the affected requirements. One function's pass does not silently become another's.

The accountable manufacturer decides whether the investigation and justified repeated checks support progression on the corrected baseline. The revised verification record supports the stated implementation milestone. Intended-user response to a blocked state, clinical suitability and risk acceptability still need their appropriate evidence and authorized decisions. Neither a test count nor a consultant signature transfers that responsibility.

Reviewer Lens. I would follow supplier and manufacturer records into the tested build, separating component response, aggregation and display. This keeps a narrow supplier success useful without allowing it to conceal an integration question the manufacturer owns.

Consultant Lens. Clear ownership makes the next decision practical: who retains the two failures, who produces and reviews correction evidence, who follows security and human-use effects, and who signs progression? Allocating those jobs does not confer release authority on an adviser.

Advice that makes closure reviewable

The consultant may receive raw outputs and manifests, or only a summary saying “all tests passed.” Those are different evidentiary starting points. An end-to-end traceability assessment states which was available and follows a requirement from input condition through observed result, anomaly and correction to the decision on the same baseline. It cannot certify behavior the adviser has neither observed nor reconstructed, and it does not replace the manufacturer's test report.

Its scope first distinguishes the failed initial build from the corrected build and revision-2 record. Availability of interrupted-state records, raw outputs, manifests and anomaly dispositions determines how strongly the assessment can describe closure. From that starting point, six connected questions make the advice specific.

Where does the evidence stop following the transaction?

Follow receipt, crop-quality decisions, permitted or blocked output, displayed identity, interruption and recovery. At every arrow, the question is what observation establishes the connection. Two well-tested components may have an untested handoff between them.

For the interrupted-display finding, “interface testing incomplete” loses the most useful information. The observed condition was that a rejected current package could expose an earlier package's suggestion after interruption. That wording identifies both a reconstructable trigger and its consequence, giving the manufacturer a concrete path to investigate.

Which installation difference could change the result?

Compare the tested and proposed runtime, dependencies, supported hardware, installation options, endpoints, resource limits and settings. The question is which difference could affect the observed behavior. A different environment is not automatically unacceptable: analysis and targeted evidence may justify it. Equally, unchanged model parameters do not justify a changed execution path.

A useful recommendation names that path: establish output interpretation and interruption-state behavior on the intended runtime with the corrected package. If evidence shows those affected behaviors meet their justified criteria, the equivalence argument can advance. “Perform more testing” leaves both the affected behavior and the decision unspecified.

What could the correction disturb?

A stricter category-map check may change loading, backward compatibility, error reporting and exports. Clearing and matching display state may affect valid recovery, continuity of review and audit records. These consequences explain the earlier definition of regression testing: repeat previously passing behavior that the change could disturb, alongside targeted correction checks.

The consultant asks why the selected challenges cover those consequences. An indiscriminate historical rerun is not a universal rule. This particular fictional revision does stipulate all 120 prescribed challenges, and their coverage rationale still matters. If an affected hazardous transition remains unchallenged, a high percentage of executed tests does not answer it.

Does the same correction change a security question?

Package verification, authentication, dependencies and permissions can be affected by a correction. The assessment identifies the security evidence for the same baseline and routes relevant changes and open questions to the appropriate specialist. A functional message match may contribute to the system argument without establishing resistance to malicious messages.

Reviewing documents cannot prove that a security control executed correctly. Assessing a record, witnessing tests within an agreed scope and recommending additional work are different contributions. The advisory conclusion should identify which contribution was actually made.

What is known about the anomaly and its reach?

A material anomaly needs an account of what was observed, how broadly its triggering condition could occur, which product states were affected, and which evidence supports its correction or other disposition. “No complaints received” does not resolve a pre-release implementation failure. “No harm demonstrated” does not establish acceptable risk. At the same time, this case supplies no clinical incident to add to the record.

If a finding remains open, the manufacturer assesses its relevance to progression and eventual release; the consultant can challenge that assessment without assuming the manufacturer's decision authority. For later investigations in the pathology or sequencing episodes, retaining the original transaction and version similarly preserves what happened rather than letting a changed system overwrite the evidence.

What record would change the decision?

For the stale-display finding, a checkable closure condition calls for corrected-baseline evidence covering interruption, delayed delivery, duplicate retries and new acquisitions while preserving identity and state. It identifies who executes the checks, who reviews them and what downstream evidence must be reconsidered. If a display correction changes visible states, human-use evidence may need renewed examination even when the software requirement passes.

A bounded advisory conclusion could therefore read:

Note

The revised verification record supports the specified implementation requirements on the corrected build within the recorded challenge set. The original map and stale-state failures require retained investigation and linked correction evidence. Clinical performance, intended-user interpretation, complete security and release need their own evidence and decisions.

That statement gives the successful software result its due while specifying the work it cannot close. Raw records that support the named build and transitions strengthen the closure claim; a summary without those links warrants a narrower account of what was reviewed.

Reviewer Lens. Across these checks I would compare the conclusion with the exact build, requirement-to-observation links, retained failure history, environment and justified regression scope. Evidence is enough for the stated question when those links can be followed; a polished summary cannot supply a missing transition.

Consultant Lens. Record retrieval should precede a blanket request for new work. Once the missing link is located, the adviser can sequence an environment comparison, targeted observation and owner-specific follow-up, then state a closure condition the manufacturer can actually judge.

What could replace “the system passed testing”?

What two sentences would state the actual verification result and one remaining evidence question, while giving the positive result its proper credit?

Pause — which result and which open question belong together?

A useful positive result with a precise limit

The corrected build met the criteria for all 120 prespecified functional and integration challenges, with the original failures and targeted correction evidence retained in the revised verification record. Clinically representative performance or intended-user recognition and action on a blocked state still needs evidence suited to that separate question.

The first sentence identifies what passed and on which baseline. The second does not undo that success; it prevents a software milestone from being stretched into a claim about a population or human interaction it did not examine.

Transfer: the same parameters on another runtime

After revision 2, a site proposes a runtime using reduced-precision arithmetic. Model parameters, intended use and input profile are unchanged. On a small set of ordinary development images, the highest category matches. No comparison is supplied at operating boundaries or for quality, interface and recovery behavior.

What evidence question determines the next installation decision?

Pause — what has the ordinary-image comparison left untested?

Locating the effect of the changed calculation

The supplied comparison has not established installation equivalence. Reduced precision can change numerical outputs, and a highest-category match on ordinary images does not examine operating boundaries or the quality, display and recovery states. The original corrected-build result remains true for its tested configuration; it does not automatically transfer to the proposed runtime.

The next question is which numerical, decision and state behaviors the changed execution could affect, and what prospective criteria and targeted evidence would establish those behaviors before accepting the installation. Missing evidence is not proof of a defect. If the justified affected comparisons meet their criteria on the proposed runtime, they can support the bounded equivalence decision; shared parameters alone cannot.

From implemented behavior to performance claims

For optional depth, Book 1A Chapters 4–6 address software architecture, controlled development and verification; Book 1B Chapters 4–6 address security architecture and secure verification. Book 1C develops whole-product and evidence-applicability methods, and Book 2 Chapters 5, 6 and 14 address European Union documentation and change consequences. Security follows the same baseline alongside the implementation work; it is not a separate result that can be attached without checking configuration.

Chapter 12 starts with an identifiable candidate and reconstructable implementation evidence. It asks which analytical, technical, clinical and workflow conclusions the available evidence can support for the product role. A passed implementation challenge is a useful input to that question, not its answer.

Can the conclusion follow the record?

Can you distinguish numerical agreement from decision equivalence, a component pass from an integrated requirement pass, and challenge-set success from a clinical performance estimate? Which records preserve identity through interruption? When could a visible-state correction reopen a human-use question?

Each distinction needs an actual record or a specific evidence request. Without it, the trace still relies on an assumption.

Pause — which record, or precisely unresolved evidence request, would support each distinction?

Source Trail

[S1] IEC 62304:2006+AMD1:2015, Edition 1.1, Scope 1.2, clean final text, printed p. 9/PDF p. 197 in the licensed combined edition. The scope distinction is used here; detailed clause-by-clause conformity is not claimed. Official identity: IEC publication record.

The verification architecture, fixtures, artifact fields, fictional requirements, VER120 episode and consultant assessment are author-designed teaching constructions. They are not quotations from a standard or claims that one universal test set is sufficient. Detailed secure-development and AI-testing propositions remain with their specialist sources; no unverified clause-level conformity claim is made here.

CHAPTER 12

Chapter 12: Integrate Analytical, Technical, and Clinical Evidence

A category suggestion appears beside one supplied cell image. How much evidence would justify saying that the product detects abnormal cells or reduces missed diagnoses? The image on screen cannot answer either question alone. Someone selected it before the software saw it, and a professional still has work to do after seeing the suggestion.

LumaCell Review remains a candidate for assisting trained professionals with externally supplied crops from adult peripheral-blood smears. Chapter 11's verification trace, joining requirements, tested configuration and observations, established the corrected candidate's implementation milestone. Its quality route controls whether an ordinary category suggestion is available. The professional checks suggestions and completes an independent review of the full smear; the laboratory retains patient interpretation and release. Here, “independent” means that the required review extends beyond the supplied crops and is not replaced by them. It does not mean that a second member of staff must universally perform it. No device release occurs in this chapter.

The manufacturer now has to join implementation evidence to a new crop evaluation without losing sight of that task. A report can establish something useful while leaving the next link unobserved. Following those links will let us give a positive result its proper credit, identify the further question created by a broader claim, and leave the human-use results of Chapter 13 for that chapter.

What changed between a correct output and a supported claim?

Imagine three statements on the cover of an evidence review: the software works as specified; selected images agree with their references; the evaluation images came from outside development. Each sounds favorable. Which further conclusion would become unsafe if we treated it as equivalent to, respectively, adequate diagnostic performance, detection of all relevant findings in a specimen, or representativeness of the intended deployment?

What is the missing distinction in each pair, and what practical consequence follows from confusing it?

Reasoning through the three distinctions

A product can implement the chosen operations correctly while those operations remain insufficient for its clinical role. **Verification** of the implementation therefore helps answer whether specified requirements were met; it does not settle whether the chosen product and use meet the intended need. That wider **validation** question can remain open after a software pass.

Agreement on selected images leaves anything never selected outside the comparison. If the claim becomes “finds all clinically relevant cells,” the evidence must reach back before image selection. Similarly, an external source may differ from deployment in patients, acquisition, reference methods or workflow. Separation from development protects one relationship; representativeness concerns another.

Recall the **pathology queue episode**: already imported cases moved earlier on screen, while scanner import continued to dominate total slide-to-review time. The local queue result was real, but its start event came after a major part of the claimed task. In the **sequencing review episode**, suggestions displayed as designed while specimen-identity reconciliation still constrained report release. A display observation and a release decision belong to different work. These are the earlier fictional observations, not additional studies.

Reviewer lens. The first unsupported word often appears after a perfectly accurate result. Locate the report's unit and endpoint, then locate the claim's first demand for an unobserved population, action or outcome.

Consultant lens. Before asking for another study, put its proposed question beside the claim with product, clinical and statistical owners. A new study is useful when it closes an identified link; another favorable report on the same narrow unit may leave that link untouched.

Which question does the claim's verb create?

Take the **proposed intended-use contract** from Chapter 2: population, input, output, user, action, setting and permitted reliance. Now change only the verb in its claim. “Suggests a category for a supplied crop” asks about a crop, a category reference, the accepted and no-result routes, and an identified product configuration. Its count begins with the supplied input.

“Detects abnormal cells in a smear” starts earlier. To count a missed cell, the comparison must include relevant cells available before selection, including those never sent to the classifier. A larger collection of already selected crops still cannot show what selection omitted. “Reduces missed diagnoses” reaches later: it asks about patient-linked professional decisions and an appropriate comparison of those decisions.

A crop, a smear finding and a patient decision are different units, even if they appear in the same development program.

That is why a **performance claim** is more than a favorable description of software. Each asserted kind of success requires evidence for its elements and for the links between them. Calling the product assistive does not observe the professional completing the retained review. The display, instructions and actual workflow must make that limitation workable. Conversely, reducing the claim to the easiest measurable output is no solution if it leaves no useful role. The manufacturer needs a useful bounded role, a product or workflow revision, evidence for a broader role, or a decision not to advance.

The earlier episodes make the required records concrete. In the pathology queue episode, prioritizing imported cases and shortening slide-to-review time begin at different timestamps. In the sequencing review episode, displaying a candidate variant and improving report release span different reconciliation actions and actors. An advisory change to either claim, before or after use, needs the relevant transaction history. Replacing “suggests” with “improves” supplies none of it.

Reviewer lens. A verb such as “detects,” “reduces” or “replaces” should lead you to a counted unit and a source-to-action path. If the supplied record stops before that path ends, identify the missing observation precisely.

Consultant lens. The decision is between enforceable scope, changed product or workflow, targeted evidence, and no advance. Show owners which claim element each option changes. Scope narrowing earns its place only when the resulting use and communication actually respect the narrower role.

What can each report contribute to the same claim?

Suppose the corrected software passes its implementation challenges. Why is a crop comparison still needed? Because “it did what we specified” leaves open whether its output agrees with an appropriate reference on the intended inputs. Before either question, there is another: why should that output help the medical task at all? The relationship between the output and a clinical condition or physiological state is the **clinical or scientific association**. A clinically accepted relationship can explain why the task matters, but it does not demonstrate the installed product's behavior.

Follow the crop onward. The gate, classifier, label mapping and display must implement their specified operations. Chapter 11's corrected verification challenge record, VER120 revision 2, supplies bounded evidence at this link. The new 600-crop evaluation later in this chapter compares the candidate's route with references on a defined set. That observes device output. It still does not observe what a professional does with a plausible wrong ordinary suggestion or an unavailable output while

reviewing the full smear. A further assertion that the workflow improves an outcome needs a suitable utility or comparative observation.

The Medical Device Coordination Group's 2020-1 guidance distinguishes association/scientific validity, **technical or analytical performance**—generating the intended output reliably from the input—and **clinical performance** in relation to the intended clinical use. Its vocabulary helps ask which evidence component a report addresses. It is nonbinding guidance; its Medical Devices Regulation and In Vitro Diagnostic Medical Devices Regulation terminology does not assign LumaCell a jurisdictional route. Nor do the five questions here require the same study type or legal path in every setting. [S1]

The joins matter in both directions. A correct image transformation cannot establish that the reference represents the intended clinical task. A literature association cannot show that this installed configuration works. A selected-crop result cannot reveal professional reliance or patient benefit. Equally, a faster workflow would not excuse a consequential wrong route. Integration means keeping the favorable and adverse observations attached to the claim they bear on.

In the pathology queue episode, computation checks, imported-case timing and a total slide-to-review comparison would observe different portions of the path. In the sequencing review episode, suggestion-reference agreement, specimen-link behavior and release decisions would do the same. For an investigator, these distinctions locate the record that could explain a bottleneck. They prevent “the software worked” from ending an investigation into work the software result never measured.

Reviewer lens. At each join, ask what has been observed and what is still assumed. Two reports can be individually credible while leaving their connecting use assumption untested.

Consultant lens. Clinical, engineering, statistics and human-factors owners can map existing reports to the questions they actually answer. Assign the material unfilled links to future work, preserving the value of the evidence already available.

Which population does the percentage describe?

Before choosing the headline, imagine repeating the comparison. Among whom, under what input and product conditions, would which event count as success? Chapter 9 called that target quantity the **estimand**. Its population, unit, selection process, reference, product state, treatment of non-results and summary have to be stated before the observed number can answer a question. A category match on a crop does not become disease sensitivity merely because the crop came from a patient.

Start with the unit. If one adult supplies several related crops, those rows are not several independent people. A professional-performance study can repeat both cases

and readers. The later exhibit selects one index crop per adult, removing that particular duplicate-adult problem. It leaves the selection question intact: which adults and crops entered, and how do they relate to intended use?

Suppose each of six reference categories is assigned the same number of places. An uncommon category then contributes as much to the pooled result as a common one. This **fixed quota design** is useful for ensuring that each route is examined. Its pooled proportion describes the constructed mixture, however, rather than automatically reproducing routine category frequencies. Change the weights assigned to categories with different success rates and the pooled target changes.

Sites raise the same issue. Three sites can each supply equal numbers of adults without having equal category composition. The category totals and site totals alone do not tell us their joint distribution. Without the category-by-site table, we cannot calculate site-specific category behavior or construct justified routine-use weights from these data. Independence from development and independence between adults also do not establish a common success probability across categories or sites. Those distinctions will matter when an interval is attached to the result.

Now trace entry states. Restricting the analysis to quality-accepted crops with resolved references defines a legitimate selected condition. It removes the evidence needed to count all initially received packages that failed the gate, lacked a reference or could not produce an output. If the claim needs routine yield, the protocol must also preserve eligibility, rejected and unresolved routes, reference gaps and deviations in an all-received accounting. Keeping convenient pairs alone cannot recover those missing denominators afterward.

The same choice appears in the recurring episodes. Total slide-to-review time starts before scanner import; queue-only time starts later. Report release counts specimens and reconciliation states; a display comparison can count variant suggestions. The earlier observations show why the start event and unit can change the meaning of flawless arithmetic.

Reviewer lens. Rebuild the denominator from source selection through reference and output states. Then ask whether quota weights, repeated units or excluded paths change the population named in the headline.

Consultant lens. Statistics and data owners should agree on the target quantity and sampling or weighting plan before access to decisive outcomes. If the intended claim includes all received packages, give that question its own data path rather than adding its language to a selected-crop percentage.

What observation would settle the remaining question?

An owner asks for an external study. External to what, and to answer what? A retrospective crop set from sources not used for fitting or product choice can compare a fixed candidate on independent material. It may contain no record of live receipt, interrupted delivery or professional response. Its strength at the first question does not supply observations at the others.

A prospective observational study can capture those paths as they occur. It could show where time is spent or which actions follow an unavailable output. To attribute an improvement to the product, however, the design needs an appropriate comparison. A randomized or paired comparison may help when its population, assignment, tasks, endpoints and analysis fit the claim. The adjectives “external,” “prospective” and “randomized” describe features whose value depends on the unanswered question; they are not a universal ranking of clinical usefulness.

For a **reader study**, the observation includes professionals working with cases. The case mix, reference and reader population determine whose work is represented. Repeated readers and cases connect observations; task order, learning and recall can change performance on a second encounter. The analysis and design must address those relationships. A preference questionnaire instead observes what people report preferring. A simulated critical-task study can expose identity or reliance difficulties under specified conditions. Neither observation by its name establishes patient benefit.

A future prospective import-to-review record in the pathology queue episode might locate delay, while a suitable comparison would be needed to credit its reduction to the aid. Following specimen reconciliation in the sequencing review episode might similarly locate delay without establishing that displayed suggestions improve release decisions. The earlier observations motivate these questions; they do not supply the proposed studies.

The practical aim is the smallest defensible set of methods answering the consequential claims. A randomized outcome trial is not justified merely by its prestige if a different strategy fits the bounded role and applicable requirements. Nor is a pile of favorable reports sufficient because it is small. Optional study-design detail in Book 1C can help choose the methods; the local task remains to attach them to the same identifiable artificial intelligence product and role.

Reviewer lens. Credit the feature that answers the uncertainty: separation from development, observation of prospective states, an appropriate comparison, or direct observation of reliance. Do not credit the adjective for observations it does not contain.

Consultant lens. Sequence device-alone, human-use and comparative work around the claim. Preserve which data remain independent of subsequent fitting or product choices

so that a later development decision does not quietly consume the intended evaluation evidence.

Does a study-conduct standard choose the success rule?

Suppose a protocol cites ISO 20916:2019 and then treats its endpoint, reference and acceptance threshold as settled. The missing step is the source's scope. ISO 20916 addresses good study practice for regulatory clinical-performance studies of in vitro diagnostic devices using human specimens. Its scope excludes analytical-performance studies and does not decide whether a particular device's technical specifications are adequately addressed. [S2] A conduct framework can guide how an applicable study is planned and performed without choosing what LumaCell must demonstrate.

The manufacturer therefore still has to justify the question, method, acceptance logic and applicability. If the study question or legal setting changes, the applicable conduct and permission basis must be assessed again. Specimen acquisition and use, ethical review, privacy and site arrangements have their own applicable requirements. “Retrospective” or “leftover specimen” describes a circumstance; neither phrase supplies a blanket exemption.

This separation explains the relevance of the U.S. Food and Drug Administration's non-recognized listing for ISO 20916. The FDA's stated rationale concerns differences in human-subject protections: it identifies an inconsistency between the standard's treatment of interventional and non-interventional studies and FDA expectations for informed consent and ethics review, including studies using leftover specimens. [S3] Thus a protocol cannot assume that citing the standard settles the applicable U.S. protections. The database entry is a bounded regulatory-status fact, not the controlling legal assessment of this fictional study or a universal prohibition on using the standard's methods.

Clinical and regulatory owners must establish the arrangements for the actual product, study and jurisdiction. Merely involving human material does not justify importing a different, non-IVD framework. The changed fact that matters is the study's actual scope and applicable legal setting, not the presence of a familiar standard number.

Reviewer lens. Read the scope against the kind of study being proposed. Then identify the endpoint, permission and jurisdictional decisions still needing product-specific support.

Consultant lens. Before commissioning work, have qualified owners record its applicable conduct and permission basis. Standards conformance, evidence sufficient for a claim, and legal authorization need distinct reasons even when the same protocol contributes to all three.

Can the old result follow a changed product?

The model weights retain their name, but the image preparation changes. Has the study result followed the product? To answer, reconstruct what actually produced the tested output. LumaCell's candidate includes the evaluation classifier, identified preparation procedure, original quality gate and specified package aggregation, operating point, initial presentation (the professional display), identified runtime and dependencies, under the proposed acquisition profile. A use study also includes the instructions and workflow encountered by the person.

At the engineering record join, these are M1/Q0/P1/U1/R1 under AP1, tied to the corrected verification build BUILD-V2 and VER120 revision 2. These identifiers point to Chapter 11's corrected candidate. They do not make its earlier failed build interchangeable with it or turn the implementation milestone into a release. The **version baseline** is the particular identified version against which the evidence was established, not simply a reusable model label.

Compare that tested path with the proposed path field by field. If the relevant content and conditions match, the evidence may apply directly. If a material difference has a justified relationship to the tested behavior, a **targeted evidence bridge** can support reuse for that affected question. If the required relationship does not hold, the old result is inapplicable to that claim. If the comparison cannot yet be resolved, its disposition remains unresolved. Each branch depends on what is known about the changed output or use link; a documentation discrepancy alone proves neither harmlessness nor changed behavior.

A bridge needs its question and acceptance criteria before its result is interpreted. Average numerical similarity can miss a threshold crossing that changes the displayed route. A bench check that preserves outputs can still omit changed visibility that affects professional reliance. Several individually small changes can interact, so the assessment retains their combined path and chronology. "Same weights" answers only one field in that comparison.

For a hypothetical change in the pathology queue episode, altered import-to-queue logic could preserve a narrower queue-position finding while reopening total-time applicability. In the sequencing review episode, changed variant presentation could preserve underlying analytical evidence while reopening specimen-link or release-use questions. These are examples of prospective change assessment, not new events in those cases.

Reviewer lens. Find the content-bound tested and proposed products, then locate the changed link. The bridge must reach the output or use affected by that link; a convenient average over another behavior is insufficient.

Consultant lens. Engineering, clinical and use owners can separate direct matches, differences and cumulative changes. For each material gap, specify the targeted comparison and the owner authorized to dispose of it. That preserves evidence where justified without declaring every change harmless or every prior result lost.

Can one sentence survive the evidence trace?

Imagine a summary page colored green because several reports passed. The reviewer still needs to know which sentence those reports support. Start with the proposed crop-suggestion claim, identify the candidate that produced its outputs, follow the observed units into the comparison, and then ask where the result can travel. Those linked questions form a **claim-to-evidence and configuration matrix**.

The following table names the eleven links needed to trace that sentence. Use it to locate inspectable records and decisions; it does not replace the underlying study report.

Field	What the reader needs to recover
Claim elements	The exact population, input, output, action, user and claimed benefit
Configuration and status	The evaluated and proposed products; whether each is a candidate, accepted baseline or manufacturer release; market authorization recorded separately
Evidence questions	What must be established for this particular claim element
Study or test	The identified method and controlled report that address the question
Population and sites	How units were sampled, relevant groups and setting, and separation from development
Reference	How comparison judgments were established, reviewed or adjudicated, including uncertainty and unresolved states
Results and uncertainty	Numerators, denominators, endpoints, intervals and their limitations
Configuration match	Direct match, justified bridge, mismatch or unresolved comparison
Transport limits	Conditions to which the observed result cannot yet be extended
Residual gaps	Material unanswered questions and why they affect the decision

Field	What the reader needs to recover
Supported conclusion	The useful, accurately bounded statement justified by the preceding links

Consider the configuration/status row first. A passed study on a candidate does not itself change that candidate into a manufacturer release. An accepted internal baseline and a release are also different statuses, and market authorization is a separate decision. Next, population, reference and result describe what was actually observed. The transport row then asks whether the proposed sentence reaches beyond those observations. The residual-gap row identifies which unobserved links still affect a decision. Only after that trace can the last row state a supported conclusion.

Thus the table does more than collect eleven completed cells. If a proposed sentence moves from selected crops to all received specimens, the transport row exposes the move even when the result row is favorable. If a justified new comparison closes that gap, the supported conclusion can change. It should express the useful result actually supported, rather than automatically choose the most restrictive wording. Text must carry these reasons even in monochrome or an accessible rendering; a traffic-light color alone cannot explain the decision.

Reviewer lens. Read through every field between claim and conclusion. An attractive numerator attached to the wrong population or configuration is a broken link, not a minor formatting problem.

Consultant lens. Ask each owner for the record pointer and decision their field needs. Draft the supported sentence only after the whole trace can be followed without borrowing an unanswered question.

What exactly entered the 600-crop evaluation?

The manufacturer now evaluates the corrected candidate from Chapter 11. The **600-crop evaluation**, recorded as EVAL600, follows the corrected verification build and VER120 revision 2. Its exact candidate identity is M1/Q0/P1/U1/R1 under AP1. This is a new device-alone evaluation, separate from Chapter 9's 1,000-unit method fixture and from the earlier development collections. Its status remains candidate.

The proposed role is category assistance for externally supplied candidate images from adult peripheral-blood morphology review. The routes are five common white-cell categories and the evaluable **other/unclassified route**, a supported output outside those common categories. The product does not locate every cell, produce a complete differential count, diagnose disease or declare abnormal cells absent. The laboratory still completes the independent whole-smear review described at the opening. Whether

intended professionals understand and carry out that review with this presentation is a separate question about actual use.

Selection takes one prespecified index crop from each of 600 distinct fictional adults. New evaluation sites C, D and E supply 200 adults each; no adult or site contributed to development. Selection also fixes 100 reference examples in each of the six categories. These category quotas ensure every route is examined, while deliberately departing from natural prevalence. Only the separate category and site totals are supplied. There is no category-by-site table, so equal category composition within each site and site-specific category rates cannot be inferred.

All 600 selected crops meet the **proposed acquisition profile** and **original quality gate** acceptance conditions. Every **reference** is resolved. The assessors established it without seeing the model suggestion: **reference blinding** prevents that suggestion from guiding its own comparison, although the reference can still be wrong. There are no product non-results within this selected set. The record supplies no initially received denominator from which to calculate gate rejection, missing-reference or other unavailability rates in routine use.

Read the next table as a comparison of routes on selected adult/crop units. The middle column counts the index crops assigned to that reference category; the right column counts candidate outputs agreeing with that reference. It counts neither correctly diagnosed patients nor every abnormal cell in their smears.

Reference category or route	Selected index crops	Candidate routes agreeing with the reference
Neutrophil	100	98
Lymphocyte	100	98
Monocyte	100	97
Eosinophil	100	98
Basophil	100	96
Other/unclassified	100	97

Follow the last row into its fuller reference record. “Other” does not describe one biological kind. Of its 100 examples, 30 are in the immature white-cell reference group, 30 in the abnormal white-cell group, 20 in the nucleated-red-cell group and 20 in the artifact/non-target group. The candidate routes 29, 29, 19 and 20, respectively, to other/unclassified. The remaining three receive ordinary suggestions: one lymphocyte suggestion in the immature group, one neutrophil suggestion in the abnormal group and one monocyte suggestion in the nucleated-red-cell group. These are stipulated

reference routing groups, not a complete taxonomy or a disease-specific evaluation. Their different error types will matter when the remaining use question is identified.

The prospective teaching rule requires at least 95% reference-concordant routing overall **and** at least 95% in every one of the six categories. Both conditions must hold. The overall denominator is 600; each category denominator is 100. A category below the threshold would fail the milestone even if the pooled result passed. These are fictional point-estimate criteria, not recommended clinical thresholds; uncertainty and error consequences still require interpretation.

The corrected implementation verification record is available alongside this evaluation. No comparative reader-performance, complete-smear detection, time-saving or patient-outcome study is supplied. Those absences identify questions the crop table cannot answer, rather than changing what its observed routes mean.

Reviewer lens. Before judging a percentage, recover the adult/crop unit, site and category allocation, reference masking, output states and exact corrected candidate. The initially received population and the professionals' responses are not additional columns hidden in this table.

Consultant lens. Data, clinical and statistics owners should preserve the allocation and individual reference/route records. Assign routine quality yield and professional reliance their own future questions, so that the crop numerator is not made to stand for either.

What decision follows from those observations?

The overall result has to be calculated and both parts of the prospective rule applied. Beyond that calculation, three conclusions need to be kept distinct: the result supported by this evaluation, the broader claims it cannot support, and the next evidence question raised by the three other-route references receiving ordinary suggestions.

Now consider the proposed wording: “LumaCell detects abnormal blood cells with 97.3% accuracy and reduces missed diagnoses.” Where does it first leave the observed unit, and which additional unsupported steps follow?

Does the milestone pass, what sentence is justified, and what missing observation prevents the proposed broader wording?

Working through the count, uncertainty and claim

What passed, and under which interpretation?

Adding the six concordant counts gives $98 + 98 + 97 + 98 + 96 + 97 = 584$. The overall proportion is therefore $584/600 = 97.33\%$. Since each category contains 100 crops, its concordant count also gives its percentage. The lowest is the basophil result, $96/100 = 96\%$. Overall and every category meet the invented 95% point requirement, so the corrected candidate passes this device-alone selected-crop milestone. If even one category had fallen below 95%, the overall percentage could not rescue the joint rule.

Can the result now carry an interval? Applying Chapter 9's **Wilson interval** arithmetically to $584/600$ gives approximately 95.71%–98.35% for a two-sided 95% interval under a binomial working model. The model treats each binary success as an independent draw with one common underlying success probability. Under that model, intervals constructed across repeated samples would cover that fixed probability about 95% of the time. This is not a 95% probability that a particular crop is correct.

The common-probability premise is doing work. Six hundred distinct adults remove repeated-adult counting; they do not establish that neutrophil, basophil and other-route crops, or crops from different sites, have the same chance of agreement. Moreover, the six category quotas and three site quotas fix a mixture rather than reproduce routine frequencies. The quoted Wilson numbers are consequently **working-model arithmetic, not established design-valid coverage for this quota sample or routine use**. Passing the point rule does not establish that extra interpretation of the bounds.

This does not make every uncertainty analysis impossible. A real protocol would define the intended category/site target quantity and sampling basis, then justify a stratified or other design-appropriate method. Here, the needed joint category-by-site information and routine weights are not supplied. We cannot invent them in order to produce a population interval.

For the other/unclassified row, $97/100$ gives a corresponding Wilson working-model interval of approximately 91.55%–98.97%, again with the extra common-probability assumption. Its four internal groups contain only 30, 30, 20 and 20 examples and may have different success probabilities. That pooled arithmetic neither establishes disease sensitivity nor removes the three references that received ordinary suggestions.

A supported result sentence keeps the observation visible: “In the fictional 600-crop evaluation of selected, quality-accepted index crops, the corrected candidate's routing agreed with the specified reference for 584 of 600 crops across six equally represented categories.” Its accompanying record retains the category counts, exact configuration, selection limits and working-model caveat. Resolved references assigned without model suggestions strengthen the comparison process but do not establish infallibility.

The proposed marketing sentence changes several things at once. The denominator contains selected crops, not every abnormal cell available in a smear. The endpoint is six-category routing agreement, not disease detection. The pooled 97.33% cannot be assigned to the 30-example abnormal-white-cell reference group, much less to a disease endpoint. Finally, no comparison of missed diagnoses was observed. Each change asks for a different observation, so changing a label on the same percentage cannot support it.

The three wrong ordinary suggestions still matter even though they do not overturn this stipulated point milestone. An ordinary label may look plausible to a professional reviewing the smear. Will the professional still complete the retained review, recognize the discrepancy and act appropriately? What happens when an output is unavailable? Chapter 13 needs to examine those **reliance** and **fallback** paths. This chapter contains no user outcome with which to settle them.

Reviewer lens. Record the positive point-estimate decision accurately. Keep the Wilson model separate from what the quota design establishes, and keep crop routing separate from disease detection. Those are different judgments about the same exhibit.

Consultant lens. Preserve the three distinct error types for a prospectively justified use evaluation. If the manufacturer wants a broader population estimate, have a statistician define and justify the sampling and analysis that target it; a more confident description of the current pooled number will not supply that evidence.

Can the completed record preserve both progress and open questions?

The manufacturer's clinical/evidence owner now records the result, with statistical and configuration review, in the **LumaCell crop-evidence matrix** (MATRIX-EVAL600-1). Its status is a passed fictional device-alone milestone with integration gaps open. Read the following rows as the path from proposed claim to authorized conclusion, rather than as eleven independent approvals.

Matrix field	What the 600-crop record establishes
Claim element	Image-level six-category routing assistance for supplied, quality-accepted crops meeting the acquisition profile; no exhaustive locating, abnormal-cell detection or patient diagnosis claim
Configuration and status	Corrected candidate M1/Q0/P1/U1/R1 under AP1, tied to BUILD-V2 and VER120 revision 2; candidate at this snapshot
Evidence questions	Device-alone routing agreement overall and per category under the predefined selected-input conditions; implementation, human reliance and routine non-results remain distinguishable questions

Matrix field	What the 600-crop record establishes
Study or test	The 600-crop evaluation supplies the selected-crop comparison; the corrected verification record supplies bounded implementation evidence. Neither replaces the other
Population and sites	One index crop from each of 600 distinct adults; 200 from each site C, D and E; no adult or site used in development; six deliberately balanced reference categories of 100 each
Reference	Resolved references established without model suggestions; retain the method and richer outside-common-category composition. Reference adequacy for a broader diagnostic claim is not supplied
Results and uncertainty	Overall $584/600 = 97.33\%$; category counts 98/98/97/98/96/97 of 100. Wilson working-model bounds are 95.71%–98.35% overall and 91.55%–98.97% for other/unclassified: arithmetic illustrations, not established design-valid coverage for the fixed quotas. Retain the three ordinary-route outputs and the need for a justified uncertainty plan
Configuration match	Direct match to the corrected candidate with the initial presentation in this exercise. Later display, preparation, gate or runtime changes need explicit applicability decisions; unchanged classifier weights alone do not settle them
Transport limits	Enriched, quality-accepted index crops; no natural-prevalence estimate, all-received package yield, complete-smear detection, routine rejection burden or site-specific performance result
Residual gaps	User response to plausible ordinary suggestions on outside-common-category references; completion of retained whole-smear review; fallback, broader clinical claims and integration with residual risk
Supported conclusion	The invented overall/per-category point milestone passes. Clinical/evidence owner records the bounded conclusion and exclusions; clinical and human-factors owners take reliance into Chapter 13; product owner does not treat the result as release

Trace the middle rows before relying on the last. The corrected verification record addresses specified implementation behavior, while the crop report supplies observed category routes. The 600-unit allocation, masked resolved references and outputs determine the result's meaning. The transport row then stops that result becoming an all-received yield or complete-smear detection estimate. The remaining human response appears in the gap row. With those links intact, the last row can credit the milestone without pretending the other questions were observed.

The other-route row offers a concrete handoff to the use team. One immature white-cell reference received a lymphocyte suggestion; one abnormal white-cell reference received a neutrophil suggestion; one nucleated-red-cell reference received a monocyte suggestion. None of the artifact/non-target examples received an ordinary suggestion in this exhibit. Send those event types and their context forward. Sending only “97 of 100” with a request to validate oversight would conceal the situations professionals need to encounter.

The configuration row also provides a later change reviewer with an anchor: which behavior produced this comparison, and which part of the proposed product differs? Chapter 14 will use that question to distinguish a retained device-alone comparison from human-use evidence that needs renewal. No later change outcome is required to understand the present record.

Reviewer lens. Follow both reports through the corrected candidate, selected population, fixed-quota result and exclusions. A conclusion using a smear or patient denominator needs evidence beyond the crop denominator in this matrix.

Consultant lens. Give each open link an owner and observable decision: statistics for a broader population estimate, clinical and use owners for the plausible wrong suggestions, and engineering for changed-configuration comparisons. Retain the passing bounded result while that work remains open.

Figure 12.1 — Evidence Must Match the Claim and Configuration. Evidence integrates only through explicit claim and configuration alignment; a strong result cannot silently travel.

Book 1D • Chapter 12

Evidence Must Match the Claim and Configuration

A strong result cannot silently travel across products, populations, sites, workflows, or releases.

Key takeaway

Evidence integrates only through explicit claim and configuration alignment; a strong result cannot silently travel.

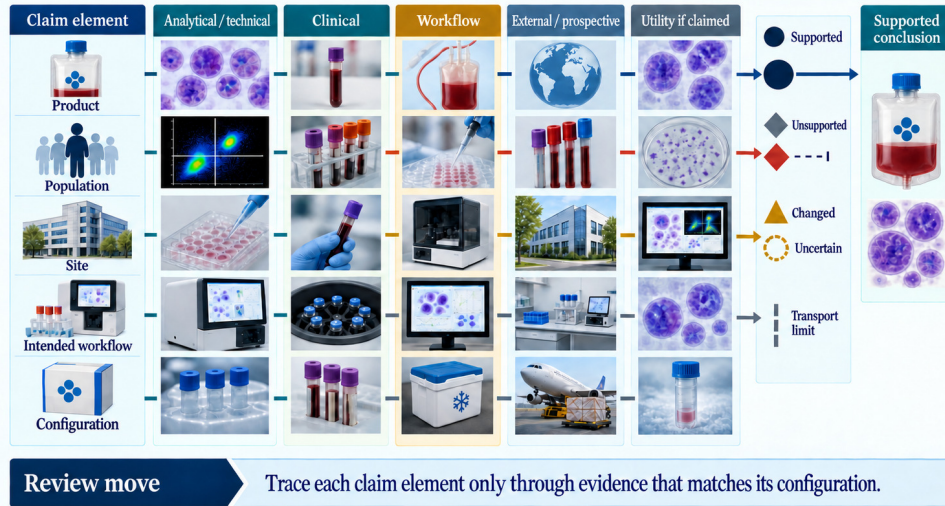


Figure 12.1. Evidence Must Match the Claim and Configuration.

Who can turn the result into a development decision?

The completed matrix gives the clinical/evidence owner a defensible narrow conclusion. It does not transfer the manufacturer's responsibility to the table's author or to whichever specialist signed a report. The manufacturer owns the claim, evidence strategy and integrated conclusion. Clinical and statistical expertise assess the meaning of endpoints and uncertainty; engineering establishes configuration applicability; human-factors and risk owners assess use consequences; regulatory owners determine the applicable route.

Those contributions permit a positive decision now: record the selected-crop pass and continue the evidence program with its open questions identified. Broader claims and release remain separate product decisions. A favorable layer cannot erase a material gap in another, but an open use question does not turn observed crop agreement into no progress.

Reviewer lens. A useful sign-off identifies the owner of the bounded conclusion and separately records the open use and transport questions. “Evidence passed” without that distinction leaves the decision's scope unclear.

Consultant lens. Propose the next permitted development step, its prerequisites and the owner authorized to change claim or release status. The advice makes that decision reviewable; the advisory memo itself is not the manufacturer's authorization.

What should an evidence-readiness assessment let the owner do?

A consultant receiving the two reports needs to reconstruct the present argument before recommending another study. The **evidence-readiness assessment** should make the supported conclusion and next evidence need specific enough for a decision-maker to act. Reviewing documents does not itself certify clinical validity or regulatory sufficiency. Its value lies in explaining which inspected record supports which decision and what would settle the remaining uncertainty.

Which record was inspected, and when was the rule chosen?

Begin with the actual claim and candidate configuration. Put the protocol, analysis plan, executed report and configuration record in chronological order. If only a summary was available, identify the underlying method and results as uninspected. A familiar report title is not evidence that its methods were recovered.

For LumaCell, the corrected verification record concerns specified implementation challenges; the 600-crop report concerns routing on selected, quality-accepted index crops. Keeping them separate shows how they contribute together. Relabeling either as the complete clinical evidence package would hide the remaining links.

This exercise stipulates prospective acceptance criteria. In actual work, the reviewer would inspect the chronology that supports that statement. A rule chosen after seeing the decisive result answers a different evidential question. The same review checks whether the evaluation member list remained protected from later fitting or product-choice decisions. **Data independence** concerns that separation from development influence; outside-site labels alone cannot establish it.

Which difference is known, and which impact remains unresolved?

For every material study-to-candidate comparison, show the affected link and its present disposition: direct applicability, a justified bridge, unresolved difference or inapplicability. These outcomes should follow from the record, not from a preferred readiness verdict.

For example, a shared model label does not resolve different preparation content. Conversely, a discrepancy in documentation does not prove that different executable content actually ran. State what is known, why that difference could affect the claimed output or use, and what content-bound comparison or targeted evidence would resolve it. Calling an unknown impact “none” loses precisely the uncertainty the owner needs to decide.

Is this an observed error or an unanswered question?

The three ordinary suggestions for outside-common-category references are observed crop-level errors. Their effects on professional action remain unknown in this exhibit. Those are different entries: investigate the observed behavior and evaluate the unobserved use consequence. Similarly, no comparative time-saving study is supplied, but that gap is not an observation that the product saves no time.

The pathology queue episode contains both the favorable queue placement and the continuing scanner-import bottleneck as findings. The effect of a proposed end-to-end change would be a new question. The sequencing review episode contains suggestion/display behavior and specimen-reconciliation work at different stages; combining them into one release-speed result would lose the stage boundary. If discrepancies after use motivate a broader claim, preserve the original timestamps and actor/action trail, investigate the missing link and plan the appropriate comparison.

This distinction changes the recommended work. An unfavorable finding calls for investigation and interpretation of what happened. A gap calls for evidence answering what remains unknown. A claim exceeding both needs revision. Treating every gap as failure dismisses useful evidence; treating every gap as reassurance leaves the claim unsupported.

Which word asks the evidence to do more?

Read the proposed sentence one verb at a time and ask which record supports that verb at its stated unit. “Detects,” “rules out,” “reduces,” “replaces” and “improves” may change the role and question. They cannot substitute for “suggests” merely because they read more confidently.

The corrected LumaCell sentence should preserve the contribution of the 600-crop evaluation while avoiding unobserved complete-smear detection or improved diagnosis. The consultant can propose the wording and show its evidence trace. The manufacturer retains ownership of the final intended purpose and information supplied with the product.

What changes if the owner chooses a different path?

One option is to retain the bounded candidate-image role and generate the remaining human-use evidence. The practical condition is that actual use and communication respect the retained whole-smear review. Narrower words on paper would not help if users still rely on suggestions as a replacement for it.

Another option is to pursue a broader detection claim with a different product and evaluation strategy. That brings acquisition and coverage into the evidence question: a larger selected-crop dataset cannot count omitted cells. A third option is to redesign the output or workflow before further evaluation. Such a redesign can reopen

implementation and human-use evidence because it changes what the product does or what the person encounters. The assessment should show these consequences so that the owner chooses a path with its required work visible.

What observation would close the selected gap?

For the reliance question, “add training” is an activity, not a closure observation. A justified workflow evaluation would specify representative users, tasks, information, relevant errors and fallback conditions, with appropriate acceptance logic and analysis. The result must address whether the intended work is performed under those conditions.

For a configuration gap, “same model” is similarly incomplete. Closure needs evidence of direct match or an adequate bridge between the evaluated and proposed products for the affected outputs and use. For a transport gap, the answer may be targeted external evidence or a justified limit on the claim. Which answer is adequate depends on the actual differences, claim and risk. These closure conditions make it possible to recognize a changed evidential state when the work returns.

What would a usable recommendation say?

Note

The supplied records support progression of the bounded candidate-image evidence program, including the passing fictional device-alone milestone. They do not support complete-smear abnormality detection, reduced missed diagnoses or release readiness. The next material question is how intended users complete the retained review when ordinary suggestions are wrong and when output is unavailable.

That conclusion preserves a useful result and attaches the next decision to the missing observation. It makes neither document review nor caution a substitute for reasoning about the actual claim.

Reviewer lens. Across these advisory steps, check source access, chronology, findings versus unanswered questions, unit and configuration. A proposed closure is meaningful when it says what observation would settle the particular gap.

Consultant lens. Sequence the owners' options: retain the crop pass, investigate the three observed misroutes, and commission a justified use comparison if the reliance claim requires one. A broader detection claim needs evidence on selection and whole-smear coverage. In the two recurring episodes, the import and reconciliation boundaries similarly determine what a prospective end-to-end comparison must include.

Could the manufacturer assign the requested work without guessing?

A material finding needs an observation, its consequence for the claim, a specific evidence request and an accountable decision owner. Consider how those four parts would connect for one issue in the supplied LumaCell record.

Which observation would you report, why does it affect the claim, what evidence would address it, and who decides the disposition?

One connected advisory finding

The corrected candidate gave three ordinary suggestions to crops whose references were outside the common categories. Those crop-level errors raise a question about how professionals respond while completing the full smear review; they are not observed missed diagnoses. A prospectively specified evaluation should expose representative users to the relevant wrong-suggestion and no-output tasks under justified conditions. The manufacturer owns the interpretation and resulting claim decision, with clinical and human-use expertise informing it.

Each sentence does a different job. Together they name the unit, configuration, practical consequence and assignable work. “More validation needed” would leave the owner to invent those missing connections.

Which PathSight result could follow the changed image path?

The **PathSight Cloud** capstone in Books 1A and 1B supplies another applicability question. Those two renderings describe the same case state, not successive events. Use only the supplied snapshot here.

Its model is M3.2, with claimed support for Scanners A and B. The validation set is dominated by Scanner A. Scanner B applies proprietary sharpening, and normalization was upgraded after validation. The commercial weights are described as unchanged. The bench comparison examines average color values but does not cover weakly stained tissue near low-confidence boundaries.

What would have to be compared before the earlier evidence could support the claimed scanner reach on the changed path? There are two distinct applicability questions. The snapshot supplies neither exact component versions nor deployment status, so neither can be assumed in framing them.

What are the two applicability questions, and why does unchanged weight content fail to settle them?

Following the image through the two differences

First, the normalization upgrade changes what reaches the model even when its weights stay fixed. The evaluated and proposed end-to-end paths need an applicability assessment. Second, the Scanner A-dominated evaluation leaves a separate question about Scanner B's proprietary sharpening and the intended A/B claim. The missing weak-stain, low-confidence boundary assessment matters because an average color comparison can conceal a behavior change in those difficult inputs.

Targeted evidence should reach those differences. Demonstrating relevant behavior on the changed normalization path and the underrepresented scanner/boundary conditions could justify specified reuse; average similarity alone leaves the questions open. The snapshot does not prove that every prior result is invalid, and it establishes no legal release history for either configuration.

Reviewer lens. Compare claimed scanner reach with the actual validation mixture, then compare evaluated processing with the changed path. The missing boundary condition is a concrete evidence gap, not a reason to substitute an unspecified demand for more validation.

Consultant lens. Ask the manufacturer to reconstruct both paths and select the targeted Scanner B/boundary assessment. Record exactly which prior result can be reused after that assessment, keeping any unsupplied deployment or release status unresolved.

What if fewer crops reach the unchanged classifier?

Suppose LumaCell's candidate categorization remains unchanged, but a proposed acquisition update selects fewer crops to accelerate review. No evidence about omitted findings is supplied. The proposed conclusion is that the new workflow preserves complete-smear diagnostic performance.

Can the 600-crop evaluation carry that conclusion into the new workflow, and which question changes despite the unchanged categorization?

Separating retained crop evidence from omitted information

The supplied evaluation concerns crops already selected and received. It therefore cannot establish the clinical consequences of a process that now omits more information. A conditional crop-routing result may remain reusable for a justified matching condition, but that is a narrower conclusion than preserving complete-smear diagnostic performance.

The manufacturer needs to examine selection coverage, the retained complete review and the new workflow. If professional review reliably reaches the omitted findings, that

fact would be material to the broader question; the unchanged categorizer does not supply it. Information that never enters the classifier cannot be recovered by pointing to how well it categorizes received inputs.

Reviewer lens. Keep the possibly reusable selected-crop comparison distinct from the whole-smear coverage and diagnostic-performance claim. The changed selection process creates the missing link.

Consultant lens. Acquisition, clinical and human-use owners should plan the selection-coverage comparison and examine whether retained review still covers omitted candidates before proposing the broader wording.

Where does the next evidence question lead?

Optional depth is available in Book 1C on references, estimands, study design, analytical and clinical validation, and evidence transport; in Book 1A Chapters 8 and 10–13 on evidence architecture, modalities, statistics and submission coherence; in Book 1B Chapters 8–10 on integrating security evidence; and in Book 2 Chapters 7–13 on European Union performance evaluation and studies. The local method already connects the claim, evidence question and configuration. An actual jurisdictional decision additionally needs its applicable sources, product facts and qualified assessment.

The next chapter takes up the person using this identifiable product under stated conditions. A human signature added to a model result would not answer that question. The observation must concern the work the person actually performs with the product.

Can the conclusion keep its original unit?

A useful evidence matrix lets the reader move from a supported result to the exact remaining question. It distinguishes independence from representativeness, an implementation milestone from clinical adequacy, and a crop error from an observed patient consequence. When the configuration changes, it identifies the affected link instead of treating all evidence as either permanent or lost.

What supported conclusion and still-open claim link would you retain for LumaCell, and why do they need different denominators?

Source Trail

[S1] MDCG 2020-1, Guidance on Clinical Evaluation / Performance Evaluation of Medical Device Software, March 2020, nonbinding guidance; cover disclaimer and Sections 3.3–4.4. The chapter uses the distinction between evidence components, not a case-specific conformity determination.

[S2] ISO 20916:2019 official scope, Scope 1. The study-conduct scope and exclusions are distinct from device-specific endpoints and national study requirements.

[S3] FDA non-recognized standards record: ISO 20916:2019, rationale concerning human-subject protections; checked 2026-09-24. This is a regulatory-status source, not a substitute for the applicable regulations.

The matrix, EVAL600 counts, prospective teaching criteria, worked interpretations, and transfer are author-designed fictional constructions. Statistical methods are retrieved from Chapter 9. No jurisdiction-specific release conclusion is made.

CHAPTER 13

Validate the Human–AI Team

A morphology-review professional opens the next work item. The current identity and source crop are available, and LumaCell Review's corrected candidate may display an ordinary category suggestion. The professional checks that the image belongs to this item, examines the cell and decides whether the suggestion is defensible. A supported suggestion can be accepted. A questionable one may need correction or escalation. The complete smear still needs review beyond the supplied candidates; the laboratory retains patient interpretation and release.

Now suppose the ordinary suggestion is plausible but wrong. The professional might correct it, accept it, or let it end a search that should continue. The software's output record cannot distinguish those actions. If the manufacturer relies on the professional's check, the evidence must reach what the person could see, what they did and whether the work conditions allowed the required review and continuation. That arrangement of people, information and product behavior is the human–AI team considered here.

Chapter 11's corrected implementation and Chapter 12's passing selected-crop comparison are useful evidence. They leave this interaction to be observed. A prospective plan will turn the human tasks into observable actions, and a later fictional evaluation will supply bounded findings. Neither a task result nor a comparative reader result automatically becomes a claim of patient benefit.

What action is hidden inside “human oversight”?

The 600-crop evaluation from Chapter 12, comparing selected accepted crops with resolved references, passed its fictional device-alone milestone, yet three outside-common-category references received ordinary suggestions. Those were crop-level observations. The remaining question follows the suggestion to the person expected to check it.

A phrase such as “human in the loop” leaves the control unspecified. Which information reaches the person, which action is expected, when can they act, and what could follow if they do not?

Follow one plausible wrong suggestion

What source and identity check should precede the final category decision? What review remains after that crop is dispositioned? Where does a disagreement go when the assigned professional cannot resolve it?

The control is the action the person can actually complete

The three ordinary suggestions matter because their consequences depend on the next professional action. The current-item and source check precede a defensible category disposition. Complete-smear review continues beyond the supplied crops. A disagreement outside the professional's ability or assigned role follows an accepted escalation route. A trained person beside a screen does not establish that any of those actions occurred.

The interface can also defeat the intended control. A hidden source makes comparison difficult; wording that implies the review is complete can discourage the retained work. The action therefore needs both an opportunity to occur and an observation showing what happened.

The pathology queue episode asks a related question: does the professional distinguish earlier queue placement for an imported case from completion of slide review? The sequencing review episode leaves specimen-identity reconciliation and report release after the candidate suggestion. These earlier fictional observations locate human work to examine; they supply no new use study here.

Reviewer Lens. Visible information, an assigned actor, an observable action and a work-state record make a proposed human control assessable. A general competence statement cannot show that the action was possible or performed on the correct item.

Consultant Lens. Clinical, interface and site owners can agree on crop checking, full review and escalation before selecting evaluation tasks. A step the design cannot make observable is then a specific planning problem, rather than an assumption hidden under “oversight.”

Which human question is being evaluated?

Imagine an early interface sketch whose current-item and history areas might be hard to distinguish. This is a hypothetical design question, not a diagnosis of the later case. A few users could try the sketch while the team watches, asks questions and changes the layout. That is formative evaluation: learning about a design while revision is expected. Coaching or immediate redesign can serve that purpose. A favorable questionnaire afterward would still leave uncoached performance on a finished product untested.

Once the interface and intended training are specified, human-factors validation asks whether intended users can carry out the important tasks on the final or production-equivalent interface under justified conditions. “Production-equivalent” matters because the tested information and actions must represent what users would encounter. Observed actions, understanding that needs direct inquiry, and later explanations all contribute. A person can complete the final action correctly after a hesitation that still warrants analysis.

A comparative reader study asks another question: do readers using the product perform differently from readers without it under a specified comparison? The endpoint might be category decisions or time to a defined action. A claim about fewer missed diagnoses reaches further, to patient-linked professional decisions, an appropriate comparator and an observation period. Preference, task completion and a reader difference each leave that further claim to be established.

These methods can form one evidence program while retaining different endpoints. An interface people like may invite unchecked acceptance. A favorable average reader result may coexist with a serious difficulty during an uncommon recovery task. In pathology, exploring a queue display, observing uncoached worklist use and comparing total review time answer different questions. In sequencing, display readability, correct specimen reconciliation and comparative release decisions likewise need their own observations.

IEC 62366-1:2015+AMD1:2020 connects use conditions and foreseeable problems to interface requirements, planned evaluations and analysis of use errors and difficulties. Its summative evaluation addresses the final or production-equivalent interface: the completed design is examined through the selected use scenarios. That process relationship explains why an early coached exercise cannot simply inherit the later conclusion. The standard does not prescribe this chapter's training duration or user count, or establish zero use risk. [S1]

Reviewer Lens. The interface, training and behavior actually evaluated give “validated” or “improved” a scope. Identifying those conditions is more useful than accepting the study's preferred label.

Consultant Lens. Exploratory sessions can settle design questions before a production-equivalent use evaluation. Claim-driven reader or outcome comparisons can then be planned for the questions they add. The sequence gives each method a particular gap to close.

Who has the information and authority to complete the task?

The person preparing a smear, the professional assigning a morphology category and the person authorizing a report may have different duties. “Laboratory staff” does not specify who supplies source material, changes a suggestion, escalates uncertainty or releases the patient result. Clinical competence and familiarity with a particular interface are also different. Product training does not create clinical expertise; clinical experience does not demonstrate knowledge of this product's controls.

The actor then needs a setting in which the expected action is feasible. After interruption, can the reviewer reach the original source? Does competing work make prior history more noticeable than the current identity? Is qualified fallback available when assistance is unavailable? The plan need not reproduce every inconvenience. It needs a reason for including the conditions that could change consequential actions.

Allowed help changes that evidence. Developer coaching during the task tests an assisted interaction. Practicing the exact evaluation item immediately beforehand can test recall of the item. Instructions, training content, permitted help, source access and the interval between training and use therefore belong in the evaluated conditions. If training is a safety control, intended sites need a capability they can actually obtain and deliver, beyond a one-time explanation from the study team.

In the pathology episode, the queue user and scanner-import operator may be different people. A whole-route task needs their handoff. In sequencing, viewing a suggestion, reconciling specimen identity and authorizing a report may belong to different roles. A later discrepancy should be investigated against those responsibilities rather than attributed to an undifferentiated group called staff.

Reviewer Lens. A completed task can still have limited applicability if recruited users, authority, help or source access differ from intended use. Those mismatches identify which part of the proposed conclusion remains unsupported.

Consultant Lens. A role map joining source access, training and escalation authority gives clinical and site owners a basis for recruitment and simulation. It also reveals a task that no recruited participant is actually authorized to perform.

What distinguishes justified acceptance from blind agreement?

Watch one decision. The professional confirms the current item, opens the source crop and compares morphology with the suggestion. When the source supports it, accepting after review is appropriate reliance. When the source conflicts, an override changes the proposed category while retaining both the initial suggestion and final disposition. When uncertainty or authority prevents a defensible decision, the professional escalates the item. Complete-smear review continues whichever crop route was taken.

This sequence turns reliance into observable actions: source opened or omitted, category accepted or changed, reason retained, unresolved work handed off or lost, complete review performed or skipped. An override button demonstrates capability. It does not demonstrate that someone recognized a reason to use it. Overriding every correct suggestion would also fail to show good judgment. The scenarios need correct and plausible wrong suggestions among ordinary work, without a facilitator announcing which is which.

Automation bias describes a possible influence of the suggestion on attention or judgment. A plausible ordinary label might displace checking of contradictory morphology, or candidate-screen completion might be mistaken for complete-smear review. That is a hypothesis to examine through behavior, not a diagnosis of a person from one click. Defaults, ordering, confidence labels and wording that looks final can change the information the user attends to.

An earlier pathology queue position can appropriately prompt timely review of an imported case while leaving slide assessment to the professional. A sequencing suggestion can be accepted after source and specimen-link checks, or corrected and escalated when evidence conflicts. Records of the source available and the action taken at that moment permit a use conclusion; model acceptance alone does not.

Reviewer Lens. Both justified acceptance and justified correction belong in the action trail. The task should distinguish attention to source evidence from mere agreement with the suggestion.

Consultant Lens. Clinical and interface owners can define observable acceptance, override and escalation before inspecting outcomes. Scenarios can then expose excessive reliance as well as unnecessary rejection of useful assistance.

What work remains when every candidate row is complete?

A category recorded for each supplied crop does not establish that the smear has been reviewed. The retained laboratory procedure extends beyond those candidates. “Independent complete review” names that task boundary; it does not impose a universal requirement for a second person. The interface must keep crop disposition, full-smear completion and laboratory release distinguishable, and the evaluation needs to observe the required review and its completion record before release.

The external acquisition and selection process is not LumaCell's locating function. A relevant cell omitted before classification cannot appear in the classifier's crop result. The retained review may address that limitation, but instructions saying it exists do not prove it works or guarantee detection. Source access, training, screen order and local staffing must make the work possible. Compressing or removing the review changes the use condition even if every supplied crop retains the same suggestion.

The pathology parallel is a queued case still needing slide assessment. The sequencing parallel is a set of displayed variants still needing specimen-linked review and the report decision. The consequential observation follows the downstream action on the right item, beyond a correct intermediate display.

Reviewer Lens. A task and its record should distinguish candidate completion from complete-smear review and patient-result release. Without that distinction, the study can appear to observe the whole task while stopping at the last software row.

Consultant Lens. Clinical, workflow and site owners can specify the retained review's start, completion evidence and handoff. A proposed shortcut then becomes an assessable change in the product's use conditions.

How does unfinished work reach the right person?

An ordinary output still requires professional review. A quality rejection blocks an ordinary usable suggestion. An unresolved quality state means the required decision is incomplete, while a service outage may prevent output altogether. None of those missing-output states means normal or negative. The user needs to recognize the current state and retain responsibility for the affected item.

Fallback is the alternative work route. With no suggestion, the assigned professional uses the necessary source material to carry out the laboratory's qualified complete morphology review. The work item stays open until disposition is recorded. If the category or blocked state cannot be resolved within that role, escalation conveys current identity, status and the source issue to the designated responsible laboratory professional. Sending a message alone does not complete that transfer: the recipient must accept responsibility, or the defined alternative route must take effect.

Return to ordinary use requires a verified current state. Refreshing a screen is insufficient. A retry of the same attempt and a new acquisition must remain distinguishable in the history. Chapter 11's corrected software checks address whether prior information is exposed as current; a use evaluation asks whether the person interprets current identity and history correctly when the software presents them as specified.

Authentication and service failures also need an authorized alternative. An undocumented security bypass would change the conditions being evaluated. Book 1B develops that security design; the local question is whether people can complete the permitted continuation when the ordinary route is unavailable.

A scanner-import delay in the pathology episode calls for an owned slide-review route. Missing sequencing suggestions leave specimen reconciliation and report disposition with an identified professional. To determine whether fallback occurred, a future investigation would need the item state, recipient and eventual disposition.

Reviewer Lens. Comparing the displayed state with the action, available alternative and accepted handoff tests fallback as work. A functioning button cannot demonstrate that the professional completed it.

Consultant Lens. Clinical, site, service and security owners can examine a blocked path and an interruption path together before evaluation. Responsibility and return-to-normal evidence then have a defined place instead of being improvised by participants.

Which information belongs at this decision?

At category disposition, the professional needs current identity, source image, suggestion and the meaning of unavailable or historical information. A long model description elsewhere cannot replace that timely signal. A procurement decision may instead need a summary of intended role and evidence limits. The audience's action determines what useful information looks like.

The joint FDA, Health Canada and United Kingdom Medicines and Healthcare products Regulatory Agency transparency principles ask who needs information, why, what, where, when and how it is communicated. These are nonbinding principles, not a universal labeling template or proof of comprehension. [S2] For a blocked package, the operator needs to know that no current suggestion is available and which authorized route continues review. Technical support needs state and incident records; receiving them does not give support clinical disposition authority.

A graphic or confidence score may aid understanding, distract from source checking, or imply certainty the evidence does not support. Its effect on action and escalation needs observation. In pathology, queue placement should remain identifiable as placement

among imported cases. In sequencing, a suggestion should remain visually distinct from a reconciled report decision. If complaints later arise, the information available at the decision time is evidence to inspect before assuming a new instruction will solve the problem.

Reviewer Lens. The actual decision point determines where to assess comprehension and behavior. A complete document library can coexist with an unclear current-item display.

Consultant Lens. Mapping audience, message and timing for correct, wrong-suggestion and blocked states gives targeted tests a purpose. The question becomes which misunderstanding could change a clinical action, rather than whether the company has enough explanatory material.

Whose work does the average leave out?

An ordinary crop decision may be quick. A rejection, uncertain category or interruption can consume more time and move work to a supervisor or fallback team. Workload therefore includes the demands, switching and rework borne by the relevant actors. A mean across easy tasks can hide those demands.

The clock must match the question: active source review, elapsed waiting, rework and time to a completed laboratory action are different measures. Switching between current and prior history, repeated alerts and source reopening can make a task harder even when its final category is correct. An interview adds context about those demands; it does not replace the observed consequential action.

The fictional plan supplies no speed target. Removing source checking or whole-smear review to improve an average would change the intended workflow. The pathology episode demonstrates the timing boundary: improved queue placement coexists with import-dominated total time. In sequencing, a quick suggestion can coexist with slow specimen reconciliation. Start and end events, along with the person carrying the rework, are needed before calling an intermediate time a workflow benefit.

Reviewer Lens. Ordinary, blocked and interrupted paths may place different demands on different people. Comparing those paths with the claimed endpoint reveals whether the headline mean measured the intended work.

Consultant Lens. Clinical and operations owners can choose relevant event timestamps and workload observations prospectively. Preserving a necessary check, even if it slows one step, keeps the evaluated workflow consistent with the proposed role.

What must the plan make observable?

An interrupted work item offers a concrete test of planning. Who resumes review, which identity and source appear, what should the person check, and what happens if uncertainty remains? The human–AI validation plan connects those questions to users, product conditions, scenarios and the evidence needed for a decision. It is an index to a protocol and its risk links.

A critical task is an action whose omission or incorrect performance could cause serious harm, including compromised care. Its importance explains why the plan observes the action rather than merely asking whether the user liked the screen. The success definition must be fixed before scoring. Yet scoring does not exhaust the observations: a use difficulty is trouble encountered while performing a task, which may remain important even when the person eventually succeeds. A close call involves recovery from a situation in which a use problem could have led to harm. An observed hesitation needs its actual circumstances recorded before being given a stronger interpretation.

The first rows below identify the person and what they encounter. The middle rows specify expected checking and continuation under workload. The last rows connect local conditions to the evidence question and remaining limits. Completing the fields alone supplies no validation result.

Field	Decision-bearing content
Users and context	Intended profiles, roles, recruitment basis, and relevant settings
Tasks and decision points	Observable actions, information needs, and consequences
Information presentation	Interface, labels, instructions, confidence, and state distinctions
Training	Content, delivery, availability, competence assumptions, and retention interval
Reliance and override	Expected checking, acceptance, disagreement, and correction behavior
Automation-bias hypotheses	Plausible ways suggestions could distort attention or judgment
Workload	Relevant demands, interruptions, timing, and redistribution of work
Fallback	Required alternative path and its availability
Escalation	Trigger, recipient, information, and clinical work disposition

Field	Decision-bearing content
Site conditions	Resources, supported configuration, local responsibilities, and exclusions
Estimand and acceptance	Separate clinical-performance quantities from use-risk evaluation logic
Residual limitations	Uncovered conditions, observed difficulties, remaining risks, and follow-up

Follow the interrupted item across the table. User and site fields specify who can act at the station. Display, training and workload determine what that person encounters. Task, reliance, fallback and escalation fields define correct work on the item. The evidence and acceptance field determines how those observations inform the use-risk decision; it is not a disease-sensitivity calculation. The last field keeps unexamined conditions visible. The same trace applies to a correct suggestion: appropriate checking and acceptance also need an observation.

Each selected scenario needs a rationale linked to the concern it tests. The recording method also needs room for unexpected difficulty. Otherwise a correctly completed scripted action can erase the confusion that preceded it.

FDA's August 2026 nonbinding guidance distinguishes observation of task performance, neutral knowledge questions where understanding is not directly observable, and interviews after the tasks. It calls for analysis of use errors, close calls and difficulties. A later explanation can help investigate an action; it cannot replace or overwrite the observed sequence. An aggregate task count is therefore insufficient to supply a universal acceptance threshold. [S3]

Reviewer Lens. One consequential task should be traceable through actor, source, interface, training, expected action, observation and unresolved risk. A plan with every field filled can still fail that trace if it never says what the observer should watch.

Consultant Lens. Human-factors, clinical and site owners can justify scenarios and success definitions before evaluation while retaining a route for unexpected problems. That route needs both investigation responsibility and a manufacturer decision owner.

Two proposed ways to examine the human control

Both fictional proposals concern the corrected candidate and retained complete-review workflow. Its engineering configuration is M1/Q0/P1/U1/R1 under the proposed acquisition profile (AP1), bound to corrected BUILD-V2: the evaluation classifier, original quality gate, identified preparation procedure, initial presentation and identified runtime. At this point A and B are **unexecuted alternatives**. Their value must be judged from the work and observation they propose.

Proposal A recruits experienced laboratory professionals. After the proposed training, they handle ordinary correct suggestions, clearly blocked packages and an unavailable system. The investigator demonstrates fallback immediately before the blocked tasks. A final questionnaire asks whether suggestions were useful. The proposal includes neither plausible wrong ordinary suggestions nor a task observing whether complete-smear review continues.

Proposal B recruits intended laboratory professionals with a proposed range of morphology-review experience. They use the production-equivalent interface after proposed training and a declared retention interval. Tasks include correct suggestions, plausible wrong common-category suggestions, an outside-common-category reference receiving an ordinary suggestion, quality rejection, unresolved status, interruption/recovery and retained full review. Facilitators do not coach completion. Actions, neutral knowledge responses where understanding is not directly visible, and post-task interviews are recorded.

Neither proposal includes a comparative clinical-performance endpoint or patient-outcome study. Neither has results or establishes a universal participant count.

Which proposal reaches the missing action?

Which design better addresses the reliance question left by the 600-crop evaluation? What must still be specified before it can serve as an acceptable validation plan? Could the weaker design still answer a useful development question, and why would a favorable usefulness questionnaire leave the wrong-suggestion question open?

A design decision with conditions

Which proposal would you develop for the missing use evidence, which bounded purpose remains for the other, and which concrete protocol detail is still needed?

From the preferred proposal to a plan, script and later observations

Why the broader task set matters

Proposal B reaches the consequential gap because it includes plausible wrong suggestions and the retained complete review. Recognizing an obvious block does not establish that a professional will challenge a plausible ordinary label. Correct suggestions remain necessary too: source checking followed by defensible acceptance differs from blind agreement or indiscriminate override.

B still needs participant profiles and sample justification, exact scenarios and success definitions, training and interval, environment, observation methods, analysis, risk links and handling of unexpected difficulty. Those details must justify the selected coverage. Calling the proposal validation does not supply them.

A can support formative design learning about ordinary and blocked states. Coaching and immediate rehearsal limit what its observations could establish about unassisted intended use. Its questionnaire can reveal perception without demonstrating action when a plausible suggestion is wrong. The useful choice preserves that formative purpose while developing B's protocol.

Reviewer Lens. The scenarios and allowed help must address the missing reliance behavior. Both designs remain proposals at this decision point; a favorable outcome cannot be assumed for either.

Consultant Lens. Human-factors and clinical owners can retain A for exploratory feedback while specifying B's participants, critical actions, environment, training interval and analysis before commissioning the intended-use evaluation.

What does the worked plan fix before anyone is observed?

The worked human-use plan, PLAN-HAI-1, develops B for the fictional initial 60-user evaluation, HAI60. It fixes the corrected verification build with initial presentation U1 under the proposed acquisition profile. The ordinary task is current-item identification, source checking, accepting or changing a suggestion, and complete review. The same actions remain relevant when a suggestion is wrong or unavailable.

The initial use-training program, TR-H1, supplies 45 minutes of instruction and guided practice on **separate material**. Participants do not rehearse the evaluation items. A recorded seven-day interval precedes uncoached testing, allowing performance to be examined after time has passed rather than immediately after practice. Neither number is a universally recommended value or evidence of enduring routine training effectiveness.

The plan includes 60 intended morphology-review professionals. Each encounters all seven scenario types in a recorded, prespecified order rotation. Rotating order helps the

team examine whether sequence affects observation; it does not by itself prove that all learning or order effects are controlled. These are author-chosen setup values, without a general recruitment-adequacy, scenario-coverage or clinical-use claim.

The human-factors lead owns the plan with clinical input. Software confirms configuration; quality/risk reviews control links; the designated manufacturer owner decides readiness and later evidence acceptance. No results have yet been supplied at this planning point.

Plan field	PLAN-HAI-1 entry
Users and context	Intended laboratory morphology-review professionals, 60 in this teaching record, with the relevant experience spread documented; production-representative local review station. A real recruitment/coverage rationale must justify the sample rather than copy 60
Tasks and decision points	Identify the current package/source, inspect evidence, confirm or change a category, recognize non-results, undertake the retained complete review, and escalate or record unresolved work according to the scenario
Information presentation	Corrected M1/Q0/P1/U1/R1 under AP1; production-equivalent current identity, suggestion, source view and history. No facilitator hints or answer-revealing case labels
Training	TR-H1, 45-minute instruction/practice with non-evaluation material; recorded seven-day interval; no coached refresh immediately before the difficult evaluation tasks
Reliance and override	Require source checking and a defensible final action; the ability to click an override is not itself the endpoint. Preserve initial suggestion and final reviewer disposition
Automation-bias hypotheses	A plausible ordinary suggestion can displace attention to source morphology or the retained whole-smear review; a prior item's history can compete with current identity
Workload	Include defined interruption and competing-work conditions with recorded timing/task order. Observe hesitations, switching and rework; no time-saving endpoint or speed target is supplied
Fallback	When an ordinary result is unavailable, preserve the work item and use the laboratory's complete morphology-review route; do not treat non-result as negative

Plan field	PLAN-HAI-1 entry
Escalation	If the task cannot be resolved within the assigned role, notify the designated responsible laboratory professional with current identity/status; retain responsibility until handoff is accepted
Site conditions	Required source material, supported local configuration, training, assigned roles and fallback are present for evaluation. Later site installation/staffing must be checked separately
Estimand and acceptance	Evaluate specified use-risk behaviors, not diagnostic sensitivity. Record task success/failure, close calls, difficulties, neutral knowledge responses and interviews. Any omitted/incorrect critical action is a task failure; analyze all safety-relevant observations and their causes/mitigations before a manufacturer residual-risk disposition. An unresolved safety-relevant finding cannot be closed by a high aggregate success count
Residual limitations	Purposive scenarios and selected professionals; no comparative clinical benefit, routine error rate or demonstrated coverage of every workload/site. Repeated tasks within one participant are not independent new users

Display, training and workload change what a professional can do. Reliance, fallback and escalation specify what to watch under correct, wrong and unavailable suggestions. The final rows explain why a perfect scored count would still need problem analysis and would not become a disease or release claim. Local resources supplied for this evaluation also leave later site installation and staffing to be checked separately.

The following scenario matrix makes the actions concrete. Its last column describes possible findings requiring analysis; it reports no participant outcome.

Scenario	Observe the critical action	A finding requiring analysis
Ordinary reference-concordant suggestion	Check current identity/source and complete the required review	Accepting without the required source/review action
Plausible wrong common-category suggestion	Examine source and make the supported final category decision	Repeating the suggestion despite contrary source evidence
Outside-common-category reference shown as ordinary	Use the complete taxonomy and escalation/review route as appropriate	Treating the limited model categories as the limit of the laboratory review

Scenario	Observe the critical action	A finding requiring analysis
Quality rejection	Recognize blocked work and initiate the defined alternative route	Treating rejection as a completed negative/normal result
Unresolved quality status	Keep the current item accountable and act on the unavailable decision	Assuming missing status is acceptance or losing the work item
Interruption/recovery	Confirm current identity before acting after resumption	Acting on another item's information, or difficulty distinguishing current from prior work
Retained complete review	Complete the required review beyond supplied candidates	Substituting candidate-screen completion for complete-smear review

The first row is not passed merely by accepting the correct suggestion: current-item, source and complete-review actions are also required. The next two ask whether source evidence can overcome a plausible ordinary label, including a reference outside the model's common categories. The quality rows test different non-result dispositions. Interruption tests identity on resumption, and the last row reaches work beyond the candidate screen.

Success definitions precede observation. An unexpected difficulty remains recorded even after the correct action is completed. Seven scenarios for each of 60 people produce repeated task observations within people—420 tasks, not 420 independent users or routine clinical encounters.

A prospective pathology version would follow an imported queued case into actual slide review and retain a blocked or import-delayed alternative. A sequencing version would observe justified acceptance and correction while specimen identity remains attached through reconciliation. These are planning transfers, without new executed studies or a claim that the episodes' existing records satisfy this plan.

Reviewer Lens. Actor, visible source/state, scenario, predefined action and observation record should connect across both tables. Counting seven tasks cannot repair a missing user profile or multiply the independent-participant count.

Consultant Lens. Human-factors and clinical owners can refine scripts, realism, task definitions and recruitment rationale before execution, while preserving unexpected observations for the manufacturer's eventual decision.

How does the interruption script test the intended behavior?

The interruption-use script, USE - INT - 01, uses controlled simulated work items. They are neither additional patients nor monitoring receipts. H-PREV is a completed prior item visible in history. H-CUR is the current accepted-quality item with source material available. Their recorded identities differ. The corrected software should display the current response as current after recovery; the person resumes ordinary work without being told that identity is the test.

Read the five steps in time order. The first fixes the two identities. The neutral instruction starts ordinary work. The interruption controls when the current response arrives. Observation preserves what the screen and person did, before scoring and later inquiry interpret the record.

Script step	Setup, instruction or observation
Starting state	H-PREV is completed and visible only as prior-attempt history. H-CUR is the next identified item, with accepted quality and source material available. Both identities are distinct and recorded in the fixture.
Neutral instruction	“Review the next prepared work item and record your disposition using the usual procedure.” Do not tell the participant to inspect the identity display or warn which difficulty is being studied.
Controlled interruption	After H-CUR's processing acknowledgment but before its response reaches the display, pause response delivery and invoke the supported interruption/recovery sequence. After recovery, release the held current response. The restored U1 view shows H-CUR as current and H-PREV as separately identified history; no stale suggestion is supplied as current.
Observation checkpoints	Record the screen and actions before interruption, at resumption and before final disposition: which identity/source the participant checks, any switching between current and history, what is acted on, and whether the required review is completed. No correctness cue or rescue is given during the task.
Scoring and inquiry	Acting on prior information as current, or omitting the required identity/review action, is a critical-task failure. Visible uncertainty or repeated checking that impedes completing the identity task is retained as a difficulty even after correct completion; no arbitrary delay threshold defines failure. After the task, replay the relevant moment and ask, “Tell me what you were considering here.” Record the explanation without substituting it for observed actions.

The held response belongs to the current item. After recovery it is released, with the current item still identified as current and the completed item separately identified as

history. No stale suggestion is inserted as current. Chapter 11's earlier V44 defect had exposed prior information as current before correction; this script deliberately evaluates another question on the corrected software.

What changes the scoring branch? Acting on prior information as current, or omitting the required identity or review action, is a critical failure under this fictional protocol. Moving between history and current identity before correctly completing the check can be a retained difficulty. No arbitrary duration converts hesitation into failure. The observation must establish which behavior occurred; the software pass alone establishes neither.

The neutral post-task question helps investigate what the participant was considering. Asking it during the evaluated action, or warning the participant to check identity, would change the task by providing help.

Reviewer Lens. The screen chronology and identity assertions must be checked before scoring. Otherwise the test might recreate the corrected software defect or cue its own answer. Recovery before action and wrong-item action must remain distinguishable in the record.

Consultant Lens. Engineering and human-factors owners can first check the controlled pause/release and recording points technically. Neutral inquiry afterward can then explore uncertainty without turning the task itself into coached practice.

What does the later execution actually show?

Only now does the fictional record supply execution. The initial 60-user evaluation follows developed Proposal B on the corrected candidate. It uses the initial training's separate-material practice and seven-day interval, all seven scenario types in rotated recorded order, and uncoached source checking, possible override, escalation and complete-review tasks. This is observed use work, without an independent comparative reader study.

The summary reports **no observed failures of the predefined critical-task behaviors** among the 60 participants. It also reports **two distinct recovered identity-confirmation difficulties** after interruption. Both people correctly complete the check; neither is reported to act on a wrong item. The next table separates the supplied observations from explanations and risk decisions that are still absent.

Finding	Observation	Current disposition and closure owner
HAI-ID-1	First participant hesitates after interruption, then confirms current identity correctly	Retain as a use difficulty; human-factors/engineering owners investigate observation, participant explanation and presentation; clinical/quality-risk assess consequence
HAI-ID-2	A second participant shows the same class of difficulty and recovers	Link to the shared investigation without assuming an identical cause; preserve the individual observation
Overall use-evidence status	No predefined critical-task failures observed; two difficulties have no supplied cause or residual-risk disposition at this snapshot	HAI60 not closed. Manufacturer decides corrective work and evidence acceptance after analysis; neither recovery nor 0/60 alone closes the findings

The two individual rows preserve two observations. The overall row gives their current status: zero predefined critical failures, two retained difficulties, and no supplied cause or residual-risk disposition. Populated result fields do not populate the missing analysis or manufacturer acceptance. The positive task observation remains useful while the use-risk argument stays open.

For the first difficulty, the participant alternates between the current identity area and prior history, then correctly checks the current source and completes the required review. The recorded action is recovered checking, not a wrong-item action. The second participant's similar difficulty does not establish the same cause. Each screen/action sequence and later neutral explanation must remain available before attributing the behavior to layout, training, workload or anything else.

Why does zero observed failure leave uncertainty?

Consider a separate arithmetic caution, not a new acceptance rule. Imagine one specified binary failure and 60 independent people, each with the same chance p of making it in this defined evaluation. One person's chance of avoiding it is $1 - p$. Independence allows multiplication across people, giving $(1 - p)^{60}$ as the probability that none makes the specified failure.

For an illustrative one-sided 95% upper bound, find the value of p for which observing zero failures would still have probability 5%, or 0.05. A larger event chance would make this all-zero observation still less likely under the model. Taking the sixtieth root reverses multiplication across the 60 people; subtracting that no-failure chance from one returns the failure chance:

$(1 - p)^{60} = 0.05 \Rightarrow p = 1 - 0.05^{(1/60)} \approx 0.0487$, or about 4.87%.

This is a one-sided 95% working-binomial upper-bound calculation. Its repeated-sampling coverage concerns the common event chance under the stipulated model. It is neither a routine-use error-rate estimate nor an acceptable residual-risk level or interface-approval threshold. The scenarios were selected for purpose, users can differ, and repeated tasks within a person do not add independent users. Most directly, the calculation concerns an absent predefined critical failure; the two observed difficulties are a different part of the evidence and remain present.

The evaluation also supplies no improved diagnostic sensitivity, time saving or reduction in missed diagnoses. Those claims need their own justified designs and observations.

Reviewer Lens. The positive no-observed-critical-failure result deserves credit for the evaluated tasks. Both recovered difficulties must also remain, with no invented wrong-item action or causal diagnosis. The binomial illustration cannot decide their risk disposition.

Consultant Lens. The two individual observations, neutral explanations and interface/recovery records can go to human-factors and engineering investigation. Clinical and risk owners assess possible consequences; the manufacturer decides what change or further evidence is needed for closure.

Figure 13.1 — Validate the Human-AI Team in the Real Workflow. Training, interface, reliance, workload, override, fallback, escalation and site conditions can change the complete system outcome.

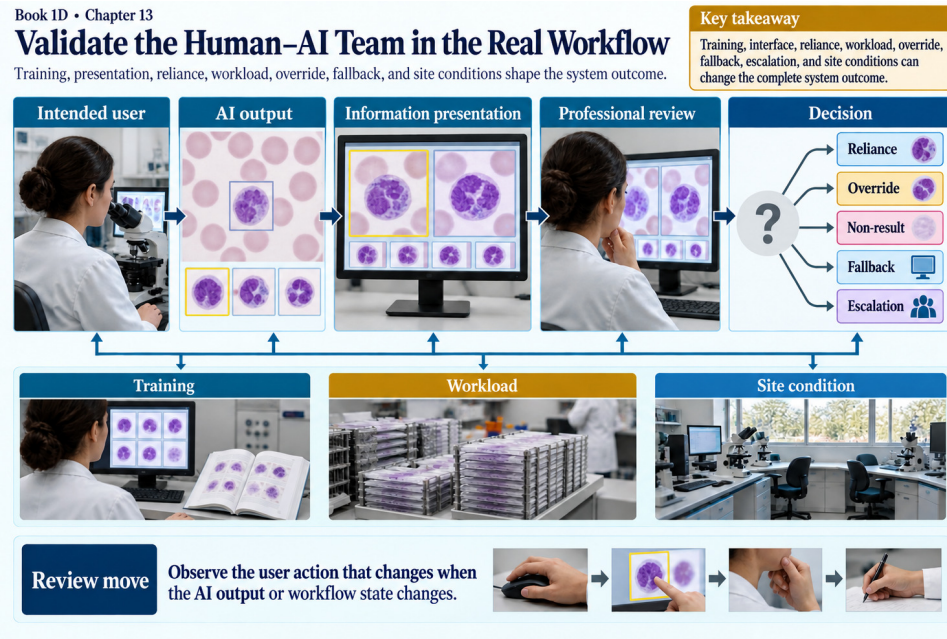


Figure 13.1. Validate the Human-AI Team in the Real Workflow.

Who owns the investigation and the next decision?

The initial 60-user evaluation joins a defined candidate, training and tasks to observed professional behavior. The manufacturer owns the use specification, design controls, validation plan, analysis of observed problems and residual use-risk assessment. Human-factors expertise contributes alongside clinical, engineering, security, quality and site knowledge.

An unexplained use observation calls for its circumstances and consequences to be investigated. Another classifier percentage would answer a different question. The resulting evidence supports a manufacturer decision about whether the interface and training remain supportable or need change and renewed evaluation. Residual risk remains an accountable judgment about what is left after controls, rather than a task-count label.

A site's training certificate does not transfer responsibility for interface design to its users. Manufacturer design also cannot supply a site's absent resources or qualified roles. Product control and evidence disposition belong to the manufacturer; professional interpretation and patient release belong to the laboratory; local operation must satisfy the specified site conditions. Records must connect those responsibilities.

Reviewer Lens. A task-count statistic cannot close an unexplained safety-relevant use finding. Nor can “user training” close a possible design issue without an investigation that supports that response.

Consultant Lens. A handoff connecting the observation, possible consequence, investigative owner and authorized decision owner makes the next work assignable. Assessing that handoff does not authorize the adviser to accept the manufacturer's residual risk.

How can the use-evidence review lead to a specific next action?

A workflow and reliance evidence review follows a claimed human control from person and work item to observed action and consequence. It identifies what evidence supports the control, what remains unresolved and who can close the gap. A model score or preference survey cannot supply the missing action record, and the review does not approve residual risk.

Follow the initial use findings into an accountable review

Which retained action did the study examine?

The role map starts with the person seeing a suggestion and the person making the final decision. It then connects required checks to available information and to the workflow beyond the device. LumaCell's complete review is consequential because it reaches smear material beyond the externally supplied crops.

A category-screen assessment alone leaves that retained action unobserved. The precise finding would be that the protocol does not evaluate completion of full review under the intended workflow. That identifies an action and a missing observation; “human oversight insufficient” leaves both unspecified.

Which observed behavior needs an explanation?

For a wrong or unavailable suggestion, the review follows the risk hypothesis through interface, expected action, scenario, observation and disposition. Every wrong suggestion does not necessarily cause harm. Every correction is not necessarily harmless: delay or another consequence may still matter. The actual action must be preserved before either interpretation is made.

Here the two recovered identity-confirmation difficulties remain observations even though the predefined critical actions were completed. The initial record contains switching or hesitation, followed by recovery. Whether layout, training, workload or another factor contributed remains unknown. Preserving both individual sequences prevents a shared investigation from becoming an assumed shared cause.

Would the same person encounter the same conditions?

Training, instructions, interface, staffing, source access, escalation and fallback define the evaluated workflow. A changed default or shorter training program could affect use with the same classifier. The comparison must identify known differences and unresolved ones without calling every difference a proven defect.

A proposed evidence bridge needs to reach the affected behavior. A screenshot can establish visual similarity while leaving task performance open. A software test can establish a visible state while leaving comprehension open. If the changed condition concerns understanding or action, the bridge needs evidence that answers that question.

Can the intended site supply the evaluated capability?

Training has content, delivery, maintenance and access. The evaluated program must be compared with what intended users can actually receive, when they need it. A recommendation to retrain users needs an investigation supporting training as the response; it cannot substitute for addressing a design-related difficulty.

Fallback needs the same operational check. If an outage disables both the AI and the proposed alternative, that alternative cannot provide continuation for this failure. Its existence in ordinary conditions is insufficient. A different, available authorized route would change that assessment.

Which study could close this particular gap?

The initial use evaluation concerns specified behaviors. Its immediate open question is analysis of two difficulties and the adequacy of the use-risk evidence. The 600-crop evaluation's favorable agreement cannot supply that analysis. The use observations likewise cannot establish fewer missed diagnoses.

A reader-performance or patient-outcome claim needs the corresponding comparison and analysis. An unstructured satisfaction survey answers another question. The pathology episode's earlier queue result remains valid for its narrower start point, while a total-time claim needs import-to-review measurement. The sequencing episode's displayed suggestions and completed specimen reconciliation remain different observations. An adviser can request the missing action record or justified comparison without rewriting the intermediate observation as a clinical benefit.

What would make the identity finding ready for disposition?

Closure needs the recorded actions, participant explanations and investigation of the relevant interface and interruption conditions. It then needs a justified decision about design improvement and additional evaluation. The two identities were presented correctly as current and history; the investigation must explain the observed difficulty without inventing the earlier stale-current software defect.

Human-factors and engineering owners investigate, with clinical and risk owners examining possible consequences. The manufacturer holds decision authority. If a change follows, the affected verification and validation evidence must be identified. The consultant can assess whether that package is complete and well reasoned; the manufacturer decides its disposition and any progression or release.

A conclusion that credits the observation and preserves the open work

Note

The initial 60-user evaluation contributes evidence about specified behaviors on the evaluated candidate and workflow. No predefined critical-task failure was observed. Two recovered identity-confirmation difficulties remain for analysis and documented disposition. The supplied summary does not establish absence of use risk, comparative clinical benefit, or readiness for another interface, training program or staffing arrangement.

These limits define the conclusion itself. They cannot be detached from the favorable observation without changing what the evidence supports.

Reviewer Lens. Role map, task/action chain, workflow comparison and closure request must agree with the supplied record. Both an invented wrong-item action and an unexplained assurance that recovery made the difficulty harmless would overstate it.

Consultant Lens. Source-preserving observation review leads to neutral inquiry, investigation of relevant interface/training/workload conditions and clinical risk disposition. Each step needs an owner and an observable completion condition. A proposed correction or targeted evaluation follows the affected behavior rather than preceding its explanation.

What must the next owner be able to act on?

An advisory handoff needs six connected elements: role clarity, a use-risk observation, a workflow match to check, training or fallback evidence, an unresolved clinical question and a closure owner. Each element should identify an actual relationship rather than a general request for more validation.

Make the handoff specific

Which observed behavior and record would anchor those six elements? What consequence remains a question, which evaluated conditions need comparison, and who should investigate and decide?

One source-preserving handoff

The first recorded difficulty anchors the handoff: current/history checking was hesitant but recovered before the correct source check and complete review. The consequence question is whether similar confusion could affect a final action. Evaluated training, source access and history presentation need comparison with proposed use, along with the availability of the continuation route. Human-factors and engineering investigate, clinical/risk owners assess consequences, and the manufacturer dispositions the finding.

“Careless users” would invent a cause. An added warning would remain a proposed control whose effectiveness needs evidence. Neither can replace the observed sequence or its investigation.

What changes when paired reader results become available?

A **separate** fictional methods exercise supplies 120 paired reader-case assessments. Each analysis unit has a correct/incorrect status under assistance and without assistance. The table groups pairs by those two statuses. It is not the initial 60-user evaluation and supplies no additional LumaCell release evidence. The source gives neither reader count nor case count nor an independence structure beyond the 120 pairs.

The first row agrees across conditions, as does the last. The middle rows identify changes in opposite directions. Preserving those pairs will matter when comparing totals.

Paired result	Assessments
Correct in both conditions	95
Correct only with assistance	10
Correct only without assistance	5
Incorrect in both conditions	10

What is the correct proportion under each condition, and what is the paired net difference? What does a positive net leave visible about losses? Would this table alone establish clinical superiority?

Follow both outcomes in each pair

Which rows contribute to each correct total, which cancel in the difference, and which design facts would be needed before a clinical comparison could be concluded?

The paired calculation and its limits

Assisted correctness combines 95 correct in both conditions with 10 correct only with assistance: $105/120 = 87.5\%$. Unassisted correctness combines the same 95 with 5 correct only without assistance: $100/120 = 83.33\%$. The shared 95 cancel in the difference. The paired net is $(10 \text{ gains} - 5 \text{ losses})/120 = 5/120$, or 4.17 percentage points. Five assessments became incorrect with assistance even though the net change is favorable.

That observed difference does not by itself establish clinical superiority. Protocol, reference, case selection, reader/case dependence, order and recall controls, uncertainty, prospective hypothesis and clinical relevance remain necessary. Repeated readers or cases could connect assessments; treating the 120 pairs as 120 independent patients or readers would invent a design fact. The paired result also supplies no explanation for the interruption difficulty in the separate use evaluation.

Chapter 9's denominator discipline now applies to a human-performance comparison. A future paired pathology total-time study would need consistent start/end events and import conditions. A sequencing report-decision comparison would need specimen identity and the matching reference, beyond suggestion agreement. These are designs still to specify, not additional episode studies.

Reviewer Lens. Reproducing the totals is the first check; identifying what the paired unit represents is the next. Both are needed before the calculation can support its proposed endpoint and analysis.

Consultant Lens. Clinical and statistical owners can define comparator, reader/case sampling, order controls, uncertainty and meaningful benefit prospectively. That gives a later paired difference a claim it could legitimately inform.

From evaluated use to a controlled service

Book 1A Chapters 12–14 offer optional depth in reader studies, statistics, human factors and reliance. Book 1C develops device-alone, reader-alone and reader-plus-device evidence. Book 1B addresses authentication, availability, degraded operation and secure fallback; Book 2 addresses European Union (EU) human–AI and evidence-sufficiency questions. The local task method and evidence distinctions already support this chapter's reasoning.

Chapter 14 asks whether the specified product and human workflow can be transferred into controlled deployment with usable information and a monitoring protocol. A study configuration is not yet an installed service. Its evaluated conditions must remain identifiable when that next decision is made.

Can the control be described as an action?

What task is hidden inside “oversight”? What separates a user's opinion from observed behavior, an observed difficulty from a diagnosed cause, and successful scenarios from a comparative clinical claim? Which evidence would need reconsideration if a default, training program or retained review step changed?

Which use observation is supported, which use question remains open, and which different claim would need a different study?

Source Trail

[S1] IEC 62366-1:2015+AMD1:2020, Edition 1.1, clean final text, Clauses 5.1–5.9, printed pp. 15–21/PDF pp. 149–155 in the licensed combined edition. Used for the use-specification, scenario, evaluation, training, and observed-problem framework; paraphrased without reproducing the standard's text.

[S2] FDA, Health Canada, and MHRA: Transparency for Machine Learning-Enabled Medical Devices—Guiding Principles, 2024, “Who” through “How.” Nonbinding principles, not a complete labeling specification.

[S3] FDA: Applying Human Factors and Usability Engineering to Medical Devices, issued August 3, 2026, originally February 3, 2016; Sections 8.1.5–8.1.7 and Appendix A, printed pp. 24–27 and 30–31. The current edition is distinguished from the historical issue. Recommendations are not presented as universal numerical acceptance requirements.

The workflow plan, proposals A/B, HAI60, binomial caution, paired-count exercise, and advisory examples are author-designed teaching constructions. They do not certify a real product or establish a jurisdiction-specific study route.

CHAPTER 14

Chapter 14: Software/Model Release Baseline, Deployment, Transparency, and Monitoring Design

The installer reports success. A laboratory professional opens the worklist and finds a new candidate-cell crop. What makes it reasonable to act on the suggestion that appears? The current package and source must be identified, the installed product must match the evidence being relied on, and the professional must be able to check or correct the category. Complete smear review and qualified fallback must still be available. The laboratory retains patient interpretation and release. A successful file copy answers a much smaller question.

Three records help locate that gap. The manufacturer release records the controlled package and internal evidence the manufacturer has accepted for supply. The installed configuration records the contents and settings actually running at the laboratory. Local readiness concerns the people, sources, fallback and owners who will use that installation under the specified conditions. A favorable record for one does not supply the other two. National legal authority for clinical supply or use is a further, separate question.

LumaCell begins this chapter with its corrected but unreleased candidate and initial presentation, U1. The first site records must be judged in that state. Later fictional events will supply different evidence; they cannot make the earlier record complete retrospectively.

What did installation leave unfinished?

What remained unresolved at the end of Chapter 13's initial 60-user evaluation? Which two product questions could a successful local installation still leave unanswered?

Pause — which issue belongs to the manufacturer, and what can the installation record actually establish?

Keeping the useful installation result within its scope

Two people recovered from identity-confirmation difficulties after interruption. Their correct completion is a positive observation, but Chapter 13 supplied neither an investigated cause nor a residual-risk disposition. Copying the product or running a local sample cannot perform that investigation, establish broader clinical benefit, accept residual risk or confer national authority. It also cannot establish that another site's package and workflow are equivalent.

The manufacturer decides what controlled product package its evidence supports. The site must identify what it installed and whether its people and operational conditions can use it as specified; clinical use also depends on applicable authority. That is why a successful installation can remain useful evidence while other decisions stay open.

The pathology queue episode gives a familiar boundary: installing a queue aid does not resolve the scanner-import bottleneck or establish total slide-to-review benefit. In the sequencing review episode, launching the suggestion service does not complete specimen reconciliation or report release. Before use or during a later investigation, installed identity needs to connect to the responsible person's actual work.

Reviewer Lens. I would first identify which decision each record answers. The local sample result may be sound; the problem arises if it is offered to close an unresolved manufacturer use finding. Keeping the records separate permits a positive narrow finding without treating it as clinical-start approval.

Consultant Lens. Product, clinical and site owners can agree the evidence and authority for each decision before a start date is promised. Otherwise the last completed installation task can acquire an unintended meaning simply because it sits at the end of the schedule.

Which product does the evidence describe?

Two installations show the same classifier name. One could nevertheless contain a different file. Or the classifier computation could genuinely be unchanged while a new display changes what the professional thinks is current. The name alone cannot distinguish those situations.

Follow the crop instead. Its profile and identity must fit the package; the original quality gate supplies required crop decisions and their package aggregation; preparation supplies model input; the operating point determines the output route; runtime and interface make the output current and usable. These are logical jobs, not a specified execution order. At this stage, the actual gate input representation and its placement relative to the identified preparation procedure remain unsupplied.

The controlled version baseline therefore includes the evaluation classifier, gate, preparation, operating rule, runtime, presentation, dependencies, input contract and delivered instructions. At an engineering join those are M1, Q0, P1, R1, the applicable presentation, and AP1, together with their actual content and settings. A manifest bound to contents lets a reviewer compare what was evaluated with what actually shipped.

That comparison must also reach the evidence: implementation checks, the selected-crop evaluation, human-use observations, risk and security decisions, and dispositions of anomalies. A release checklist points to those records; it cannot recreate them. Earlier failures and their corrections remain in the history even after a later finding closes. If instructions or presentation change, the user's task may change with unchanged classifier weights, so the relevant evidence needs an applicability decision.

A future pathology import-adapter change could alter scanner-receipt-to-queue behavior without renaming the ranking component. A sequencing display change could alter report review with unchanged suggestion logic. In either episode, exact installed content and delivered instructions would be needed before attributing a later discrepancy to the model or workflow.

Reviewer Lens. I would reconstruct one crop's route to the current display from content-bound records, then follow its evidence and anomaly history. A release nickname is a useful locator. It is not the observation that proves the installed product is the one those records describe.

Consultant Lens. Engineering, quality, clinical and use owners can compare the candidate and proposed package field by field. For each material difference, the record should distinguish direct evidence, a justified bridge and the decision owner. That makes retained evidence as visible as work that must be renewed.

What would a normal sample miss?

A normal package runs successfully at the site. That establishes the path it exercised, but it has not yet challenged rejection, interruption, local resource limits or the ability of every intended professional to complete the work. Choosing the next check starts with the condition that could break the transfer.

Deployment equivalence asks whether the installed arrangement preserves the conditions on which the evidence depends. Compare the content manifest, supported operating environment, runtime and dependencies, interface settings, input profile, identity path, security controls and clinical workflow. Some differences can be checked through bytes or settings. Others need an observed local task or a justified compatibility assessment.

The local checks should therefore reach consequential normal, blocked and recovery paths under the supported load. They are transfer checks chosen for those risks; performing them at a laboratory does not make them a new clinical validation study. Likewise, asking every local user to reproduce the manufacturer's whole clinical evidence would answer a different question from whether this installation meets its supported conditions.

The comparison also needs a future change route. A later operating-environment, acquisition-setting, network-policy or work-schedule change can affect the evidence link. Manufacturer and site owners need to know who reports it and who assesses its effect. A pathology deployment would follow the installed import/queue chain and scanner conditions; a sequencing deployment would follow specimen linkage and suggestion/report paths. Old and new conditions must remain identifiable when earlier evidence is carried forward.

Reviewer Lens. I would compare evaluated and installed conditions before judging the local test selection. A normal pass is useful, but the rejected and recovery paths often answer the transfer question it leaves open. The request should concern those affected conditions rather than an undefined demand to “validate the site.”

Consultant Lens. A content comparison, a targeted local observation and a named change-notification owner can be assigned to each supported condition. This gives both parties a maintainable agreement: which evidence supports initial transfer and what fact would reopen it later.

Can the site perform the supported work?

Suppose the bytes match exactly, but an interrupted review leaves the source smear inaccessible. Or only part of the intended staff has completed training. The installation may be correct while the local workflow is not ready. People, material, access, escalation and maintenance are part of the supported arrangement.

The manufacturer needs to distinguish required conditions from optional site choices. Current-item identity, complete smear review and qualified fallback are consequential requirements here. The site records how it provides them and who owns each. A training attendance list is evidence of attendance; where the task matters, it may need an observation that the intended person can perform it. None of these local records can settle a known manufacturer-wide interface finding.

This coordination is the narrow point supported by International Electrotechnical Commission (IEC) 80001-1:2021. Its scope concerns organizational risk management before, during and after connected health-system use, considering safety, effectiveness and security. [S1] Connecting a product makes the manufacturer's conditions and the organization's operation meet. That scope explains why both parties need to

coordinate; it does not establish conformity of this installation or transfer one party's responsibility to the other.

In the pathology episode, scanner/import staffing remains a site condition even if the queue software matches its release. In sequencing, someone still needs the authority and resources to reconcile the specimen and authorize the report. A later delay or complaint should be traced to the relevant role and interface before a cause is assigned.

Reviewer Lens. I would ask whether each intended role can reach its source, training, escalation and fallback at the claimed operating hours. The hours matter: an arrangement that exists only during the day cannot support an unexamined overnight claim. That question stays separate from whether the manufacturer has closed its own evidence findings.

Consultant Lens. Site clinical and deployment owners, with manufacturer support, can turn required conditions into local checks and named responsibilities. The aim is to demonstrate the supported work, not to ask the site to certify broad classifier performance.

Who needs the information at this moment?

At the worklist, the professional needs current identity, source, state meaning and the action to take when no ordinary suggestion is available. Before installation, the clinical owner needs the role and claim limits, the deployment owner needs supported configurations and checks, and security/support owners need dependencies, maintenance and problem routes. Sending one manual to a shared mailbox does not establish those different handoffs.

Each handoff should retain the product and information version, the recipient and unresolved questions. Where the control depends on understanding, task evidence matters too. Receipt is not comprehension. If people repeatedly confuse current work and history, another acknowledgment form cannot establish that the design or explanation has improved.

The joint Food and Drug Administration (FDA), Health Canada and United Kingdom Medicines and Healthcare products Regulatory Agency (MHRA) transparency principles give placement and timing an explicit role in communication. They are nonbinding principles, not evidence that a particular interface is understood. [S2] In the pathology example, the operator needs queue-versus-import status where they act, while administrators need supported scanner/import conditions. In sequencing, the reviewer needs suggestion and specimen-link states at the decision, while technical owners need update and reconciliation records. A change notice must reach the person whose task changes.

An operational information transfer can therefore be inspected: what did this actor receive, for which version, before which decision, and what showed it could be used? It does not automatically supply a sufficient assurance case, legal permission or a specialist safety/security conclusion. Those need their own applicable methods and evidence.

Reviewer Lens. I would match a consequential instruction to its recipient, version and decision time, then inspect behavior or unresolved questions where receipt is insufficient. A complete document library is weaker evidence for that specific task than an identifiable, timely handoff the person could act on.

Consultant Lens. Clinical, deployment, security and support owners can map the information needed for their own actions and the route for questions. Choosing a targeted comprehension or task check at that point makes the handoff useful before installation rather than merely documenting that a file was sent.

What has the monitoring record actually observed?

Start with the first newly received package. Its record can say when it arrived, whether its required crops met the quality rule, whether processing produced a valid current output, and whether professional review later closed. These are four observations. A missing field could mean work was never done, a message was delayed or a person has not yet acted. The received package is the starting unit; none of those fields is automatically a clinical-accuracy label.

Once that unit is clear, a monitoring protocol can specify its question, numerator, denominator, receipt window, reporting cutoff, site/profile strata, reference source and delay, trigger, owner and response. “Monitor accuracy” needs suitable independent clinical references joined to the correct product and patient or task. “Monitor drift” needs a defined change and a reason it matters to a risk or product decision. The label does not provide either observation.

What might change? Input profiles, rejection counts, output categories, professional overrides, clinical outcomes and service/security events describe different things. An override records a professional action, not necessarily the correct reference answer. A category shift could reflect case mix, acquisition, workflow, product behavior or recording. Source and configuration context make those alternatives investigable. An urgent identity or service problem also needs a route that does not wait for the periodic chart.

The records need permission as well as a useful design. Chapter 5's distinction still applies: authority for service monitoring does not silently permit fitting a model, external sharing or indefinite retention. In the pathology episode, post-use queue data need import timestamps and case ownership to explain delay. In sequencing,

suggestion counts need reconciliation states before they can support any report-release argument.

Reviewer Lens. I would reconstruct the receipt-to-state chain, or the source-to-reference chain for a clinical claim, before accepting a metric's label. A well-counted operational measure can be useful without becoming accuracy or benefit. The missing relationship should determine the finding.

Consultant Lens. Quality, clinical, data and site owners can begin with a small set of questions that leads to action. Defining the observations, collection rights, delays and response owners before the first cohort arrives avoids a dashboard whose attractive trends have no accountable interpretation.

Is a late response new work?

A package arrives once, but a required response may be delivered twice. The first received acquisition creates an attempt; a delivery retry stays with it. A newly acquired and resubmitted source creates a new linked attempt. The link preserves why work recurred without turning a retry into another patient or hiding a genuinely different acquisition.

A receipt cohort groups attempts first received during a declared window. At a fixed later census—the reporting cutoff—each has one package-quality disposition. If a missing response arrives after that point, the original census remains and a dated reconciliation records the update. Unresolved older attempts stay in a backlog until resolved; they do not become next window's new arrivals.

Those rules protect more than a total. Under the known-rejection precedence, a rejected package may also have a missing crop response to investigate. Its quality category remains rejected. An accepted package may still lack usable current output after a processing failure. A closed review records completion, not clinical truth. If those states are merged, the denominator can change merely because the dashboard starts excluding unfinished work.

The parallel is concrete in pathology: resending a scanner message does not create a second new slide-review case, while a new scan may be a linked new acquisition. In sequencing, a repeated message for one specimen/analysis differs from a new specimen run. Their old unresolved work needs a backlog record rather than the appearance of a new surge.

Reviewer Lens. I would reconstruct one first receipt, retry, reacquisition, cutoff state and late update before trusting a pooled proportion. Both a frozen view and an updated view can be legitimate. The problem is presenting one as the other without identifying when the information was available.

Consultant Lens. Stable receipt and linkage fields, frozen snapshots and named backlog owners are worth deciding before metric code is written. They let operations explain a later change in counts from preserved events rather than repairing history to fit the current screen.

What should a rising fraction cause someone to do?

Suppose more new receipts are rejected. A prespecified investigation trigger can send that observation to review. It cannot say whether acquisition, case mix, the gate or recording caused the rise. The next evidence is the receipt ledger, site/profile split, missing-response records and configuration history.

A trigger must therefore name the event or comparison, its owner, available records, response expectation and escalation. Some events need attention without crossing a proportion: an unresolved identity problem may require immediate containment, while a suspected security incident or outage needs its applicable technical and clinical urgency assessment. Periodic analysis and urgent triage are different routes.

Their decisions also differ. Opening an investigation does not itself decide product acceptance, corrective and preventive action, or legal reporting. Those may follow under their own rules once the relevant facts are examined. If the analytical trigger changes later, its effective version and affected comparisons must remain visible. An inconvenient earlier signal is not erased by recalculating the past under a more convenient rule.

A future pathology timing trigger could open import-path review; a sequencing specimen-link complaint could need immediate triage rather than a wait for the suggestion-rate chart. Neither would diagnose cause or authorize a product change simply by firing.

Reviewer Lens. The useful question is what the signal instructs an owner to examine. I would distinguish that bounded action from cause, corrective action and reporting decisions. A protocol that names only a threshold has not yet specified what happens when it is crossed.

Consultant Lens. Analytical, clinical, security and quality owners can agree both periodic and urgent routes prospectively. A controlled revision process then permits the protocol to improve without losing which rule governed an earlier window.

Continue the clinical work or restore older software?

If the aid is unavailable, the work item still belongs to someone. LumaCell's qualified fallback may use the laboratory's complete morphology-review route, with source access and an assigned responsible professional. Staffing and authority must exist at the time of failure. A manual that names an unavailable person does not provide the route.

Restoring an earlier software package is a different operation: rollback. That older package may contain the defect that led to replacement, lack current security support or conflict with current data structures, runtime or instructions. A proposed target needs current risk and compatibility review, a controlled decision, handling of state and data, communication and checks that the restored system works. Suspending the aid while continuing qualified clinical review may be preferable to restoring an unsupported package.

The support plan connects these choices. Who receives vulnerability, defect, compatibility and end-of-support notices? Who assesses local impact and decides when use must be held or changed? Book 1B supplies further security methods, but this handoff still needs product-specific owners. A pathology import failure or sequencing reconciliation-service failure similarly calls first for a viable continuation of the task, before any old component is proposed as recovery.

Reviewer Lens. I would ask whether the clinical alternative survives the outage that triggered it and whether the proposed older package remains supported, compatible and authorized. A familiar installer is not evidence that returning to it is an acceptable current decision.

Consultant Lens. Manufacturer support and site clinical/deployment owners can rehearse one failure-to-fallback route and record the conditions for considering rollback, suspension or another controlled recovery. This gives an outage response a practical decision path without preapproving every previous version.

Can one receipt be followed across the decisions?

A release decision is in one folder, installation records in another, and a monitoring chart in a third. Can an investigator join them for one affected package? The release-and-monitoring record provides that connection: which product was meant to process the receipt, what the site installed, who could use it and how its later state or problem remains accountable.

The thirteen fields below are an index for finding those underlying decisions. A complete-looking form does not make them true or constitute release by itself.

Part of the record	What it must identify
Release identity	Controlled manufacturer decision and exact package
Model and software baseline	Contents, dependencies, content identities and supported combinations
Evidence trace	Applicable requirements, observations, anomalies and risk conclusions

Part of the record	What it must identify
Deployment comparison	Installed-versus-evaluated conditions and local checks
Site readiness	Users, resources, training, workflow, fallback and responsibilities
User information	Delivered instructions and controlled material versions
Transparency	Relevant audiences, limits, timing and change communication
Monitoring quantities	Observable measures, units, cohorts, strata and state rules
Reference and delay	Reference source, selection, availability, linkage and latency
Triggers and owners	Comparison rules, accountable roles and expected responses
Signal triage	Initial assessment, urgent routes and investigation records
Fallback and rollback	Available alternatives, compatibility, authority and recovery checks
Information handoff to the site	What each recipient needs, receives, understands and reports back

The first three rows locate the manufacturer's product and accepted evidence. The deployment, site and information rows then ask what the laboratory received and can do. The remaining monitoring and response rows let a later investigator reconstruct the receipt and act on a problem. For example, the reference row cannot be filled by a quality-gate count: a clinical reference may arrive late or never arrive, and its source and selection determine what comparison it supports. The record can link to specialist statistical or security work without reproducing every detail.

Reviewer Lens. I would choose one work item and follow installed identity, responsible user, quality, available output, review completion and escalation. The table becomes valuable when it reveals a missing join, rather than merely showing that thirteen headings have text beneath them.

Consultant Lens. Manufacturer and site owners can name the underlying record that would close each link. That turns the form into a usable handoff while keeping the actual release, local-use and legal-authority decisions with their owners.

Two sites receive the candidate

At this first snapshot, the manufacturer proposes release of the corrected candidate with the initial presentation: M1/Q0/P1/U1/R1 under AP1. The revised software-verification record, VER120 revision 2, supports bounded implementation. The 600-crop evaluation, EVAL600, supports selected-crop routing. The initial 60-user evaluation, HAI60, observed no predefined critical-task failure, but its two recovered identity-confirmation difficulties still lack the investigation and risk disposition needed to determine their significance. This is not yet the later released product.

The proposed acquisition profile gives the site's input comparison concrete content. The candidate crops come from an outside acquisition and locating process, not a LumaCell cell-finding function. Adult peripheral blood is collected with ethylenediaminetetraacetic acid (EDTA), which prevents clotting, and prepared under the specified smear/stain procedure. The fixed acquisition/locator stack is identified as ACQ1/LOC1. These fictional development conditions are not recommended settings for other laboratories.

For the image itself, the profile specifies red–green–blue values in a 256-by-256 pixel crop at 0.25 micrometre per pixel, stored losslessly as a portable network graphics image. The color profile is identified as CP1, and the preparation procedure performs deterministic normalization. Size and scale identify the image content supplied; lossless storage preserves pixel values; profile and preparation identities make color handling traceable. Those facts establish neither clinical adequacy of the crop nor selection of every relevant cell.

The identity-field requirements, IF-1, require order, specimen, acquisition, profile, locator, crop, timestamp and image-integrity fields. The specification names the fields; each received package supplies actual values that must be checked as current. Complete independent smear review still extends beyond the selected crops. LumaCell neither locates all cells nor autonomously reports a patient result.

Site A identifies the candidate manifest and supported environment. Its local checks are complete, all eight intended users are trained, qualified clinical fallback is available, and deployment and clinical owners are named. Those are positive local observations whose reach is part of the decision.

Site B reports “installation complete.” Its record does not identify the installed runtime content, shows training attendance for four of six intended users, and gives a shared support mailbox without naming an owner for unresolved package states. No clinical operation has begun at either site.

The proposed monitoring chart counts “all completed results this month.” It does not say whether rejected or unresolved received packages enter, whether a retry is new

work, or when status is frozen. Excluding slow or blocked work could change the displayed fraction without any change in underlying service behavior.

Which decision can these records support?

Should clinical operation begin at either site? Which evidence gaps concern the manufacturer across sites, and which concern a particular installation or local workflow?

What receipt-based definition would repair the monitoring chart? It needs an attempt, a receipt window, a cutoff, accepted/rejected/unresolved states and a separate review-completion measure. The numerical investigation trigger is a later decision; none needs to be chosen yet.

Pause — what do Site A and Site B each support, and what would the repaired denominator count?

From the initial hold to later evidence and accountable monitoring

Why the initial request remains on hold

Neither site's supplied record supports beginning clinical operation. The two recovered human-use difficulties still have no supplied cause, control decision or residual-risk disposition. That issue belongs to the manufacturer across sites. Site A's identified package, eight trained users, checks and fallback are useful readiness facts. They cannot decide the open product-use question or supply legal authority.

Site B has additional gaps that would remain even if the manufacturer issue closed tomorrow. Its runtime needs a content comparison with the manifest. Two of its six intended users lack documented readiness. An unresolved work item needs an accepted owner, which a shared mailbox has not identified. These are three different closure requests, not one missing “installation complete” signature.

The monitoring denominator begins with each new acquisition attempt, counted once at first receipt in a fixed window. Delivery retry stays with that attempt; reacquisition is a new linked attempt. At a declared cutoff, every new receipt receives exactly one quality disposition—accepted, rejected or unresolved—while missing-response investigations are retained separately. The primary rejection fraction includes all new receipts. A secondary fraction can use accepted plus rejected if that different denominator is clearly labeled. Older unresolved work stays in a backlog; professional closure remains separate from quality acceptance and clinical reference truth.

The hold does not permanently disqualify the candidate. It identifies the work that could change the decision: investigate both recovered difficulties, establish the needed control and risk disposition, and complete Site B's content, all-user readiness and

ownership records. Recovery cannot be assumed harmless, but it also does not establish that no supportable design is possible.

Reviewer Lens. I would preserve four separate findings: the manufacturer use issue, Site A's positive local evidence, Site B's additional gaps and applicable legal authority. That structure makes later partial closure visible instead of forcing every record into a single approve/reject box.

Consultant Lens. Product/quality owners can commission the human-use investigation while Site B's deployment and clinical owners complete their local work. Operations can repair the receipt ledger in parallel. The memo recommends and explains those actions; it does not authorize clinical use.

New evidence changes the record later

The following events occur after that initial hold. They supply new observations, changes and decisions. Their presence later cannot make either earlier site record sufficient. The evidence must show what changed before a different disposition can be understood.

What ordinary task could the optional aid improve?

The feasibility record, FEAS-1, first returns to the professional's work. The product owner and clinical lead observe reviewers inspecting externally supplied crops, manually entering provisional morphology categories and sometimes consulting a reference gallery before continuing full smear review. They select structured manual category entry with that gallery as the credible non-AI comparator. This identifies a category-entry and checking task; it does not measure missed disease or demonstrate a shortage of professional review.

Existing exploratory work shows that a traceable provisional suggestion can be produced on the bounded input and checked or corrected by the professional. It has not compared the aided route with manual entry for speed, interpretation or clinical outcome. The resulting positive decision is continued development of optional suggestions, with manual entry/gallery and full review retained. The next study question is whether the aid adds value in that actual workflow. Chapter 8's simpler classifier baseline, B0, is a technical comparison and supplies no answer against this manual alternative. Neither necessity for artificial intelligence, superiority nor patient benefit has been established.

Reviewer Lens. I would check that the selected comparator represents the provisional-category task that was observed. The feasibility decision can support bounded optional-aid development without turning the ability to produce a suggestion into evidence that the suggestion improves care.

Consultant Lens. Product and clinical owners can now specify a manual-versus-assisted workflow comparison before making a benefit claim. Keeping the manual route viable preserves a genuine option for both future comparison and continued work.

Which evidence must change with the display?

The release evidence-transition record, RELEASE-REL1, follows the held findings through investigation, changed design and evidence review. The next table should be read down the rows: an observed difficulty needs an explanation; that explanation motivates a change; the change needs implementation and human-use evidence; unaffected computation needs an explicit applicability decision. The named functions are fictional accountable roles that a real manufacturer would resolve to authorized people and underlying records.

Issue or evidence	New investigation, change or result	What it supports
Two recovered identity difficulties	Each observation is replayed and neutral questions ask which identity was being checked. Both people describe comparing current work with adjacent prior-attempt history; neither reports using history as current. The hesitation occurs at that comparison. Competing grouping and labels are identified as the design contributor	A shared design explanation for two retained individual observations, supporting a targeted change; not a universal cause of identity risk
Initial to revised presentation	Current identity is grouped with current state and suggestion; history is separately labeled and grouped. Classifier, gate, preparation and runtime computation stay unchanged. Revised information and training use the new screen examples	New interface content requires the controlled revised-presentation build, BUILD-U2, even with unchanged computation
Renewed implementation checks	VER120 revision 3 applies relevant suite and identity/recovery challenges to BUILD-U2; no current/prior substitution or stale usable suggestion occurs in the specified sequences, and prescribed regression checks pass	Software/configuration lead closes the tested implementation change; this does not decide human use

Issue or evidence	New investigation, change or result	What it supports
Renewed human-use evaluation	HAI60 revision 2 uses 60 new intended users, all seven scenario types, recorded order rotation and uncoached tasks. Updated training retains 45-minute preparation and a seven-day interval. No critical-task failure or unresolved difficulty is observed	Human-factors and clinical leads review behavior, neutral follow-up and remaining identity risk; accept the revised control for the bounded workflow with full review, fallback and identity-complaint monitoring retained; no zero-field-risk or universal-adequacy inference
Existing 600-crop evaluation	Configuration comparison finds no change to selected-crop computation or its input contract; the revised display changes presentation rather than category routing	Clinical/evidence lead retains MATRIX-EVAL600-1 for that device-alone claim; it cannot replace renewed use evidence or establish disease detection, throughput or patient benefit

The first row preserves what Chapter 13 actually observed: two difficulties followed by correct completion, not wrong-item actions. Replay and neutral inquiry now supply an investigated explanation about competing current/history grouping and labels. That explanation is new evidence, not a diagnosis silently inserted into the original observation.

The revised presentation, U2, places current identity with current state and suggestion and separates history. The information package, INFO-U2, and training, TR-H2, carry its changed examples. Computation in M1/Q0/P1/R1 is unchanged, yet the whole executable package is the distinct BUILD-U2. Keeping that distinction allows the next two evidence questions to be asked precisely.

First, the relevant implementation and identity/recovery checks in revision 3 show that the changed package meets the specified software behaviors and regression checks. Second, renewed use evidence asks how people act on the changed display. It uses **60 new intended users**, not the earlier participants counted again, with the same seven scenario types and recorded order rotation. Tasks remain uncoached; the updated training retains **45 minutes of preparation and a seven-day interval**. No predefined critical-task failure or unresolved difficulty is observed in that renewed fictional evaluation. Human-factors and clinical leads review behavior and remaining risk before accepting the revised control within the bounded workflow. Monitoring remains part of that disposition; 60 people cannot prove zero field risk or universal adequacy.

The final row answers what can be retained. Because selected-crop computation and its input contract did not change, the evidence lead retains the existing 600-crop comparison for that device-alone routing claim. This is an applicability decision, not a

new study. It supplies no disease-detection, throughput or patient-benefit conclusion and cannot replace the renewed human-use evidence. In a future pathology display change, an unchanged queue calculation would likewise leave the changed human action to examine. An altered sequencing reconciliation display would reopen specimen-link use evidence despite unchanged variant computation. Those are transfer examples, not new episode events.

Reviewer Lens. The important chain is observation, investigated contributor, new package, implementation result, new-user evaluation and retained computational claim. Skipping the investigation would turn observation into presumed cause; skipping renewed use would turn a software pass into a human conclusion. Each positive result earns its own scope.

Consultant Lens. Engineering, human-factors and clinical owners can document which behavior changed, which evidence was renewed and which narrow computational evidence still applies. The manufacturer then receives a reasoned residual use-risk disposition to consider before release, rather than a collection of favorable but disconnected reports.

What does the manufacturer release, and what does each site accept?

The completed release record connects that evidence transition to actual deployment. Its first rows identify the manufacturer decision and package. Its middle rows identify local content, people and information. Its last rows make later monitoring and response accountable. These are stipulated teaching excerpts with underlying reviews, not a claim that every omitted record in a real dossier has been supplied.

Required part	Completed release-record entry
Release identity	REL1 records the bounded manufacturer release LC1.0 after the stipulated technical, clinical, use, security, quality and residual-risk reviews. National market access remains a separate unresolved jurisdictional question
Model and software baseline	BUILD-U2 contains M1/Q0/P1/U2/R1 under AP1; the controlled manifest resolves content and dependencies. No online update is enabled
Evidence trace	The transition above, package-accounting and recognition/action requirements REQ-017/018, applicable risk conclusions and anomaly dispositions are linked. Earlier BUILD-V0 failures remain history
Deployment comparison	Both site records resolve to BUILD-U2. Supported configuration and local identity, non-result, interruption and recovery checks pass; these are transfer checks, not new clinical studies

Required part	Completed release-record entry
Site readiness	All eight Site A and all six Site B intended users complete revised training. Named clinical, deployment and monitoring owners accept responsibilities; hours, full review and qualified fallback are confirmed
User information	INFO-U2 covers supplied-crop scope, current/prior identity, ordinary/blocked/unresolved states, retained review and escalation. Delivered versions match the evaluated presentation
Transparency	State and identity reach users where they act; claim limits reach clinical owners; dependencies, maintenance and change information reach deployment/security owners. Receipt and use evidence stay distinct
Monitoring quantities	MON1 fixes attempts, receipt cohorts, census, quality dispositions, strata and separate operational measures before the field windows
Reference and delay	Receipt, quality and closure are operational events. No clinical reference stream or postmarket accuracy estimate is supplied; delayed statuses remain traceable
Triggers and owners	Manufacturer analytical/quality review owns the rejection trigger; named site and manufacturer reconciliation owners handle unresolved items
Signal triage	Clinical complaints, identity concerns, security and unavailable-service events follow their own urgency assessment without waiting for the 28-day comparison
Fallback and rollback	Qualified manual workflow remains available. No older package is preauthorized as safe rollback; a target needs current risk, compatibility and recovery assessment
Site information handoff	Named recipients receive the revised information and relevant technical/support records; local checks and observed training tasks support readiness. The laboratory responsible professional records site use separately from REL1

The first row is a positive manufacturer decision: REL1 releases LC1.0, the controlled BUILD-U2 configuration under the stated input profile. The underlying configuration now defines the gate/preparation interface and the stipulated applicable integration evidence checks it. Earlier chapters left that interface unsupplied to the reader; this later release premise does not reveal its engineering values, execution order or test details, and none should be inferred. The release follows the specified internal reviews. It is not national market authorization, a full real dossier or coverage under a

predetermined change control plan. The local fixed-inference package has no online-learning update.

The site rows are separately positive. Both sites now identify the revised build and pass supported local identity, non-result, interruption and recovery checks. Site A has all eight and Site B all six intended users trained with the revised material; each has named clinical, deployment and monitoring owners, full review and qualified fallback at approved hours. The laboratory responsible professional records site use separately from the manufacturer release. A manufacturer signature alone does not start clinical operation.

The delivered information matches the evaluated presentation. That is a version check; receipt and comprehension still remain different observations. The monitoring rows likewise define future work rather than report future outcomes. Operational receipt, quality and review-closure events have no supplied clinical-reference stream, so they cannot produce a postmarket accuracy estimate.

Reviewer Lens. I would separate the accepted manufacturer package, each content comparison, each site's trained users and fallback, and national authority. The revised record supports bounded progress on its actual evidence. Neither its release label nor its operational counts supply the separate legal or clinical-truth relationship.

Consultant Lens. Manufacturer and site owners can reconcile the manifest, installed contents, evaluated instructions, all intended users and monitoring fields, with a named owner for each decision. This makes the later positive state inspectable without treating a single signature as all of those acts.

Only after this documented state and any applicable external authority can the later field windows begin. No outcome from those windows is supplied here, and nothing in this later record changes the insufficiency of the initial site request.

Fix the monitoring question before the first window

Consider one receipt, A101. A required crop is known to be rejected while another response is missing. The rejection prevents ordinary output; the missing response still needs an investigation. A delivery retry can fill that response without creating another received work item. A new acquisition, such as the linked A102, is different. That distinction comes before any percentage.

The receipt-monitoring protocol, MON1, fixes **28-day first-receipt windows**. Every attempt first received during a window belongs to that cohort. Its quality census is taken **seven days after the window ends**, allowing delayed responses a specified period to arrive while retaining a defined reporting cutoff. Retry stays with the original attempt; reacquisition is a new linked receipt. Neither rule turns an attempt count into a count of independent patients.

At the census, each receipt takes exactly one quality disposition. Any known rejection of a required crop makes it rejected, even with another response missing. If no rejection is known, incomplete membership or identity, an empty required set, or a missing/unavailable required status leaves it unresolved. A complete nonempty identified set with every crop accepted gives accepted quality. Missing-response facts remain separately accountable even when rejection has already decided the package state.

These conditions determine what can change the branch. Filling a missing response with acceptance cannot cancel another known rejection. If no rejection exists, completing the required evidence with all accepted statuses can resolve an otherwise unresolved package. Acceptance still does not establish that processing produced usable current output, that professional review closed or that the clinical interpretation was correct. The protocol retains processing availability and review closure as separate fields.

The principal rejection measure is **rejected attempts divided by all new received attempts**, including unresolved ones. The separately labeled known-state measure divides rejected by accepted plus rejected. It answers the narrower question among receipts with a known accepted/rejected disposition. Unresolved divided by all new receipts is reported separately, alongside the individually owned backlog. Review closure among accepted attempts is another operational measure, with accepted attempts as its denominator. None supplies clinical truth.

The analytical trigger compares the principal rejection percentage with the preceding comparable window, pooled and separately at each site. An increase of **more than three percentage points on either comparison** opens manufacturer analytical/quality review. A move from 4% to 7% is exactly three points and does not meet this particular rule; 4% to 7.1% does. Percentage points subtract two percentages, rather than calculating a relative percentage change. A pooled result below the threshold cannot cancel a site comparison that crosses it.

The trigger opens investigation into acquisition, case mix, gate behavior, configuration or recording. It does not identify which changed or automatically decide a corrective and preventive action. Every unresolved receipt also has a named site/manufacturer support or quality reconciliation owner, with urgency assessed from the actual circumstances. Older unresolved receipts remain in backlog and do not become new arrivals next window. Complaints, possible clinical consequences, identity concerns, security events and unavailable service follow their urgent routes without waiting for this periodic comparison. There is still no clinical-reference stream for estimating post-use accuracy.

The method transfers without borrowing another episode's results. Before pathology use, first scanner/import receipt and review-closure events would be defined;

afterward, a resent message and a new scan must remain distinguishable. Before sequencing use, specimen/run identity and missing suggestion or reconciliation states would be specified; afterward, a delayed specimen link needs its owner rather than a replayed message counted as a new case. Neither queue closure nor reconciled identity automatically supplies a clinical reference for benefit or correctness.

Reviewer Lens. I would reconstruct one receipt through quality precedence, processing and review fields, both rejection denominators and the pooled-or-site trigger. Then I would compare the frozen census with later reconciliations. The protocol's meaning lives in those joins, not in a chart titled rejection rate.

Consultant Lens. Analytical/quality, site clinical and support owners can assign stable receipt/link fields, data rights, census production, backlog work and urgency routes before the first window. They should also identify what extra evidence would be needed for a later causal or clinical claim, so an operational signal has a clear path forward.

How many receipts survive the retries?

The small ledger below is a separate accounting exercise, not either later field window, a clinical dataset or an additional verification result. It follows three first receipts and one delivery retry. The first two rows show why response completeness and package quality must remain different fields.

Attempt and event	Required crop responses	Quality and receipt accounting
A101 first receipt	Accepted, rejected, missing	Rejected; also open a missing-response item. One new receipt; no ordinary suggestion
A101 delivery retry	Accepted, rejected, accepted	Still rejected; close the missing-response item. Same attempt, no new receipt, no ordinary suggestion
A102 new acquisition linked to A101	All required crops accepted	Accepted quality and a new receipt. Usable output still needs valid current identity and completed processing
A103 first receipt	Accepted, missing; no rejection known	Unresolved and one new receipt; no ordinary suggestion. This state remains at the census

For A101, the retry supplies the missing acceptance and closes that missing-response item. The other crop's rejection remains, so the package stays rejected and is counted only once. A102 contributes the second receipt because its source was acquired anew, despite its link to earlier work. A103 contributes the third and remains unresolved at the cutoff because a required response is absent with no known rejection.

The frozen census is therefore **three receipts: one accepted, one rejected, one unresolved**. Principal rejection is $1/3$. Known-state rejection is $1/(1 + 1) = 1/2$. The unresolved fraction is $1/3$. The difference between the first two fractions comes from whether the unresolved receipt is included; neither fraction is a second observation of clinical performance. A101 was never simultaneously in the rejected and unresolved quality categories. Its missing-response item was separate.

After the census, a retry for A103 supplies a rejected status. The original **1/1/1** view remains frozen and a dated updated **1/2/0** view is appended. No new acquisition occurred, so neither the retry nor its resolution adds a next-window receipt. No usable-processing, professional-closure or clinical-truth outcome is supplied for these attempts; the changed quality view cannot establish any of them.

Reviewer Lens. The invariant is three first receipts through the retry, linked reacquisition and late resolution. I would inspect both legitimate views with their dates and labels. Overwriting the original census would erase the information available when the comparison was made.

Consultant Lens. Ledger and service owners can preserve first-receipt identity, the new-acquisition link, the separate missing-response ticket and dated reconciliation. Those records let a later dashboard reproduce both views without inventing new work or concealing the old unresolved state.

Figure 14.1 — Release Freezes Identity; Monitoring Starts Before Launch. Release freezes identity and responsibilities; monitoring must be designed before launch with denominators, references, thresholds, latency, owners and actions.

Book 1D • Chapter 14

Release Freezes Identity; Monitoring Starts Before Launch

Deployment equivalence, site qualification, operational information, and monitoring need named owners.

Key takeaway

Release freezes identity and responsibilities; monitoring must be designed before launch with denominators, references, thresholds, latency, owners, and actions.

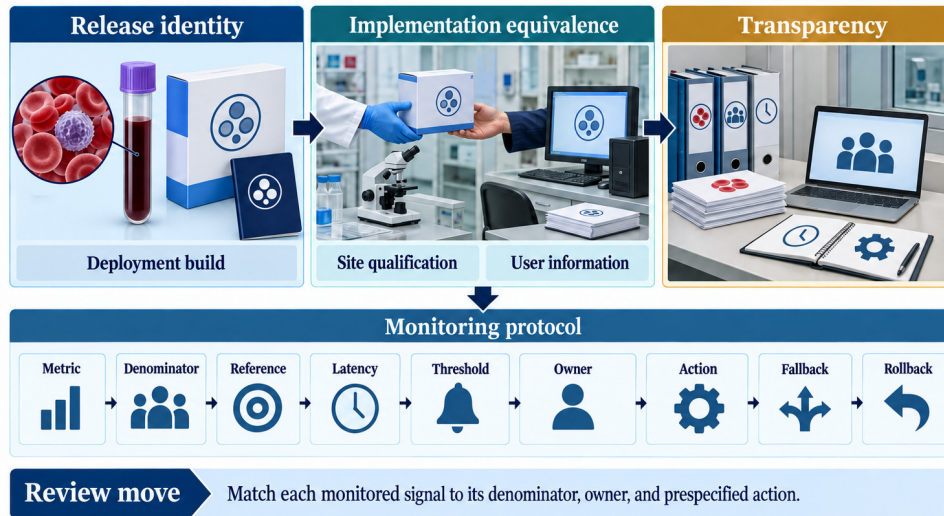


Figure 14.1. Release Freezes Identity; Monitoring Starts Before Launch.

Advice that can change a specific decision

The release-and-monitoring challenge memo makes a requested clinical start reviewable: which missing record affects which decision, who can obtain it, and what a satisfactory follow-up would show. It can recommend a hold and a route to closure. It cannot itself release the package, authorize the site, choose monitoring rules after seeing their outcomes or settle national market access.

The following memo returns to the initial deployment request. Its closure conditions must be intelligible from the evidence available then; the later story can be compared with them without pretending that the adviser already knew its outcomes.

How the initial memo keeps its findings and owners distinct

What would close each initial finding?

The worked advice, ADVICE-REL-1, separates the manufacturer-wide use observations from Site B's installation and training gaps. Site A's eight trained users remain positive local evidence, but they cannot explain the two recovered difficulties. Each row below identifies what its own missing relationship requires; there is no single approve-deployment box.

Memo part	Evidence-specific finding and follow-up
Baseline gaps	The initial 60-user evaluation has two recovered but unexplained identity difficulties. Site B lacks adequate runtime-content identity. Product-use closure and local configuration are different missing records
Deployment assumptions	Site A cannot close the manufacturer use issue. Site B's installation message and four attendance records cannot establish readiness for all six intended users
Information gaps	Obtain evaluated identity/state instructions, delivered version and task-performance evidence. Receipt alone cannot close a confusing display
Operational monitoring	Replace completed-results counting with a prospective receipt/census protocol. Reconstruct the A101–A103 example, including rejection with a missing-response item and the later dated reconciliation
Owners	Product/quality commissions use investigation and evidence disposition; Site B deployment resolves content; its clinical owner resolves training/fallback; named site/manufacturer owners reconcile missing states
Closure conditions	Investigate and disposition both use difficulties; supply implementation and renewed human-use evidence for resulting package changes. Resolve installed content, every intended user's readiness and operational monitoring separately. The memo supplies neither manufacturer release nor market authorization

The ownership row turns a general concern into work. Product/quality owns the commission for the use investigation; Site B's deployment owner identifies what is installed; its clinical owner addresses users and fallback; site and manufacturer owners accept missing-state reconciliation. The last row says what later evidence would have to establish. The revised-presentation investigation, release and site records can subsequently be checked against that condition, but those outcomes were not known when the initial hold was recommended.

This is why a hold can permit progress. A training signature cannot replace evidence about current/prior identity, and a software pass cannot investigate the observed use difficulty. Each completed relationship can change its corresponding finding. Continued operation also needs the ledger's data-use and retention permissions, a route for delayed references if clinical performance is later claimed, and outage support. Operational counts supply no clinical reference; the manufacturer release preauthorizes no old installer as safe rollback.

A proposed pathology site with an installed queue aid but an unresolved import handoff would need import-owner and case-link records before extended hours could be supported. The reviewer would also check whether the recorded queue timing reaches the claimed endpoint. In sequencing, workflow expansion would need a specimen-reconciliation owner and a missing-suggestion route, while review would compare delivered version with the actual report-decision record. These are prospective applications, not new incidents in those episodes.

Reviewer Lens. I would test the initial hold and any later closure against the underlying use observations, installed identity, all intended users, source/fallback and reproducible receipt ledger. A favorable row is valuable for its own question; it cannot replace another row's missing evidence.

Consultant Lens. Owners can investigate the product, repair local gaps, deliver information and establish monitoring independently before a changed disposition is recommended. Sequencing those concrete closures gives the manufacturer choices without turning the consultant's memo into approval authority.

Would three signatures answer three different questions?

What observable closure condition would you use for product release, site readiness and monitoring readiness? Each needs an underlying record rather than the same phrase, “management approval.”

Pause — what would each of the three records have to show?

Three different observations behind the decisions

For product release, the record needs a documented disposition of both use observations and a controlled whole-package decision. For each site, it needs comparison of installed content, readiness of all intended users, source access and staffed fallback with supported conditions. For monitoring, it needs prospective receipt/census rules, named reconciliation owners and an urgent-event route before field observation starts.

Those are different observations to which signatures can point. None itself supplies national clinical-use authority, which remains with the applicable jurisdiction. A form repeating “management approval” for every row would conceal rather than resolve the differences.

Transfer: the same installation overnight

After local acceptance, Site B proposes overnight operation with the same installed baseline. The daytime escalation role is unavailable at night, and qualified fallback is staffed only during the original hours.

Does the unchanged software preserve the deployment argument? Which control has changed?

Pause — what would need to be established before the longer hours could be accepted?

When unchanged bytes meet changed work

The evidence supported a site arrangement with available escalation and qualified fallback. Overnight operation removes those conditions. The manifest can establish unchanged software, but it cannot establish that an unresolved item has an available responsible professional or that full review can continue.

Manufacturer and site owners therefore need an impact assessment, named overnight escalation, an adequate review/fallback arrangement and checks of the changed task before accepting expanded hours. This is not proof that overnight use can never be supported or that the product has a new defect. Providing and checking the missing operational controls could change the decision.

Reviewer Lens. I would compare the accepted hours with the proposed ones and ask who can handle identity concerns, unresolved packages and full review overnight. Those were part of the supported use, so software identity answers only one part of the question.

Consultant Lens. Site clinical, deployment and manufacturer-support owners can design and test the overnight control path. Its acceptance should be recorded separately from the unchanged package identity, so a later reviewer can see what actually changed and why use remained supportable.

What carries into field observation?

The receipt-monitoring protocol, MON1, precedes the field observations in Chapter 15. Its fixed questions and accounting rules let the reader distinguish a signal from a conclusion chosen after the numbers were seen.

For optional depth, Book 1C addresses transfer and reopening evidence; Book 1A develops monitoring and evidence coherence; Book 1B Chapters 5–8 cover release, dependencies, security testing, maintenance, disclosure and support. Book 2 Chapters 6, 11, 14 and 15 address European Union documentation and lifecycle questions.

Relevant security and support work remains part of release and site readiness alongside the local baseline, transfer and monitoring methods.

Can the receipt still be followed?

Which installed product does the record identify beyond its classifier name? How does it separate quality acceptance, processing availability, professional closure and clinical correctness? Can unresolved work be followed across windows without a second new receipt? Is fallback actually available when needed?

Pause — which record and responsible role would support each answer? A bare “yes” leaves the release-to-monitoring argument unexamined.

Source Trail

[S1] IEC 80001-1:2021 official scope and publication record, organizational risk management for connected health systems. The September 24, 2026 check identifies published Edition 2 at stage 90.92, to be revised; this is not withdrawal. Only the scope-level coordination point is attributed here; detailed conformity requires the applicable full-text analysis, and revision status must be rechecked before later use.

[S2] Joint transparency guiding principles, “Where” and “When,” for placement and lifecycle timing of information.

The release/deployment artifact, MON1 rules and triggers, U2 change, renewed fictional evidence, REL1, and site records are author-designed teaching constructions. Their counts and time periods are not universal requirements. Detailed manufacturer information, assurance-case, health-software product-safety, and cybersecurity claims remain routed to their specialist sources and Books 1A/1B/2 rather than asserted through a scope citation.

CHAPTER 15

Chapter 15: Drift, Incidents, Retraining, PCCP, CAPA, and Retirement

A candidate-crop package reaches a LumaCell site. Its quality state is rejected, so the professional receives no ordinary suggestion and continues through the laboratory's complete-smear-review route. The receipt is countable. Its cause is still a question: did acquisition produce a difficult crop, did the quality gate behave unexpectedly, or is a required record wrong? Nothing in the rejection alone tells us whether a patient was harmed.

Now the next monitoring window shows a larger rejected share. What decision should follow? First the manufacturer needs to know whether the comparison counts the same kind of work under the same rules. Then it can examine changed workload, inputs, product behavior and professional practice. Meanwhile, a credible identity or service concern may require immediate protection. Later in this chapter, a supplier proposal and a support notice will add decisions of their own. Their arrival will not answer the earlier causal question.

The useful output is a **lifecycle change packet**: a record connecting the observed signal to its denominator, urgent actions, investigation, risk, evidence and accountable disposition. It also connects any proposed change to data eligibility, authority, deployment and support. The packet may justify keeping the current product while assigning considerable investigation and transition work. Its value lies in those supported decisions, rather than in the number of completed fields.

What exactly did monitoring count?

Chapter 14's receipt-monitoring protocol fixes how received work enters a comparison before anyone sees the next result. Its engineering record is MON1. Suppose a package is first delivered, retried after an interruption, then replaced by a newly acquired linked package. How many attempts enter the next receipt cohort? What happens to unresolved work from an older cohort? If one required crop is rejected while another response is missing, which package-quality state applies? What action follows an increase in the primary rejection measure?

Follow the attempts through the census

A first received package creates one attempt. A delivery retry belongs to that attempt; reacquisition creates a new attempt linked to the earlier one. The monitoring contract groups attempts first received during a fixed 28-day window and records their quality states seven days after the window ends. That later census allows a consistent observation point. It does not turn an old attempt into a new receipt: older unresolved work remains in its own backlog, with dated reconciliation and an owner.

At census, a known rejection for any required crop makes the package **rejected**, including when another required response is missing. Without a known rejection, an empty or incomplete required set, or a missing or unavailable required status, makes quality **unresolved**. **Accepted** requires identified, complete, nonempty membership and acceptance of every required crop. Thus retrieving a missing response can resolve an unresolved state; it cannot erase an already known rejection. Missing-response events inside rejected packages still need records. Processing outages also need records, because accepted quality does not establish usable current output, completed professional review or clinical truth.

The primary measure divides rejected attempts by **all new receipts**, retaining unresolved attempts in the denominator. A pooled **or site-specific** increase of **more than three percentage points** against the preceding comparable window opens analytical review. Exactly three points would not satisfy this particular trigger. Neither outcome settles cause, incident classification or whether a product update is warranted; other urgent routes remain available.

A reviewer can test the chart by reconstructing one first receipt, its quality-state precedence and census, then tracing old backlog outside the new-receipt denominator. The chart is interpretable only if those rules survive the aggregation.

The consultant's corresponding question is practical: can the site and manufacturer ledger owners retrieve those event links and the frozen census? If not, reconciliation precedes a claim about the trend.

What changed, and what would explain it?

For the rejected package, three statements do different jobs. “The ledger records a rejection” reports an observation. “A changed stain setting caused it” proposes an explanation that acquisition and product records could test. “Replace the gate” proposes an intervention whose effects need evidence. Keeping those statements separate prevents a plausible explanation from acquiring the authority of a tested finding merely through repetition.

What does someone mean when they call the change **drift**? Inputs may have shifted—for example, more crops now come from one supported acquisition condition. Outputs may be distributed differently—for example, more packages are rejected. Performance against a suitable reference may also have changed. These are three questions with different evidence. The first recalls distribution shift, a difference in input conditions relative to earlier experience; observing it does not establish the third. A rejection ledger without linked clinical references cannot measure diagnostic deterioration or establish patient harm.

Why investigate an operational observation when harm has not been shown? ISO 14971 asks the manufacturer to collect production and post-production information, review its relevance to safety, and act on safety-relevant findings. A previously unrecognized hazardous situation or changed acceptability of an existing risk would make the risk record need reassessment. The standard does not diagnose this rejection or choose a national reporting deadline. [S1] The investigation therefore starts with coding and cutoff, then compares site and case mix, the acquisition profile, installed content, user practice, available references and relevant infrastructure or security conditions. Each comparison should separate explanations: a changed cohort rule, for instance, would call for a different response from changed behavior in the same installed package under the same conditions.

The pathology queue episode supplies a useful reminder: its recorded improvement concerns queue placement after import, while scanner import still dominates total slide-to-review time. Here the new question is which start and finish events a later trend would use. The sequencing review episode similarly separates displayed variant suggestions from specimen reconciliation needed for report release. A monitoring plan defined before use can preserve scanner-to-review and specimen-to-report links. Review after use can then locate a changed stage without treating an already observed local result as a new whole-workflow outcome.

A reviewer asks which statements describe records, which remain hypotheses and which are decisions. For a hypothesis, what observation would strengthen it, and what observation would contradict it?

The consultant can turn that distinction into an investigation with owners for cohort integrity, site/acquisition differences, installed versions and workflow consequences. A change request follows the supported branch; it should not predetermine what the investigation must find.

Which work needs protection before the cause is known?

Suppose a professional reports that a suggestion may belong to the wrong patient. Waiting for a 28-day chart would leave the current item exposed to the very consequence under investigation. The site may need to hold affected work and use its qualified manual route while identity and exposure are checked. This is **containment**: a temporary, bounded protective action before the final cause is known. Unavailable fallback or a credible security compromise can likewise change the urgency, even when the rejection trend has not crossed a threshold.

How broad should the hold be? That depends on which configurations, work items and users might share the possible problem and what can continue safely. The record needs that scope, the temporary decision-maker and the condition or time for review. Evidence that confines a problem to one installation may support a narrower action than evidence of a shared component fault. Later findings can revise the explanation; the original record should still show what was known when protection was chosen.

A complaint, a suspected clinical incident and a possible security event may enter different procedures. A larger rejection percentage supplies none of those classifications by itself. AAMI TIR97, the Association for the Advancement of Medical Instrumentation's postmarket cybersecurity guidance, connects security information to the affected products and their current risk. Evidence of unauthorized access or an exploitable dependency would therefore add questions a rejection count cannot answer. The guidance neither classifies this fictional signal as an attack nor sets a universal reporting deadline. [S2]

In the pathology queue episode, imagine an import outage with no staffed fallback: someone must decide where slides go even if the queue's statistics look ordinary. In the sequencing review episode, imagine a suspected specimen-link error: a report hold and reconciliation may be needed before suggestion rates are compared. These conditions illustrate urgency; neither is a newly reported event in the episodes.

The reviewer follows the potentially affected unit to its possible consequence, temporary control and responsible owner. Does the chosen review timing leave that consequence uncontrolled?

The consultant helps site clinical and manufacturer quality/security owners connect the hold, fallback and reevaluation. Applicable complaint or reporting criteria still determine their own classifications.

Did we fix an item or remove a recurring cause?

A required response fails to arrive. Retrieving it or reprocessing the item may repair that particular record: a **correction**. Holding affected work can contain exposure while the transfer path is investigated. If evidence then identifies a recurring transfer defect, an action directed at that cause may prevent recurrence: **corrective action**. The recovered item does not show that the transfer defect has disappeared; the cause-directed action needs its own result.

What would count as that result? The team needs evidence that recurrence stopped under the relevant conditions, that affected work was reconciled and that the action did not introduce an adverse effect. ISO 13485's corrective-action provisions connect nonconformity review, cause, proportionate action and effectiveness. Its design-change provisions require review, verification, appropriate validation and approval before implementation. [S3] A quality signal can start the investigation without yet establishing a nonconformity. **Corrective and preventive action (CAPA)** names the broader quality-system route; it is not a synonym for every analytics ticket or model update.

For the pathology queue episode, an import constraint points toward investigating source-to-queue transfer before assuming that ranking weights need refitting. For the sequencing review episode, specimen-link evidence points toward reconciliation controls before assuming a variant-model defect. If later post-use evidence establishes a recurring defect, it can support the relevant cause-directed route. A better model metric alone does not show that this cause was removed, that the intervention was effective or that the released product may be changed.

A reviewer can follow one affected item through correction, cause evidence, action, checks for new problems and effectiveness. Which link is still proposed, and which has an observed result?

The consultant makes the handoffs explicit: the ticket owner repairs the item, the investigator tests cause, quality/risk owners choose the route and a later assessor evaluates effectiveness. Complaint and security records remain linked where relevant.

What changes along the whole product path?

A gate determines which inputs reach a classifier. Changing that selection can send unfamiliar crops to unchanged weights. The product's earlier result may then answer only what happens to the old selected inputs. Image preparation can similarly change pixels, a display default can change what a professional notices, and lost supplier support can remove a dependency of safe operation. In each case the useful question is where the proposed before/after path differs, rather than whether the classifier's name stayed the same.

The change description follows component and content identity through the input envelope, intended purpose, ordinary outputs and non-results, professional task, dependencies, instructions and monitoring. For each affected link, can earlier evidence still answer the question directly? Does a specific comparison justify a bridge? Or has the question reopened? Chapter 11's package checks, Chapter 12's selected-crop evaluation and Chapter 13's human-use observations concern different links. A claim about one cannot fill an unobserved link in another.

Suppose a proposal promises fewer rejections. More suggestions would help only if newly admitted inputs receive appropriate handling and the resulting errors, non-results, review burden and fallback remain acceptable. That requires a comparator and criteria specified before the new results. In the pathology queue episode, an import-adapter change could alter who enters the queue without changing ranking. In the sequencing review episode, foregrounding a suggestion could alter reliance without changing variant computation. Those are prospective change questions. A later discrepancy must be joined to the version actually used before either proposed mechanism becomes a causal finding.

The reviewer follows one newly affected input through admission, computation, state display and professional action. At which link is old evidence still direct, where is a bridge justified, and where is fresh evidence needed?

The consultant asks supplier, software, clinical and human-use owners to agree on that path and comparator. Their component, integrated and use questions then have a reasoned order before the manufacturer considers release.

May these records be used to fit another classifier?

An operations team holds records to monitor an installed service. That explains possession, not permission to fit a new classifier. The proposed use must meet its permission and retention conditions. The manufacturer also needs source-to-crop lineage, the links through which each input was derived, and patient/specimen grouping so related observations stay together. A suitable reference-label method, relevant population coverage and protected development/evaluation separation determine whether permitted data can answer the scientific question.

Why cannot a professional override simply become the training target? It may reflect a correct correction, a local convention, a misunderstood display or influence from the product's own suggestion. The reference standard, the specified method for assigning the comparison target, must resolve that problem. The team must also know which records have already informed candidate selection. Repeatedly inspecting a field cohort during development uses information from it; calling some of its rows "test" later does not restore data independence.

ISO/TS 24971-2:2026 gives risk guidance for machine learning in medical devices and distinguishes controlled retraining from continuously optimizing parameters during patient use. Its scope excludes devices using large language models or other generative artificial intelligence. The specification discusses broader monitoring considerations, but this chapter's predictive classifier remains fixed during use; no online updates occur. [S4] A proposed retraining study needs an identified candidate, controlled experiment, suitable independent evaluation, risk questions and a stop rule. If investigation instead locates an acquisition, identity or workflow problem, changing that link may be better supported than changing weights.

The pathology team would need permissions and a defensible clinical-priority reference for scanner/import cases; elapsed time alone does not label clinical priority. The sequencing team would need a specimen-linked, independently assessed variant target; an accepted suggestion is not automatically that reference. Logs can identify where to investigate while remaining ineligible for the proposed fitting use.

The reviewer asks which records are permitted and fit for the proposed use, which have informed model choice and which comparison remains protected. These are separate gates.

The consultant connects data stewardship, reference and machine-learning owners before experiment planning. A no-retraining option remains available while the causal question is open.

Which plan could govern the proposed modification?

Suppose a manufacturer wants specified future modifications to proceed through a route settled in advance. The changes, their evaluation and their implementation must be bounded: what may vary, within which conditions, and what evidence must be produced? This is the purpose of a **predetermined change control plan (PCCP)**. An internal draft helps planning. Whether a particular plan provides a regulatory route depends on its actual authorization.

The first question is **authority**: which jurisdiction's decision applies to this device, and which plan version does that decision identify? A plan saved in the manufacturer's folder does not establish that it was authorized with the device.

The second question is **scope**: does the described modification include the actual before/after behavior, population, data, function and operating limits? A title such as “threshold update” leaves those details unanswered. A change in an input envelope can matter even when a team gives it a familiar maintenance label.

The third question is **protocol**: did development, evaluation and implementation follow the prescribed methods and satisfy the prospective criteria? A change can match the description yet fail this test. Passing an aggregate metric does not cancel a failed

required condition. Several changes also need consideration together, because their combined effects on the product, users and monitoring may differ from each change considered alone.

IMDRF N90, the International Medical Device Regulators Forum's final document published August 6, 2026, describes this architecture through a change description, change plan and impact assessment. It links identified versions and prospective evaluation criteria to implementation, communication and individual/cumulative effects. These are nonbinding international principles, not national requirements or permission in every market. [S5] The authority, scope and protocol questions still have to be answered for the proposed product.

A prospective plan for the pathology queue episode might address an import-path change and specify queue-to-review checks. A later scanner proposal would need comparison with that exact scope. A sequencing specimen-link presentation change could require different evidence from a variant-scoring change. These illustrations do not give either episode an authorized plan.

The reviewer needs the authorized version, the actual modified product and the completed protocol records. Which required condition, if changed or failed, would remove the basis for proceeding under the plan?

The consultant first asks regulatory, product and evidence owners for the authority record, then maps each proposed change to its scope and tests. Missing records leave coverage unassessed.

What does the applicable national change rule ask?

Consider the United States as a labeled jurisdictional example. The cited 2024 compilation of 21 U.S.C. 360e-4 distinguishes a plan **approved** with a device in the premarket approval route, or PMA, from a plan **cleared** in the premarket-notification route commonly called 510(k). Each provides its respective submission exception for a change consistent with that plan. The authority comes from the applicable mechanism and device record, not from internal approval of a document. [S6]

FDA's August 2025 guidance gives nonbinding recommendations for artificial-intelligence-enabled device software. It recommends planned modification descriptions, development/evaluation/implementation methods and impact assessment. Its consistency test has **two required parts**: the change is specified in the authorized description **and** it is implemented according to the modification protocol. A matching description with a skipped required evaluation does not satisfy that test. Either an omitted change or a protocol departure sends the manufacturer to the applicable ordinary change assessment. The guidance supplies neither device authorization nor a universal sample size. [S7]

What does that ordinary assessment ask? Under the applicable **510(k) rule**, 21 CFR 807.81(a)(3), a modification to an already marketed device requires premarket notification if it could significantly affect safety or effectiveness, or makes a major change to intended use. Paragraph (b)(1)(ii) supplies the exception for consistency with a cleared PCCP. Thus an outside-plan change needs evaluation of its actual potential effect and use; being outside the plan alone does not answer that test. [S8]

For an **approved PMA device**, 21 CFR 814.39(a) generally requires FDA review and approval of a supplement before a change affecting safety or effectiveness. Indications, labeling and performance/design specifications are among the possible affected features. Paragraph (b) separately addresses changes that do not affect safety or effectiveness, with required periodic reporting, and changes consistent with an approved PCCP. Other specified alternatives have their own conditions. This is not the 510(k) test with a different form name. [S8]

For a hypothetical gate modification, evidence that its added inputs could significantly affect safety or effectiveness matters to the 510(k) branch; for an applicable PMA, an effect on safety or effectiveness engages its supplement framework. Evidence of no such effect, a qualifying exception, or consistency with the applicable authorized plan changes the assessment. A manufacturer cannot select among those branches from “weights unchanged.” It needs the actual device route, change effects and relevant conditions. Neither route has been assigned to LumaCell.

The submission question does not suspend quality-system duties. The Quality Management System Regulation took effect February 2, 2026. Current Part 820 combines applicable incorporated ISO 13485:2016 requirements with U.S. provisions; controlling U.S. law prevails where there is a conflict. Its complaint-record provisions require review/evaluation/investigation records for covered complaints, including a documented rationale where an earlier similar investigation makes another unnecessary. Closing an analytics ticket therefore does not itself close complaint handling. Other reporting criteria still need their own assessment. Older section numbers in the 2025 PCCP guidance cannot substitute for this current text. [S9]

Health Canada's April 2026 guidance principally addresses new or amended applications for Class II–IV machine-learning-enabled medical devices. It distinguishes changes according to an authorized PCCP from changes needing outside-plan assessment, while other licence-amendment and postmarket obligations can remain. It connects protocols to data, risk, evaluation, updating and monitoring, with collective and workflow effects in the impact assessment. A U.S. plan does not establish the required Canadian decision. This is a guidance comparison; no Canadian class or legal disposition is selected here. [S10] European qualification, change and postmarket details remain a Book 2 handoff, without an artificial-intelligence-law date or permission claim in this exercise.

For LumaCell, the jurisdiction, device authority record and authorized plan have not been supplied. **Coverage is unassessed.** That is neither a clearance nor proof that the change is prohibited or that a plan does not exist. The reviewer asks whether any claimed permission can be traced to authority, scope and protocol. The consultant asks the regulatory owner to obtain that chain before promising rollout; clinical and product owners still assess whether the evidence supports the changed product.

How would the site receive, recover from or retire the change?

Suppose a future candidate meets its development criteria. How will the next received item remain attributable to what actually ran? Deployment planning identifies the controlled package, eligible sites and environments, installation/configuration checks, affected users, instructions, communication and version-linked monitoring. A pause condition is set before rollout. Evidence that the installed content or site conditions differ from those accepted would stop that progression even if the earlier evaluation passed.

Clinical fallback determines how professional work continues or is held through the qualified manual route. **Rollback** instead returns software to an earlier identified version. The old package may retain a defect, depend on unsupported components or use an incompatible data/state format. ISO/TS 24971-2 treats that return as a risk decision: returning should not introduce new risks or increase existing ones. [S4] The target, compatibility, state recovery, current risk and authorizing owner therefore need assessment before the old installer is used.

Support adds another decision. In AAMI TIR97, **end of guaranteed support (EOGS)** ends the promise of full support; some limited or best-effort service may continue. **End of support (EOS)** ends service support. [S2] The actual intervening terms matter: continued vulnerability response and compatibility help would support different options from their absence. Neither milestone demonstrates a replacement's adequacy or sets the integrated device's retirement date. A transition, continued-use limitation or retirement decision needs affected-site inventory, record and credential handling, replacement checks, fallback and confirmation that obsolete configurations cease operating. Applicable legal duties depend on the product and setting.

For the pathology queue episode, support planning would preserve which import adapter served each site while alternatives are evaluated. Prior use would not make an old adapter an acceptable rollback target today. Sequencing report operations would preserve specimen-link state and a qualified report route during migration. Later complaints could then be traced to the installed version and handoff rather than attributed retrospectively to an undifferentiated model label.

The reviewer follows one item through rollout, pause, fallback and any proposed rollback, then asks what supplier support actually remains at the claimed date.

The consultant joins manufacturer support, deployment and security owners with site clinical owners. A supported transition, including records and retirement where needed, can be evaluated before an expiring dependency forces an improvised product choice.

Can the lifecycle packet join the decisions?

The author-designed lifecycle change packet has fourteen fields because one decision crosses several kinds of records. Its first rows establish what happened, what needs protection and what the investigation actually found. Its middle rows ask whether a particular change has eligible data, adequate evidence and applicable authority. Its final rows ask how sites could receive it, recover or leave service. The table is an index to the underlying records, not a regulator-prescribed form. Following the rows in that order prevents a supplier proposal from entering as an already proven corrective action.

Field	What must remain reconstructable
Signal and denominator	Observation, cohort, state, cutoff, configuration, and missingness
Triage	Urgency, affected scope, initial protective action, and routes
Investigation	Competing explanations, evidence, uncertainty, and findings
Risk review	Hazards, controls, consequences, and residual-risk questions
CAPA	Applicable quality-system route, cause-directed action, and effectiveness
Change description	Actual product, data, workflow, supplier, or support modification
PCCP coverage	Identified jurisdiction and authorized plan, scope, protocol, and conditions
Data and retraining eligibility	Rights, provenance, reference, partitions, and intended use
Verification and validation	Retained, bridged, reopened, and new evidence
Regulatory assessment	Applicable route, obligations, unresolved questions, and owners

Field	What must remain reconstructable
Deployment and communication	Controlled rollout, recipients, instructions, and monitoring
Rollback	Acceptable target, compatibility, state handling, and recovery evidence
Retirement or end of support	Support scope, dates, dependencies, migration, and communication
Decision and owner	What is decided now, conditions, rationale, and accountable authority

Consider the link between investigation and decision. An investigation may establish that two records describe different receipt cohorts; the disposition may then require a corrected comparison rather than a model change. A useful calculation can also coexist with an unapproved candidate. The reviewer follows one finding to its underlying evidence and decision authority. The consultant identifies the next missing record and owner. Neither needs to manufacture a passing outcome to make the packet useful.

What do the two receipt windows show?

The next records concern the fictional released LumaCell baseline from Chapter 14. Its engineering identity is LC1.0: evaluation classifier M1, original quality gate Q0, identified preparation procedure P1, revised presentation U2 and identified runtime R1, under acquisition profile AP1, packaged as BUILD-U2. **That installed product stays fixed across both windows.** There is no model, gate or product update between them. The receipt-monitoring protocol MON1 keeps its 28-day first-receipt cohort and census seven days later. Delivery retry remains within an attempt; reacquisition is a new linked attempt.

Each row below is a cohort of new receipts. Accepted, rejected and unresolved partition **package quality at census**, so their sum must equal new received. The last column answers a different question: how many quality-accepted attempts had a closed professional review? It is conditional on acceptance, not a fourth quality state and not a count of clinically correct decisions. These distinctions determine the denominators in the first decision.

Receipt cohort	New received	Accepted	Rejected	Unresolved	Accepted attempts with review closed
W0	1,000	900	80	20	850
W1	1,200	1,000	150	50	910

The 20 unresolved attempts in the earlier window, W0, do not join W1 as new receipts when someone works on them later. Their subsequent status belongs in a dated backlog record. Quality rejection can coexist with a missing required response, and accepted quality can coexist with unavailable processing. The exhibit gives **no separate event counts** for those situations. A complete partition of quality states does not mean every operational incident has been counted here.

There are no clinical references, diagnosed causes, patient harms, incident classifications or product updates in these inputs. Recall the pathology distinction between imported cases and all scanner receipts, and the sequencing distinction between displayed suggestions and all specimen-linked runs. A monitoring comparison answers the question defined by its start event and denominator. It cannot silently expand from an internal stage to the complete workflow.

The reviewer first checks the version and census, then re-sums each quality row. Where do processing availability and professional closure remain separately attributable?

The consultant asks site and manufacturer data owners for the version-linked attempt ledger and earlier backlog. Those are the records needed to investigate the coming comparison.

First decision: does the pooled comparison open review?

For each window, what fraction of all new receipts was rejected? What fraction of known quality outcomes—accepted or rejected—was rejected? What share of new receipts remains unresolved? Among quality-accepted attempts, what share had a closed professional review?

Does the primary pooled comparison meet the monitoring contract's trigger? What is one supported observation, and which two tempting conclusions would require evidence absent from this table? The site breakdown comes after this decision and its worked reasoning.

Compare the pooled measures and the trigger

What the pooled calculations support

The primary measure asks about all new received attempts, so unresolved attempts stay in its denominator:

- W0: $80/1,000 = 8.0\%$ rejected.
- W1: $150/1,200 = 12.5\%$ rejected.

Their difference is $12.5\% - 8.0\% = 4.5$ percentage points. That is **more than three**, so the pooled rule opens analytical review. This is a difference between

percentages, not a percentage increase relative to the earlier value. The required action is to investigate. The calculation identifies no cause and authorizes no update.

The known-state measure conditions on quality being resolved at census. Its denominators are $900 + 80 = 980$ and $1,000 + 150 = 1,150$. Rejection among those known outcomes is therefore $80/980 = 8.16\%$ and $150/1,150 = 13.04\%$. Removing unresolved receipts makes this a different question. Substituting these percentages for the primary measure would change the monitoring contract after seeing results.

Unresolved quality uses all receipts: $20/1,000 = 2.0\%$ in the earlier window and $50/1,200 = 4.17\%$ in the later one. Closed review instead uses accepted attempts: $850/900 = 94.44\%$ and $910/1,000 = 91.0\%$. The lower closure share describes the state of professional work at the relevant observation; it does not say whether any completed judgment was correct. None of these calculations recovers the absent counts of missing responses inside rejected packages or processing outages among accepted packages.

The exhibit supports a pooled rejection increase beyond the prespecified review trigger and a higher unresolved share. It does not establish classifier deterioration, patient harm, the reason for rejection or the benefit of a replacement gate. The next evidence request is concrete: preserve the two cohort snapshots, reconcile open work and obtain the site/source breakdown needed to investigate the change.

A reviewer can reproduce the conclusion by matching each numerator to its eligible denominator and checking the point difference. Clinical outcomes remain a separate evidence question.

The consultant assigns the site/acquisition and backlog breakdown, including clinical-urgency assessment of unresolved work. Engineering now has an investigation question, not a diagnosed defect to fix.

New evidence: how are the same receipts distributed across sites?

The investigation now supplies site rows for the **same** two cohorts. For a within-site rejection rate, the numerator and denominator must both come from that site and window. The rows are not four additional datasets. The final column still counts closed professional review among accepted attempts.

Window and site	New received	Accepted	Rejected	Unresolved	Accepted review closed
W0, A	800	750	40	10	710

Window and site	New received	Accepted	Rejected	Unresolved	Accepted review closed
W0, B	200	150	40	10	140
W1, A	600	550	30	20	500
W1, B	600	450	120	30	410

Do the two sites recombine to each earlier pooled row? What is each site's rejection fraction in each window? Then consider a controlled arithmetic comparison: hold the earlier receipt mix fixed at 80% Site A and 20% Site B while using the later within-site rates. Multiply each later site rate by its earlier share of receipts and add the contributions. This **standardized comparison** asks what the later pooled proportion would be under the old mix. It leaves the actual later counts untouched.

What would agreement or disagreement between that old-mix comparison and the earlier pooled rate tell us about the investigation? Would either result settle why a site rejects its inputs or whether the current burden is acceptable?

Separate site rates, receipt mix and unresolved work

What the site comparison changes

The split reconciles. Earlier receipts total $800 + 200 = 1,000$, with $40 + 40 = 80$ rejected. Later receipts total $600 + 600 = 1,200$, with $30 + 120 = 150$ rejected. Accepted, unresolved and review-closure counts also add to their earlier pooled totals. We are explaining the same observation.

Within Site A, the rejected fractions are $40/800 = 5\%$ and $30/600 = 5\%$. Within Site B, they are $40/200 = 20\%$ and $120/600 = 20\%$. Both recorded primary rates remain unchanged. The weighting changes: A/B supplies 80%/20% of earlier receipts, but 50%/50% of later receipts. The later pooled rate gives more weight to the site whose recorded rate is higher.

The earlier weights are 0.80 for A and 0.20 for B, summing to one. Multiplying the later rate for each site by that site's earlier weight gives its contribution to the old-mix comparison:

$$0.80 \times 5\% + 0.20 \times 20\% = 4\% + 4\% = 8\%.$$

Under the old site mix, the later rejection proportion is 8%, equal to the earlier pooled rate. The actual 12.5% arises arithmetically because half of later receipts, rather than one fifth, comes from the higher-rate site. This is a positive finding about the supplied site strata. It does not explain why work shifted, why Site B rejects 20%, whether case mix within either site changed or what the consequences were. The original pooled

trigger was correctly opened. Explaining its arithmetic does not retrospectively cancel it. Neither site-specific rejection rate rose by more than three points, but that does not close other investigative or urgent routes.

Does the same explanation cover unresolved quality? Site A moves from $10/800 = 1.25\%$ to $20/600 = 3.33\%$ (rounded); Site B remains at $10/200 = 5\%$ and $30/600 = 5\%$. Applying the old weights to the later unresolved rates gives:

$$0.80 \times (20/600) + 0.20 \times (30/600) \approx 3.67\%.$$

The earlier pooled unresolved rate was $20/1,000 = 2\%$. Holding site mix fixed therefore does **not** remove this increase. Site A's within-site unresolved share has changed as well as the receipt composition. Its reason remains unknown. Earlier unresolved backlog still requires dated reconciliation outside the later new-receipt count; review closure remains a separate workflow state.

The packet can now credit two findings: site composition accounts for the pooled rejection difference at this level of grouping, and the unresolved comparison remains higher after holding that mix fixed because Site A changed. Those findings direct work toward Site B's persistent burden, Site A's unresolved attempts, acquisition and actual workflow consequences. They do not establish “no drift,” clinical acceptability or patient harm.

The same method could help the pathology queue episode if a future pooled scanner-to-queue delay reflected more work from a slower import site. A separate increase in unresolved imports would still need investigation. In the sequencing review episode, a changed specimen-run mix could affect pooled suggestion availability while one laboratory's reconciliation backlog grew. These are prospective analysis patterns, not additional observations in either episode.

A reviewer checks both weighted calculations. Does the conclusion explain pooled arithmetic at the supplied grouping, or quietly claim that Site B's burden and clinical consequences have been settled?

The consultant assigns Site A's unresolved investigation, Site B's persistent rejection review and manufacturer cohort/version reconciliation separately. Each route closes on its own documented evidence and decision.

A proposed gate and two support dates arrive

Only now does a new proposal enter the record. The supplier offers a **replacement quality gate**, Q1, intended to admit more low-contrast inputs rejected by the original gate, Q0. Classifier M1 computation would remain unchanged. The replacement is **not installed**, and the exhibit supplies no comparative whole-product, subgroup, human-use or deployment result for it. The site calculation did not identify low contrast as a cause or demonstrate that admitting such crops is safe.

A separate supplier notice gives fictional dates for the original gate: end of guaranteed support at $W1 + 12 \text{ months}$, and end of support at $W1 + 24 \text{ months}$. The intervening service terms remain unknown. These dates are case inputs, not legal deadlines, recommended notice periods or evidence that the replacement works.

Do the monitoring findings justify immediate installation of the replacement gate? What evidence would the proposed changed input path need, and what decision must the support notice start now even if replacement evaluation remains incomplete?

Separate the replacement decision from the support transition

The supplied evidence does **not** justify immediate Q1 deployment. A gate that accepts formerly rejected low-contrast crops changes what reaches M1, even with unchanged weights. That can affect wrong suggestions, non-results, review load and fallback. The next evidence request includes component/data identity, the actual before/after input and output envelope, risk review, gate and package verification, performance on newly admitted conditions and relevant groups, and human-use/deployment checks where affected. The regulatory owner also needs the actual jurisdictional change route and any authorized plan before implementation.

The support notice creates parallel work. Manufacturer and site owners need the actual interim terms, dependency inventory, current original-gate risk and feasible replacement, limited-use or retirement options. A qualified clinical fallback does not require pretending that any old installer remains acceptable. Neither support date demonstrates the replacement's adequacy or supplies permission to roll it out.

The reviewer asks whether the proposed gate addresses a supported cause and whether newly admitted crops have adequate whole-product and use evidence. A supplier schedule answers neither question.

The consultant opens two owned tracks: replacement development/evaluation and original-gate support transition. Different progression criteria keep the latter from becoming an improvised release criterion for the former.

What the initial packet can say today

The following completed example is a **proposed initial packet**. It is not a finished investigation, accepted risk decision or approved replacement release. The first rows can contain calculated findings. Later rows must retain the missing evidence, owner and decision condition. Reading that distinction across the fourteen fields is the point of the table.

Field	Worked initial entry
Signal and denominator	MON1 W1: 150/1,200 rejected, compared with 80/1,000 in W0. Preserve each receipt cohort and cutoff, plus W0's separate unresolved backlog. Report accepted-review closure separately.
Triage	Open the required pooled analytical review. Assess unresolved work and any associated clinical concern through the site's defined route; do not wait for a completed statistical explanation where urgency requires action. No incident, attack, or patient harm is established by the exhibit.
Investigation	Record the site-mix reconstruction of rejection. Investigate the persistent Site B burden, Site A's increased unresolved share, backlog aging, acquisition/profile conditions, and actual workflow consequences. Link received-attempt and version records.
Risk review	Clinical and quality/risk leads assess whether rejection, delay, unresolved status, or altered fallback creates a new or changed risk. The arithmetic does not establish acceptability or prove a particular harm.
CAPA	Determine the applicable quality-system route from the investigation. Do not label Q1 as effective corrective action before a cause and an effectiveness question are established. Keep any complaint or security record distinct and linked.
Change description	Q1 would replace Q0's learned quality decision and admit some low-contrast inputs previously rejected. M1/P1/U2/R1 remain proposed as unchanged. Q1 is not installed; assess the accepted-input and workflow consequences, not just the component label.
PCCP coverage	Unassessed: jurisdiction, device authority record, authorized plan version, change description, protocol, and conditions are not supplied. Regulatory lead obtains and assesses them; no coverage conclusion is made.
Data and retraining eligibility	Request provenance, permissions, reference strategy, and controlled development/evaluation separation for Q1's evidence and newly accepted input conditions. This Q1 packet does not authorize M2 retraining or repurpose monitoring records.

Field	Worked initial entry
Verification and validation	Plan gate/interface and package-state verification; assess M1 behavior on newly accepted conditions, relevant strata, wrong suggestions, non-results, and changed human-use/fallback burden. Identify retained evidence and justify each bridge. No new passing result is supplied.
Regulatory assessment	Regulatory and quality owners assess the actual change and any separately triggered reporting or field-action question under the selected jurisdiction. Neither REL1 nor MON1 supplies that determination.
Deployment and communication	No Q1 rollout is authorized by this packet. A later accepted candidate needs a controlled baseline, eligible-site conditions, instructions/training where affected, installation checks, and a version-linked monitoring plan.
Rollback	Assess an identified target against current risk, compatibility, state handling, and support. Retaining Q0 is not an indefinite safe-use conclusion; clinical fallback remains a distinct option.
Retirement or end of support	Record Q0 EOGS at W1 plus 12 months and EOS at W1 plus 24 months. Obtain the intervening support scope and develop a supported transition or retirement plan with affected sites. The dates do not establish Q1 adequacy.
Decision and owner	Proposed disposition: do not implement Q1 from the supplied evidence; continue the defined investigations and change evaluation. The manufacturer product owner retains the progression/release decision with the required clinical, quality/risk, software/security, human-factors, and regulatory contributions. Sites retain their local clinical-work decisions.

The top rows join counting to action. The all-received denominator makes the trigger reproducible. Triage keeps unresolved work and urgent concerns visible while investigation records the bounded composition finding. Persistent Site B burden and Site A's unresolved increase remain open. Risk review asks which consequences matter; the corrective-action row needs cause and effectiveness evidence before a supplier offer can be called a remedy.

The middle rows follow the proposed gate through eligible evidence, its own behavior, classifier behavior on added inputs and the professional's review/fallback path. They then ask for jurisdiction and plan records. The manufacturer's earlier release record, REL1, concerns the existing LC1.0 product; it does not authorize this modification. The final rows anticipate controlled deployment, assessed rollback and support

transition. The present no-implementation proposal is compatible with continuing all those investigations. It does not falsely declare them complete.

A later packet revision should be able to show exactly which new record changed a row and who accepted the next progression. Replacing all open entries with “approved” would lose both the useful site-mix finding and the unresolved product question.

For this worked packet, the reviewer distinguishes completed calculations from proposed actions in every row, particularly where cause, eligible data, new-product results and plan authority remain absent.

The consultant gives each open row an owner, a record to obtain and the decision that record could support. Site investigations, candidate evaluation and support transition retain separate completion conditions.

Figure 15.1 — A Signal Opens an Investigation. Monitoring detects a question; it does not authorize learning or change without investigation, evidence, risk and release control.

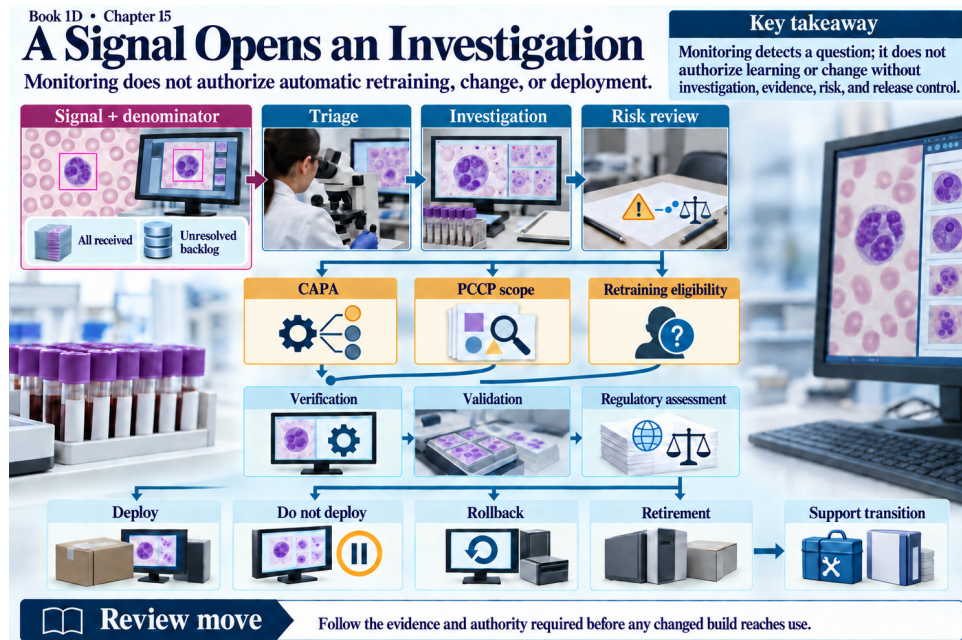


Figure 15.1. A Signal Opens an Investigation.

Who owns the resulting decisions?

The manufacturer owns the integrated product disposition: investigation and triage, quality/risk routing, data use, evidence for the changed package, regulatory assessment, release/rollout and support strategy. The site owns local clinical work and escalation under its deployment arrangement, supplies observations and can hold local use when warranted. The supplier contributes component evidence and support terms. Offering a replacement does not make it an accepted LumaCell product.

A consultant can reconstruct and challenge these links while leaving CAPA, plan coverage, retraining, rollout, rollback and retirement decisions with their accountable owners. A useful handoff states the bounded present finding, the unresolved question and the person responsible for deciding it.

How would the consultant challenge the monitoring and change argument?

The **monitoring and change-control challenge memo** turns the packet into advice about three decisions: interpreting the later window, assessing the uninstalled gate and preparing for the original gate's support horizon. For each, which supplied fact supports the current position, which fact would change it and which owner must obtain that evidence? The following worked example makes those questions concrete.

Read the worked advisory chain

Which part of the signal is explained?

The primary rejected share rises from 8.0% to 12.5%, crossing the pooled trigger. The actual A/B mix changes, while the old-mix later rejection comparison is 8.0%. That explains the pooled arithmetic at the supplied site level. It supplies no cause for an individual rejection. The earlier unresolved backlog remains outside later new receipts, Site A's unresolved increase remains visible, and Site B's persistent burden still needs review. Both “the classifier deteriorated” and “nothing needs attention” would exceed what the investigation supports.

Which exact change would a plan need to cover?

The regulatory owner needs the jurisdiction, device decision, authorized plan version, specified modifications and evaluation/implementation protocol with its conditions. None is supplied for this proposed replacement, so coverage remains unassessed. The fictional manufacturer release of LC1.0 does not supply national authority for a different gate.

The comparison is between a gate that rejects certain low-contrast crops and a gate proposed to admit more of them upstream of the same classifier. A plan that mentions

threshold adjustment or retraining may or may not describe that actual change. Even a matching description would leave the protocol question: were the required methods followed and conditions met? Changing that fact changes whether the plan can support the proposed progression.

Where does the added crop reopen evidence?

An added crop creates a connected series of questions. Does the new gate accept it for the intended reason? How does the classifier route it, including wrong ordinary suggestions and other/unclassified? Does the package show the correct current state? Can the professional still review, correct and use fallback appropriately? Component, integrated, selected-input, subgroup and human-use evidence address different parts of this path. The earlier LumaCell 600-crop evaluation, EVAL600, was retained for unchanged selected-crop computation in the released baseline. It did not study these newly admitted inputs. Unchanged weights cannot extend its scope automatically, and an improved pooled number cannot close every affected link.

Which other routes still need decisions?

An analytics review does not classify a particular report as a complaint, establish a cause-directed corrective action, diagnose a security issue or settle national reporting. Each responsible owner applies the relevant facts and criteria and preserves the linked record. The current table establishes no incident, attack or harm. A later urgent report would add evidence requiring its own assessment, even if the pooled arithmetic had been explained.

The support notice asks software/security and product owners what help and vulnerability response remain between the two milestones. That evidence affects the original gate's continued-use options independently of whether its proposed replacement ever passes evaluation.

What evidence would move each option forward?

Investigating while retaining the original gate under its currently supported conditions is one near-term option, with site differences and unresolved work actively assessed. Other prospective options include controlled development of the offered gate, another supported component, a justified local limitation or retirement. Retention needs current risk and support evidence. A replacement needs whole-product evidence and applicable authority. Retirement needs continuity of work plus record and credential handling. A fact that supports one option does not necessarily support the others.

What observation would close each assigned question?

For unresolved receipts, the memo identifies site/manufacture reconciliation owners, attempt links, backlog age and clinical urgency. For Site B it assigns acquisition, case-mix, workflow and consequence questions; for Site A it assigns the reason unresolved quality increased. For the proposed gate it identifies supplier evidence, the added-input definition, reopened evaluations and manufacturer progression authority. For support it identifies affected installations, both dates, actual interim terms, recipients, migration criteria and qualified fallback. Sending a notice does not demonstrate that the transition can work.

A bounded advisory conclusion**Note**

The later window opens the prespecified pooled review. Changed site composition explains the supplied pooled rejection difference, while Site B's persistent burden, Site A's unresolved increase, backlog reconciliation and clinical significance remain open. The replacement gate is an uninstalled proposal requiring whole-product evidence and the applicable jurisdictional assessment. The original gate's support dates start transition planning; they do not establish replacement readiness.

The conclusion credits what the investigation learned and states which decisions that finding cannot yet support. A later revision can then show exactly what evidence changed the advice.

In the pathology queue episode, this advisory method asks whether the recorded post-import benefit answers a full scanner-to-review question before recommending a queue update. The reviewer tests the actual case/time denominator. In the sequencing review episode, the consultant assigns specimen-link reconciliation and support questions before recommending a refit; the reviewer asks whether report-release outcomes were observed. Those local questions preserve the separate episodes and use their logs to direct investigation without treating operational records as clinical truth.

Across the memo, the reviewer tests the chain from signal and weighted explanation through unresolved exceptions, exact change, plan records and applicable evidence. A skipped link is a reason to narrow the conclusion or obtain a record.

The consultant sequences manufacturer, supplier and site contributions across investigation, candidate evaluation and support transition. Each handoff names a record and the owner who can accept the next decision.

Can two ledger rows keep the decisions separate?

What would a ledger row for the proposed gate say about its current state, proposed state, reopened evidence, owner and condition for progression? How would a row for the support transition differ in each of those fields?

Compare the gate row with the support row

The proposed-gate row starts at **uninstalled proposal** and points toward a controlled whole-package candidate. Progression requires evidence for newly admitted inputs, integration and affected use, together with the applicable change authority. The manufacturer product owner coordinates supplier, clinical, software, quality and regulatory contributions.

The support row starts at the current original gate and its stated end-of-guaranteed-support/end-of-support dates. It points toward a supported transition or justified retirement. Progression needs actual interim terms, affected-site inventory, fallback/migration checks and owner acceptance. If both rows simply say “retrain and monitor,” they have lost the difference between a component proposal and a support-life decision.

Separate branch: may the offered field records support retraining?

After the gate analysis, consider a separate proposal. A team names a future retrained LumaCell classifier M2 and offers field records collected under permission limited to service monitoring. No training right, independently constructed reference, protected evaluation partition or authorized plan record is supplied. Nor is there a finding that the current classifier's weights caused the later-window signal.

What can the team conclude about starting fitting with those records or claiming that the future classifier is covered by a plan? Which missing fact would change each answer?

Resolve eligibility before judging the new model

The supplied monitoring permission does not grant the different use of fitting a classifier. The manufacturer first needs legal and contractual permission for that use, traceable source/patient/specimen lineage, a suitable reference-label method and controlled development/evaluation separation. Those facts would establish whether the proposed work is eligible to start; they would not establish how the new classifier performs.

If those gates were later cleared, the future classifier would still be a controlled candidate needing risk and product evidence. Nothing supplied establishes that

retraining addresses the cause or fits an authorized plan. An absent authorization record means coverage cannot be concluded; it does not prove that the record is nonexistent.

The reviewer keeps this retraining branch separate from the replacement-gate proposal and checks both permission and scientific fitness before crediting a model comparison.

The consultant assigns data stewardship, clinical reference, machine-learning and regulatory owners to those distinct gates. A non-retraining option remains open while cause is unresolved.

Retrieve PathSight: what does an included plan actually establish?

The same frozen PathSight Cloud snapshot used earlier supplies the next question. It identifies M3.2 weights, a packet that includes a PCCP for threshold and scanner expansion, normalization changed after validation, and a default display that collapses low-confidence cases. Two positive low-score slides were delayed in a **high-volume simulation**. The snapshot supplies no denominator, exact delay time or clinical consequence. This is the earlier snapshot, not a new field episode.

Do unchanged weights and an included plan settle whether the normalization and display changes are covered? What can the simulation observation establish about possible clinical consequences, and what record is missing?

Follow the changed path and the missing authority

The needed records are the jurisdiction and exact authorized plan, its scope and protocol, and the whole configuration that was evaluated. A plan title mentioning threshold or scanner expansion does not show that changed normalization or low-confidence presentation is covered. The unchanged weights do not keep those other product links fixed.

The two delayed positives are a bounded simulation observation. They justify investigation of visibility and workflow, but without the denominator, exact time or clinical consequence they supply neither a field incident rate nor proof of patient harm. LumaCell's receipt counts do not fill those missing PathSight facts.

The reviewer traces preprocessing, low-confidence display and professional action against the evaluated configuration and plan. The simulation's missing denominator remains missing.

The consultant asks PathSight product, human-use and regulatory owners for a plan-scope assessment and a targeted low-confidence workflow investigation. Each has a distinct question and evidence requirement.

Carry the method into the next decision

Optional depth lies in Book 1C's signal-to-question and renewed-evidence reasoning; Book 1A Chapters 7, 8 and 15 on monitoring and integrated review; Book 1B Chapters 7–11 on incidents, remediation, resilience and support; and Book 2 Chapters 14–16 on European lifecycle and evidence reuse. The local method here supplies the receipt, investigation, change and retirement reasoning needed for this task. A relevant security event still needs timely assessment and an accountable response through its own route.

Chapter 16 replaces the familiar case with an unfamiliar product. The transferable skill is reconstructing what the records support and who can decide the next step.

Reasoning check

When does a monitoring trigger justify investigation, and what additional evidence supports a cause-directed action? Which changed input, display or dependency can reopen a product question despite unchanged weights? What makes offered retraining data eligible? How do authority, scope and protocol each affect a plan-coverage conclusion? What current evidence could justify continued operation, rollback or retirement?

For each conclusion, which record makes it inspectable and which decision-maker is accountable for accepting it?

Source Trail

[S1] ISO 14971:2019, 10.1–10.4, printed pp. 15–16/PDF pp. 23–24 in the licensed edition; collection, review, and action on production/post-production information.

[S2] AAMI TIR97:2019 (R2023), Sections 2.5–2.6, 6 opening, 6.2, and 7.1; printed pp. 2, 7–8, 12, and 19–20/PDF pp. 12, 17–18, 22, and 29–30. Consensus security guidance; no universal notice interval is asserted.

[S3] ISO 13485:2016, 7.3.9 and 8.5.2, printed pp. 16 and 25/PDF pp. 24 and 33; design change and corrective action.

[S4] ISO/TS 24971-2:2026, Scope, 4.4, and 9–10.4; printed pp. 1, 5, and 9–11/PDF pp. 7, 11, and 15–17. Preserve the predictive-ML scope and LLM/generative-AI exclusion.

[S5] IMDRF N90 FINAL:2026, Sections 2.2, 4, and 5–5.1.3, pp. 6 and 8–13; nonbinding international principles, published August 6, 2026, official status checked September 25, 2026.

[S6] 21 U.S.C. 360e-4, subsections (a)–(b), statutory approved/cleared PCCP mechanisms. The cited compilation is identified; current application requires the applicable legal record.

[S7] FDA PCCP guidance, August 2025, Sections V.A–V.F, VII.B, and VIII; nonbinding recommendations. The official final-guidance page was checked September 25, 2026; the downloaded guidance was issued August 18, 2025.

[S8] 21 CFR 807.81, (a)(3) and (b)(1)(ii), and 21 CFR 814.39, (a)–(b); distinct applicable change routes, checked September 25, 2026. The eCFR is current but unofficial; actual applicability still needs the device record.

[S9] Current 21 CFR Part 820, 820.1, 820.7, 820.10, and 820.35; FDA QMSR implementation page for the February 2, 2026 effective-date explanation, checked September 25, 2026. Codified text and explanatory page remain distinct.

[S10] Health Canada MLMD guidance, April 1, 2026, scope, policy statements, PCCP change protocol/impact assessment, and postmarket monitoring; official page checked September 25, 2026. Guidance-only comparison, not a Canadian legal determination.

The packet, MON1 arithmetic, site standardization, Q1/M2 proposals, supplier dates, and advisory conclusions are author-designed fictional teaching material. No reported rate is a clinical truth estimate.

CHAPTER 16

Chapter 16: Sealed Development Capstone and Advisory Handoff

What development decision can you justify when the product is unfamiliar? This capstone asks you to use the reasoning developed in the book without the familiar LumaCell case.

The complete packet below is the evidence for your initial decision and advisory handoff. Both are due before the separate instructor guide is opened. Record headings identify the exhibits without diagnosing them, and the learner packet contains no worked answer. Their order is administrative; it does not rank their importance.

Your own notes from earlier chapters may be used. The first attempt excludes another person's analysis, and a team discussion follows each participant's individual commitment. The assessment concerns the reasoning you can generate before a supplied answer becomes available.

The Decision Meeting

A manufacturer has scheduled a decision meeting for CurveRead CRP revision C1. The product team wants to make C1 available to its two existing laboratory sites at the start of the next operating week.

The clinical lead asks for a concise decision that can be implemented. The software lead wants to know which work can proceed if the overall proposal is not ready. The laboratory representatives need to know what, if anything, changes for their current operation. The quality lead asks that observations, assumptions, and requested evidence remain distinguishable.

You are reviewing the supplied packet. You are not being asked to operate the laboratories, approve a manufacturer's release, issue a regulatory determination, or invent missing study results.

CurveRead CRP and every record below are fictional teaching constructions. The compatibility settings, numerical bands, counts, criteria, roles, and decisions are not recommendations for a real CRP assay or medical device. The exercise concerns artificial intelligence (AI) and machine learning (ML) product development and evidence; it does not teach reagent, analyzer, or clinical-treatment design.

Record A: Product Description

CurveRead CRP is a fixed-release ML software component for a specified laboratory assay/analyzer arrangement. An optical reaction curve is a time-ordered record of the assay's light response. The software maps that curve to a numerical C-reactive protein (CRP) concentration suggestion in milligrams per liter (mg/L).

The supported profile in the case is adult serum, the specified assay family AF1, analyzer configuration AX1, and a defined acquisition/time profile TP1. AF1, AX1, and TP1 are identifiers for this case's permitted assay, analyzer, and sampling conditions, not general clinical standards. The software does not accept other specimen types or analyzer profiles. The underlying assay chemistry, reagent formulation, and analyzer hardware are unchanged in the proposal.

The intended user is a qualified laboratory professional. The professional reviews the numerical suggestion and the product's applicable flags under the laboratory's procedure before a laboratory result is released. The software does not independently diagnose a condition, recommend treatment, or directly issue a patient-facing report.

The stated numerical range is 2–100 mg/L. For the study exhibits, the lower range is 2–10 mg/L inclusive and the upper range is greater than 10–100 mg/L. These ranges are study strata in this fictional case, not clinical decision thresholds.

The existing controlled package is C0. It contains model K1, transformation T1, runtime R1, and user interface V1. Here a *runtime* is the execution software used to run the calculation. The case stipulates a recorded manufacturer release decision and site-use decisions for C0 within this bounded role. Their complete supporting files are not reproduced. No jurisdiction-specific marketing authorization or predetermined change control plan (PCCP) is supplied.

Record B: Proposed Package

The proposed package C1 contains K1, transformation T2, runtime R2, and interface V1. The K1 parameter content is unchanged.

T1 requires the specified complete time-point series under TP1. If an interval is missing, it returns an unavailable-estimate state for that curve.

T2 permits a single missing interior interval within a defined part of the acquisition sequence. It estimates the missing time-point value by a specified interpolation operation before applying K1. *Interpolation* here means estimating a value inside the time sequence, not measuring it anew. Other unsupported patterns remain unavailable. The permitted missing-interval position and interpolation method are recorded in the engineering change description.

Runtime R2 is the supplier's maintained replacement for R1. A comparison report exists for K1 on complete development curves. It reports matching numerical suggestions within the declared arithmetic tolerance. It does not include the T2 missing-interval path.

The proposed release note says: “Model unchanged; improved availability with no change to the intended use.” The accompanying draft summary also says: “C1 maintains performance throughout the supported range.”

Neither C1 nor an intermediate T2 package has been used for patient-result production. All C1 results in this packet come from segregated development or evaluation activities.

Record C: Evaluation Protocol And Summary

The protocol identifies 220 distinct adult patients, one prespecified curve per patient, from two sites. All curves are within the stated specimen, assay, analyzer, and concentration scope. The evaluation includes both complete curves and curves with the single missing interval addressed by T2.

A separately specified reference measurement procedure provides a numerical reference for every record. The reference operators do not have access to CurveRead suggestions. There are no missing reference results in this exhibit. The packet does not supply the full reference-method performance report.

For the exercise only, a *returned* suggestion is a numerical output rather than an unavailable-estimate state. It is within the agreement band when the unsigned distance between it and the reference—its absolute difference—is no more than 2 mg/L. Do not interpret that band as an appropriate real clinical acceptance limit.

The prospectively recorded C1 milestone has three simultaneous conditions:

- At least 90% of returned numerical suggestions are within the agreement band.
- At least 95% of all eligible received curves yield a numerical suggestion.
- In each prespecified concentration-range stratum, at least 90% of eligible received curves yield a numerical suggestion.

The protocol calls for descriptive counts in the full cohort and both strata. It also requires identification and investigation of material errors and unavailable results. No condition states that a favorable pooled value overrides another condition.

The supplied summary is:

Reference range	Eligible received curves	Numerical suggestions returned	Returned suggestions within band
2–10 mg/L	40	30	29

Reference range	Eligible received curves	Numerical suggestions returned	Returned suggestions within band
Greater than 10–100 mg/L	180	180	165
Total	220	210	194

All ten unavailable estimates are in the lower-range row. They are retained in the receipt count. No patient harm, clinical management decision, or comparative time-saving result is supplied.

Record D: Development History

The source records have patient-level identifiers and a recorded permission for the specified segregated product-development and evaluation uses. This permission is a stipulated case fact, not a general legal rule. No broader sharing or future use is implied.

Before the 220-record evaluation was formally executed, the team compared three candidate transformation rules. It used a 52-record subset to choose the missing-interval positions and interpolation rule for T2. The final T2 rule was then fixed before the 220-record run.

The dataset lineage record shows that those same 52 patients and curves are included in the 220-record evaluation. Their numerical references were visible during the earlier transformation comparison.

The final evaluation report describes all 220 records as “independent test cases because K1 weights were not retrained.” The report contains no separately protected replacement cohort or analysis excluding the 52 records. No result for a different cohort may be inferred from the supplied table.

The remaining 168 records were not used in the described transformation selection. The packet does not establish their concentration-range counts or the full history of any other development access.

Record E: Implementation And Use Records

The C1 engineering record contains passing unit tests: small, constructed-input checks of the specified interpolation calculation in isolation. It also contains passing K1 numerical comparisons for complete curves under R2.

An *integrated* test follows connected product steps rather than only one calculation. The supplied list covers ordinary output, unsupported assay profile, invalid identity, and complete-curve interruption/recovery. The list does not contain a test that follows a missing-interval curve through T2, K1, the flag, professional review, and export.

The on-screen V1 display includes an estimate-status field. The export template used by Site North contains that field. Site South's current export template contains patient identity, numerical value, unit, and review status, but not estimate status.

The proposed C1 use procedure asks the laboratory professional to distinguish an estimate generated from an interpolated curve from one generated from a complete curve, and to follow the stated review path. The packet contains no evaluation of that procedure with intended users.

An earlier C0 user study evaluated V1 with complete-curve results and unavailable-estimate states. It did not include numerical suggestions generated after interpolation.

The two laboratory representatives report that their current review includes the exported record. The packet does not establish whether every intended reviewer also sees the on-screen status before making the relevant decision.

Record F: Supplier And Site Records

The R1 supplier has issued a notice identifying a security vulnerability relevant to the product's use of R1. The manufacturer has opened an assessment. The notice states that R2 addresses the identified issue; the packet does not include the manufacturer's completed verification of that claim.

No exploitation, unauthorized access, or patient consequence has been confirmed. The manufacturer has not yet completed assessment of exposure or compensating controls at either site. This absence of a confirmed event is a supplied fact, not a statement that continued operation is safe.

Site North's configuration record resolves to C0 and identifies its local owners. Site South's record names C0 but does not resolve the installed R1 build. Both sites have qualified alternatives for conducting their laboratory work under their existing procedures; use of those alternatives requires the responsible laboratory professional's decision.

The supplier states that R1 will reach end of guaranteed support (EOGS) in 90 days. It has not announced an end of support (EOS) date in this packet. The product team proposes retaining a C0 installer as the rollback package for C1.

No final decision on current-site containment, R2-only packaging, full C1 release, or retirement has been recorded.

Record G: Change And Regulatory Records

The manufacturer's internal change form calls C1 a “minor model-preserving update.” It contains an engineering signature but no completed clinical, human-factors, security, quality/risk, or regulatory disposition.

A planning document includes the phrase “future updates may include signal preprocessing improvements.” It is a manufacturer-authored draft. The packet contains no national authority decision granting or accepting that plan.

The proposed rollout would install C1 at both sites during a single maintenance window. Its monitoring draft counts completed numerical results and support requests. It does not define an eligible receipt cohort, unavailable-result denominator, reference linkage, follow-up delay, or concentration-range analysis.

The rollback section identifies the stored C0 installer. It does not address the R1 notice, the actual installed build at Site South, configuration compatibility, or criteria for selecting rollback rather than the qualified alternative workflow.

Record H: Requested Outputs

The manufacturer asks for a decision package that distinguishes what can be decided now from what requires additional evidence. It wants the work divided by accountable function, not a general instruction to “validate everything.”

The consultancy asks for an advisory handoff based on the same packet. The consultant has access to these exhibits only and has not executed tests or inspected the complete manufacturer files.

The next meeting is intended to choose actions, owners, and conditions for progression. It is not evidence that the product has already met those conditions.

Your Initial Decision Package

Your initial package answers the following ten questions in separate sections, before feedback is available.

1. **Problem frame.** Which decisions are being requested, and does the evidence support the same answer for all of them?
2. **Claim and configuration.** What are the relevant current and proposed product states, and which statements are being supported or challenged?
3. **Material facts and unknowns.** What is observed in the supplied record, what information is missing, and which explanations are your own hypotheses?
4. **Risk and requirements.** Which consequential product or use questions arise, and which requirements or controls do they affect?

5. **Data and model evidence.** What do the supplied quantitative results mean, and where do they apply? The response includes the calculations supporting its interpretation.
6. **Verification and validation.** Which evidence can be retained, which requires a bridge, and which has not been supplied?
7. **Workflow and security.** What do the actual use and operational conditions require you to address? The response remains within the events supplied by the exhibits.
8. **Release or change decision.** What decision is supported now, and what separable work, if any, may proceed?
9. **Monitoring or recovery.** What follow-up and recovery approach is operationally interpretable at the level the packet permits?
10. **Owners and next actions.** Which functions are accountable for the next actions, and what observable conditions would permit progression?

The package can be concise. Its test is whether a manufacturer can act on the decision and reconstruct the reasoning behind it. Length cannot supply support for a conclusion.

Your Advisory Handoff

The separate advisory handoff is approximately 500–800 words. It answers:

- What decision is being considered?
- What evidence can be reconstructed from the packet?
- Which gaps are material to the decision?
- What options are available, and what are their consequences?
- What observable conditions would close the identified questions?
- Which decisions remain with the manufacturer and laboratories?
- Which specialist or jurisdictional routes cannot be settled from this packet?

The handoff and technical package express the same decision and warranted confidence. A management audience does not change what the evidence supports.

The handoff distinguishes reading a reported test from personally verifying it. It does not certify clinical validity, security, regulatory sufficiency or release. Its conclusion must reflect the limits of access to the evidence; a generic disclaimer cannot qualify an otherwise unsupported approval.

Record Your Commitment

The commitment recorded before access to the instructor guide answers five questions:

- What is your release recommendation, and what is its scope?

- What are your three highest-priority actions, in order?
- Which one conclusion have you deliberately refused to make?
- Which single additional record is most likely to change your immediate decision?
- How confident are you in the decision, and why?

The initial version remains part of the record. A revision after discussion retains both versions and explains what changed. That history makes improvement in the reasoning visible.

In a team exercise, comparison of the initial priority orders precedes debate about the final decision. Different functions may identify different urgent questions. Evidence and consequences provide the basis for reconciling them; equal numbers of action items for each discipline are not the objective.

Assessment Standard

Your response is assessed for accurate use of the supplied facts, correct calculations, justified conclusions, explicit uncertainty, and an actionable handoff within the stated authority. Different wording and defensible alternatives are permitted. The case-specific rubric and worked assessment are withheld until after commitment.

Feedback Boundary

The initial assessment ends here. The written decision and advisory response are retained before any feedback is received.

The instructor material is maintained separately and is not linked from this learner packet. A facilitator should provide it only after the individual decision and advisory handoff have been recorded. In independent study, use a separately released feedback packet after completing the same commitment. The feedback is not an additional source of facts for the initial decision; its changed-case exercise introduces a distinct later state.

This separation is a publication requirement as well as a teaching instruction. A learner-facing edition must exclude the instructor material and case-specific rubric from its content, navigation, search results, captions, and alternative text until the feedback stage. The editable authoring repository is not itself a controlled assessment-delivery system.

Completing The Book

After feedback, which earlier artifact would benefit from a correction based on the specific weakness this capstone revealed? A weakness in configuration reasoning may lead to a revised trace or evidence row; a weakness in interpretation, to a revised result and conclusion; a weakness in operational ownership, to a handoff someone can execute. The choice is yours.

The correction records what changed and why. It can be a small, precise improvement to the chosen artifact without rewriting every earlier answer.

The capability developed across this book is the ability to keep the diagnostic purpose, the actual product, the evidence, and the accountable decision connected as conditions change. The capstone is a checkpoint in that capability, not proof that every future device or jurisdiction can be handled without specialist help.

WORKING TERMINOLOGY

Working Terminology

These short explanations describe how terms are used in this book. They are not verbatim standards definitions and do not replace jurisdiction-specific definitions. Follow the linked chapter for the method and its source trail. Version and record labels identify particular objects in a fictional exhibit; their meaning belongs with that exhibit. The chapters use plain names to keep those objects recognizable.

Acceptance criterion. A specified condition used to judge an identified result or activity. It needs a question, unit, method, and role in the overall decision; it is not interchangeable with a monitoring trigger. See Chapter 9.

Adjudication. A defined process for resolving reference disagreements or uncertain cases. It should preserve the original observations and unresolved states rather than manufacture certainty. See Chapter 6.

All-received measure. A measure whose denominator retains every eligible input in the defined receipt cohort, including inputs without an ordinary result. Its numerator must still be specified. See Chapter 9.

Analytical or technical performance. The evidence question concerning whether the product generates the intended output from its inputs. Apply the precise terminology required by the device and jurisdiction; this is not automatically a patient-outcome claim. See Chapter 12.

Baseline. An identified reference state. A model-development comparator, evaluated product configuration, and released package are different kinds of baseline. See Chapter 8.

Bias. A term with several technical and broader meanings. An observed subgroup difference requires interpretation; it is not automatically proof of unfairness or a particular causal mechanism. See Chapter 10.

Bridge. A justified argument, supported by appropriate evidence, connecting an evaluated condition to a changed condition. A statement that a change is small is not a bridge. See Chapter 12.

Calibration, predictive. Correspondence between a model's stated probabilities and observed frequencies under a defined assessment. This use is distinct from calibration of a measurement system or analyzer. See Chapter 9.

Candidate. An identified development object being considered for further evaluation or release. Being the best candidate in a comparison does not establish product readiness. See Chapter 8.

Clinical performance. Performance relevant to the intended clinical condition, population, and use. Use the applicable regulatory definition; selected-input agreement must not silently become a broader clinical endpoint. See Chapter 12.

Configuration. The identifiable combination of components and settings that realizes the supplied function. It can include preprocessing, model, gate, runtime, interface, dependencies, and relevant information—not just weights. See Chapter 3.

Correction and corrective action. A correction addresses an identified problem; corrective action addresses its cause to prevent recurrence. Closing an individual work item does not necessarily demonstrate effective cause-directed action. See Chapter 15.

Data eligibility. Whether specified records may be used for a particular proposed purpose under the applicable permissions and controls. Eligibility does not establish representativeness or adequacy. See Chapter 5.

Data lineage. The reconstructable history linking records, transformations, versions, and uses. A dataset filename alone does not establish lineage. See Chapter 5.

Deployment equivalence. A justified relationship between the evaluated product arrangement and the installed, supported arrangement. It includes relevant operational conditions, not merely successful installation. See Chapter 14.

Distribution shift. A change in a specified input or output distribution. Such a change does not by itself establish unintended clinical performance deterioration. See Chapter 15.

Estimand. The precise quantity an evaluation aims to establish, including the population, unit, outcome, and treatment of relevant events or missingness. The same numerator can mean different things with a different estimand. See Chapter 9 for the measurement plan and Chapter 12 for claim integration.

Evaluation independence. Separation of evaluation evidence from the development decisions it is intended to assess. Development includes choices about preprocessing, selection, and operating points, not only fitting model weights. See Chapter 7.

Fallback. An identified alternative clinical or operational pathway when the normal function cannot be used as intended. It must be available under the triggering conditions. See Chapter 13.

Human–AI team. The defined arrangement of users, product, information, tasks, and workflow that performs the intended work. A person being present does not establish an effective control. See Chapter 13.

Inference. Applying an identified trained model and its relevant input/output processing to an input. Repeating inference with one artifact differs from repeating training. See Chapter 8.

Leakage. Information exposure that undermines the intended separation or interpretation of development and evaluation. Its effect depends on what was exposed and how it influenced decisions. See Chapter 7.

Non-result. A state in which the expected ordinary output is not available. Rejected, unresolved, and unavailable states may require different handling; none is automatically a negative clinical result. See Chapter 9.

Operating point. A selected decision rule or threshold at which model output is used for the intended action. Changing it can alter the balance of errors, non-results, and workload. See Chapter 9.

Out-of-distribution condition. An input condition materially unlike the distribution used for a specified assessment. It may still satisfy the stated input contract; an explicit contract violation is a different question. Detection and response need their own evidence. See Chapter 10.

PCCP. A predetermined change control plan describing bounded changes and their evaluation and implementation method. Jurisdiction, authority, scope, protocol, and version determine what can be concluded from it. See Chapter 15.

Quality disposition, LumaCell package. Any known required-crop rejection determines rejected/blocked; otherwise incomplete required membership or status determines unresolved; acceptance requires a complete, nonempty identified required set with all crops accepted. Missing responses remain separately traceable even in rejected packages. Quality acceptance establishes neither available processing, completed review nor clinical correctness. See Chapter 11 and the mixed-state receipt example in Chapter 14.

Reference standard or reference method. The specified basis against which the relevant output is assessed. Its quality, independence, uncertainty, and fit to the question must be established; a label is not truth merely because it is stored in a dataset. See Chapter 6.

Regression testing. Testing selected to establish that a change has not adversely affected relevant previously implemented behavior. Test count alone does not justify the selection. See Chapter 11.

Reproducibility. A property that must be defined for the object and conditions being repeated. Matching an aggregate score, reconstructing a training run, and preserving individual product decisions are different questions. See Chapter 8.

Retraining. A new controlled development activity that produces a candidate model from specified data and procedures. It does not automatically authorize deployment or preserve evaluation independence. See Chapter 15.

Rollback. Returning to an earlier controlled configuration. Its current risk, compatibility, and recovery conditions must be assessed; availability of an old installer is insufficient. See Chapter 15.

Source authority. The kind of support a source can provide, such as law, a normative standard provision, nonbinding guidance, research evidence, or author synthesis. These categories are not interchangeable. See Source and Citation Navigation and the chapter source trails.

Traceability. Reconstructable connections among purpose, requirements, risks, configuration, evidence, and decisions. A link is useful only when it leads to the relevant object and result. See Chapter 11.

Validation and verification. Terms that require their object and context. This book distinguishes implementation evidence from evidence that the complete product and use satisfy the intended purpose; neither word alone describes a sufficient evaluation. See Chapter 11 for implementation, Chapter 12 for the evidence claim, and Chapter 13 for human use.

Cross-book terminology boundaries

Calibration. Metrological calibration concerns a measurement relationship under stated conditions (Book 1C, Chapter 7). Predictive-score calibration concerns scores and observed outcome frequencies under a stated evaluation design (Book 1D, Chapter 9).

Precision and reproducibility. Measurement precision describes dispersion; classification precision concerns positive predictions. State whether reproducibility means changed measurement conditions, repeat execution or repeat training. Preserve the relevant configuration and conditions.

Traceability. Name the relationship: requirements to evidence, data provenance and lineage, or metrological traceability. In Book 1C, CRM means certified reference material; it has a different expansion in the consulting-business context.

Configuration and PCCP status. A candidate, an accepted baseline, manufacturer release and market authorization are separate states. A predetermined change control plan can be proposed, submitted or authorized; identify the actual state and exact change coverage.

Initially open and reopened evidence. An unanswered question is not evidence of a change. Record a documented change before describing evidence as reopened by it.

ABBREVIATIONS

Abbreviations

Use this as a quick expansion index, then return to the chapter that explains the term in its task. These entries are not new legal definitions or product specifications. Fictional configuration, record and study labels are identifiers, not general abbreviations; their meaning belongs to the relevant exhibit. Numbered S labels are chapter-local citations, explained in Source and Citation Navigation.

Short form	Expansion or reading
AAMI	Association for the Advancement of Medical Instrumentation
AI	artificial intelligence
AMD1	Amendment 1 in a cited standard's edition string
API	application programming interface
CAPA	corrective and preventive action
CFR	Code of Federal Regulations (U.S.)
CI	confidence interval
CPU	central processing unit
DP	digital pathology
EDTA	ethylenediaminetetraacetic acid
EOGS	end of guaranteed support
EOS	end of support
EU	European Union
FDA	U.S. Food and Drug Administration
GDPR	General Data Protection Regulation (EU)
GMLP	good machine learning practice
GPU	graphics processing unit

Short form	Expansion or reading
HDO	healthcare delivery organization
HF	human factors
ICML	International Conference on Machine Learning
IEC	International Electrotechnical Commission
IMDRF	International Medical Device Regulators Forum
ISO	International Organization for Standardization
IVD	in vitro diagnostic
IVDR	In Vitro Diagnostic Medical Devices Regulation (EU)
LLM	large language model
MDCG	Medical Device Coordination Group (EU)
MDR	Medical Devices Regulation (EU)
MHRA	Medicines and Healthcare products Regulatory Agency (UK)
ML	machine learning
MLMD	machine-learning-enabled medical device (the cited Canadian guidance's term)
NGS	next-generation sequencing
NIST	National Institute of Standards and Technology (U.S.)
OOD	out of distribution
PCCP	predetermined change control plan
PDF	Portable Document Format
PMA	premarket approval (U.S. pathway)
PNG	Portable Network Graphics
QMS	quality management system
QMSR	Quality Management System Regulation (U.S.)
RGB	red–green–blue color channels

Short form	Expansion or reading
SOP	standard operating procedure
TIR	technical information report
TR	technical report
TS	technical specification
UI	user interface
WBC	white blood cell
WSI	whole-slide image

For example, the GMLP row expands a short form encountered in a source citation; the linked chapter still explains what those principles can support. An expansion gives a name, not authority for a product decision.

For ISO/TS, ISO/TR and AAMI TIR citations, read the organization and document-type entries together. For the distinct uses of *calibration*, *precision*, *traceability* and *configuration*, use Working Terminology and its cross-book boundary notes.

SOURCE NAVIGATION

Source and Citation Navigation

The bracketed source labels in Chapters 1–15, such as [S1], lead to that **chapter's own Source Trail**. Labels restart in each chapter: Chapter 2's [S1] is not Chapter 1's [S1]. Use the linked trail for the exact document, cited passage, check date where recorded, and the limit of the claim it supports. A standard, guidance document, law, research paper and fictional teaching construction have different authority. None of the links below turns a case exercise into a real product study or a jurisdictional decision.

Source trails by reading order

Define the product: Chapter 1 · Chapter 2 · Chapter 3 · Chapter 4

Control data and model realization: Chapter 5 · Chapter 6 · Chapter 7 · Chapter 8

Evaluate the complete product: Chapter 9 · Chapter 10 · Chapter 11 · Chapter 12 · Chapter 13

Release and change: Chapter 14 · Chapter 15

Chapter 16 is a separately authored fictional transfer exercise. Its numbers, profiles, permissions and records are supplied case facts, not claims established by an outside source. The learner packet presents its decision and advisory questions before the separately distributed worked assessment. That feedback is deliberately not linked from this learner navigation.

For terms that change meaning across contexts, see Working Terminology; for short-form expansions, see Abbreviations. Return to Contents for the learner reading order.